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## COMPOSITIONS AND METHODS FOR SPLICING MODULATION OF UNC13A


### Abstract

Antisense oligonucleotides for splicing modulation of UNC13A (e.g., inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA), compositions including the antisense oligonucleotides, and methods of use are described. Also disclosed are pharmaceutical compositions including one or more antisense oligonucleotides and methods of treating an UNC13A-associated disease or a disease associated with TDP-43 dysfunction in a subject by administering the antisense oligonucleotides to the subject.



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## Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application Ser. No. 63/561,593, filed Mar. 5, 2024, entitled "COMPOSITIONS AND METHODS FOR SPLICING MODULATION OF UNC13A", the entire contents of which are incorporated herein by reference.

FIELD
[0002] The invention relates to compositions (e.g., antisense oligonucleotides) for splicing modulation and/or modulation of gene expression of UNC13A.
REFERENCE TO AN ELECTRONIC SEQUENCE LISTING
[0003] The contents of the electronic sequence listing (T083370053US01-SEQ-CBD.xml; Size: 3,049,839 bytes; and Date of Creation: Mar. 4, 2025) is herein incorporated by reference in its entirety.
BACKGROUND
[0004] Amyotrophic lateral sclerosis (ALS), also known as Lou Gehrig's disease, and frontotemporal dementia (FTD) are rare neurodegenerative diseases known as motor neuron diseases (MND). ALS is a terminal MND characterized by upper and lower motor neuron loss, which results in rapid deterioration of muscle function, and eventually death due to respiratory failure. FTD is characterized by progressive neuronal loss predominantly involving the frontal and temporal lobes, and generally presents as a behavioral or language disorder. Some patients with ALS also develop frontotemporal dementia (FTD). There is currently no known cure of ALS or FTD.
[0005] UNC13A is an essential protein for synaptic transmission and transmission at the neuromuscular junction (NMJ). Single nucleotide polymorphisms (SNPs), e.g., intronic SNPs, in UNC13A, have been shown to increase the risk of ALS and FTD. The SNPs have been shown to increase cryptic splicing and alter direct binding of RNA-binding protein TDP-43, which reduces the ability of TDP-43 to repress cryptic exon inclusion in UNC13A during RNA splicing. TDP-43 dysfunction has been shown to result in inclusion of a cryptic exon in an UNC13A mature mRNA. Cryptic exon inclusion in an UNC13A mature mRNA results in reduced UNC13A protein expression due to aberrant RNA degradation or the presence of premature stop codons.
[0006] Additional neurodegenerative diseases involving TDP-43 pathology include Alzheimer's disease and Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE).
SUMMARY
[0007] ALS and FTD patients could benefit from inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA. There is a need for compositions and methods for treating ALS and FTD, e.g., by inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell in a subject with ALS or FTD.
[0008] The present disclosure, in some aspects, provides antisense oligonucleotides for splicing modulation and/or modulation of gene expression of UNC13A. Dual targeting antisense oligonucleotides comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region allow for effective splicing modulation of an UNC13A cryptic exon while reducing off-target risks. In some embodiments, the UNC13A RNA transcript may be within a cell, e.g., a cell within a subject, such as a human subject. Compositions

comprising such antisense oligonucleotides and methods of using such (e.g., for treating an UNC13A-associated disease, such as ALS or FTD) are also provided.

[0009] Some aspects of the present disclosure provide antisense oligonucleotides comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region, wherein the first target region and/or the second target region each comprises an UNC13A sequence, wherein the antisense oligonucleotide modulates splicing of an UNC13A cryptic exon. In some embodiments, the first target region or the second target region comprises a nucleotide sequence as set forth in SEQ ID NO: 313 or SEQ ID NO: 315. In some embodiments, the UNC13A cryptic exon is in intron 20 of an UNC13A pre-mRNA.

[0010] In some embodiments, the first target region and the second target region are in intron 20 of an UNC13A pre-mRNA. In some embodiments, the first target region is adjacent to the second target region in an UNC13A pre-mRNA. In some embodiments, the first target region is downstream of the second target region in an UNC13A pre-mRNA. In some embodiments, the first target region is more than 1 nucleoside downstream of the second target region. In some embodiments, the first target region is downstream of the cryptic exon and the second target region is in the cryptic exon.

[0011] In some embodiments, the first target region comprises a nucleotide sequence as set forth in SEQ ID NO: 315 and/or the second target region comprises a nucleotide sequence as set forth in SEQ ID NO: 313. In some embodiments, the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 313.

[0012] In some embodiments, the first antisense sequence and/or the second antisense sequence is 6-15 nucleosides in length. In some embodiments, the first antisense sequence and the second antisense sequence is 6-15 nucleosides in length. In some embodiments, the first antisense sequence and/or the second antisense sequence is 8-15 nucleosides in length. In some embodiments, the first antisense sequence comprises at least 6 contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises at least 6 contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence comprises at least 7 contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises at least 7 contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA,

ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC, and/or wherein the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA.

[0013] In some embodiments, the first antisense sequence and the second antisense sequence are immediately adjacent to each other. In some embodiments, one or more nucleobases are between the first antisense sequence and the second antisense sequence. In some embodiments, one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 3′ end of the first antisense sequence is complementary to one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 3′ end of the second target region, and/or one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 5′ end of the second antisense sequence is complementary to one or more (e.g., 1, 2, 3, 4, 5, or more) nucleobases at the 5′ end of the first target region. In some embodiments, a spacer is between the first antisense sequence and the second antisense sequence. In some embodiments, the spacer is a C3 spacer (e.g., a C3 spacer shown in Table 16). In some embodiments, the spacer is a C6 spacer (e.g., a C6 spacer shown in Table 16).

[0014] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0015] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0016] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 497 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 249, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0017] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 506 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 259, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0018] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0019] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 198 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0020] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first

antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0021] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).
[0022] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 534, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0023] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 534, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).
[0024] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 535, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0025] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 535, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).
[0026] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0027] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).
[0028] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0029] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0030] In some embodiments, an antisense oligonucleotide provided herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first

antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0031] In some embodiments, an antisense oligonucleotide provided herein comprises a nucleobase sequence of any one of SEQ ID NOs: 1-190 and 338-494. In some embodiments, an antisense oligonucleotide provided herein comprises a nucleobase sequence of any one of SEQ ID NOs: 1-182, 184-190, 338-434, 442-494.

[0032] In some embodiments, an antisense oligonucleotide provided herein comprises one or more modified nucleosides. In some embodiments, the one or more modified nucleosides comprise a non-bicyclic 2′-modified nucleoside, a 2′-4′ bridged nucleoside, or combinations thereof. In some embodiments, each nucleoside of the antisense oligonucleotide is a modified nucleoside.

[0033] In some embodiments, the non-bicyclic 2′-modified nucleoside is a 2′-O-methoxyethyl (2′-MOE) modified nucleoside, a 2′-O—N-methylacetamide (2′-O-NMA) modified nucleoside, a 2′-O-Methyl (2′-O-Me) modified nucleoside, or a 2′-fluoro (2′-F) modified nucleoside, and/or the 2′-4′ bridged nucleoside is a locked nucleic acid (LNA, 2′-4′ methylene bridge), an ethylene-bridged nucleic acid (ENA, 2′-4′ ethylene bridge), a constrained ethyl nucleic acid (cEt, 2′-4′ ethylene bridge), a 2′-O, 4′-C-Spirocyclopropylene bridged nucleic acid (scpBNA), an amido-bridged nucleic acid (AmNA), or a 2′-O, 4′-C-aminomethylene bridged nucleic acid (BNA (NC)). In some embodiments, the antisense oligonucleotide comprises one or more 2′-MOE modified nucleosides and one or more LNAs. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more modified nucleosides selected from ENA, scpBNA, 2′-O-NMA, AmNA, and BNA (NC). In some embodiments, each nucleoside of the antisense oligonucleotide is a modified nucleoside, wherein the antisense oligonucleotide comprises one or more modified nucleosides selected from ENA, scpBNA, 2′-O-NMA, AmNA, and BNA (NC), and wherein the remaining modified nucleosides are 2′-MOE modified nucleosides, LNAs, or a mix of 2′MOE modified nucleosides and LNAs.

[0034] In some embodiments, an antisense oligonucleotide provided herein comprises one or more modified internucleoside linkages. In some embodiments, each internucleoside linkage of the oligonucleotide is a modified internucleoside linkage. In some embodiments, the modified internucleoside linkage is a phosphorothioate linkage. In some embodiments, each internucleoside linkage of the oligonucleotide is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotide comprises a mix of phosphodiester linkages and phosphorothioate linkages.

[0035] In some embodiments, the antisense oligonucleotide is a phosphorodiamidate oligomer (PMO).

[0036] In some embodiments, each cytosine of the oligonucleotide is not a 5-methyl-cytosine. In some embodiments, each cytosine of the oligonucleotide is a 5-methyl-cytosine. In some embodiments, one or more of the cytosines of the oligonucleotide is a 5-methyl-cytosine.

[0037] In some embodiments, the antisense oligonucleotide is linked to a lipid. In some embodiments, the lipid is a C16 lipid. In some embodiments, the C16 lipid is linked the 5′ nucleoside of the antisense stand via a DNA linker. In some embodiments, the DNA linker is a phosphodiester d(TCA) linker, wherein each internucleoside linkage is a phosphodiester linkage.

[0038] In some embodiments, an antisense oligonucleotide provided herein comprises a structure as provided in Table 5. In some embodiments, an antisense oligonucleotide provided herein comprises a structure as provided in Table 11. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of ASOs 1-331 or 350-814. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of SEQ ID NOs: 569-925, 927-957, 960-970, 973-1039, 1042-1165, 1170-1350, and 1355-1362. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of ASOs 407, 408, 420, 488, 613, 614, and 796-806. In some embodiments, an antisense oligonucleotide provided herein comprises a structure of any one of SEQ ID NOS: 569-880, 882-925, 927-957, 960-970, 973-

1039, 1042-1155, 1165, 1170-1350, 1355-1362 and ASOs 407, 408, 420, 488, 613, 614, and 796-806. In some embodiments, the antisense oligonucleotide inhibits inclusion of the UNC13A cryptic exon into an UNC13A mature mRNA. In some embodiments, the antisense oligonucleotide induces a secondary structure between the first target region and the second target region in the UNC13A pre-mRNA.

[0039] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iG][iA][iG][iT][iTs][i.sup.5Cs] (SEQ ID NO: 569), wherein iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0040] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iA][iG][iT][iTs][i.sup.5C] (SEQ ID NO: 570), wherein iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0041] In some embodiments, the antisense oligonucleotide comprises the following structure: [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] (SEQ ID NO: 571), wherein iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0042] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iG][iT][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iG][iT][iTs][i.sup.5C] (SEQ ID NO: 572), wherein iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0043] In some embodiments, the antisense oligonucleotide comprises the following structure: [iTs][iG][iT][iT][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iA][iG][iT][iTs][i.sup.5Cs] (SEQ ID NO: 573), wherein iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0044] In some embodiments, the antisense oligonucleotide comprises the following structure: [nmaTs][nmaGs][nmaTs][nmaTs][nma.sup.5Cs][nmaAs][nmaAs][nmaTs][nma.sup.5Cs][nmaAs][nmaTs][nmaTs][nma.sup.5Cs][nmaGs][nmaGs][nmaGs][nmaAs][nmaTs][nmaAs][nmaA][nmaG][nmaA][nmaG][nmaT][nmaTs][nma.sup.5C] (SEQ ID NO: 969), wherein nmaA, nmaG, nma.sup.5C, and nmaT are 2′O-NMA modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage; and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage.

[0045] In some embodiments, the antisense oligonucleotide comprises the following structure: [nma.sup.5Cs][nmaGs][nmaAs][nmaAs][nmaTs][nma.sup.5Cs][nmaTs][nmaAs][nma.sup.5Cs][nma.sup.5Cs][nma.sup.5Cs][nmaAs][nma.sup.5Cs][nmaAs][nmaTs][nma.sup.5Cs][nmaTs][nmaGs][nmaTs][nmaTs][nma.sup.5Cs][nmaAs][nmaAs][nmaTs][nma.sup.5Cs][nmaA] (SEQ ID NO: 1115), wherein nmaA, nmaG, nma.sup.5C, and nmaT are 2′O-NMA modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0046] In some embodiments, the antisense oligonucleotide comprises the following structure:

[nmaAs][nma.sup.5Cs][nmaGs][nmaAs][nmaAs][nmaTs][nma.sup.5Cs][nmaTs][nmaAs][nma.sup.5Cs][nma.sup.5Cs][nma.sup.5Cs][nmaAs][nma.sup.5Cs][nmaAs][nmaTs][nma.sup.5Cs][nmaTs][nmaGs][nmaTs][nmaTs][nma.sup.5Cs][nmaAs][nmaAs][nmaTs][nma.sup.5C] (SEQ ID NO: 1116), wherein nmaA, nmaG, nma.sup.5C, and nmaT are 2′O-NMA modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0047] In some embodiments, the antisense oligonucleotide comprises the following structure: [lGs][nmaAs][nmaAs][nmaTs][nma.sup.5Cs][nmaTs][nmaAs][nma.sup.5Cs][nma.sup.5Cs][nmaAs][nmaTs][nma.sup.5Cs][nma.sup.5Cs][nmaAs][nmaTs][nmaGs][nmaTs][nmaAs][nma.sup.5Cs][iT] (SEQ ID NO: 1355), wherein lA, lG, l.sup.5C, and lT are 2′-4′ methylene bridge (LNA) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; nmaA, nmaG, nma.sup.5C, and nmaT are 2′O-NMA modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0048] In some embodiments, the antisense oligonucleotide comprises the following structure: [lGs][nmaAs][nmaAs][nmaTs][nma.sup.5Cs][nmaTs][nmaAs][nma.sup.5Cs][nma.sup.5Cs][nma.sup.5Cs][nmaAs][nmaTs][nma.sup.5Cs][nma.sup.5Cs][nmaAs][nmaTs][nmaGs][nmaTs][nmaAs][nma.sup.5Cs][lT] (SEQ ID NO: 1356), wherein lA, lG, l.sup.5C, and lT are 2′-4′ methylene bridge (LNA) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; nmaA, nmaG, nma.sup.5C, and nmaT are 2′O-NMA modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage.

[0049] In some embodiments, the antisense oligonucleotide is in the form of a pharmaceutically acceptable salt.

[0050] Some aspects of the present disclosure provide compositions comprising an antisense oligonucleotide provided herein. In some embodiments, a composition provided herein further comprises a pharmaceutically acceptable carrier.

[0051] Some aspects of the present disclosure provide methods of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell, comprising contacting the cell with an antisense oligonucleotide or composition provided herein.

[0052] Some aspects of the present disclosure provide methods of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, comprising administering to the subject an antisense oligonucleotide or composition provided herein. In some embodiments, the method comprises administering to the subject a therapeutically effective amount of an antisense oligonucleotide or composition provided herein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject is human. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0053] Some aspects of the present disclosure provide methods of treating a disease associated with abnormal UNC13A expression in a subject, comprising administering to the subject an antisense oligonucleotide or composition provided herein. In some embodiments, the method comprises administering to the subject a therapeutically effective amount of an antisense oligonucleotide or composition provided herein. In some embodiments, the disease is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is amyotrophic lateral sclerosis (ALS). In some embodiments, the neurodegenerative disease is frontotemporal dementia (FTD). In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE). In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0054] Some aspects of the present disclosure provide methods of treating a disease associated with TDP-43 dysfunction in a subject, comprising administering to the subject an antisense oligonucleotide or composition provided herein. In some embodiments, the method comprises administering to the subject a therapeutically effective amount of an antisense oligonucleotide or composition provided herein. In some embodiments, the disease is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is amyotrophic lateral sclerosis (ALS). In some embodiments, the neurodegenerative disease is frontotemporal dementia (FTD). In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE). In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0055] Some aspects of the present disclosure provide pharmaceutical compositions for use in a method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject comprising the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition comprises a therapeutically effective amount of an antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0056] Some aspects of the present disclosure provide pharmaceutical compositions for use in a method of treating a disease associated with abnormal UNC13A expression in a subject comprising the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition comprises a therapeutically effective amount of the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0057] Some aspects of the present disclosure provide pharmaceutical compositions for use in a method of treating a disease associated with TDP-43 dysfunction in a subject comprising the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition comprises a therapeutically effective amount of the antisense oligonucleotide provided herein. In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0058] Some aspects of the present disclosure provide antisense oligonucleotides for use in a method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. In some embodiments, the method comprises administering to the subject an effective amount of an antisense oligonucleotide provided herein. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0059] Some aspects of the present disclosure provide antisense oligonucleotides for use as a medicament. In some embodiments, the medicament is administered to a subject. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0060] Some aspects of the present disclosure provide antisense oligonucleotides for use in a method of treating a disease associated with abnormal UNC13A expression in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. In some embodiments, the method comprises administering to the subject an effective amount of an antisense oligonucleotide provided herein. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0061] Some aspects of the present disclosure provide antisense oligonucleotides for use in a method of treating a disease associated with TDP-43 dysfunction in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. In some embodiments, the method comprises administering to the subject an effective amount of an antisense oligonucleotide provided herein. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0062] Some aspects of the present disclosure provide a use of an antisense oligonucleotide provided herein for the manufacture of a medicament for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject. In some embodiments, the composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0063] Some aspects of the present disclosure provide a use of an antisense oligonucleotide provided herein for the manufacture of a medicament for treating a disease associated with abnormal UNC13A expression in a subject. In some embodiments, the composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0064] Some aspects of the present disclosure provide a use of an antisense oligonucleotide provided herein for the manufacture of a medicament for treating a disease associated with TDP-43 dysfunction in a subject. In some embodiments, the composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the subject is human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

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## Description

DETAILED DESCRIPTION

[0065] Generally, nomenclatures used in connection with cell and tissue culture, molecular biology, immunology, microbiology, genetics and protein and nucleic acid chemistry and hybridization described herein are those well-known and commonly used in the art. Certain methods and techniques provided herein are generally performed according to methods well-known in the art and as described in various general and more specific references that are cited and discussed throughout the present specification unless otherwise indicated. Enzymatic reactions and purification techniques are performed according to manufacturer's specifications, as commonly accomplished in the art or as otherwise described herein. The nomenclatures, laboratory procedures and techniques of analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well-known and commonly used in the art. Standard techniques are used for chemical syntheses, chemical analyses, pharmaceutical preparation, formulation, delivery, and treatment of patients.

[0066] Any of the methods for gene therapy available in the art can be used in the methods provided herein. For general reviews of the methods of gene therapy, see Goldspiel et al. (1993) Clin. Pharmacy 12:488-505; Wu and Wu (1991) Biotherapy 3:87-95; Tolstoshev (1993) Ann. Rev. Pharmacol. Toxicol. 32:573-596; Mulligan (1993) Science 260:926-932; Morgan and Anderson (1993) Ann. Rev. Biochem. 62:191-217; and May (1993) TIBTECH 11 (5): 155-215. Methods commonly known in the art of recombinant DNA technology which can be used are described in Ausubel et al. (eds.), Current Protocols in Molecular Biology, John Wiley &Sons, NY (1993); and Kriegler, Gene Transfer and Expression, A Laboratory Manual, Stockton Press, NY (1990). Detailed description of various methods of gene therapy are disclosed in US Patent Publication No. US20050042664.

Definitions

[0067] In order that the present disclosure may be more readily understood, certain terms are first defined. Unless otherwise defined herein, scientific and technical terms used in connection with the present disclosure shall have the meanings that are commonly understood by those of ordinary skill in the art. The meaning and scope of the terms should be clear, however, in the event of any latent ambiguity, definitions provided herein take precedent over any dictionary or extrinsic definition. Unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. The use of “or” means “and/or” unless stated otherwise. The use of the term “including”, as well as other forms of the term, is not limiting.

[0068] In addition, it should be noted that whenever a value or range of values of a parameter are recited, it is intended that values and ranges intermediate to the recited values are also part of this disclosure.

[0069] As used herein, the singular forms “a”, “an” and “the” include plural referents unless the context clearly dictates otherwise. “And” as used herein is interchangeably used with “or” unless expressly stated otherwise. The terms “comprising, “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value recited or falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited.

[0070] The term “about” or “approximately,” as applied to one or more values provided herein, refers to a value that is similar to a stated reference value. In some embodiments, the term “about” or “approximately” refers to a range of values that fall within and include 15%, 14%, 13%, 12%, 11%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, or less in either direction (greater than or less than) of the stated reference value unless otherwise stated or otherwise evident from the context. In some embodiments, “about” or “approximately” can be understood as about 2 standard deviations from the mean. In some embodiments, “about” or “approximately” means up to and including ±10%

(e.g., ±10%, ±9%, ±8%, ±7%, ±6%, ±5%, ±4%, ±3%, ±2%, ±1%, or less). In some embodiments, "about" or "approximately" means±5%. When "about" or "approximately" is present before a series of numbers or a range, it is understood that it can modify each of the numbers in the series or range.

[0071] The term "administering" or "administration of," as used herein, means to provide an agent, e.g., antisense oligonucleotide to a subject. In some embodiments, "administering" or "administration of" means to provide an antisense oligonucleotide to a subject in a manner that is physiologically and/or pharmacologically useful (e.g., to treat a condition in the subject). Non-limiting examples of routes of administration include intravenous, intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, intrathecal routes, or intracerebroventricular routes. In some embodiments, the route of administration is intrathecal, or intracerebroventricular routes. In some embodiments, the route of administration is intrathecal. In some embodiments, the route of administration is intracerebroventricular. In some embodiments, the route of administration is subcutaneous.

[0072] The term "antisense oligonucleotide (ASO)," as used herein, refers to a single-stranded oligonucleotide which comprises one or more (e.g., 1, 2, or more) regions of complementarity each capable of hybridizing with at least a part, for example, a target region of a target sequence (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence), and which can modulate the processing of the transcription product of a target gene and/or the expression (e.g., mRNA and/or protein level) of the target gene. In some embodiments, a target region of a target nucleotide sequence can have a length of at least 8 nucleosides, for example, a length of 8, 9, 10, 11, 12, 13, 14, 15, or more nucleosides.

[0073] In some embodiments, an antisense oligonucleotide disclosed herein modulates splicing of an UNC13A pre-mRNA. In some embodiments, an antisense oligonucleotide disclosed herein modulates splicing of an UNC13A cryptic exon present in the UNC13A pre-mRNA (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, splicing modulation of UNC13A pre-mRNA and/or an UNC13A cryptic exon present in the UNC13A pre-mRNA results in modulation of UNC13A gene expression (e.g., mRNA and/or protein level).

[0074] In some embodiments, an antisense oligonucleotide disclosed herein targets two target regions of a target sequence (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence). In some embodiments, an antisense oligonucleotide disclosed herein targets two target regions of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, the two target regions of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence) have 0-200 nucleobases (e.g., 0-200, 0-150, 0-100, 0-50, 0-20, 0-10, 0-5, 5-200, 5-150, 5-100, 5-50, 5-20, 5-10, 10-200, 10-150, 10-100, 10-50, 10-20, 20-200, 20-150, 20-100, 20-50, 50-200, 50-150, 50-100, 100-200, 100-150, or 150-200 nucleobases) in between.

[0075] In some embodiments, an antisense oligonucleotide comprises a first antisense sequence and a second antisense sequence, each targeting one of the two target regions of a target sequence (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence). In some embodiments, an antisense oligonucleotide comprises a first antisense sequence and a second antisense sequence, each targeting one of the two target regions of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, the first antisense sequence is immediately adjacent to the second antisense sequence (i.e., having no nucleobases in between). In some embodiments, one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more) nucleobases are between the first antisense sequence and the second antisense sequence.

[0076] For the purposes of this disclosure, in an antisense oligonucleotide disclosed herein, the first antisense sequence is at the 5′ end of the antisense oligonucleotide and the second antisense sequence is at the 3′ end of the antisense oligonucleotide. Further, in an antisense oligonucleotide disclosed herein, the first antisense sequence targets the first target region of a target sequence (e.g., UNC13A pre-mRNA) and the second antisense sequence targets the second target region of a target sequence (e.g., UNC13A pre-mRNA). As such, the first target region is 3′ to the second target region in the target sequence (e.g., UNC13A pre-mRNA), and the second target region is 5′ to the first target region in the target sequence (e.g., UNC13A pre-mRNA).

[0077] The term "at least" prior to a number or series of numbers is understood to include the number adjacent to the term "at least", and all subsequent numbers or integers that could logically be included, as clear from context. For example, the number of nucleotides or nucleosides in a nucleic acid molecule must be an integer. For example, "at least 19 nucleosides of a 21-nucleotide nucleic acid molecule" means that 19, 20, or 21 nucleosides have the indicated property. When at least is present before a series of numbers or a range, it is understood that "at least" can modify each of the numbers in the series or range.

[0078] The term "Amyotrophic lateral sclerosis (ALS)" or "ALS," as used herein, refers to a neurodegenerative disease characterized by progressive upper and lower motor neuron loss.

[0079] Two types of the disease, sporadic ALS (sALS) and familial ALS (fALS) have been described. sALS is the most common form of ALS, accounting for around 90% of all ALS cases, while fALS is rarer and genetic, as it occurs in a series of generations in families. Both forms of ALS show rapid clinical deterioration after onset of the disease, leading to death within a few years often due to respiratory failure. The exact cause of ALS is unknown and there is no known cure.

[0080] The term "frontotemporal dementia" or "FTD," as used herein, refers to a progressive neurodegenerative disease of humans which has been found to have significant overlap with ALS at clinical, genetic, and pathologic levels. Burrell J R, Halliday G M, Kril J J, et al. The frontotemporal dementia-motor neuron disease continuum. Lancet. 2016; 388 (10047): 919-931. FTD is characterized by damage to neurons in the frontal and temporal lobes of the brain. Three types of the disease, behavioral variant frontotemporal dementia (bvFTD), primary progressive aphasia (PPA), and movement disorders, have been described, with bvFTD being the most common and involving changes in personality, behavior, and judgment. Some patients suffer from both Frontotemporal Dementia and ALS. Thus, these diseases are on one molecular and clinical spectrum. The exact cause of FTD is unknown and there is no known cure.

[0081] While causes of ALS and FTD are generally known, single nucleotide polymorphisms (SNPs), e.g., intronic SNPs, in UNC13A have been shown to increase the risk of ALS and FTD. In some instances, the SNPs increase cryptic splicing and alter direct binding of RNA-binding protein TDP-43, which reduces the ability of TDP-43 to repress cryptic exon inclusion in UNC13A during RNA splicing. TDP-43 dysfunction has been shown to result in inclusion of a cryptic exon in an UNC13A mRNA. Cryptic exon inclusion in an UNC13A mature mRNA results in reduced UNC13A protein expression due to aberrant RNA degradation or the presence of premature stop codons. In some embodiments, antisense oligonucleotides disclosed herein inhibit inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restore UNC13A expression in a cell in a subject with ALS or FTD.

[0082] The term "Alzheimer's disease," as used herein is a progressive degenerative brain disease and the most common form of dementia. Alzheimer's disease is associated with the abnormal accumulation of amyloid-β (AB) plaques and neurofibrillary tangles of tau protein aggregates. Alzheimer's disease is characterized by a progressive pattern of cognitive and functional impairment. TDP-43 pathology also plays a role in the progression of Alzheimer's disease. Meneses et al., TDP-43 Pathology in Alzheimer's Disease. Mol Neurodegener. 2021 Dec. 20; 16(1):84. There is currently no known cure for Alzheimer's disease.

[0083] The term "Limbic-Predominant Age-Related TDP-43 Encephalopathy" or "LATE," as used herein is a late-onset neurogenerative disease characterized by amnestic dementia resembling Alzheimer's disease dementia. LATE typically affects persons 80 years of age or older. LATE is characterized by TDP-43 inclusions predominated in the limbic regions of the brain, including the amygdala and hippocampus. Nelson et al., Limbic-predominant age-related TDP-43 encephalopathy (LATE): consensus working group report, Brain, Volume 142, Issue 6, June 2019, Pages 1503-1527. There is currently no known cure for LATE.

[0084] The term "biological activity" means any biological property of a molecule, whether present naturally in vivo, or provided or enabled by recombinant means. Biological activities include, but are not limited to, binding to a receptor, inducing cell proliferation, inhibiting cell growth, inducing other cytokines, inducing apoptosis, and enzymatic activity.
[0085] The term "contiguous" in the context of an oligonucleotide refers to nucleosides, nucleobases, sugar moieties, or internucleoside linkages that are immediately adjacent to each other. For example, "contiguous nucleobases" means nucleobases that are immediately adjacent to each other in a sequence.
[0086] The term "complementary," as used herein, refers to the capacity for base pairing between two nucleobases or two nucleobase sequences. In particular, complementary is a term that characterizes an extent of hydrogen bond pairing that brings about binding between two nucleobases or two nucleobase sequences. For example, if a base at one position of a nucleobase sequence (e.g., of an antisense oligonucleotide described herein) is capable of hydrogen bonding with a base at the corresponding position of another nucleobase sequence (e.g., target gene sequence, pre-mRNA sequence, or mRNA sequence), then the bases are considered to be complementary to each other at that position. The nucleic acid molecules (e.g., antisense oligonucleotide) whose nucleobase sequence is complementary may comprise one or more modified nucleosides and modified internucleoside linkages. The nucleic acid molecules (e.g., antisense oligonucleotide and target sequence) whose nucleobase sequence is complementary may also comprise nucleobase analogous that result in bases at certain positions not being complementary, but the nucleobase sequences of the two molecules must be sufficiently complementary over the entire length to result in a desired biological activity (e.g., modulation of gene expression).
[0087] Complementary sequences include Watson-Crick base pairs or non-Watson-Crick base pairs (e.g., Wobble base pairs (e.g., G-U/T, A-G) and Hoogsteen base pairs) and may include natural or modified nucleosides or nucleoside mimics. For example, in some embodiments, for complementary base pairings, adenosine-type bases (A) are complementary to thymidine-type bases (T) or uracil-type bases (U), that cytosine-type bases (C) are complementary to guanosine-type bases (G), and that universal bases such as 3-nitropyrrole or 5-nitroindole can hybridize to and are considered complementary to any A, C, U, or T. Inosine (I) has also been considered in the art to be a universal base and is considered complementary to any A, C, U or T.
[0088] Complementarity is independent of modifications in the sugar of a nucleoside. For example, 2′-modified A, as defined herein, are complementary to U (or T) and identical to A for the purposes of determining identity or complementarity.
[0089] The term "perfectly complementary" or "fully complementary" means that all (100%) of the nucleobases, nucleosides, or nucleotides in a contiguous sequence of a first nucleotide sequence (e.g., an antisense oligonucleotide) will hybridize with the same number of nucleobases, nucleosides, or nucleotides in a contiguous sequence of second nucleotide sequence (e.g., a target sequence such as an UNC13A pre-mRNA). The contiguous sequence may comprise all or a part of a first or second nucleotide sequence. The term "partially complementary" means that in a hybridized pair of nucleobase, nucleosides, or nucleotide sequences, at least 70%, but not all, of the bases in a contiguous sequence of a first nucleotide sequence (e.g., an antisense oligonucleotide) will hybridize with the same number of bases in a contiguous sequence of a second nucleotide sequence (e.g., a target sequence such as an UNC13A pre-mRNA). The term "sufficiently complementary" or "substantially complementary" means that in a hybridized pair of nucleobase, nucleosides, or nucleotide sequences, at least 85%, but not all, of the bases in a contiguous sequence of a first nucleotide sequence (e.g., an antisense oligonucleotide) will hybridize with the same number of bases in a contiguous sequence of a second nucleotide sequence (e.g., a target sequence such as an UNC13A pre-mRNA). The terms "complementary," "fully complementary," "partially complementary," and "sufficiently/substantially complementary" herein are used with respect to the nucleobase, nucleosides, or nucleotide matching between an antisense oligonucleotide and a target

sequence (e.g., an UNC13A pre-mRNA).

[0090] The term "control" or "reference," when referring to a substance, means a composition used as a standard or a point of comparison against which other test results are measured. In some embodiments, a "control" or "reference" is a composition known to not contain analyte ("negative control") or to contain analyte ("positive control"). A positive control can comprise a known concentration of analyte. "Control," and "positive control," may be used to refer to a composition comprising a known concentration of analyte. A "positive control" can be used to establish assay performance characteristics and is a useful indicator of the integrity of reagents (e.g., analytes). In some embodiments, an appropriate "control" or "reference" is where only one element is changed in order to determine the effect of the one element. In some embodiments, a control is a level of a target gene (e.g., in a cell or in a subject) before treatment (e.g., with an antisense oligonucleotide described herein).

[0091] The term "control" or "reference" also means a baseline level of a measurement depending upon the context, in which the term is used. A baseline level of a measurement is a standard or a point of comparison against which the measurement is compared. In some embodiments, a "control" or a "reference" refers to a level of a measurement for certain biological activity or substance in a cell, a tissue, an organ, or a subject, e.g., the expression level of a gene, copy number of mRNA for such gene, or level of protein encoded by such gene, without treatment of the cell, the tissue, the organ, or the subject, with an agent, e.g., antisense oligonucleotide. In some embodiments, a "control" or a "reference" refers to a level of an average measurement for a certain biological activity or substance in a cell, a tissue, an organ, or a subject, e.g., certain enzyme activity of the liver, among a group of healthy subjects, e.g., the general population within in certain geographic or demographic limits or any other limits that may be appropriate for the study of certain disease or disorder, that does not have certain disease or disorder, e.g., liver disease.

[0092] The term "reference" may also be used in "reference sequence." The term "reference sequence" refers to a sequence, e.g., a nucleic acid sequence or an amino acid sequence, used as a basis for sequence comparison. In certain embodiments, a reference sequence is an RNA sequence, e.g., human UNC13A pre-mRNA sequence, upon which the design of the antisense oligonucleotide is based.

[0093] The term "cross-reactive" means the ability of a binding molecule (e.g., an antisense oligonucleotide) to bind a target molecule (e.g., gene sequence, pre-mRNA sequence, or mRNA sequence) other than that against which it was designed or generated. For example, the binding molecule is capable of specifically binding to more than one target molecule of a similar type or class (e.g., mRNA variants or mRNA homologous from closely related species) with similar affinity. Generally, a binding molecule will bind its target molecule with an appropriately high affinity but can bind to the same target molecule of another species or display a low affinity for non-target molecules. In some embodiments, an antisense oligonucleotide that is cross-reactive against human and non-human primate UNC13A comprises a region of complementarity to human and non-human primate UNC13A gene sequence, pre-mRNA sequence, or mRNA sequence. Individual binding molecules are generally selected to meet two criteria: (1) tissue staining appropriate for the known expression of the target or (2) similar staining pattern between human and toxicology species (mouse and cynomolgus monkey) tissues from the same organ. These and other methods of assessing cross-reactivity are known to one skilled in the art.

[0094] The term "effective amount" or "therapeutically effective amount," as used herein, refers to that amount of an antisense oligonucleotide to produce a molecular (e.g., reducing inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a subject), biological, pharmacological, therapeutic (e.g., treatment of an UNC13A associated disease or a disease associated with TDP-43 dysfunction in a subject), or preventive result. The amount administered will likely depend on such variables as the overall health status of the patient, the relative biological efficacy of the compound delivered, the formulation of the drug, the presence and types of excipients in the formulation, and the route of administration. Also, it is to be understood that the initial dosage administered can, in some instances, be increased beyond the above upper level to rapidly achieve the desired blood-level or tissue level, or the initial dosage can, in some instances, be smaller than the optimum.

[0095] The terms “hybridize” and “hybridization” refer to the pairing of complementary compounds (e.g., an antisense oligonucleotide and its target nucleic acid). While not limited to a particular mechanism, the most common mechanism of pairing involves hydrogen bonding, which may be Watson-Crick, Wobble, Hoogsteen or reversed Hoogsteen hydrogen bonding, between complementary nucleobases.

[0096] The term “internucleoside linkage,” as used herein, means a covalent linkage between adjacent nucleosides in an oligonucleotide (e.g., antisense oligonucleotide described herein). An internucleoside linkage may be a natural phosphodiester internucleoside linkage, or may be a modified (non-natural) internucleoside linkage. Modified internucleoside that may be used in an antisense oligonucleotide disclosed herein include, but are not limited to, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates comprising 3′ alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates comprising 3′-amino phosphoramidate and aminoalkylphosphoramidates, mesyl phosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates having normal 3′-5′ linkages, 2′-5′ linked analogs of these, and those having inverted polarity wherein the adjacent pairs of nucleoside units are linked 3′-5′ to 5′-3′ or 2′-5′ to 5′-2′; see U.S. Pat. Nos. 3,687,808; 4,469,863; 4,476,301; 5,023,243; 5,177,196; 5,188,897; 5,264,423; 5,276,019; 5,278,302; 5,286,717; 5,321,131; 5,399,676; 5,405,939; 5,453,496; 5,455,233; 5,466,677; 5,476,925; 5,519,126; 5,536,821; 5,541,306; 5,550,111; 5,563,253; 5,571,799; 5,587,361; and 5,625,050. In some embodiments, in any one of the antisense oligonucleotides disclosed herein, all of the internucleoside linkages are stereorandom.

[0097] The term, “nucleoside,” as used herein, refers to a compound comprising a nucleobase moiety and a sugar moiety. Nucleosides include, but are not limited to, naturally occurring nucleosides (as found in DNA and RNA) and modified nucleosides. Nucleosides may be linked to a phosphate moiety. The term “nucleoside” encompasses a natural nucleoside and chemically modified nucleosides (e.g., with modifications in the base and/or sugar moiety).

[0098] The term “nucleotide,” as used herein, refers to a compound comprising a nucleoside linked to a phosphate group. As used herein, “linked nucleosides” may or may not be linked by phosphate linkages and thus includes, but is not limited to “linked nucleotides.” As used herein, “linked nucleosides” are nucleosides that are connected in a continuous sequence (i.e., no additional nucleosides are present between those that are linked). The term “nucleotide” encompasses a natural nucleotide and chemically modified nucleotides (e.g., with modifications in the base, sugar moiety, and/or phosphate group).

[0099] The term “nucleobase,” as used herein, refers to nitrogen-containing compounds that can be linked to a sugar moiety to create a nucleoside that is capable of incorporation into an oligonucleotide, and wherein the compound is capable of bonding with a complementary naturally occurring nucleobase of another oligonucleotide or nucleic acid. Nucleobases may be naturally occurring or may be modified. As used herein a “naturally occurring nucleobase” is adenine (A), thymine (T), cytosine (C), uracil (U), and guanine (G). The term “nucleobase” encompasses 5′-methylated bases (e.g., 5′-methyl cytosine or 5′-methyl guanine).

[0100] The term “modified internucleoside linkage” refers to a linkage between two nucleosides (e.g., in an oligonucleotide) that is not the natural phosphodiester linkage. Non-limiting examples of modified internucleoside linkages include phosphorothioates, phosphorodiamidates, phosphotriesters, methyl phosphonates, short chain alkyl or cycloalkyl intersugar linkages or short chain heteroatomic or heterocyclic intersugar linkages.

[0101] The term “nucleoside modification” or “modified nucleoside” means a nucleoside that has one or more modifications to the nucleoside, including modifications to the nucleobase moiety and/or the sugar moiety. Any of the modified chemistries or formats of nucleosides described herein can be combined with each other. Non-limiting examples of modified nucleosides includes 2′-fluoro (2′-F), 2′-O-methyl (2′-O-Mc), 2′-O-methoxyethyl (2′-MOE), 2′-O-aminopropyl (2′-O-AP), 2′-O-dimethylaminoethyl (2′-O-DMAOE), 2′-O-dimethylaminopropyl (2′-O-DMAP), 2′-O-dimethylaminoethyloxyethyl (2′-O-DMAEOE), 2′-O—N-methylacetamide (2′-O-NMA), locked nucleic acid (LNA, methylene-bridged nucleic acid), unlocked nucleic acid (UNA), ethylene-bridged nucleic acid (ENA), 2′-O, 4′-C-Spirocyclopropylene bridged nucleic acid (scpBNA), amido-bridged nucleic acid (AmNA), or 2′-O, 4′-C-aminomethylene bridged nucleic acid (BNA (NC)), and (S)-constrained ethyl-bridged nucleic acid (cEt) modified nucleosides. Further non-limiting examples of modified nucleosides include a conformationally restricted nucleoside, an abasic nucleoside, a 2′-amino-modified nucleoside, a morpholino nucleoside, a phosphoramidate, a non-natural base comprising nucleoside, a tetrahydropyran modified nucleoside, a 1,5-anhydrohexitol modified nucleoside (HNA), a cyclohexenyl modified nucleoside (CeNA), a nucleoside comprising a phosphorothioate group, a nucleoside comprising a methylphosphonate group, a nucleoside comprising a 5′-phosphate, a nucleoside comprising a 5′-phosphate mimic, a thermally destabilizing nucleoside, a glycol modified nucleoside (GNA).
[0102] The term “2′-modified nucleoside” refers to a nucleoside having a sugar moiety modified at the 2′ position, meaning the sugar moiety comprises at least one 2′-substituent group other than H or OH. Non-limiting examples of 2′-modified nucleosides include: 2′-O-methoxyethyl (2′-MOE), 2′-O-Methyl (2′-O-Me), 2′-fluoro (2′-F), LNA (2′-4′ methylene bridge), ENA (2′-4′ ethylene bridge), or cEt (2′-4′ ethylene bridge), 2′-deoxy, 2′-O-aminopropyl (2′-O-AP), 2′-O-dimethylaminoethyl (2′-O-DMAOE), 2′-O-dimethylaminopropyl (2′-O-DMAP), 2′-O-dimethylaminoethyloxyethyl (2′-O-DMAEOE), 2′-O—N-methylacetamide (2′-O-NMA), 2′-O, 4′-C-Spirocyclopropylene bridged nucleic acid (scpBNA), amido-bridged nucleic acid (AmNA), or 2′-O, 4′-C-aminomethylene bridged nucleic acid (BNA (NC)) modified nucleosides. In some embodiments, any one of the 2′-modified nucleosides described herein are high-affinity modified nucleosides and a modified antisense oligonucleotide has increased affinity to target sequences, relative to an unmodified antisense oligonucleotide. In some embodiments, at least one modified nucleoside is a 2′-modified nucleoside. In some embodiments, the 2′-modified nucleoside is a 2′-MOE modified nucleoside or an LNA or combinations thereof. In some embodiments, the 2′-modified nucleoside is a 2′-MOE modified nucleoside.
[0103] The term “modified oligonucleotide” or “modified antisense oligonucleotide” refers to oligonucleotides or antisense oligonucleotides that comprise one or more modified nucleosides and/or one or more modified internucleoside linkages. In some embodiments, a “modified oligonucleotide” or “modified antisense oligonucleotide” comprises a mix of modified nucleosides and unmodified nucleosides and/or a mix of modified internucleoside linkages and unmodified modified internucleoside linkages. In some embodiments, each nucleoside of a modified oligonucleotide” or “modified antisense oligonucleotide” is a modified nucleoside, and/or each internucleoside linkage of a modified oligonucleotide” or “modified antisense oligonucleotide” is a modified internucleoside linkage.
[0104] The term “region of complementarity,” as used herein, refers to a nucleobase sequence, e.g., of an antisense oligonucleotide, that is sufficiently complementary to a cognate nucleobase sequence, e.g., of a target nucleic acid, such that the two nucleobase sequences are capable of annealing to one another under physiological conditions (e.g., in a cell). In some embodiments, a region of complementarity is fully complementary (e.g., 100% complementarity) to a cognate nucleobase sequence of target nucleic acid. In some embodiments, a region of complementarity is partially complementary to a cognate nucleobase sequence of target nucleic acid (e.g., at least 80%, 90%, 95% or 99% complementarity). In some embodiments, a region of complementarity contains

1, 2, 3, 4, or 5 mismatches compared with a cognate nucleobase sequence of a target nucleic acid.

[0105] The term "sequence identity," as used herein, refers to the extent that sequences are identical (independent of chemical modification) on a nucleobase-by-nucleobase basis or an amino acid-by-amino acid basis over a window of comparison. Thus, a "percentage of sequence identity" may be calculated by comparing two optimally aligned sequences over the window of comparison, determining the number of positions at which the identical nucleic acid base (e.g., A, T, C, G, I) or the identical amino acid residue (e.g., Ala, Pro, Ser, Thr, Gly, Val, Leu, Ile, Phe, Tyr, Trp, Lys, Arg, His, Asp, Glu, Asn, Gln, Cys and Met) occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison (i.e., the window size), and multiplying the result by 100 to yield the percentage of sequence identity. Optimal alignment of sequences for aligning a comparison window may be conducted by computerized implementations of algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package Release 7.0, Genetics Computer Group, 575 Science Drive Madison, Wis., USA) or by inspection and the best alignment (i.e., resulting in the highest percentage homology over the comparison window) generated by any of the various methods selected. Reference also may be made to the BLAST family of programs as for example disclosed by Altschul et al., Nucl. Acids Rcs. 25:3389, 1997.

[0106] The term "restore" when referring to expression of a given gene (e.g., UNC13A), means that the expression of the gene, as measured by the level of RNA transcribed from the gene or the level of polypeptide, protein or protein subunit translated from the mRNA in a cell, group of cells, tissue, organ, or subject in which the gene is transcribed, is returned to normal levels (e.g., baseline level of gene expression in a subject that does not have an UNC13A-associated disease and/or abnormal UNC13A expression) when the cell, group of cells, tissue, organ, or subject is treated with an antisense oligonucleotide disclosed herein. In some embodiments, when a cell, group of cells, tissue, organ, or subject is treated with an antisense oligonucleotide disclosed herein, expression (e.g., protein and/or mRNA expression) of a target gene (e.g., UNC13A) is restored by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90% or at least 95% relative to a control, e.g., baseline level of gene expression prior to treatment.

[0107] The term "decrease" when referring to expression of TDP-43, means that the expression of the gene, as measured by the level of RNA transcribed from the gene or the level of polypeptide, protein or protein subunit translated from the mRNA in a subject in which the gene is transcribed, is reduced relative to expression of TDP-43 in a healthy subject or a subject that does not have an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer's disease, or LATE). For example, a subject described herein having decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100%, has decreased expression of TDP-43 relative to expression of TDP-43 in a subject that does not have an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer's disease, or LATE). Expression of TDP-43 gene or protein may be measured by the level of RNA transcribed from the gene or the level of polypeptide, protein or protein subunit translated from the mRNA in a subject in which the gene is transcribed.

[0108] The term "single nucleotide polymorphism" or "SNP" is a genetic variation caused by changes in a single nucleobase. Without being bound by any particular theory, SNPs in UNC13A (e.g., intronic SNPs) typically increase cryptic splicing and alter direct binding of RNA-binding protein TDP-43, which reduces the ability of TDP-43 to repress cryptic exon inclusion in UNC13A during RNA splicing. SNPs may be associated with one or more diseases and may serve as genetic biomarkers for the one or more diseases. For example, SNPs can be used to predict drug exposure or effectiveness of a treatment in a subject. Non-limiting examples of SNPs associated with ALS include rs12608932, rs12973192, rs56041637, rs116169349, and rs62121687.

[0109] The term "subject," as used herein, refers to a mammal. In some embodiments, a subject is non-human primate, or rodent. In some embodiments, a subject is a human. In some embodiments, a subject is a patient, e.g., a human patient that has or is suspected of having a disease. In some

embodiments, the subject is a human patient who has or is suspected of having an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) and/or a disease associated with TDP-43 dysfunction in a subject. In some embodiments, the UNC13A-associated disease and/or a disease associated with TDP-43 dysfunction in a subject is a neurodegenerative disease (e.g., amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), Alzheimer's disease or Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE). In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. SOD-1 gene mutations have been described, for example, in Miller et al., N Engl J Med 2022; 387:1099-110 and Ruffo et at., Genes 2022, 13, 537. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.
[0110] The term “specificity” means the ability to inhibit the target RNA without manifest effects on other genes of the cell. The consequences of inhibition can be confirmed by examination of the outward properties of the cell or organism or by biochemical techniques such as RNA solution hybridization, nuclease protection, Northern hybridization, reverse transcription, gene expression monitoring with a microarray, antibody binding, enzyme linked immunosorbent assay (ELISA), Western blotting, radioimmunoassay (RIA), other immunoassays, and fluorescence activated cell analysis (FACS).
[0111] The term “symptom” as used herein, refers to any manifestation or indication of an underly disease. A symptom can be any biochemical, cellular, genetic, histological, and/or physiological observation, measurement, and/or test result in a subject that deviate from those of a control or reference. For example, a symptom may be an elevated level of a liver enzyme, such as aminotransferases, as compared to a normal reference range.
[0112] The term “target sequence,” as used herein, refers to a nucleoside sequence whose expression or activity is to be modulated. In some embodiments, the target sequence is a contiguous portion of the nucleoside sequence of a gene, a cDNA, a pre-mRNA, or an mRNA molecule formed during the transcription of a target gene, e.g., UNC13A gene, such an unprocessed UNC13A pre-mRNA transcript. In some embodiments, a target sequence disclosed herein is in an UNC13A pre-mRNA. In some embodiments, a target sequence disclosed herein is in an intron of an UNC13A pre-mRNA. In some embodiments, a target sequence disclosed herein is in intron 20 of UNC13A pre-mRNA. In some embodiments, a target sequence disclosed herein is in or near (e.g., within 100 nucleobases upstream or downstream of) a cryptic exon in intron 20 of UNC13A pre-mRNA.
[0113] The term “treat,” “treatment,” as used herein, mean the methods or steps taken to provide relief from or alleviation of the number, severity, and/or frequency of one or more symptoms of a disease (e.g., UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) or a disease associated with TDP-43 dysfunction) in a subject. As used herein, “treat” and treatment” may include the prevention, management, prophylactic treatment, and/or inhibition of the number, severity, and/or frequency of one or more symptoms of a disease (e.g., UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) or a disease associated with TDP-43 dysfunction) in a subject.
[0114] The term “variant” means a molecule (e.g., nucleic acid or polypeptide) that differs from a given molecule (e.g., a reference nucleic acid or polypeptide) in sequence (nucleic acid or amino acid respectively) by the addition (e.g., insertion), deletion, or conservative substitution of nucleic acids or amino acids, respectively, but that retains the biological activity of the given molecule. Changes in the reference nucleic acid sequence of the variant may be silent. That is, they may not alter the amino acid sequence encoded by the nucleic acid. Alternatively, changes in the nucleoside sequence of the variant may alter the amino acid sequence of a polypeptide encoded by the reference polynucleotide. Such nucleoside changes may result in amino acid substitutions, additions, deletions, fusions, and truncations in the polypeptide encoded by the reference sequence. The term

“variant” encompasses fragments of a variant unless otherwise defined. A variant may be 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, 90%, 89%, 88%, 87%, 86%, 85%, 84%, 83%, 82%, 81%, 80%, 79%, 78%, 77%, 76%, or 75% identical to the reference sequence. The degree of homology (percent identity) between a native and a variant sequence can be determined, for example, by comparing the two sequences using freely available computer programs commonly employed for this purpose on the world wide web (e.g., BLASTn with default settings).

[0115] Definitions of specific functional groups and chemical terms are described in more detail below. The chemical elements are identified in accordance with the Periodic Table of the Elements, CAS version, *Handbook of Chemistry and Physics*, 75.sup.th Ed., inside cover, and specific functional groups are generally defined as described therein. Additionally, general principles of organic chemistry, as well as specific functional moieties and reactivity, are described in *Organic Chemistry*, Thomas Sorrell, University Science Books, Sausalito, 1999; Smith and March, *March's Advanced Organic Chemistry*, 5.sup.th Edition, John Wiley & Sons, Inc., New York, 2001; Larock, *Comprehensive Organic Transformations*, VCH Publishers, Inc., New York, 1989; and Carruthers, *Some Modern Methods of Organic Synthesis*, 3.sup.rd Edition, Cambridge University Press, Cambridge, 1987.

[0116] It is also to be understood that compounds that have the same molecular formula but differ in the nature or sequence of bonding of their atoms or the arrangement of their atoms in space are termed “isomers”. Isomers that differ in the arrangement of their atoms in space are termed “stereoisomers”.

[0117] Stereoisomers that are not mirror images of one another are termed “diastereomers” and those that are non-superimposable mirror images of each other are termed “enantiomers”. When a compound has an asymmetric center, for example, it is bonded to four different groups, a pair of enantiomers is possible. An enantiomer can be characterized by the absolute configuration of its asymmetric center and is described by the R- and S-sequencing rules of Cahn and Prelog, or by the manner in which the molecule rotates the plane of polarized light and designated as dextrorotatory or levorotatory (i.e., as (+) or (−)-isomers respectively). A chiral compound can exist as either individual enantiomer or as a mixture thereof. A mixture containing equal proportions of the enantiomers is called a “racemic mixture”.

[0118] The term “TDP-43,” as used herein, means an RNA binding protein encoded by the TARDBP gene. TDP-43 has important role in RNA-processing activities and splicing regulatory function as a repressor of cryptic exon inclusion during RNA splicing. A key feature of neurodegenerative diseases, amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), Alzheimer's disease, and Limbic-Predominant Age-Related TDP-43 Encephalopathy (LATE) is mislocalization of TDP-43 from the nucleus and aggregated in the cytoplasm in human brain, neuronal cell lines motor neurons, and glial cells. It has been shown that TDP-43 dysfunction results in the inclusion of a cryptic exon in UNC13A mRNA and reduction of UNC13A protein expression. Diseases associated with TDP-43 dysfunction in a cell include the neurodegenerative diseases of ALS, FTD, Alzheimer's disease, and LATE.

[0119] The term “UNC-13 homolog A,” used interchangeably with the term “UNC13A,” refers to the well-known gene and polypeptide, also known in the art as Munc 13-1.

[0120] The term “UNC13A” includes human (*Homo sapiens*) UNC13A, the amino acid and nucleotide sequences of which may be found in, for example, GenBank Accession No. Gene ID: 23025, and NCBI Accession Nos. NG_052872.1: 4991 . . . 92009 (SEQ ID NO: 318), NC_000019.10: 17601336 . . . 17688354 complement (Reference GRCh38.p14 Primary Assembly) (SEQ ID NO: 318), NM_001080421.3 (SEQ ID NO: 307), and NP_001073890.2 (SEQ ID NO: 308). The term “UNC13A” also includes cynomolgus monkey (*Macaca fascicularis*) UNC13A, the amino acid and nucleotide sequences of which may be found in, for example, GenBank Accession No. GI: 102123626 and NCBI Accession Nos. NC_052273.1:17161646 . . . 17254007 complement (Reference MFA1912RKSv2 Primary Assembly) (SEQ ID NO: 319), XM_045380772.1 (SEQ ID

NO: 309) and XP_045236707.1 (SEQ ID NO: 310). Additional examples of UNC13A RNA sequences are readily available using, e.g., GenBank, UniProt, and OMIM, and the *Macaca* genome project web site. Exemplary UNC13A nucleotide and amino acid sequences may also be found in Table 1, SEQ ID NOs: 307-310.

[0121] The term "UNC13A," as used herein, also refers to naturally occurring DNA sequence variations of the UNC13A gene. Numerous sequence variations within the UNC13A gene (e.g., SNPs rs12608932, rs12973192, rs56041637, rs116169349, rs62121687) have been identified and may be found at, for example, NCBI dbSNP and UniProt (see, e.g., ncbi.nlm.nih.gov/snp).

[0122] Further information on UNC13A can be found, for example, at ncbi.nlm.nih.gov/gene/23025. The entire contents of each of the foregoing Accession numbers and the Gene database numbers are incorporated herein by reference as of the date of filing this application.

[0123] Table 1 below summarizes exemplary amino acid sequences of UNC13A proteins and DNA sequences of UNC13A genes of human and cynomolgus monkey.

TABLE-US-00001 TABLE 1 Exemplary Amino Acid and DNA Sequences of UNC13A Sequence SEQ ID NO Human

GCCCCCGGTGCTGAACCAAGATGGCCGGTGGCGGCCGGGCC 307 UNC13A
CCGGCGTGAGCCAAGCGCGGGCTGCAGCCGGGAGATGCCCC nucleotide
AGCCCAGCGGCCGCTGAGCCCGACCCGACAGAGCCGGCCCG (NM_001080421.3)
GCCGCCTCCGGCCCACCTGCGAGCTCGGAGACATGTCTCTGC
TTTGCGTTGGAGTCAAAAAAGCCAAGTTTGATGGTGCCCAAG
AGAAATTCAACACGTACGTGACCCTGAAAGTGCAGAATGTC
AAGAGCACGACCATCGCGGTGCGGGGCAGCCAGCCCAGCTG
GGAGCAGGATTTCATGTTCGAGATTAACCGTCTGGATTTGGG
ACTGACGGTGGAGGTGTGGAATAAGGGTCTCATCTGGGACA
CAATGGTGGGCACTGTGTGGATCCCACTGAGGACCATCCGCC
AGTCCAATGAGGAGGGCCCTGGAGAGTGGCTGACGCTGGAC
TCCCAGGTCATCATGGCAGACAGTGAGATCTGTGGCACCAA
GGACCCCACCTTCCACCGCATCCTCCTGGACACGCGCTTTGA
GCTACCCTTAGACATTCCTGAAGAGGAGGCTCGCTACTGGGC
CAAGAAGCTGGAGCAGCTCAATGCTATGCGGGACCAGGATG
AATATTCGTTCCAAGATGAGCAAGACAAGCCTCTGCCTGTCC
CCAGCAACCAGTGCTGCAACTGGAATTATTTTGGCTGGGGTG
AGCAGCACAACGATGACCCCGACAGTGCAGTGGATGATCGT
GACAGTGACTACCGCAGTGAAACGAGCAACAGCATCCCGCC
GCCCTATTATACTACGTCACAACCCAACGCCTCAGTCCACCA
ATATTCTGTTCGCCCACCACCCCTGGGCTCCCGGGAGTCCTA
CAGTGACTCCATGCACAGTTACGAGGAGTTCTCTGAGCCACA
AGCCCTCAGCCCCACGGGTAGCAGCCGCTATGCCTCTTCCGG
GGAGCTGAGCCAGGGAAGCTCTCAGCTGAGCGAGGACTTCG
ACCCTGACGAGCACAGCCTGCAGGGCTCCGACATGGAGGAT
GAGCGGGACCGGGACTCCTACCACTCCTGCCACAGCTCGGTC
AGCTACCACAAAGACTCGCCTCGCTGGGACCAGGATGAGGA
AGAGCTGGAGGAGGACCTGGAGGACTTCCTGGAGGAGGAGG
AGCTGCCTGAAGATGAGGAGGAGCTGGAGGAGGAGGAGGA
GGAGGTGCCTGACGATTTGGGCAGCTATGCCCAGCGTGAAG
ACGTAGCTGTGGCTGAGCCCAAAGACTTCAAACGCATCAGC
CTCCCGCCAGCTGCCCCAGGGAAGGAGGACAAGGCCCCAGT
GGCACCCACCGAGGCCCCCGACATGGCCAAGGTGGCCCCCA
AGCCAGCCACGCCCGACAAGGTGCCTGCAGCTGAGCAGATC
CCTGAGGCTGAGCCACCCAAGGACGAGGAGAGTTTCAGGCC

GAGAGAGGATGAGGAAGGCCAGGAGGGGCAGGACTCCATG
TCCAGGGCCAAGGCCAACTGGCTGCGTGCCTTCAACAAGGT
GCGGATGCAGCTGCAGGAGGCCCGGGGAGAAGGAGAGATGT
CTAAATCCCTATGGTTCAAAGGCGGCCCAGGGGGCGGTCTCA
TCATCATCGACAGCATGCCAGACATCCGCAAGAGGAAACCT
ATCCCACTCGTGAGCGACTTGGCCATGTCCCTGGTCCAGTCC
AGGAAAGCGGGCATCACCTCGGCCTTGGCCTCCAGCACGTTG
AACAACGAGGAGCTGAAAAACCACGTTTACAAGAAGACCCT
GCAAGCCTTAATCTACCCCATCTCGTGCACGACGCCACACAA
CTTCGAAGTGTGGACGGCCACCACGCCCACCTACTGCTACGA
GTGCGAGGGGCTGCTGTGGGGCATCGCGAGGCAGGGCATGC
GCTGCACCGAGTGCGGTGTCAAGTGCCACGAGAAGTGCCAG
GACCTGCTCAACGCCGACTGCCTGCAGCGGGCTGCGGAGAA
GAGCTCCAAGCACGGGGCGGAGGACCGGACACAGAACATCA
TCATGGTGCTCAAGGACCGCATGAAGATCCGGGAGCGCAAC
AAGCCCGAGATCTTCGAGCTCATCCAGGAGATCTTCGCGGTG
ACCAAGACGGCGCACACGCAGCAGATGAAGGCGGTCAAGCA
GAGCGTGCTGGACGGCACGTCCAAGTGGTCCGCCAAGATCA
GCATCACCGTGGTCTGCGCCCAGGGCTTGCAGGCAAAGGAC
AAGACAGGATCCAGTGACCCCTATGTCACCGTCCAGGTCGG
GAAGACCAAGAAACGGACAAAAACCATCTATGGGAACCTCA
ACCCGGTGTGGGAGGAGAATTTCCACTTTGAATGTCACAATT
CCTCCGACCGCATCAAGGTGCGCGTCTGGGACGAGGATGAC
GACATCAAATCCCGCGTGAAACAGAGGTTCAAGAGGGAATC
TGACGATTTCCTGGGGCAGACGATCATTGAGGTGCGGACGCT
CAGCGGCGAGATGGACGTGTGGTACAACCTGGACAAGCGAA
CTGACAAATCTGCCGTGTCGGGTGCCATCCGGCTCCACATCA
GTGTGGAGATCAAAGGCGAGGAGAAGGTGGCCCCGTACCAT
GTCCAGTACACCTGTCTGCATGAGAACCTGTTCCACTTCGTG
ACCGACGTGCAGAACAATGGGGTCGTGAAGATCCCAGATGC
CAAGGGTGACGATGCCTGGAAGGTTTACTACGATGAGACAG
CCCAGGAGATTGTGGACGAGTTTGCCATGCGCTACGGCGTCG
AGTCCATCTACCAAGCCATGACCCACTTTGCCTGCCTCTCCTC
CAAGTATATGTGCCCAGGGGTGCCTGCCGTCATGAGCACCCT
GCTCGCCAACATCAATGCCTACTACGCACACACCACCGCCTC
CACCAACGTGTCTGCCTCCGACCGCTTCGCCGCCTCCAACTT
TGGGAAAGAGCGCTTCGTGAAACTCCTGGACCAGCTGCATA
ACTCCCTGCGGATTGACCTCTCCATGTACCGGAATAACTTCC
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AAAGATTCTGAAGGCTGTATTGAAAAGACACCTATAACTGTG
GACATGGCTATAATCCCAGCATTTTGGGGAGGCTGAGGCTGG
TGGATCATTTAAGGTCAGGAGTTTGAGACCAGCCTGGCCAAC
GTGGTGAAACCCTGTCTCTGCTAAAAATACAAAAATTAGCTG
GGTGTGGTGGTGCATGCCTGTAATCCCAGCTACACAGGAGAC
TGAGGTGGGAGAATTGCTTGAACCCTGGAGGCAGAGGTTGC
AGTGAGCCAAGATCACACCACTGCACTCCAGCCTGGGTAAC
AGACCAATACTCCGTTCCCTCCCCTCTACCCCCCCCAAAAAA
AAAAAAAATCCTGCGCTTAGATGAGCTCCATGGCTGCTGAGT
ACAGTTGTCCCAGTTGCACACTGCCCAAGGGTTTGGCATCGC
TAAGAAGGCCACGTGCAAATCCTAGACATTGAGTGTTGTATG
TTTATGACCTTGGTGTCCCAACATGTGAATGGCCCAAGTGTC
TGGAAGAAGTGGCGCCACTTTCTAATTTGCTTGAAGGTGTTG
CGTGTCCCCTAAATTCAGACAGATGCAGGTAACTGGAGGTTC
TGAACCAAAGGCTAAAATGCAAATTCTCATAAAGGATTTGG
GAAGCTGTAGCCAGGGATAAGCTTATGTGACTGTAATATGG
GCTGAGGAGCAGATGTGAATTTCGGACCATGCCATGCCTGA
GGGTGGGGGTCGGGTGGATGGATGATTCAGGGTTGTAACCC
ATAGAGCCCAAAGGGGAAGTTATCTGTGACCTGTGGTGAGG
GTGACCTGGAACATTTTTGGATGGCTGGAAAGAAATGGGGA
AATTGAGCTGCCTGAGATAGTCAGGATATTTCCCAAAAGATT
CCTTAGGAGTCTTTCTGGTCAAGACGTGTGTGTGTGTGTGTG
TGTGTGTTTAGGGTTCCCCAAGAATGGCTCAGGCATGTGAAG
GAAACAAGCTTCAGGGAATATTTGTTGAATGAGTTTTCCTGC
CTCCCAGGCTAGAACTGTTTTTGCAATTTCCACCCTCTTTTCT
TTCCCTCCCAGAGAACTCCTATTCATCCTTTGAAACCCATCAT
GAAAATCCCTCTTGGAGAAAACCCTCCTTCCTTCCCCTCAGG
ACTTTCCCAGCCTCCATCCCTCCTCCAGTCCAGCCTGATGCCA
TGGGACTGGGGTTTCTCTCTTCAGCTCTGTTTCTCCCAGACTG
GGGTCTGAGGCCTCTCAGGACCCCCAACTTTACCTAGCACAG
GTTGGGCACAGGTGGATGGCAGGGAGTCTGTGCCTAGTGGA
ATTATGTGTTGGGGCAGTGTCAGTGCGAGAATACACATCTGC

ATGTGTATCTGTCCATGTCTGTCCATACCAACCTTCCCCTTCC
ACACGGACCTGGGCACATAGGAGGTGTCTGAGCCCAACACA
TGGGACAGAGAGTGGACACGGCTGAGACACGGACAGAGAC
AAGACAAGGAGTCCAGGCGGCTGAAAGCCTTTTGAAATCAG
GAAGTTCCTGGATTGGCAGAACAAAGCCCAGAGAGGAGCAG
GGCTTTCCCCATCGCCACCCAGCAAGTGGCCACAGAGCCCAG
CCTTGGATGACACCTCCAGGATTCTGAACCCTGGACCTCACT
TTATGCAAGGAGCTGGCCGCAGATTTCCATGAATCAGGGAA
ACAGCACAAGAGGGTTGGCCTGTGGCAGGGCAGGGATTAAA
GGGGTAACATTGAGGGATGCCTCAGAGTCAAAGTCCCCTGA
CCAAGAGGAATAAAGAGTAGAAGGCACAGAGAGGGTGAGA
TGACGCCCTGATGAGAACGGAGAGAGACATAGAGGTGGAAA
TATACAGAGAAAGAGAAATGCAGAGACCAGGGCAGGGGGT
GTCGGGGAAAGTAAAGAGGGTGTCCCGAGGAAAGAAGGCTC
TGTCCATTGTCACCAGTCTGTCCTCGAATGATTTGCATAAAA
TGAGGAGGTGCCTGTCCATGCCCCCAATTCCTCTCTCAGGTG
CCAGAGCCTGAGACCTCACCATGCCCCCATCAGAGATGCAA
AAAACTAAACACCCAACTAGAAATCCTTGGGACCTCTCTCGG
CTGGGATCTTAGAGCCTTTCTGTCCCCGACCCCTACCCCATGT
GCTGTCGATTTTGCAGATGGGGACAACCTGGGGCCTCCCGGA
ACTCTGCCACCCTGGGGAAGTTGGGGGGAGGGCGTTGGTCC
CGGATCACAACCCCGTCTGCTCCCCAGAATCCTTTCCTAAGA
ATCGTTGAGGACCAAAGTCGTCTTTGCTGACACGTGTTGCTT
TTCTCTTTGCCTTTTATTGTTTTAGAGAAAAATCAAGTTGACT
GTGTCAAGTAACACCCCACCCCCTTACCCCCGTCCAGCCATA
GTGGCTCTCTGGAGACACAGGTCACAGGCGGAGGGTCCCCT
GATCATCCCTGACCACACAGCCAGGGGTGTTTGACCCCTGTC
CACCCCTGTCTCGTGCTCCCTCAGACCCCCACAAACCGGCCA
AGCAGTCCGGGGATGCTTCCCCTCCACACAACTTCTCTTAGC
ATGTGATTGCAGATGTGAAATCAAAGCGTTGTTTGTTTTTTGT
TTTGTTTTGATTCCACCCCGTCGGTCTGGTGTCTGCACAGACG
CCTTCATTTCTCTGTAAATATGTGACTTGGAACAAATGTTTAA
CACAAACGAGAAGTGGTCATGAATGCATGGTGTTGAGATGT
TTTGCACTATTCTGACTTTTTGGTCTCTGTAAAAATATTTTAT
TAACAGCAGACATTAAAAAAAGAAAAACCACA Cynomolgus
MSLLCVGEKFNTYVTLKVQNVKSTTIAVRGSQPSWEQDFMFEI 310 monkey
NRLDLGLTVEVWNKGLIWDTMVGTVWIPLRTIRQSNEEGPGE UNC13A
WLTLDSQVIMADSEICGTKDPTFHRILLDTRFELPLDIPEEEARY Protein
WAKKLEQLNAMRDQDEYSFQDEQDKPLPVPSNQCCNWNYFG (XP_045236707.1)
WGEQHNDDPDSAVDDRDSDYRSETSNSIPPPYYTTSQPNASVH
QYSVRPPPLGSRESYSDSMHSYEEFSEPQALSPTGSSRYASSAEL
SQGSSQLSEDFDPDEHSLQGSEMEDERDRDSYHSCHSSVSYHK
DSPRWDQDEEELEEDLEDFLEEEELPEDEEELEEEEEVPDDLGS
YAQREEVAVAEPKDFKRISLPPAAPGKEDKTPLAPTEAPDTAKV
APKPATPEEVPAAEQIPEAEPPKAEESFRPREDEEGQEGQDSMSR
AKANWLRAFNKVRMQLQEARGEGEMSKSLWFKGGPGGGLIII
DSMPDIRKRKPIPLVSDLAMSLVQSRKAGITSALASSTLNNEELK
NHVYKKTLQALIYPISCTTPHNFEVWTATTPTYCYECEGLLWGI
ARQGMRCTECGVKCHEKCQDLLNADCLQRAAEKSSKHGAEDR
TQNIIMVLKDRMKIRERNKPEIFELIQEIFAVTKTAHTQQMKAV

KQSVLDGTSKWSAKISITVVCAQGLQAKDKTGSSDPYVTVQVG
KTKKRTKTIYGNLNPVWEENFHFECHNSSDRIKVRVWDEDDDI
KSRVKQRFKRESDDFLGQTIIEVRTLSGEMDVWYNLDKRTDKS
AVSGAIRLHISVEIKGEEKVAPYHVQYTCLHENLFHFVTDVQNN
GVVKIPDAKGDDAWKVYYDETAQEIVDEFAMRYGVESIYQAM
THFACLSSKYMCPGVPAVMSTLLANINAYYAHTTASTNVSASD
RFAASNFGKERFVKLLDQLHNSLRIDLSMYRNNFPASSPERLQD
LKSTVDLLTSITFFRMKVQELQSPPRASQVVKDCVKACLNSTYE
YIFNNCHELYSREYQTDPAKKGEVPPEEQGPSIKNLDFWSKLITL
IVSIIEEDKNSYTPCLNQFPQELNVGKISAEVMWNLFAQDMKYA
MEEHDKHRLCKSADYMNLHFKVKWLYNEYVTELPAFKDRVPE
YPAWFEPFVIQWLDENEEVSRDFLHGALERDKKDGFQQTSEHA
LFSCSVVDVFSQLNQSFEIIKKLECPDPQIVGHYMRRFAKTISNV
LLQYADIISKDFASYCSKEKEKVPCILMNNTQQLRVQLEKMFEA
MGGKELDAEASDILKELQVKLNNVLDELSRVFATSFQPHIEECV
KQMGDILSQVKGTGNVPASACSSVAQDADNVLQPIMDLLDSNL
TLFAKICEKTVLKRVLKELWKLVMNTMEKTIVLPPLTDQTMIG
NLLRKHGKGLEKGRVKLPSHSDGTQMIFNAAKELGQLSKLKDH
MVREEAKSLTPKQCAVVELALDTIKQYFHAGGVGLKKTFLEKS
PDLQSLRYALSLYTQATDLLIKTFVQTQSAQVHGGKGTRFTLSE
DIYPEKGLGVEDPVGEVSVHVELFTHPGTGEHKVTVKVVAAND
LKWQTSGIFRPFIEVNIIGPQLSDKKRKFATKSKNNSWAPKYNES
FQFTLSADAGPECYELQVCVKDYCFAREDRTVGLAVLQLRELA
QRGSAACWLPLGRRIHMDDTGLTVLRILSQRSNDEVAKEFVKL KSDTRSAEEGGAAPAP

[0124] As used herein, “intron 20” of UNC13A refers to the intron between exon 20 and exon 21 of UNC13A pre-mRNA (e.g., as described in Brown et al., Nature. 2022 March; 603(7899): 131-137). Intron 20 of UNC13A is also sometimes known in the art as intron 20-21 (see, for example, Ma et al. TDP-43 represses cryptic exon inclusion in the FTD-ALS gene UNC13A. Nature 603, 124-130 (2022)). In some embodiments, intron 20 of UNC13A corresponds to positions 45511-46798 of an UNC13A gene sequence as set forth in accession numbers: NC_000019.10:17601336 . . . 17688354 complement and NG_052872.1:4991 . . . 92009 (human UNC13A). In some embodiments, intron 20 of UNC13A comprises the nucleotide sequence set forth in SEQ ID NO: 311. One skilled in the art would understand that in an UNC13A pre-mRNA, the intron 20 sequence is SEQ ID NO: 311 with each thymine replaced with uracil (see SEQ ID NO: 325).

TABLE-US-00002 (SEQ ID NO: 311) <u>gtgagggtcattgctcggcccctcccatgccacttccactcaccattcct</u>
<u>gcctgcccagctcttcctctttctggccacaccatccacactctcctggc</u>
<u>cctctgagactgcccgccatgccattccctttacctggaaaactcctccc</u>
<u>tatccatcaaagtccagattcagggtcacctcctctgggaagcccacctt</u>
<u>ggcctccaggttgactctcactactcatcatcaggttcttccttctattc</u>
<u>cag*ccctaaccactcaggattgggccgtttgtgtctgggtatgtctcttc*</u>
<u>*cagCTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGAACTAGTTTG*</u>
<u>*TTGAATAAATGCTGGTGAATGAATGAATGATTGAACAGATGAATGAGTGA*</u>
<u>*TGAGTAGATAAAAGGATGGATGGAGAGATGG*</u>gtgagtacatggatggata
gatggatgagttggtgggtagattcgtggctagatggatgatggatggat
ggacagatggatggatatatgattgaactattgaaagtatagatgtatgg
atgggtgaatttgggggtaattgttagatgatggatgagtatagatgaat
gatggatggataacttgatgagtggatagatagattgctggatagatgat
tgactgggtggatagatgaaatgttggatgagcagattaagttgtattgg
atgggatggatggaagtgtggttgagttattagaaggaagattgagtaga
taggtgaatttgttgatagtcagatgggtagataggtagatggatggatg

gatggatggatgtataggcagatggacaaatggatgaatgggtgggtgga
tgaatggaaggatgtgtggttgaactattgcaagtattgataattgggtt
cataatttctgaatatttagatggatggttgtgagtggctggtggacaga
cgaaaaatggatggttggataaattgatgggtggatggatggttggttgt
atgaaagaatgaatgattgggtaggtggattaagttgcggatcaatgtat
gggatggatgaatggatggatggatggatgtgtggttgaattactgaaag
gttggaagagtggatgggtgaaatttggggtagttagatgggtgggtgtg
tggatggataaaagagtagatgaatgaattaatgaataaacaggcagatg
gatgatgtaagctgccccagaccctgggacctctgacccccggcgacccc ttgcactctccatgacactttctctcccatggtggcag
(SEQ ID NO: 325) gugagggucauugcucggccccucccaugccacuuccacucaccauuccu
gccugcccagcucuuccucuuucuggccacaccauccacacucuccuggc
ccucugagacugcccgccaugccauucccuuuaccuggaaaacuccuccc
uauccaucaaaguccagauucagggucaccuccucugggaagcccaccuu
ggccuccagguugacucucacuacucaucaucagguucuuccuucuauuc
cag*cccuaaccacucaggauugggccguuugugucuggguaugucucuuc*
*cag*CUGCCUGGGUUUCCUG*GAAAGAACUCUUAUCCCCAGGAACUAGUUUG*
*UUGAAUAAAUGCUGGUGAAUGAAUGAAUGAUUGAACAGAUGAAUGAGUGA*
*UGAGUAGAUAAAAGGAUGGAUGGAGAGAUGG*gugaguacauggauggaua
gauggaugaguuggugggguagauucguggcuagauggaugauggauggau
ggacagauggauggauauaugauugaacuauugaaaguauagauguaugg
augggugaauuuggggguaauuguuagaugauggaugaguauagaugaau
gauggauggauaacuugaugaguggauagauagauugcuggauagaugau
ugacuggguggauagaugaaauguuggaugagcagauuaaguuguauugg
augggauggauggaagugugguugaguuauuagaaggaagauugaguaga
uaggugaauuuguugauagucagauggguagauagguagauggauggaug
gauggauggauguauaggcagauggacaaauggaugaaugggugggugga
ugaauggaaggauguguggguugaacuauugcaaguauugauaauuggguu
cauaauuucugaauauuuagauggaugguugugaguggcugguggacaga
cgaaaaauggaugguuggauaaauugauggguggauggaugguugguugu
augaaagaaugaaugauuggguagguggauuaaguugcggaucaauguau
gggauggaugaauggauggauggauggaugugugguugaauuacugaaag
guuggaagaguggaugggugaaauuugggguaguuagauggguggguguguu
ggauggauaaaagaguagaugaaugaauuaaugaauaaacaggcagaugg
augauguaagcugccccagacccugggaccucugacccccggcgaccccu
ugcacucuccaugacacuuucucucccaugguggcag
[0125] The term “cryptic exon,” as used herein, refers to intronic sequences that are erroneously included in mature mRNA as a result of a splicing defect. Cryptic exons may introduce frameshifts or stop codons, among other changes in the resulting mRNA and can arise through single nucleotide polymorphisms (SNPs) in acceptor/donor splice sites or the activity of spliceosome or RNA-binding proteins such as TDP-43. Cryptic exons can be found in patient tissues from people affected by ALS, FTD, Alzheimer's disease, and LATE. In some embodiments, cryptic exons can be found in patient tissues from people affected by frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression. In some

embodiments, a cryptic exon disclosed herein is 128 nucleosides in length. In some embodiments, a cryptic exon disclosed herein is 178 nucleosides in length. In some embodiments, a cryptic exon disclosed herein is 431 nucleosides in length.

[0126] In some embodiments, a cryptic exon disclosed herein comprises the sequence set forth in SEQ ID NO: 312 and SEQ ID NO: 326 (see also, nucleosides 304-431 in SEQ ID NO: 311 and SEQ ID NO: 325; capitalized sequence):

TABLE-US-00003 (SEQ ID NO: 312)
CTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGAACTAGTTTGTTG
AATAAATGCTGGTGAATGAATGAATGATTGAACAGATGAATGAGTGATGA
GTAGATAAAAGGATGGATGGAGAGATGG (SEQ ID NO: 326)
CUGCCUGGGUUUCCUGGAAAGAACUCUUAUCCCCAGGAACUAGUUUGUUG
AAUAAAUGCUGGUGAAUGAAUGAAUGAUUGAACAGAUGAAUGAGUGAUGA
GUAGAUAAAAGGAUGGAUGGAGAGAUGG

[0127] In some embodiments, a cryptic exon disclosed herein comprises the sequence set forth in SEQ ID NO: 334 and SEQ ID NO: 335 (see also, nucleosides 254-431 in SEQ ID NO: 311 and SEQ ID NO: 325; italicized sequence):

TABLE-US-00004 (SEQ ID NO: 334)
CCCTAACCACTCAGGATTGGGCCGTTTGTGTCTGGGTATGTCTCTTCCAG
CTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGAACTAGTTTGTTG
AATAAATGCTGGTGAATGAATGAATGATTGAACAGATGAATGAGTGATGA
GTAGATAAAAGGATGGATGGAGAGATGG (SEQ ID NO: 335)
CCCUAACCACUCAGGAUUGGGCCGUUUGUGUCUGGGUAUGUCUCUUCCAG
CUGCCUGGGUUUCCUGGAAAGAACUCUUAUCCCCAGGAACUAGUUUGUUG
AAUAAAUGCUGGUGAAUGAAUGAAUGAUUGAACAGAUGAAUGAGUGAUGA
GUAGAUAAAAGGAUGGAUGGAGAGAUGG

[0128] In some embodiments, a cryptic exon disclosed herein comprises the sequence set forth in SEQ ID NO: 336 and SEQ ID NO: 337 (see also, nucleosides 1-431 in SEQ ID NO: 311 and SEQ ID NO: 325; underlined sequence):

TABLE-US-00005 (SEQ ID NO: 336)
GTGAGGGTCATTGCTCGGCCCCTCCCATGCCACTTCCACTCACCATTCCT
GCCTGCCCAGCTCTTCCTCTTTCTGGCCACACCATCCACACTCTCCTGGC
CCTCTGAGACTGCCCGCCATGCCATTCCCTTTACCTGGAAAACTCCTCCC
TATCCATCAAAGTCCAGATTCAGGGTCACCTCCTCTGGGAAGCCCACCTT
GGCCTCCAGGTTGACTCTCACTACTCATCATCAGGTTCTTCCTTCTATTC
CAGCCCTAACCACTCAGGATTGGGCCGTTTGTGTCTGGGTATGTCTCTTC
CAGCTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGAACTAGTTTG
TTGAATAAATGCTGGTGAATGAATGAATGATTGAACAGATGAATGAGTGA
TGAGTAGATAAAAGGATGGATGGAGAGATGG (SEQ ID NO: 337)
GUGAGGGUCAUUGCUCGGCCCCUCCCAUGCCACUUCCACUCACCAUUCCU
GCCUGCCCAGCUCUUCCUCUUUCUGGCCACACCAUCCACACUCUCCUGGC
CCUCUGAGACUGCCCGCCAUGCCAUUCCCUUUACCUGGAAAACUCCUCCC
UAUCCAUCAAAGUCCAGAUUCAGGGUCACCUCCUCUGGGAAGCCCACCUU
GGCCUCCAGGUUGACUCUCACUACUCAUCAUCAGGUUCUUCCUUCUAUUC
CAGCCCUAACCACUCAGGAUUGGGCCGUUUGUGUCUGGGUAUGUCUCUUC
CAGCUGCCUGGGUUUCCUGGAAAGAACUCUUAUCCCCAGGAACUAGUUUG
UUGAAUAAAUGCUGGUGAAUGAAUGAAUGAUUGAACAGAUGAAUGAGUGA
UGAGUAGAUAAAAGGAUGGAUGGAGAGAUGG

[0129] As used herein, the term “UNC13A-associated disease,” is a disease or disorder that is caused by, or associated with, abnormal UNC13A expression and/or activity. The term “UNC13A-associated disease” includes a disease, disorder or condition that would benefit from inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell. In some embodiments, a subject having an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) may benefit from inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell in the subject. In some embodiments, an UNC13A-associated disease is a motor neuron disease that is characterized by neuronal loss. An UNC13A-associated disease may be neurodegenerative disease. In some embodiments, an UNC13A-associated disease is ALS. In some embodiments, an UNC13A-associated disease is FTD. In some embodiments, an UNC13A-associated disease is Alzheimer's disease. In some embodiments, an UNC13A-associated disease is LATE. In some embodiments, an UNC13A-associated disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0130] Further details regarding signs and symptoms of the various diseases or conditions are provided herein and are well known in the art.

Compositions

UNC13A Antisense Oligonucleotides

[0131] Some aspects of the present disclosure provide antisense oligonucleotides that target UNC13A and/or modulates UNC13A splicing and/or expression. In some embodiments, an antisense oligonucleotide disclosed herein comprises sequences complementary to an UNC13A sequence (e.g., UNC13A gene sequence or pre-mRNA sequence). In some embodiments, an antisense oligonucleotide disclosed herein targets more than one (e.g., two or more) target regions of UNC13A (e.g., UNC13A gene or pre-mRNA). In some embodiments, an antisense oligonucleotide disclosed herein is designed for splicing modulation of an UNC13A pre-mRNA (e.g., to inhibit the inclusion of a cryptic exon present in UNC13A pre-mRNA). In some embodiments, an antisense oligonucleotide disclosed herein is designed for modulation (e.g., restoring) of UNC13A gene expression (e.g., RNA expression and/or protein expression) and/or function.

[0132] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more regions of complementarity to one or more (e.g., one, two, or more) target regions in a human UNC13A sequence (e.g., as set forth in SEQ ID NO: 318). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more regions of complementarity to one or more (e.g., one, two, or more) target regions in in an intron (e.g., intron 20 such as the intron 20 as set forth in SEQ ID NO: 311) of a human UNC13A sequence (e.g., as set forth in SEQ ID NO: 318). In some embodiments, an antisense oligonucleotide disclosed herein comprises regions of complementarity to one or more (e.g., one, two, or more) target regions in intron 20 (e.g., the intron as set forth in SEQ ID NO: 311) of an UNC13A sequence (e.g., as set forth in SEQ ID NO: 318), wherein each target region each comprises an UNC13A sequence that modulates splicing of an UNC13A cryptic exon (e.g., an UNC13A cryptic exon present in intron 20 such as the cryptic exon as set forth in SEQ ID NO: 312, SEQ ID NO: 334, or SEQ ID NO: 336). In some embodiments, an antisense oligonucleotide described herein inhibits inclusion of an UNC13A cryptic exon (e.g., an UNC13A cryptic exon present in intron 20) into an UNC13A mature mRNA.

[0133] In some embodiments, an antisense oligonucleotide disclosed herein targets one or more (e.g., one, two, or more) target regions in intron 20 (e.g., the intron 20 as set forth in SEQ ID NO: 311) of an UNC13A sequence in or near a cryptic exon (e.g., the cryptic exon as set forth in SEQ ID NO: 312, SEQ ID NO: 334, or SEQ ID NO: 336). “Near” a position or a sequence, as used herein, means within 125 (e.g., 125, 120, 115, 110, 105, 100, 95, 90, 85, 80, 75, 70, 65, 60, 55, 50, 45, 40, 35, 30, 25, 20, 15, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1) nucleosides upstream (5′ end) or downstream (3′ end) of the position or sequence. "Upstream" of a position or a sequence, as used herein, means at the 5′ side of the position or sequence. "Downstream" of a position or sequence, as used herein, means at the 3′ side of the position or sequence. For example, a target region "near" a cryptic exon in intron 20 of UNC13A sequence means the target region is within 125 (e.g., 125, 120, 115, 110, 105, 100, 95, 90, 85, 80, 75, 70, 65, 60, 55, 50, 45, 40, 35, 30, 25, 20, 15, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1) nucleosides upstream (e.g., counting towards the 5′ end from the 5′ terminal nucleotide of the cryptic exon) or downstream (e.g., counting towards the 3′ end from the 3′ terminal nucleotide of the cryptic exon) of the cryptic exon in intron 20 of UNC13A sequence.

[0134] As used herein, a "target region" is represented by a nucleotide sequence in a reference sequence. For example, for target regions 1-4 provided in Table 2, the reference sequence used is human UNC13A sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009). For the purposes of the present disclosure, when referring to antisense oligonucleotides that target a target region in a reference sequence (e.g., in human UNC13A sequence as set forth in NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; SEQ ID NO: 318), such antisense oligonucleotides encompass antisense oligonucleotides that target the target region in the reference sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009), and antisense oligonucleotides that target a corresponding target region in another UNC13A sequence (e.g., gene sequence, pre-mRNA sequence, or a variant or a homologue of human UNC13A, e.g., from a closely related species, such as a cynomolgus monkey).

[0135] To identify a target region in another UNC13A sequence that corresponds to a certain target region in the reference sequence (e.g., SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009), such other UNC13A sequences can be aligned to the reference sequence and the corresponding target region can be determined. For example, in some embodiments, a target region in human UNC13A comprises a nucleotide sequence as set forth in SEQ ID NO: 327 (CTGGGTATGTCTCTTCCAGCTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGA ACTAGTTTGTTGAATAAATGCTGGTGAATGAATGA). Target region 1 corresponds to nucleosides at positions 45795-45886 of SEQ ID NO: 318 (human UNC13A; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009). Target region 1 of SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; upper sequence) is aligned to the corresponding target region of the cynomolgus UNC13A sequence (nucleosides at positions 50846-50936 of cynomolgus UNC13A SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement; lower sequence)) as follows:

TABLE-US-00006

CTGGGTATGTCTCTTCCAGCTGCCTGGGTTTCCTGGAAAGAACTCTTATCCCCAGGAACTAGTT
||||||||||| |||||||||||| ||||||||||||| ||||||||||||||||
CTGGGTATGTCTCCTCCAGCTGCCTGGGGTTCCTGGAAAGAACTG-
TATCCCCAGGAACTAGTT TGTTGAATAAATGCTGGTGAATGAATGA (SEQ ID NO: 327) ||||||| |||||||||||||||| TGTTGAATGAATGCTGGTGAATGAATGA (SEQ ID NO: 330)

[0136] As such, positions 50846-50936 of cynomolgus UNC13A, SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement) is the corresponding target region 1. When referring to antisense oligonucleotides targeting target region 1 of an UNC13A sequence, the present disclosure encompasses antisense oligonucleotides targeting target region 1 of NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009 (SEQ ID NO: 318) and antisense oligonucleotides that target the corresponding target region 1 of NC_052273.1: 17161646 . . . 17254007 complement (SEQ ID NO: 319).

[0137] In determining a corresponding target region of another sequence in relation to a reference sequence, the other sequence and the reference sequence are optimally aligned over the window of comparison, which comprises sufficient number of nucleosides for the alignment. Optimal alignment of sequences for aligning a comparison window may be conducted by computerized implementations of algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package Release 7.0, Genetics Computer Group, 575 Science Drive Madison, Wis., USA) or by inspection and the best alignment (i.e., resulting in the highest percentage homology over the comparison window) generated by any of the various methods selected. Reference also may be made to the BLAST family of programs as for example disclosed by Altschul et al., Nucl. Acids Res. 25:3389, 1997.

[0138] One or more insertions in another UNC13A sequence from a closely related species, such as a cynomolgus monkey, relative to the human reference sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009) are represented as position "N-M," wherein N represents the corresponding position immediately before the insertions in the reference sequence, and M represents the position of the nucleoside within the one or more insertions. For example, in the following alignment between an area of the sequence set forth in SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; upper sequence) and the sequence set forth in SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement; lower sequence),

TABLE-US-00007 TG**A**----TGGGTG ||| |||||| TGATGGGTGGGTG (SEQ ID NO: 331)

[0139] The insertions TGGG in SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement) follows the position 46487 (bolded A in the alignment above) in SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009). Accordingly, the positions of insertions TGGG are designated as 46487-1, 46487-2, 46487-3, and 46487-4, respectively.

[0140] One or more deletions or mismatches in another UNC13A sequence from a closely related species, such as a cynomolgus monkey, relative to the human reference sequence (SEQ ID NO: 318; NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009) may result in the lack of a "corresponding position." In these instances, as long as the deletions/mismatches (e.g., no more than 5, 4, 3, 2, 1 mismatches) are within a stretch of sequences of sufficient length (e.g., at least 20, 25, 30, 35, or 40 nucleosides) that aligned between the two sequences, the mismatched positions are represented as the position in the reference sequence that directly aligned even if the nucleoside in the reference is different. The position for the deletion in the other sequence is "skipped."

[0141] For example, the nucleosides at positions 46024-46036 of the human UNC13A sequence SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009; upper sequence) are aligned to the nucleoside 51070-51082 of the cynomolgus UNC13A sequence SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement; lower sequence) as follows:

TABLE-US-00008 GATAT**A**TGATTGA (SEQ ID NO: 332) ||||| ||| |||
GATAT**G**TGACTGA (SEQ ID NO: 333)

[0142] The bolded A at position 46029 of SEQ ID NO: 318 (NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009) aligns with the bolded G at position 51075 of SEQ ID NO: 319 (NC_052273.1: 17161646 . . . 17254007 complement). When referring to antisense oligonucleotides that target an UNC13A sequence at a target region comprising the nucleoside at position 46029, the present disclosure encompasses antisense oligonucleotides that target a target region comprising the nucleoside at position 46029 of NC_000019.10: 17601336 . . . 17688354 complement and NG_052872.1: 4991 . . . 92009 (SEQ ID NO: 318) and antisense oligonucleotides that target a target region comprising the nucleoside at position 51075 of NC_052273.1: 17161646 . . . 17254007 complement (SEQ ID NO: 319).

[0143] In some embodiments, an antisense oligonucleotide provided herein comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region, wherein the first target region and/or the second target region each comprises an UNC13A sequence. In some embodiments, each of the first antisense sequence and the second antisense sequence comprises a region of complementary to an UNC13A sequence as set forth in SEQ ID NO: 318. In some embodiments, the region of complementary consists of at least 6 (e.g., at least 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 7 (e.g., at least 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 8 (e.g., at least 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length.
[0144] In some embodiments, each of the first target region and the second target region is in intron 20 of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, each of the first antisense sequence and the second antisense sequence comprises a region of complementary to intron 20 of an UNC13A sequence, wherein the intron 20 is as set forth in SEQ ID NO: 311 or SEQ ID NO: 325. In some embodiments, the region of complementary consists of at least 6 (e.g., at least 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 7 (e.g., at least 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length. In some embodiments, the region of complementary consists of at least 8 (e.g., at least 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) nucleotides in length.
[0145] In some embodiments, each of the first target region and the second target region is near a cryptic exon in intron 20 of an UNC13A sequence (e.g., gene sequence or pre-mRNA sequence). In some embodiments, each of the first target region and the second target region comprises an UNC13A sequence, wherein targeting such sequences modulates splicing of an UNC13A cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, targeting the UNC13A sequences using an antisense oligonucleotide described herein that modulate splicing of an UNC13A cryptic exon inhibits inclusion of the cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA) in a mature UNC13A mRNA.
[0146] In some embodiments, the first target region and second target region are adjacent (e.g., within 300 nucleobases) to each other. In some embodiments, the first target region is downstream of the second target region. In some embodiments, the first target region is more than one nucleotide (2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50, etc.) downstream of the second target region. In some embodiments, the first target region and the second target region are both in the cryptic exon (e.g., the cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, the first target region is downstream of the cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA) and the second target region is in the cryptic exon (e.g., a cryptic exon in intron 20). In some embodiments, the first target region and the second target region are both downstream of the cryptic exon (e.g., a cryptic exon in intron 20 of an UNC13A pre-mRNA). In some embodiments, the 5′ end of the first target region and 3′ end of the second target region have 1-300 nucleobases (e.g., 1-300, 3-300, 5-300, 7-300, 10-300, 1-250, 3-250, 5-250, 7-250, 10-250, 1-200, 3-200, 5-200, 7-200, 10-200, 1-150, 3-150, 5-150, 7-150, or 10-150 nucleobases) in between each other.
[0147] Non-limiting examples of target regions in an UNC13A pre-mRNA that may be targeted by an antisense oligonucleotide described herein are provided in Table 2. Such target regions are near the cryptic exon in intron 20 of an UNC13A sequence. One of ordinary skill in the art would appreciate that the target regions overlap and are not discrete regions. In some embodiments, antisense oligonucleotides disclosed herein that comprise antisense sequences targeting target regions 1 and 3 are particularly effective in skipping the cryptic exon, inhibiting inclusion of the UNC13A cryptic exon into an UNC13A mature mRNA.
TABLE-US-00009 TABLE 2 Exemplary target regions in intron 20 of UNC13A pre-

mRNA Target SEQ regions Sequence ID NO: Target
CUGGGUAUGUCUCUUCCAGCUGCCUGGG 313 region 1
UUUCCUGGAAAGAACUCUUAUCCCCAGG AACUAGUUUGUUGAAUAAAUGCUGGUG
AAUGAAUGA Target GAAUAAAUGCUGGUGAAUGAAUGAAUG 314 region 2
AUUGAACAGAUGAAUGAGUGAUGAGUA GAUAAAAGGAUGGAUGGAGAGAUGGGU
GAGUACAUGGAUGGA Target GAGAUGGGUGAGUACAUGGAUGGAUAG 315 region 3
AUGGAUGAGUUGGUGGGUAGAUUCGUG GCUAGAUGGAUGAUGGAUGGAUGGACA
GAUGGAUGGAU Target AUGGAUGGAUGGACAGAUGGAUGGAUA 316 region 4
UAUGAUUGAACUAUUGAAAGUAUA ‡It is to be understood that if the target sequence is a gene sequence (DNA), each uracil base (U) in any one of the target region sequences provided in Table 2 may be replaced with a thymine base (T).

[0148] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of any one of SEQ ID NO: 313-316. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of in SEQ ID NO: 313 or SEQ ID NO: 315. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of any one of SEQ ID NO: 313-316. In some embodiments, the second antisense sequence targets the second target region comprising the nucleotide sequence of SEQ ID NO: 313 or SEQ ID NO: 315.

[0149] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence targets the first target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 316, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 315, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 315, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 315, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 314, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 314, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence targets a first target region comprising the nucleotide sequence of SEQ ID NO: 313, and the second antisense sequence targets a second target region comprising the nucleotide sequence of SEQ ID NO: 313.

[0150] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 314. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 315. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 316. In some embodiments, the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 316.

[0151] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 316. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 316. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 316 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 315. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 314 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 314 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 314. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 314 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 314 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313. In some embodiments, the first antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313 and the second antisense sequence comprises a region of complementarity of at least 6 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313.

[0152] In some embodiments, the first antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313 and the second antisense sequence comprises a region of complementarity of at least 8 (e.g., 8, 9, 10, 11, 12, 13, 14, 15, or more) nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO 313.

[0153] For the purposes of the present disclosure, a region of complementary need not be 100% complementary to that of its target to be specifically hybridizable or specific for an UNC13A sequence. In some embodiments, the region of complementarity is at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% complementary to a target region in a target sequence. In some embodiments, the target region is a region of consecutive nucleosides in the target sequence. In some embodiments, the region of complementarity comprises a nucleoside sequence that contains no more than 1, 2, 3, 4, or 5 base mismatches compared to the complementary portion of a target sequence. In some embodiments, the region of complementarity comprises a nucleoside sequence that has up to 3 mismatches over 15 nucleosides, or up to 2 mismatches over 10 nucleosides.

[0154] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence is 6-15 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15) nucleosides in length. In some embodiments, the second antisense sequence is 6-15 (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15) nucleosides in length. In some embodiments, the first antisense sequence is 8-15 (e.g., 8, 9, 10, 11, 12, 13, 14, or 15) nucleosides in length. In some embodiments, the second antisense sequence is 8-15 (e.g., 8, 9, 10, 11, 12, 13, 14, or 15) nucleosides in length. In some embodiments, the first antisense sequence is 10-14 (e.g., 10, 11, 12, 13, or 14) nucleosides in length and the second antisense sequence is 10-14 (e.g., 10, 11, 12, 13, or 14) nucleosides in length. In some embodiments, the first antisense sequence and the second antisense sequence are of the same length. In some embodiments, the first antisense sequence and the second antisense sequence are of different lengths. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35). In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length. In some embodiments, the antisense oligonucleotide is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length.

[0155] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity to a first target region comprising the nucleotide sequence of SEQ ID NOS: 313-316, wherein the region of complementarity is at least 6 nucleosides (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14 nucleosides) in length. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity to a first target region comprising the nucleotide sequence of SEQ ID NOs: 313-316, wherein the region of complementarity is at least 8 nucleosides (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14 nucleosides) in length. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity to a first target region comprising the nucleotide sequence of SEQ ID NO: 313 or 315, wherein the region of complementarity is at least 8 nucleosides (e.g., at least 8, at least 9, at least 10, at least 11, at least

12, at least 13, or at least 14 nucleosides) in length. In some embodiments, the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity to a second target region comprising the nucleotide sequence of SEQ ID NOs: 313-316, wherein the region of complementarity is at least 6 nucleosides (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14 nucleosides) in length. In some embodiments, the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity to a second target region comprising the nucleotide sequence of SEQ ID NOs: 313-316, wherein the region of complementarity is at least 8 nucleosides (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14 nucleosides) in length. In some embodiments, the region of complementarity includes 1, 2, 3 or more mismatches. In some embodiments, the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity to a second target region comprising the nucleotide sequence of SEQ ID NO: 313 or 315, wherein the region of complementarity is at least 8 nucleosides (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14 nucleosides) in length. In some embodiments, the region of complementarity includes 1, 2, 3 or more mismatches.

[0156] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NOs: 313-316 and the second antisense sequence of an antisense oligonucleotide disclosed herein is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NOs: 313-316. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NOs: 313-316 and the second antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NOs: 313-316.

[0157] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 315.

[0158] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 314.

[0159] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementary of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 315 and the second antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313.

[0160] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region. In some embodiments, the first antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementary of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a first target region comprising the nucleotide sequence of SEQ ID NO: 313 and the second antisense sequence of an antisense oligonucleotide disclosed herein is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises a region of complementarity of 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides in length to a second target region comprising the nucleotide sequence of SEQ ID NO: 313.

[0161] Non-limiting examples of the nucleobase sequences of an antisense oligonucleotide described herein, the first antisense sequence and/or the second antisense sequence of an antisense oligonucleotide described herein, are provided in Table 3. Non-limiting examples of the nucleobase sequences of an antisense oligonucleotide described herein, the first antisense sequence and/or the second antisense sequence of an antisense oligonucleotide described herein, are provided in Table 9.

[0162] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-245. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 246, 247, 274, 288, and 495-515. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 246, 247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-245. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 246, 247, 274, 288, and 495-515. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of

SEQ ID NOs: 191-245. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 246, 247, 274, 288, and 495-515. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC. In some embodiments, the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 516-535. In some embodiments, the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA. In some embodiments, the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOS: 516-535. In some embodiments, the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 516-535. In some embodiments, the second antisense sequence comprises the nucleobase sequence of any one of CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA.

[0163] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-245 and the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-245 and the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 191-245 and the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306.

[0164] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, and 495-515 and the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, and 516-535. In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence. In some embodiments, the first antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC and the second antisense sequence is 6-15 nucleosides (e.g., 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 6 (e.g., at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA. In some embodiments, the first antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 191-247, 274, 288, and 495-515 and the second antisense sequence is 8-15 nucleosides (e.g., 8, 9, 10, 11, 12, 13, 14, or 15 nucleosides) in length and comprises at least 8 (e.g., at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, or at least 14) contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, and 516-535. In some embodiments, the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 191-247, 274, 288, 495-515, AGCCACGA, ACGAATC, CGAATCT, GAATCTA, AATCTAC, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TCACCCA, ACGAATCTA, GCCACGA, GAATCTAC, CGAATCTA, CGAATCTAC, GAATCTACC, TCTACCC, ATCTACC, GTCGCCG, GGGGTCG, GGGTCGC, and GGTCGCC and the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 194, 199, 212, 226, 227, 246-306, 516-535, CAATCAT, GTTCAAT, TTCAATC, TCAATCA, TTTTATC, CTTTTAT, TGTACTC, and TCACCCA.

[0165] In some embodiments, in any one of the antisense oligonucleotides described herein, the first antisense sequence and second antisense sequence are immediately adjacent (e.g., having no nucleobases between the first antisense sequence and second antisense sequence) to each other. In some embodiments, in any one of the antisense oligonucleotides described herein, one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) are between the first antisense sequence and second antisense sequence. In some embodiments, the present disclosure provides antisense oligonucleotides in which one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 3′ end of the first antisense sequence is complementary to one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 3′ end of the second target region, and/or one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 5′ end of the second antisense sequence is complementary to one or more nucleobases (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) at the 5′ end of the first target region.

[0166] In some embodiments, any one of the antisense oligonucleotides described herein comprises a spacer between the first antisense sequence and second antisense sequence. In some embodiments, a spacer comprises one or more (1, 2, 3, 4, 5, 6, or more) abasic moieties. In some embodiments, the spacer is a C3 spacer (e.g., a C3 spacer as shown in Table 16). In some embodiments, the spacer is a

C6 spacer (e.g., a C5 spacer as shown in Table 16).

[0167] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0168] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 194 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0169] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 497 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 249, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0170] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 506 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 259, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0171] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0172] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of SEQ ID NO: 198 and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 253, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0173] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0174] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).

[0175] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 534, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).

[0176] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTA and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 534, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a

C6 spacer, as shown in Table 16).
[0177] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 535, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0178] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 535, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).
[0179] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0180] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTACC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).
[0181] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 264, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0182] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C3 spacer (e.g., a C3 spacer, as shown in Table 16).
[0183] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence comprising the nucleobase sequence of GAATCTAC and a second antisense sequence comprising the nucleobase sequence of SEQ ID NO: 246, wherein the 3′ end of the first antisense sequence is linked to the 5′ end of the second antisense sequence by a C6 spacer (e.g., a C6 spacer, as shown in Table 16).
[0184] In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35) and comprises at least 16 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) contiguous nucleobases of any one of SEQ ID NOs: 1-190. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35) and comprises at least 16 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) contiguous nucleobases of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises at least 16 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) contiguous nucleobases of any one of SEQ ID NOs: 1-190. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises at least 16 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) contiguous nucleobases of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29,

30, 31, 32, 33, 34, or 35) and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 1-190. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35) and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 12-35 (e.g., 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, or 35) and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 1-182, 184-190, 338-434, 442-494. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 1-190. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 338-494. In some embodiments, the antisense oligonucleotide is 16-30 (e.g., 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30) nucleosides in length and comprises a nucleobase sequence that is at least 80% identical (e.g., at least 80%, 85%, 90%, 95%, 99%, or 100% identical) to the nucleobase sequence of any one of SEQ ID NOs: 1-182, 184-190, 338-434, 442-494.

[0185] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence, wherein the first antisense sequence comprises a nucleobase sequence selected from (5′.fwdarw.3′):

TABLE-US-00010 (i) (SEQ ID NO: 191) CACGAATCTACCC (ii) (SEQ ID NO: 192) CGAATCTACCCAC (iii) (SEQ ID NO: 198) CCATCCATCATCC (iv) (SEQ ID NO: 199) GCATTTATTCAAC (v) (SEQ ID NO: 193) CCACGAATCTACC (vi) (SEQ ID NO: 194) TGTTCAATCATTC (vii) (SEQ ID NO: 204) CGAATCTACCCA (viii) (SEQ ID NO: 212) CGAATCTACC (ix) (SEQ ID NO: 196) ACGAATCTACCCA (x) (SEQ ID NO: 205) ACGAATCTACCC (xi) (SEQ ID NO: 222) TCCATCCATCATC and (xii) (SEQ ID NO: 233) CATCCATCTATCC.

[0186] In some embodiments, an antisense oligonucleotide disclosed herein comprises a first antisense sequence and a second antisense sequence, wherein the second antisense sequence that comprises a nucleobase sequence selected from (5′.fwdarw.3′):

TABLE-US-00011 (i) (SEQ ID NO: 246) CCATCCATGTACT (ii) (SEQ ID NO: 247) ATCTGTTCAATCA (iii) (SEQ ID NO: 253) GGGATAAGAGTTC (iv) (SEQ ID NO: 254) GGGGATAAGAGTT (v) (SEQ ID NO: 252) GGATAAGAGTTCT (vi) (SEQ ID NO: 261) ATCTGTTCAATC (vii) (SEQ ID NO: 271) ATCCATGTAC (viii) (SEQ ID NO: 276) CATCTGTTCAATC and (ix) (SEQ ID NO: 262) CCATCCATGTAC.

[0187] In some embodiments, an antisense oligonucleotide disclosed herein comprising a first antisense sequence at the 5′ end of the antisense oligonucleotide and a second antisense sequence at the 3′ end of the antisense oligonucleotide, wherein the first antisense sequence comprises a nucleobase sequence selected from (5′-+3′):

TABLE-US-00012 (i) (SEQ ID NO: 191) CACGAATCTACCC (ii) (SEQ ID NO: 192) CGAATCTACCCAC (iii) (SEQ ID NO: 198) CCATCCATCATCC (iv) (SEQ ID NO: 199) GCATTTATTCAAC (v) (SEQ ID NO: 193) CCACGAATCTACC (vi) (SEQ ID NO: 194) TGTTCAATCATTC (vii) (SEQ ID NO: 204) CGAATCTACCCA (viii) (SEQ ID NO: 212) CGAATCTACC (ix) (SEQ ID NO: 196) ACGAATCTACCCA (x) (SEQ ID NO: 205) ACGAATCTACCC (xi) (SEQ ID NO: 222) TCCATCCATCATC and (xii) (SEQ ID NO: 233) CATCCATCTATCC;

wherein the second antisense sequence comprises a nucleobase sequence selected from (5′.fwdarw.3′):

TABLE-US-00013 (i) (SEQ ID NO: 245) CCATCCATGTACT (ii) (SEQ ID NO: 247) ATCTGTTCAATCA (iii) (SEQ ID NO: 253) GGGATAAGAGTTC (iv) (SEQ ID NO: 254) GGGGATAAGAGTT (v) (SEQ ID NO: 252) GGATAAGAGTTCT (vi) (SEQ ID NO: 261) ATCTGTTCAATC (vii) (SEQ ID NO: 271) ATCCATGTAC (viii) (SEQ ID NO: 276) CATCTGTTCAATC and (ix) (SEQ ID NO: 262) CCATCCATGTAC.

[0188] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence selected from (5′.fwdarw.3′):

TABLE-US-00014 (i) (SEQ ID NO: 1) CACGAATCTACCCCCATCCATGTACT (ii) (SEQ ID NO: 2) CGAATCTACCCACATCTGTTCAATCA (iii) (SEQ ID NO: 16) CCATCCATCATCCGGGATAAGAGTTC (iv) (SEQ ID NO: 17) GCATTTATTCAACGGGGATAAGAGTT (v) (SEQ ID NO: 8) CCACGAATCTACCGGATAAGAGTTCT (vi) (SEQ ID NO: 9) TGTTCAATCATTCGGGATAAGAGTTC (vii) (SEQ ID NO: 14) CCACGAATCTACCGGGATAAGAGTTC (viii) (SEQ ID NO: 36) CGAATCTACCCAATCTGTTCAATC (ix) (SEQ ID NO: 82) CGAATCTACCATCCATGTAC (x) (SEQ ID NO: 93) ACGAATCTACCCACATCTGTTCAATC (xi) (SEQ ID NO: 37) ACGAATCTACCCCCATCCATGTAC (xii) (SEQ ID NO: 121) TCCATCCATCATCGGGATAAGAGTTC and (xiii) (SEQ ID NO: 159) CATCCATCTATCCGGGATAAGAGTTC

[0189] It is to be understood that, for the purposes of the present disclosure, an antisense oligonucleotide comprising nucleobase sequences (e.g., nucleobase sequences of the antisense oligonucleotide, first antisense sequence, and second antisense sequence) of the antisense oligonucleotides listed in Table 3 encompasses antisense oligonucleotides comprising such nucleobase sequences and comprising no chemical modifications (e.g., modified nucleosides and/or modified internucleoside linkages), and an antisense oligonucleotides comprising such nucleobase sequences and comprising chemical modifications (e.g., one or more modified nucleosides and/or one or more modified internucleoside linkages; e.g., those provided in Table 5), and encompasses such modified or unmodified antisense oligonucleotides unconjugated or conjugated to a targeting moiety.

[0190] It is to be understood that, for the purposes of the present disclosure, an antisense oligonucleotide comprising nucleobase sequences (e.g., nucleobase sequences of the antisense oligonucleotide, first antisense sequence, and second antisense sequence) of the antisense oligonucleotides listed in Table 9 encompasses antisense oligonucleotides comprising such nucleobase sequences and comprising no chemical modifications (e.g., modified nucleosides and/or modified internucleoside linkages), and an antisense oligonucleotides comprising such nucleobase sequences and comprising chemical modifications (e.g., one or more modified nucleosides and/or one or more modified internucleoside linkages; e.g., those provided in Table 11), and encompasses such modified or unmodified antisense oligonucleotides unconjugated or conjugated to a targeting moiety.

TABLE-US-00015 TABLE 3 Nucleobase sequences of antisense oligonucleotides provided herein ASO SEQ Antisense Sequence 1 SEQ Antisense Sequence 2 SEQ # Nucleobase Sequence (5′.fwdarw.3′) ID NO: (5′.fwdarw.3′) ID NO: (5′.fwdarw.3′) ID NO: 1 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 2 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 3 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 4 CGAATCTACCCACGAGTTCTTTCCAG 4 CGAATCTACCCAC 192 GAGTTCTTTCCAG 249 5 CGAATCTACCCACTAAGAGTTCTTTC 5

CGAATCTACCCAC 192 TAAGAGTTCTTTC 250 6
CCACGAATCTACCGATAAGAGTTCTT 6 CCACGAATCTACC 193 GATAAGAGTTCTT
251 7 CACGAATCTACCCGGATAAGAGTTCT 7 CACGAATCTACCC 191
GGATAAGAGTTCT 252 8 CCACGAATCTACCGGATAAGAGTTCT 8
CCACGAATCTACC 193 GGATAAGAGTTCT 252 9
TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC
253 10 GAATCTACCCACCGGGATAAGAGTTC 10 GAATCTACCCACC 195
GGGATAAGAGTTC 253 11 CGAATCTACCCACGGGATAAGAGTTC 11
CGAATCTACCCAC 192 GGGATAAGAGTTC 253 12
ACGAATCTACCCAGGGATAAGAGTTC 12 ACGAATCTACCCA 196 GGGATAAGAGTTC
253 13 CACGAATCTACCCGGGATAAGAGTTC 13 CACGAATCTACCC 191
GGGATAAGAGTTC 253 14 CCACGAATCTACCGGGATAAGAGTTC 14
CCACGAATCTACC 193 GGGATAAGAGTTC 253 15
CTAGCCACGAATCGGGATAAGAGTTC 15 CTAGCCACGAATC 197 GGGATAAGAGTTC
253 16 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198
GGGATAAGAGTTC 253 17 GCATTTATTCAACGGGGATAAGAGTT 17
GCATTTATTCAAC 199 GGGGATAAGAGTT 254 18
CCACCAACTCATCGGGGATAAGAGTT 18 CCACCAACTCATC 200 GGGGATAAGAGTT
254 19 GAATCTACCCACCGGGGATAAGAGTT 19 GAATCTACCCACC 195
GGGGATAAGAGTT 254 20 CGAATCTACCCACGGGGATAAGAGTT 20
CGAATCTACCCAC 192 GGGGATAAGAGTT 254 21
CACGAATCTACCCGGGGATAAGAGTT 21 CACGAATCTACCC 191 GGGGATAAGAGTT
254 22 CCACGAATCTACCGGGGATAAGAGTT 22 CCACGAATCTACC 193
GGGGATAAGAGTT 254 23 GCCACGAATCTACGGGGATAAGAGTT 23
GCCACGAATCTAC 201 GGGGATAAGAGTT 254 24
CGAATCTACCCACCCTGGGGATAAGA 24 CGAATCTACCCAC 192 CCTGGGGATAAGA
255 25 ACGAATCTACCCATCCTGGGGATAAG 25 ACGAATCTACCCA 196
TCCTGGGGATAAG 256 26 CGAATCTACCCACGTTCCTGGGGATA 26
CGAATCTACCCAC 192 GTTCCTGGGGATA 257 27
CACGAATCTACCCGTTCCTGGGGATA 27 CACGAATCTACCC 191 GTTCCTGGGGATA
257 28 CTAGCCACGAATCGTTCCTGGGGATA 28 CTAGCCACGAATC 197
GTTCCTGGGGATA 257 29 CGAATCTACCCACTAGTTCCTGGGGA 29
CGAATCTACCCAC 192 TAGTTCCTGGGGA 258 30
CTAGCCACGAATCTAGTTCCTGGGGA 30 CTAGCCACGAATC 197 TAGTTCCTGGGGA
258 31 ACGAATCTACCCACTAGTTCCTGGGG 31 ACGAATCTACCCA 196
CTAGTTCCTGGGG 259 32 ATCTAGCCACGAACTAGTTCCTGGGG 32
ATCTAGCCACGAA 202 CTAGTTCCTGGGG 259 33 GAATCTACCCACTCTGTTCAATCA
33 GAATCTACCCAC 203 TCTGTTCAATCA 260 34 CGAATCTACCCATCTGTTCAATCA
34 CGAATCTACCCA 204 TCTGTTCAATCA 260 35 GAATCTACCCACATCTGTTCAATC
35 GAATCTACCCAC 203 ATCTGTTCAATC 261 36 CGAATCTACCCAATCTGTTCAATC
36 CGAATCTACCCA 204 ATCTGTTCAATC 261 37 GAATCTACCCACTCTGTTCAATCA
33 GAATCTACCCAC 203 TCTGTTCAATCA 260 38 CGAATCTACCCATCTGTTCAATCA
34 CGAATCTACCCA 204 TCTGTTCAATCA 260 39 GAATCTACCCACATCTGTTCAATC
35 GAATCTACCCAC 203 ATCTGTTCAATC 261 40 CGAATCTACCCAATCTGTTCAATC
36 CGAATCTACCCA 204 ATCTGTTCAATC 261 41
CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT
246 42 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191
CATCCATGTACTC 248 43 CCATCCATCATCCGGGATAAGAGTTC 16
CCATCCATCATCC 198 GGGATAAGAGTTC 253 44
TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC

253 45 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 46 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 47 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 48 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 49 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 50 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 51 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 52 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 53 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 54 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 55 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 56 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 57 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 58 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 59 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 60 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 61 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 62 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 63 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 64 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 65 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 66 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 67 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 68 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 69 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 70 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 71 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 72 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 73 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 74 CACGAATCTACCCCATCCATGTACTC 3 CACGAATCTACCC 191 CATCCATGTACTC 248 75 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 76 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 77 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 78 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 79 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 80 ACGAATCTACCCCCATCCATGTAC 37 ACGAATCTACCC 205 CCATCCATGTAC 262 81 CACGAATCTACCCCATCCATGTACT 38 CACGAATCTACC 206 CCATCCATGTACT 246 82 ACGAATCTACCCCCATCCATGTACT 39 ACGAATCTACCC 205 CCATCCATGTACT 246 83 CACGAATCTACCCCCATCCATGTAC 40 CACGAATCTACCC 191 CCATCCATGTAC 262 84 CACGAATCTACCCTCCATCCATGTACT 41 CACGAATCTACCC 191 TCCATCCATGTACT 263 85

CACGAATCTACCCACCATCCATGTACT 42 CACGAATCTACCCA 207 CCATCCATGTACT 246 86 ACGAATCTACCCCATCCATGTACT 43 ACGAATCTACCC 205 CATCCATGTACT 264 87 CACGAATCTACCCATCCATGTACTC 44 CACGAATCTACC 206 CATCCATGTACTC 248 88 ACGAATCTACCCCATCCATGTACTC 45 ACGAATCTACCC 205 CATCCATGTACTC 248 89 CACGAATCTACCCCATCCATGTACTCA 46 CACGAATCTACCC 191 CATCCATGTACTCA 265 90 CCATCCATCATCGGGATAAGAGTT 47 CCATCCATCATC 208 GGGATAAGAGTT 266 91 CCATCCATCATCGGATAAGAGTTC 48 CCATCCATCATC 208 GGATAAGAGTTC 267 92 CATCCATCATCCGGGATAAGAGTT 49 CATCCATCATCC 209 GGGATAAGAGTT 266 93 CATCCATCATCCGGATAAGAGTTC 50 CATCCATCATCC 209 GGATAAGAGTTC 267 94 CCATCCATCATCGGGATAAGAGTTC 51 CCATCCATCATC 208 GGGATAAGAGTTC 253 95 CATCCATCATCCGGGATAAGAGTTC 52 CATCCATCATCC 209 GGGATAAGAGTTC 253 96 CCATCCATCATCCGGGATAAGAGTT 53 CCATCCATCATCC 198 GGGATAAGAGTT 266 97 CCATCCATCATCCGGATAAGAGTTC 54 CCATCCATCATCC 198 GGATAAGAGTTC 267 98 CCATCCATCATCCGGGGATAAGAGTTC 55 CCATCCATCATCC 198 GGGGATAAGAGTTC 268 99 CCATCCATCATCCGGGATAAGAGTTCT 56 CCATCCATCATCC 198 GGGATAAGAGTTCT 269 100 TCCATCCATCATCCGGGATAAGAGTTC 57 TCCATCCATCATCC 210 GGGATAAGAGTTC 253 101 GTTCAATCATTCGGGATAAGAGTT 58 GTTCAATCATTC 211 GGGATAAGAGTT 266 102 GTTCAATCATTCGGATAAGAGTTC 59 GTTCAATCATTC 211 GGATAAGAGTTC 267 103 GTTCAATCATTCGGGATAAGAGTTC 60 GTTCAATCATTC 211 GGGATAAGAGTTC 253 104 TGTTCAATCATTCGGATAAGAGTTC 61 TGTTCAATCATTC 194 GGATAAGAGTTC 267 105 TGTTCAATCATTCGGGATAAGAGTTCT 62 TGTTCAATCATTC 194 GGGATAAGAGTTCT 269 106 GAATCTACCCACATCTGTTCAATCA 63 GAATCTACCCAC 203 ATCTGTTCAATCA 247 107 CGAATCTACCCACATCTGTTCAATC 64 CGAATCTACCCAC 192 ATCTGTTCAATC 261 108 CGAATCTACCCACCATCTGTTCAATCA 65 CGAATCTACCCAC 192 CATCTGTTCAATCA 270 109 CACGAATCTACCCCCCATCCATGTACT 66 CACGAATCTACCC 191 CCATCCATGTACT 246 110 CACGAATCTACCCGCCATCCATGTACT 67 CACGAATCTACCC 191 CCATCCATGTACT 246 111 CACGAATCTACCCTCATCCATGTACTC 68 CACGAATCTACCC 191 CATCCATGTACTC 248 112 CACGAATCTACCCGCATCCATGTACTC 69 CACGAATCTACCC 191 CATCCATGTACTC 248 113 TGTTCAATCATTCCGGGATAAGAGTTC 70 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 114 CCATCCATCATCCCGGGATAAGAGTTC 71 CCATCCATCATCC 198 GGGATAAGAGTTC 253 115 CACGAATCTACCCCACCATCCATGTACT 72 CACGAATCTACCC 191 CCATCCATGTACT 246 116 CACGAATCTACCCTCCCATCCATGTACT 73 CACGAATCTACCC 191 CCATCCATGTACT 246 117 CACGAATCTACCCCCCCATCCATGTACT 74 CACGAATCTACCC 191 CCATCCATGTACT 246 118 CACGAATCTACCCCTCATCCATGTACTC 75 CACGAATCTACCC 191 CATCCATGTACTC 248 119 CACGAATCTACCCTTCATCCATGTACTC 76 CACGAATCTACCC 191 CATCCATGTACTC 248 120 TGTTCAATCATTCCAGGGATAAGAGTTC 77 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 121 TGTTCAATCATTCCTGGGATAAGAGTTC 78 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 122 CCATCCATCATCCCAGGGATAAGAGTTC 79 CCATCCATCATCC 198 GGGATAAGAGTTC 253 123 CCATCCATCATCCTCGGGATAAGAGTTC 80 CCATCCATCATCC 198 GGGATAAGAGTTC 253 124 CCATCCATCATCCTTGGGATAAGAGTTC 81 CCATCCATCATCC 198 GGGATAAGAGTTC 253 125 ACGAATCTACCCCCATCCATGTAC 37 ACGAATCTACCC 205 CCATCCATGTAC 262 126 ACGAATCTACCCCATCCATGTACT

43 ACGAATCTACCC 205 CATCCATGTACT 264 127 CCATCCATCATCGGGATAAGAGTT
47 CCATCCATCATC 208 GGGATAAGAGTT 266 128 CCATCCATCATCGGATAAGAGTTC
48 CCATCCATCATC 208 GGATAAGAGTTC 267 129 CATCCATCATCCGGGATAAGAGTT
49 CATCCATCATCC 209 GGGATAAGAGTT 266 130 CATCCATCATCCGGATAAGAGTTC
50 CATCCATCATCC 209 GGATAAGAGTTC 267 131 GTTCAATCATTCGGGATAAGAGTT
58 GTTCAATCATTC 211 GGGATAAGAGTT 266 132 GTTCAATCATTCGGATAAGAGTTC
59 GTTCAATCATTC 211 GGATAAGAGTTC 267 133 CGAATCTACCCATCTGTTCAATCA
34 CGAATCTACCCA 204 TCTGTTCAATCA 260 134 CGAATCTACCCAATCTGTTCAATC
36 CGAATCTACCCA 204 ATCTGTTCAATC 261 135 ACGAATCTACCCCCATCCATGTAC
37 ACGAATCTACCC 205 CCATCCATGTAC 262 136 ACGAATCTACCCCATCCATGTACT
43 ACGAATCTACCC 205 CATCCATGTACT 264 137 CCATCCATCATCGGGATAAGAGTT
47 CCATCCATCATC 208 GGGATAAGAGTT 266 138 CCATCCATCATCGGATAAGAGTTC
48 CCATCCATCATC 208 GGATAAGAGTTC 267 139 CATCCATCATCCGGGATAAGAGTT
49 CATCCATCATCC 209 GGGATAAGAGTT 266 140 CATCCATCATCCGGATAAGAGTTC
50 CATCCATCATCC 209 GGATAAGAGTTC 267 141 GTTCAATCATTCGGGATAAGAGTT
58 GTTCAATCATTC 211 GGGATAAGAGTT 266 142 GTTCAATCATTCGGATAAGAGTTC
59 GTTCAATCATTC 211 GGATAAGAGTTC 267 143 CGAATCTACCCATCTGTTCAATCA
34 CGAATCTACCCA 204 TCTGTTCAATCA 260 144 CGAATCTACCCAATCTGTTCAATC
36 CGAATCTACCCA 204 ATCTGTTCAATC 261 145 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 146 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 147 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 148 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 149 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 150 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 151 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 152 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 153 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 154 AATCTACCCACCTGTTCAATCA 83 AATCTACCCAC 213 CTGTTCAATCA 272 155 AATCTACCCACATCTGTTCAAT 84 AATCTACCCAC 213 ATCTGTTCAAT 273 156 CGAATCTACCCCTGTTCAATCA 85 CGAATCTACCC 214 CTGTTCAATCA 272 157 CGAATCTACCCATCTGTTCAAT 86 CGAATCTACCC 214 ATCTGTTCAAT 273 158 ACGAATCTACCCAATCTGTTCAATCA 87 ACGAATCTACCCA 196 ATCTGTTCAATCA 247 159 CACGAATCTACCCATCTGTTCAATCA 88 CACGAATCTACCC 191 ATCTGTTCAATCA 247 160 GAATCTACCCACCATCTGTTCAATCA 89 GAATCTACCCACC 195 ATCTGTTCAATCA 247 161 AATCTACCCACCAATCTGTTCAATCA 90 AATCTACCCACCA 215 ATCTGTTCAATCA 247 162 CGAATCTACCCACTCTGTTCAATCAT 91 CGAATCTACCCAC 192 TCTGTTCAATCAT 274 163 CGAATCTACCCACTCATCTGTTCAAT 92 CGAATCTACCCAC 192 TCATCTGTTCAAT 275 164 ACGAATCTACCCACATCTGTTCAATC 93 ACGAATCTACCCA 196 CATCTGTTCAATC 276 165 GAATCTACCCACCCATCTGTTCAATC 94 GAATCTACCCACC 195 CATCTGTTCAATC 276 166 ACGAATCTACCCATCTGTTCAATCAT 95 ACGAATCTACCCA 196 TCTGTTCAATCAT 274 167 GAATCTACCCACCTCTGTTCAATCAT 96 GAATCTACCCACC 195 TCTGTTCAATCAT 274 168 CCACGAATCTACCTCAATCATTCATT 97 CCACGAATCTACC 193 TCAATCATTCATT 277 169 CGAATCTACCCACCTCTGTTCAATCA 98 CGAATCTACCCACC 216 TCTGTTCAATCA 260 170 ACGAATCTACCCACTCTGTTCAATCA 99 ACGAATCTACCCAC 217 TCTGTTCAATCA 260 171 CGAATCTACCCAATCTGTTCAATCAT 100 CGAATCTACCCA 204 ATCTGTTCAATCAT 278 172

GAATCTACCCACATCTGTTCAATCAT 101 GAATCTACCCAC 203 ATCTGTTCAATCAT 278
173 AATCTACCCACCTGTTCAATCA 83 AATCTACCCAC 213 CTGTTCAATCA 272 174
AATCTACCCACATCTGTTCAAT 84 AATCTACCCAC 213 ATCTGTTCAAT 273 175
AATCTACCCACTCTGTTCAATC 102 AATCTACCCAC 213 TCTGTTCAATC 279 176
CGAATCTACCCCTGTTCAATCA 85 CGAATCTACCC 214 CTGTTCAATCA 272 177
CGAATCTACCCATCTGTTCAAT 86 CGAATCTACCC 214 ATCTGTTCAAT 273 178
CGAATCTACCCTCTGTTCAATC 103 CGAATCTACCC 214 TCTGTTCAATC 279 179
GAATCTACCCACTGTTCAATCA 104 GAATCTACCCA 218 CTGTTCAATCA 272 180
GAATCTACCCAATCTGTTCAAT 105 GAATCTACCCA 218 ATCTGTTCAAT 273 181
GAATCTACCCATCTGTTCAATC 106 GAATCTACCCA 218 TCTGTTCAATC 279 182
AATCTACCCACTCTGTTCAATC 102 AATCTACCCAC 213 TCTGTTCAATC 279 183
CGAATCTACCCTCTGTTCAATC 103 CGAATCTACCC 214 TCTGTTCAATC 279 184
GAATCTACCCACTGTTCAATCA 104 GAATCTACCCA 218 CTGTTCAATCA 272 185
GAATCTACCCAATCTGTTCAAT 105 GAATCTACCCA 218 ATCTGTTCAAT 273 186
GAATCTACCCATCTGTTCAATC 106 GAATCTACCCA 218 TCTGTTCAATC 279 187
GAATCTACCCATCTGTTCAATC 106 GAATCTACCCA 218 TCTGTTCAATC 279 188
GAATCTACCCATCTGTTCAATC 106 GAATCTACCCA 218 TCTGTTCAATC 279 189
GAATCTACCCATCTGTTCAATC 106 GAATCTACCCA 218 TCTGTTCAATC 279 190
GAATCTACCCATCTGTTCAATC 106 GAATCTACCCA 218 TCTGTTCAATC 279 191
GAATCTACCCATCTGTTCAATC 106 GAATCTACCCA 218 TCTGTTCAATC 279 192
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 193
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 194
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 195
CGAATCTACCCACTCATCTGTTCAAT 92 CGAATCTACCCAC 192 TCATCTGTTCAAT 275
196 CGAATCTACCCACTCATCTGTTCAAT 92 CGAATCTACCCAC 192 TCATCTGTTCAAT
275 197 ACGAATCTACCCACATCTGTTCAATC 93 ACGAATCTACCCA 196
CATCTGTTCAATC 276 198 ACGAATCTACCCACATCTGTTCAATC 93 ACGAATCTACCCA
196 CATCTGTTCAATC 276 199 CAAATCTACCCACATCTGTTCAATCA 107
CAAATCTACCCAC 219 ATCTGTTCAATCA 247 200 CGAATCTACCCACATCTATTCAATCA
108 CGAATCTACCCAC 192 ATCTATTCAATCA 280 201
CAAATCTACCCACATCTATTCAATCA 109 CAAATCTACCCAC 219 ATCTATTCAATCA 280
202 CGAATCTACCCAATCTATTCAATC 110 CGAATCTACCCA 204 ATCTATTCAATC 281 203
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CACGAATCTACCCCATCCATGTAC 116 CACGAATCTACC 206 CCATCCATGTAC 262 209
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CATTTATTCAACGGGGATAAGAGT 118 CATTTATTCAAC 220 GGGGATAAGAGT 284 211
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247 212 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192
ATCTGTTCAATCA 247 213 CGAATCTACCCACATCTGTTCAATCA 2
CGAATCTACCCAC 192 ATCTGTTCAATCA 247 214 CGAATCTACCCACATCTGTTCAATCA
2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 215
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 216
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 217
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 218
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 219
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 220

CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 221
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 222
CGAATCTACCCAATCTGTTCAATC 36 CGAATCTACCCA 204 ATCTGTTCAATC 261 223
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246 224 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191
CCATCCATGTACT 246 225 CACGAATCTACCCCCATCCATGTACT 1
CACGAATCTACCC 191 CCATCCATGTACT 246 226 CACGAATCTACCCCCATCCATGTACT
1 CACGAATCTACCC 191 CCATCCATGTACT 246 227
CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC
253 228 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198
GGGATAAGAGTTC 253 229 CCATCCATCATCCGGGATAAGAGTTC 16
CCATCCATCATCC 198 GGGATAAGAGTTC 253 230 CCATCCATCATCCGGGATAAGAGTTC
16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 231
GCATTTATTCAACGGGGATAAGAGTT 17 GCATTTATTCAAC 199 GGGGATAAGAGTT
254 232 GCATTTATTCAACGGGGATAAGAGTT 17 GCATTTATTCAAC 199
GGGGATAAGAGTT 254 233 GCATTTATTCAACGGGGATAAGAGTT 17
GCATTTATTCAAC 199 GGGGATAAGAGTT 254 234 GCATTTATTCAACGGGGATAAGAGTT
17 GCATTTATTCAAC 199 GGGGATAAGAGTT 254 235 CGAATCTACCATCCATGTAC
82 CGAATCTACC 212 ATCCATGTAC 271 236 ATCCATCCATCATTGGGGATAAGAGT 119
ATCCATCCATCAT 221 TGGGGATAAGAGT 285 237 TCCATCCATCATCGGGGATAAGAGTT
120 TCCATCCATCATC 222 GGGGATAAGAGTT 254 238
TCCATCCATCATCGGGATAAGAGTTC 121 TCCATCCATCATC 222 GGGATAAGAGTTC 253
239 TCCATCCATCATCGGATAAGAGTTCT 122 TCCATCCATCATC 222 GGATAAGAGTTCT
252 240 CCATCCATCATCCGATAAGAGTTCTT 123 CCATCCATCATCC 198
GATAAGAGTTCTT 251 241 GCATTTATTCAACGGGATAAGAGTTC 124 GCATTTATTCAAC
199 GGGATAAGAGTTC 253 242 GCATTTATTCAACGGATAAGAGTTCT 125
GCATTTATTCAAC 199 GGATAAGAGTTCT 252 243 CGAATCTACCCACCCTTTTATCTACT
126 CGAATCTACCCAC 192 CCTTTTATCTACT 286 244
CGAATCTACCCACCATGTACTCACCC 127 CGAATCTACCCAC 192 CATGTACTCACCC 232
245 CACGAATCTACCCCCATCCTTTTATC 128 CACGAATCTACCC 191 CCATCCTTTTATC
287 246 CACGAATCTACCCCTGTTCAATCATT 129 CACGAATCTACCC 191
CTGTTCAATCATT 288 247 CACGAATCTACCCGTTCAATCATTCA 130 CACGAATCTACCC
191 GTTCAATCATTCA 289 248 CGAATCTACCCACTCCTTTTATCTAC 131
CGAATCTACCCAC 192 TCCTTTTATCTAC 290 249 CGAATCTACCCACCTGTTCAATCATT
132 CGAATCTACCCAC 192 CTGTTCAATCATT 288 250
CGAATCTACCCACCTTTTATCTACTC 133 CGAATCTACCCAC 192 CTTTTATCTACTC 291
251 CGAATCTACCCACGTTCAATCATTCA 134 CGAATCTACCCAC 192 GTTCAATCATTCA
289 252 ACGAATCTACCCAGTACTCACCCATC 135 ACGAATCTACCCA 196
GTACTCACCCATC 226 253 CGAATCTACCCACTGTTCAATCATTC 136 CGAATCTACCCAC
192 TGTTCAATCATTC 194 254 CACGAATCTACCCCATCTGTTCAATC 137
CACGAATCTACCC 191 CATCTGTTCAATC 276 255 CACGAATCTACCCCCTTTTATCTACT
138 CACGAATCTACCC 191 CCTTTTATCTACT 286 256
CACGAATCTACCCCCATGTACTCACC 139 CACGAATCTACCC 191 CCATGTACTCACC 227
257 CGAATCTACCCACATTCACCAGCATT 140 CGAATCTACCCAC 192 ATTCACCAGCATT
292 258 CGAATCTACCCACCATCTGTTCAATC 141 CGAATCTACCCAC 192
CATCTGTTCAATC 276 259 ACGAATCTACCCACCATCCTTTTATC 142 ACGAATCTACCCA
196 CCATCCTTTTATC 287 260 CGAATCTACCCACGTACTCACCCATC 143
CGAATCTACCCAC 192 GTACTCACCCATC 226 261 CACGAATCTACCCGTACTCACCCATC
144 CACGAATCTACCC 191 GTACTCACCCATC 226 262
ACGAATCTACCCAATTCACCAGCATT 145 ACGAATCTACCCA 196 ATTCACCAGCATT 292

263 CGAATCTACCCACGAAACCCAGGCAG 146 CGAATCTACCCAC 192 GAAACCCAGGCAG 293 264 CCACGAATCTACCGGAAACCCAGGCA 147 CCACGAATCTACC 193 GGAAACCCAGGCA 294 265 CAATCATATATCCGGAAACCCAGGCA 148 CAATCATATATCC 223 GGAAACCCAGGCA 294 266 CAATAGTTCAATCGGAAACCCAGGCA 149 CAATAGTTCAATC 224 GGAAACCCAGGCA 294 267 TCTAGCCACGAATTTCCAGGAAACCC 150 TCTAGCCACGAAT 225 TTCCAGGAAACCC 295 268 GTACTCACCCATCGATAAGAGTTCTT 151 GTACTCACCCATC 226 GATAAGAGTTCTT 251 269 CCATGTACTCACCGGATAAGAGTTCT 152 CCATGTACTCACC 227 GGATAAGAGTTCT 252 270 TGTCCATCCATCCGGATAAGAGTTCT 153 TGTCCATCCATCC 228 GGATAAGAGTTCT 252 271 TTCATTCACCAGCGGGATAAGAGTTC 154 TTCATTCACCAGC 229 GGGATAAGAGTTC 253 272 CTACTCATCACTCGGGATAAGAGTTC 155 CTACTCATCACTC 230 GGGATAAGAGTTC 253 273 TCTCCATCCATCCGGGATAAGAGTTC 156 TCTCCATCCATCC 231 GGGATAAGAGTTC 253 274 GTACTCACCCATCGGGATAAGAGTTC 157 GTACTCACCCATC 226 GGGATAAGAGTTC 253 275 CATGTACTCACCCGGGATAAGAGTTC 158 CATGTACTCACCC 232 GGGATAAGAGTTC 253 276 CATCCATCTATCCGGGATAAGAGTTC 159 CATCCATCTATCC 233 GGGATAAGAGTTC 253 277 CCATCCATCTGTCGGGATAAGAGTTC 160 CCATCCATCTGTC 234 GGGATAAGAGTTC 253 278 CCAGCATTTATTCGGGGATAAGAGTT 161 CCAGCATTTATTC 235 GGGGATAAGAGTT 254 279 TTCATTCACCAGCGGGGATAAGAGTT 162 TTCATTCACCAGC 229 GGGGATAAGAGTT 254 280 TGTTCAATCATTCGGGGATAAGAGTT 163 TGTTCAATCATTC 194 GGGGATAAGAGTT 254 281 TTATCTACTCATCGGGGATAAGAGTT 164 TTATCTACTCATC 236 GGGGATAAGAGTT 254 282 CACCCATCTCTCCGGGGATAAGAGTT 165 CACCCATCTCTCC 237 GGGGATAAGAGTT 254 283 GTACTCACCCATCGGGGATAAGAGTT 166 GTACTCACCCATC 226 GGGGATAAGAGTT 254 284 CATGTACTCACCCGGGGATAAGAGTT 167 CATGTACTCACCC 232 GGGGATAAGAGTT 254 285 CCATCTATCCATCGGGGATAAGAGTT 168 CCATCTATCCATC 238 GGGGATAAGAGTT 254 286 CATCCATCTAGCCGGGGATAAGAGTT 169 CATCCATCTAGCC 239 GGGGATAAGAGTT 254 287 CTGTCCATCCATCGGGGATAAGAGTT 170 CTGTCCATCCATC 240 GGGGATAAGAGTT 254 288 CAATCATATATCCGGGGATAAGAGTT 171 CAATCATATATCC 223 GGGGATAAGAGTT 254 289 CCATCTAGCCACGTTCCTGGGGATAA 172 CCATCTAGCCACG 241 TTCCTGGGGATAA 296 290 CATGTACTCACCCGTTCCTGGGGATA 173 CATGTACTCACCC 232 GTTCCTGGGGATA 257 291 CTAGCCACGAATCACTAGTTCCTGGG 174 CTAGCCACGAATC 197 ACTAGTTCCTGGG 297 292 CCACGAATCTACCGCATTTATTCAAC 175 CCACGAATCTACC 193 GCATTTATTCAAC 199 293 CTAGCCACGAATCGTACTCACCCATC 176 CTAGCCACGAATC 197 GTACTCACCCATC 226 294 GAATCTACCCACTTCACCAGCATT 177 GAATCTACCCAC 203 TTCACCAGCATT 298 295 CGAATCTACCCATTCACCAGCATT 178 CGAATCTACCCA 204 TTCACCAGCATT 298 296 GAATCTACCCACATTCACCAGCAT 179 GAATCTACCCAC 203 ATTCACCAGCAT 299 297 CGAATCTACCCAATTCACCAGCAT 180 CGAATCTACCCA 204 ATTCACCAGCAT 299 298 GAATCTACCCACTTCACCAGCATT 177 GAATCTACCCAC 203 TTCACCAGCATT 298 299 CGAATCTACCCATTCACCAGCATT 178 CGAATCTACCCA 204 TTCACCAGCATT 298 300 GAATCTACCCACATTCACCAGCAT 179 GAATCTACCCAC 203 ATTCACCAGCAT 299 301 CGAATCTACCCAATTCACCAGCAT 180 CGAATCTACCCA 204 ATTCACCAGCAT 299 302 CGAATCTACCCACATTCACCAGCATT 140 CGAATCTACCCAC 192 ATTCACCAGCATT 292 303 CGAATCTACCCACATTCACCAGCATT 140 CGAATCTACCCAC 192 ATTCACCAGCATT 292 304 CGAATCTACCCACATTCACCAGCATT 140 CGAATCTACCCAC 192 ATTCACCAGCATT 292 305 GAATCTACCCACATTCACCAGCATT 181

GAATCTACCCAC 203 ATTCACCAGCATT 292 306 CGAATCTACCCACATTCACCAGCAT 182 CGAATCTACCCAC 192 ATTCACCAGCAT 299 307 CGAATCTACCCATTCACCAGCATT 178 CGAATCTACCCA 204 TTCACCAGCATT 298 308 CGAATCTACCCATTCACCAGCATT 178 CGAATCTACCCA 204 TTCACCAGCATT 298 309 CATATATCCATCTAGCCACG 183 CATATATCCA 242 TCTAGCCACG 300 310 ACGAATCTACCCATCCATGT 184 ACGAATCTAC 243 CCATCCATGT 301 311 TGTCCATCCACGAATCTACC 185 TGTCCATCCA 244 CGAATCTACC 212 312 CGAATCTACCAACTCATCCA 186 CGAATCTACC 212 AACTCATCCA 302 313 ACGAATCTACCATCCATGTA 187 ACGAATCTAC 243 CATCCATGTA 303 314 CGAATCTACCCATCTCTCCA 188 CGAATCTACC 212 CATCTCTCCA 304 315 CACGAATCTACCCATCTCTC 189 CACGAATCTA 245 CCCATCTCTC 305 316 ACGAATCTACCCATCTCTCC 190 ACGAATCTAC 243 CCATCTCTCC 306 317 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 318 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 319 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 320 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 321 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 322 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 323 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 324 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 325 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 326 CGAATCTACCATCCATGTAC 82 CGAATCTACC 212 ATCCATGTAC 271 327 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 328 CGAATCTACCCACATCTGTTCAATCA 2 CGAATCTACCCAC 192 ATCTGTTCAATCA 247 329 CACGAATCTACCCCCATCCATGTACT 1 CACGAATCTACCC 191 CCATCCATGTACT 246 330 CCATCCATCATCCGGGATAAGAGTTC 16 CCATCCATCATCC 198 GGGATAAGAGTTC 253 331 GCATTTATTCAACGGGGATAAGAGTT 17 GCATTTATTCAAC 199 GGGGATAAGAGTT 254 †When C is provided in a nucleobase sequence disclosed herein, without modification, such as in Table 3, it encompasses cytidine or 5′-methyl cytidine †Each thymine base (T) in any one of the oligonucleotide sequences provided in Table 3 may independently and optionally be replaced with a uracil base (U).

TABLE-US-00016 TABLE 9 Nucleobase sequences of antisense oligonucleotides provided herein ASO SEQ Antisense Sequence 1 SEQ Antisense Sequence 2 SEQ # Nucleobase Sequence (5′.fwdarw.3′) ID NO: (5′.fwdarw.3′) ID NO: (5′.fwdarw.3′) ID NO: 350 GCATTTATTCAACGGGGATAAGAGTT 17 GCATTTATTCAAC 199 GGGGATAAGAGTT 254 351 GCATTTATTCAACGGGGATAAGAGTT 17 GCATTTATTCAAC 199 GGGGATAAGAGTT 254 352 GCATTTATTCAACGGGATAAGAGTTC 124 GCATTTATTCAAC 199 GGGATAAGAGTTC 253 353 GCATTTATTCAACGGGATAAGAGTTC 124 GCATTTATTCAAC 199 GGGATAAGAGTTC 253 354 GCATTTATTCAACGGGATAAGAGTTC 124 GCATTTATTCAAC 199 GGGATAAGAGTTC 253 355 GCATTTATTCAACGGGATAAGAGTTC 124 GCATTTATTCAAC 199 GGGATAAGAGTTC 253 356 GCATTTATTCAACGGGATAAGAGTTC 124 GCATTTATTCAAC 199 GGGATAAGAGTTC 253 357 GCATTTATTCAACGGATAAGAGTTCT 125 GCATTTATTCAAC 199 GGATAAGAGTTCT 252 358 CATTTATTCAACGGATAAGAGTTC 338 CATTTATTCAAC 220 GGATAAGAGTTC 267 359 CATTTATTCAACGGATAAGAGTTC 338 CATTTATTCAAC 220 GGATAAGAGTTC 267 360 CATTTATTCAACGGGATAAGAGTT 117 CATTTATTCAAC 220 GGGATAAGAGTT 266 361 CATTTATTCAACGGGGATAAGAGTT 339 CATTTATTCAAC 220 GGGGATAAGAGTT 254 362 GCATTTATTCAAGGGGATAAGAGTT 340 GCATTTATTCAA 495 GGGGATAAGAGTT 254 363 GCATTTATTCAACGGGATAAGAGTT 341 GCATTTATTCAAC 199 GGGATAAGAGTT 266

364 GCATTTATTCAACGGGGATAAGAGT 342 GCATTTATTCAAC 199 GGGGATAAGAGT 284 365 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 366 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 367 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 368 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 369 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 370 TGTTCAATCATTGGGATAAGAGTTC 343 TGTTCAATCATT 496 GGGATAAGAGTTC 253 371 TGTTCAATCATTCGGGATAAGAGTT 344 TGTTCAATCATTC 194 GGGATAAGAGTT 266 372 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 373 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 374 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 375 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 376 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 377 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 378 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 379 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 380 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 381 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 382 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 383 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 384 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 385 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 386 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 387 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 388 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 389 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 390 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 391 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 392 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 393 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 394 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 395 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 396 TGTTCAATCATTCGGGATAAGAGTTC 345 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 397 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 398 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 399 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 400 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 401 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 402 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 403 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 404 TGTTCAATCATTCGGGATAAGAGTTC

9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 405 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 406 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 407 TGTTCAATCATTC[n3]GGGATAAGAGTTC 194/ TGTTCAATCATTC 194 GGGATAAGAGTTC 253 253 408 TGTTCAATCATTC[n6]GGGATAAGAGTTC 194/ TGTTCAATCATTC 194 GGGATAAGAGTTC 253 253 409 TCTGTTCAATCATTGGGGATAAGAGT 347 TCTGTTCAATCAT 274 TGGGGATAAGAGT 285 410 TCTGTTCAATCATGGGGATAAGAGTT 348 TCTGTTCAATCAT 274 GGGGATAAGAGTT 254 411 TCTGTTCAATCATGGGATAAGAGTTC 349 TCTGTTCAATCAT 274 GGGATAAGAGTTC 253 412 CTGTTCAATCATTTGGGGATAAGAGT 350 CTGTTCAATCATT 288 TGGGGATAAGAGT 285 413 CTGTTCAATCATTGGGGATAAGAGTT 351 CTGTTCAATCATT 288 GGGGATAAGAGTT 254 414 CTGTTCAATCATTGGGATAAGAGTTC 352 CTGTTCAATCATT 288 GGGATAAGAGTTC 253 415 TGTTCAATCATTCTGGGGATAAGAGT 353 TGTTCAATCATTC 194 TGGGGATAAGAGT 285 416 TGTTCAATCATTCGGGGATAAGAGTT 163 TGTTCAATCATTC 194 GGGGATAAGAGTT 254 417 TGTTCAATCATTCGGATAAGAGTTCT 354 TGTTCAATCATTC 194 GGATAAGAGTTCT 252 418 TGTTCAATCATTCGGGATAAGAGTTC 9 TGTTCAATCATTC 194 GGGATAAGAGTTC 253 419 TCCATCCATCCTTGAGTTCTTTCCAG 355 TCCATCCATCCTT 497 GAGTTCTTTCCAG 249 420 TCCATCCATCCTT[n6]GAGTTCTTTCCAG 497/ TCCATCCATCCTT 497 GAGTTCTTTCCAG 249 249 421 CTCTCCATCCATCTTCTTTCCAGGAA 357 CTCTCCATCCATC 498 TTCTTTCCAGGAA 516 422 TCCATCCATCCTTGAGTTCTTTCCAG 355 TCCATCCATCCTT 497 GAGTTCTTTCCAG 249 423 CATCCATCTATCCGGGATAAGAGTTC 159 CATCCATCTATCC 233 GGGATAAGAGTTC 253 424 CATCCATCTATCCGGGATAAGAGTTC 159 CATCCATCTATCC 233 GGGATAAGAGTTC 253 425 CATCCATCTATCCGGGATAAGAGTTC 159 CATCCATCTATCC 233 GGGATAAGAGTTC 253 426 CATCCATCTATCCGGGATAAGAGTTC 159 CATCCATCTATCC 233 GGGATAAGAGTTC 253 427 CATCCATCTATCCGGGATAAGAGTTC 159 CATCCATCTATCC 233 GGGATAAGAGTTC 253 428 ATCCATCTATCCGGATAAGAGTTC 358 ATCCATCTATCC 499 GGATAAGAGTTC 267 429 ATCCATCTATCCGGGATAAGAGTT 359 ATCCATCTATCC 499 GGGATAAGAGTT 266 430 ATCCATCTATCCGGATAAGAGTTC 358 ATCCATCTATCC 499 GGATAAGAGTTC 267 431 ATCCATCTATCCGGGATAAGAGTT 359 ATCCATCTATCC 499 GGGATAAGAGTT 266 432 CAACTCATCCATCAGAGTTCTTTCCA 360 CAACTCATCCATC 500 AGAGTTCTTTCCA 517 433 GCCACGAATCTACGGATAAGAGTTCT 361 GCCACGAATCTAC 201 GGATAAGAGTTCT 252 434 CCACGAATCTACCGGGATAAGAGTTC 14 CCACGAATCTACC 193 GGGATAAGAGTTC 253 435 CCACGAATCTACCGGGATAAGAGTTC 14 CCACGAATCTACC 193 GGGATAAGAGTTC 253 436 CCACGAATCTACCGGGATAAGAGTTC 14 CCACGAATCTACC 193 GGGATAAGAGTTC 253 437 CCACGAATCTACCGGGATAAGAGTTC 14 CCACGAATCTACC 193 GGGATAAGAGTTC 253 438 CCACGAATCTACCGGGATAAGAGTTC 14 CCACGAATCTACC 193 GGGATAAGAGTTC 253 439 CCACGAATCTACCGGATAAGAGTTCT 8 CCACGAATCTACC 193 GGATAAGAGTTCT 252 440 CCACGAATCTACCGGATAAGAGTTCT 8 CCACGAATCTACC 193 GGATAAGAGTTCT 252 441 CCACGAATCTACCGGATAAGAGTTCT 8 CCACGAATCTACC 193 GGATAAGAGTTCT 252 442 CCACGAATCTACCGGATAAGAGTTCT 8 CCACGAATCTACC 193 GGATAAGAGTTCT 252 443 CCACGAATCTACCGGATAAGAGTTCT 8 CCACGAATCTACC 193 GGATAAGAGTTCT 252 444 CACGAATCTACCGATAAGAGTTCT 362 CACGAATCTACC 206 GATAAGAGTTCT 518 445 CCACGAATCTACGATAAGAGTTCT

363 CCACGAATCTAC 501 GATAAGAGTTCT 518 446 CACGAATCTACCGGGATAAGAGT
364 CCACGAATCTAC 501 GGGGATAAGAGT 284 447 CCACGAATCTACGGGGATAAGAGT
365 GCCACGAATCTA 502 GGGGATAAGAGT 284 448 CACGAATCTACCGGATAAGAGTTC
366 CACGAATCTACC 206 GGATAAGAGTTC 267 449 CACGAATCTACCGGGATAAGAGTT
367 CACGAATCTACC 206 GGGATAAGAGTT 266 450 CCACGAATCTACGGATAAGAGTTC
368 CCACGAATCTAC 501 GGATAAGAGTTC 267 451 CCACGAATCTACGGGATAAGAGTT
369 CCACGAATCTAC 501 GGGATAAGAGTT 266 452 CACGAATCTACCGATAAGAGTTCT
362 CACGAATCTACC 206 GATAAGAGTTCT 518 453 CCACGAATCTACGGGGATAAGAGT
365 CCACGAATCTAC 501 GGGGATAAGAGT 284 454
CCACGAATCTACCGGGGATAAGAGTT 22 CCACGAATCTACC 193 GGGGATAAGAGTT
254 455 CCACGAATCTACCGGGGATAAGAGTT 22 CCACGAATCTACC 193
GGGGATAAGAGTT 254 456 CCACGAATCTACCGGGGATAAGAGTT 22
CCACGAATCTACC 193 GGGGATAAGAGTT 254 457
CCACGAATCTACCGGGGATAAGAGTT 22 CCACGAATCTACC 193 GGGGATAAGAGTT
254 458 CCACGAATCTACCGGGGATAAGAGTT 22 CCACGAATCTACC 193
GGGGATAAGAGTT 254 459 ACGAATCTACCCAGGGGATAAGAGTT 370
ACGAATCTACCCA 196 GGGGATAAGAGTT 254 460
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461 CCACGAATCTACCATAAGAGTTCTTT 372 CCACGAATCTACC 193 ATAAGAGTTCTTT
519 462 CCATCCATCATCCGGGATAAGAGTCC 373 CCATCCATCATCC 198
GGGATAAGAGTTC 253 463 TCCATCCATCATCGGGGATAAGAGTT 120 TCCATCCATCATC
222 GGGGATAAGAGTT 254 464 TCCATCCATCATCGGGGATAAGAGTT 120
TCCATCCATCATC 222 GGGGATAAGAGTT 254 465 TCCATCCATCATCGGGGATAAGAGTT
120 TCCATCCATCATC 222 GGGGATAAGAGTT 254 466
TCCATCCATCATCGGGGATAAGAGTT 120 TCCATCCATCATC 222 GGGGATAAGAGTT 254
467 TCCATCCATCATCGGGGATAAGAGTT 120 TCCATCCATCATC 222 GGGGATAAGAGTT
254 468 TCCATCCATCATCGGGATAAGAGTTC 121 TCCATCCATCATC 222
GGGATAAGAGTTC 253 469 TCCATCCATCATCGGGATAAGAGTTC 121 TCCATCCATCATC
222 GGGATAAGAGTTC 253 470 TCCATCCATCATCGGGATAAGAGTTC 121
TCCATCCATCATC 222 GGGATAAGAGTTC 253 471 TCCATCCATCATCGGGATAAGAGTTC
121 TCCATCCATCATC 222 GGGATAAGAGTTC 253 472
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GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 774
GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 775
GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 776
GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 777

GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 778
GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 779
GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 780
GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 781
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 782
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 783
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 784
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 785
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 786
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 787
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 788
GAATCTACACATCCATGTACT 482 GAATCTAC CCATCCATGTACT 246 789
GAATCTACACATCCATGTACT 482 GAATCTAC CCATCCATGTACT 246 790
GAATCTACACATCCATGTACT 482 GAATCTAC CCATCCATGTACT 246 791
GAATCTACACATCCATGTACT 482 GAATCTAC CCATCCATGTACT 246 792
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 793
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 794
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 795
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 796
GAATCTA[n3]CCATCCATGTACT 246 GAATCTA CCATCCATGTACT 246 797
GAATCTA[n6]CCATCCATGTACT 246 GAATCTA CCATCCATGTACT 246 798
GAATCTA[n3]CCCATCCATGTACT 534 GAATCTA CCCATCCATGTACT 534 799
GAATCTA[n6]CCCATCCATGTACT 534 GAATCTA CCCATCCATGTACT 534 800
GAATCTACC[n3]ATCCATGTACT 535 GAATCTACC ATCCATGTACT 535 801
GAATCTACC[n6]ATCCATGTACT 535 GAATCTACC ATCCATGTACT 535 802
GAATCTACC[n3]CATCCATGTACT 264 GAATCTACC CATCCATGTACT 264 803
GAATCTACC[n6]CATCCATGTACT 264 GAATCTACC CATCCATGTACT 264 804
GAATCTAC[n3]CATCCATGTACT 264 GAATCTAC CATCCATGTACT 264 805
GAATCTAC[n3]CCATCCATGTACT 246 GAATCTAC CCATCCATGTACT 246 806
GAATCTAC[n6]CCATCCATGTACT 246 GAATCTAC CCATCCATGTACT 246 807
GAATCTACCATCCATGTACT 455 GAATCTA CCATCCATGTACT 246 808
GAATCTACCCATCCATGTACT 461 GAATCTAC CCATCCATGTACT 246 809
TCTACCCCCATCCATGTACT 489 TCTACCC CCATCCATGTACT 246 810
ATCTACCCCATCCATGTACT 490 ATCTACC CCATCCATGTACT 246 811
GTCGCCGCCATCCATGTACT 491 GTCGCCG CCATCCATGTACT 246 812
GGGGTCGCCATCCATGTACT 492 GGGGTCG CCATCCATGTACT 246 813
GGGTCGCCCATCCATGTACT 493 GGGTCGC CCATCCATGTACT 246 814
GGTCGCCCCATCCATGTACT 494 GGTCGCC CCATCCATGTACT 246 †When C is provided in a nucleobase sequence disclosed herein, without modification, such as in Table 9, it encompasses cytidine or 5′-methyl cytidine ‡Each thymine base (T) in any one of the oligonucleotide sequences provided in Table 9 may independently and optionally be replaced with a uracil base (U). 5 [n3] is a C3 spacer as shown in Table 16. [n6] is a C6 spacer as shown in Table 16.

Modified Antisense Oligonucleotides

[0191] Some aspects of the present disclosure provide modified (e.g., chemically modified) antisense oligonucleotides. For example, any one of the antisense oligonucleotides provided herein (e.g., in Table 3 or Table 9) may comprise one or more chemical modifications including, e.g., one or more modified nucleosides and/or one or more modified nucleosides and/or modified internucleoside linkages.

[0192] In some embodiments, an antisense oligonucleotide disclosed herein, comprises a first antisense sequence and a second antisense sequence, and comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, or more) modified nucleosides and/or one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) modified internucleoside linkages. In some embodiments, a modified nucleoside comprises a modification at the 2′ position of the sugar (referred to herein as a “2′-modified nucleoside”). In some embodiments, a 2′-modified nucleoside is a not bicyclic (referred to herein as “non-bicyclic 2′-modified nucleoside”). In some embodiments, a 2′-modified nucleoside comprises a modification at the 2′ position and 4′ position of the sugar creating a bridged structure (referred to herein as a “2′-4′ bridged nucleoside”). In some embodiments, the one or more modified nucleosides comprise a non-bicyclic 2′-modified nucleoside, a 2′-4′ bridged nucleoside, or combinations thereof. In some embodiments, all modified nucleosides in an antisense oligonucleotide are 2′-modified nucleosides. In some embodiments, the non-bicyclic 2′-modified nucleoside is a 2′-O-methoxyethyl (2′-MOE) modified nucleoside, a 2′-O-Methyl (2′-O-Me) nucleoside, or a 2′-fluoro (2′-F) modified nucleoside, and/or the 2′-4′ bridged nucleoside is a locked nucleic acid (LNA, 2′-4′ methylene bridge), an ethylene-bridged nucleic acid (ENA, 2′-4′ ethylene bridge) or an constrained ethyl nucleic acid (cEt, 2′-4′ ethylene bridge). In some embodiments, the 2′-modified nucleoside is a 2′-O, 4′-C-Spirocyclopropylene bridged nucleic acid (scpBNA). In some embodiments, the 2′-modified nucleoside is a 2′-O—N-methylacetamide (2′-O-NMA) modified nucleoside. In some embodiments, the 2′-modified nucleoside is an amido-bridged nucleic acid (AmNA). In some embodiments, the 2′-modified nucleoside is a 2′-O, 4′-C-aminomethylene bridged nucleic acid (BNA (NC)). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more LNAs. In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more 2′-MOE modified nucleosides and one or more LNAs. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside.

[0193] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the following 2′-modified nucleosides: 2′-O-Mc, 2′-F, 2′-MOE, LNA, ENA, cEt, scpBNA, 2′-O-NMA, AmNA, and BNA (NC). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the following 2′-modified nucleosides: ENA, cEt, scpBNA, 2′-O-NMA, AmNA, and BNA (NC). In some embodiments, each nucleoside in an antisense oligonucleotide disclosed herein is a modified nucleoside (e.g., 2′-modified nucleoside). In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an ENA. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are cEt. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a scpBNA. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are 2′-O-NMA modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an AmNA. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a BNA (NC). In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are selected from the following 2′-modified nucleosides: 2′-O-Me, 2′-F, ENA, cEt, scpBNA, 2′-O-NMA, AmNA, and BNA (NC) and the remaining modified nucleosides are 2′-MOE modified nucleosides, LNAs, or a mix of 2′MOE modified nucleosides and LNAs. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are selected from the following 2′-modified nucleosides: ENA, scpBNA, 2′-O-NMA, AmNA, and BNA (NC) and the remaining modified nucleosides are 2′-MOE modified nucleosides, LNAs, or a mix of 2′MOE modified nucleosides and LNAs. In some embodiments, the remaining modified nucleosides are 2′-MOE modified nucleosides. In some embodiments, the remaining modified nucleosides are LNAs. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an AmNA and the remaining modified nucleosides are a mix of 2′-MOE modified nucleosides and LNAs. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an ENA and the remaining modified nucleosides are 2′-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a scpBNA and the remaining modified nucleosides are 2′-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are an AmNA and the remaining modified nucleosides are 2′-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are a BNA (NC) and the remaining modified nucleosides are 2′-MOE modified nucleosides. In some embodiments, one or more (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) of the modified nucleosides are 2′-O-NMA modified nucleosides and the remaining modified nucleosides are LNAs.

[0194] In some embodiments, an antisense oligonucleotide disclosed herein, comprises a first antisense sequence and a second antisense sequence, and comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) modified internucleoside linkages. In some embodiments, each internucleoside linkage of the antisense oligonucleotide is a modified internucleoside linkage. Non limiting examples of internucleoside linkages include phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates comprising 3′ alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates comprising 3′-amino phosphoramidate and aminoalkylphosphoramidates, mesyl phosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. It is to be understood that, in any one of the antisense oligonucleotides disclosed herein, the rest of internucleoside linkages, unless otherwise specified, are all phosphodiester internucleoside linkages. In some embodiments, each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotide comprises a mix of phosphodiester linkages and phosphorothioate linkages. In some embodiments, the antisense oligonucleotide comprises one or more phosphorodiamidate morpholinos. In some embodiments, the antisense oligonucleotide is a phosphorodiamidate morpholino oligomer (PMO).

[0195] In some embodiments, an antisense oligonucleotide disclosed herein, comprises a first antisense sequence and a second antisense sequence, wherein each cytosine of the oligonucleotide is a 5-methyl-cytosine. In some embodiments, each cytosine of the oligonucleotide is not a 5-methyl-cytosine. In some embodiments, one or more of the cytosines of the oligonucleotide is a 5-methyl-cytosine.

[0196] Table 4 lists various exemplary nucleosides with 3′-phosphate or 3′-phosphorothioate and the structures thereof.

TABLE-US-00017 TABLE 4 Exemplary nucleosides with 3′-phosphate or 3′- phosphorothioate and structures Abbre- Nucleosides viation or linkage Structure lA 2′-O,4′-C- methylene bridge adenosine- 3′-phosphate [00001]embedded image lAs 2′-O,4′-C- methylene bridge adenosine-3′- phosphorothioate [00002]embedded image lC 2′-O,4′-C- methylene bridge cytidine-3′- phosphate [00003]embedded image lCs 2′-O,4′-C- methylene bridge cytidine-3′- phosphorothioate [00004] embedded image l.sup.5C 2′-O,4′-C-methylene bridge-5′- methyl cytidine- 3′-phosphate [00005] embedded image l.sup.5Cs 2′-O,4′-C-methylene bridge-5- methylcytidine-3′- phosphorothioate [00006]embedded image lG 2′-O,4′-C- methylene bridge guanosine- 3′-phosphate [00007] embedded image lGs 2′-O,4′-C- methylene bridge guanosine-3′- phosphorothioate [00008] embedded image lU 2′-O,4′-C- methylene bridge uridine-3′- phosphate [00009] embedded image lUs 2′-O,4′-C- methylene bridge uridine-3′- phosphorothioate [00010] embedded image lT 2′-O,4′-C- methylene bridge thymidine- 3′-phosphate [00011] embedded image lTs 2′-O,4′-C- methylene bridge thymidine- 3′-phosphorothioate [00012] embedded image iA 2′-O-methoxyethyl- adenosine-3′- phosphate [00013]embedded image iAs

2′-O-methoxyethyl- adenosine-3′- phosphorothioate [00014]embedded image iC 2′-O-methoxyethyl- cytidine-3′- phosphate [00015]embedded image iCs 2′-O-methoxyethyl- cytidine-3′- phosphorothioate [00016]embedded image i.sup.5C 2′-O- methoxyethyl-5- methylcytidine- 3′-phosphate [00017]embedded image i.sup.5Cs 2′-O-methoxyethyl-5- methylcytidine-3′-phosphorothioate [00018]embedded image iG 2′-O-methoxyethyl- guanosine-3′- phosphate [00019]embedded image iGs 2′-O-methoxyethyl- guanosine-3′- phosphorothioate [00020] embedded image iU 2′-O-methoxyethyl- luridine-3′- phosphate [00021]embedded image iUs 2′-O-methoxyethyl- uridine-3′- phosphorothioate [00022]embedded image iT 2′-O-methoxyethyll-thymidine-3′- phosphate [00023]embedded image iTs 2′-O-methoxyethyl- thymidine-3′-phosphorothioate [00024]embedded image s Phosphorothioate linkage [00025]embedded image TABLE-US-00018 TABLE 10 Additional exemplary modified nucleosides Abbreviation Name Structure z AmNA [00026]embedded image R is Methyl (Me) or H b BNA(NC) [00027] embedded image R is Methyl (Me) or H ena ENA [00028]embedded image scp scpBNA [00029]embedded image nma 2′-O-NMA [00030]embedded image *"base" represents any nucleobase or derivative thereof

TABLE-US-00019 TABLE 16 Additional exemplary abbreviations Abbreviation Name Structure x C16 lipid (6-[(1- oxohexadecyl)amino] hexyl]phosphoryl) [00031]embedded image RO is 5′ terminal of oligonucleotides n3 C3 spacer [00032]embedded image n6 C6 spacer [00033] embedded image

[0197] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more modified nucleosides (e.g., one or more 2′-modified nucleosides). In some embodiments, the antisense oligonucleotide comprises a non-bicyclic 2′ modified nucleoside, a 2′-4′ bridged nucleoside, or combinations thereof (e.g., any combination of 2′-O-methoxyethyl (2′-MOE) modified nucleoside, 2′-O-Methyl (2′-O-Me) modified nucleoside, 2′-fluoro (2′-F) modified nucleosides, LNA, ENA, and cEt). In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more modified nucleosides (e.g., one or more 2′-modified nucleosides). In some embodiments, the antisense oligonucleotide comprises a non-bicyclic 2′-modified nucleoside, a 2′-4′ bridged nucleoside, or combinations thereof (e.g., any combination of 2′-MOE modified nucleoside, 2′-O-Me modified nucleoside, 2′-F modified nucleosides, LNA, ENA, cEt, AmNA, BNA (NC), scpBNA, and 2′-O-NMA). In some embodiments, the antisense oligonucleotide comprises a combination of 2′-MOE modified nucleosides and LNAs. In some embodiments, the antisense oligonucleotide comprises a combination of 2′-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) ENA. In some embodiments, the antisense oligonucleotide comprises a combination of 2′-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) AmNA. In some embodiments, the antisense oligonucleotide comprises a combination of 2′-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) BNA (NC), In some embodiments, the antisense oligonucleotide comprises a combination of 2′-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) scpBNA. In some embodiments, the antisense oligonucleotide comprises a combination of 2′-MOE modified nucleosides and one or more (e.g., 1, 2, 3, or 4) 2′-O-NMA modified nucleosides. In some embodiments, comprises a combination of LNAs and one or more (e.g., 1, 2, 3, or 4) 2′-O-NMA modified nucleosides. In some embodiments, the antisense oligonucleotide comprises a combination of 2′-MOE modified nucleosides, LNAs, and one or more (e.g., 1, 2, 3, or 4) AmNA. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside. In some embodiments, the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, 12, or more) positions of the antisense oligonucleotide are nucleosides with the same 2′ chemistry in the sugar moiety selected from, e.g., 2′-MOE modified nucleoside, 2′-O-Methyl (2′-O-Mc) modified nucleoside, 2′-fluoro (2′-F) modified nucleoside, LNA, ENA, and cEt, and the remaining nucleosides are nucleosides with a different 2′ chemistry in the sugar moiety, selected from e.g., 2′-MOE modified nucleoside, 2′-O-Methyl (2′-O-Me) modified nucleoside, 2′-fluoro (2′-

F) modified nucleoside, LNA, ENA, and cEt. In some embodiments, the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and the remaining nucleosides are nucleosides with different 2′ chemistry in the sugar moiety, e.g., 2′-MOE, 2′-O-Methyl (2′-O-Mc), 2′-fluoro (2′-F), ENA, or cEt. In some embodiments, the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides.

[0198] In some embodiments, an antisense oligonucleotide disclosed herein comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2′-4′ bridged nucleoside (e.g., LNA, ENA, or cEt). In some embodiments, the remaining nucleosides are non-bicyclic 2′ modified nucleoside (e.g., 2′-MOE modified nucleoside, 2′-O-Methyl (2′-O-Me) modified nucleoside, 2′-fluoro (2′-F) modified nucleoside). In some embodiments, at least one of the one or more motifs is located at the 5′ end of the antisense oligonucleotide. In some embodiments, at least one of the one or more motifs is located at the 3′ end of the antisense oligonucleotide. In some embodiments, at least one of the one or more motifs is located within any of positions 9-16 (counting 5′.fwdarw.3′) of the antisense oligonucleotide.

[0199] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive LNAs at the 5′ end of the antisense oligonucleotide, the 3′ end of the antisense oligonucleotide, and/or within any of positions 9-16 (counting 5′.fwdarw.3′) (e.g., 9-10, 9-12, 10-11, 10-12, 10-13, 10-15, 10-16, 11-12, 11-14, 11-16, 12-13, 12-15, and 13-15), and all of the remaining nucleosides are 2′-MOE modified nucleosides.

[0200] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside.

[0201] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein the nucleosides at one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8 or more) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-O-NMA modified nucleosides.

[0202] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside.

[0203] In some embodiments, an antisense oligonucleotide disclosed herein is 20-28 (e.g., 20, 21, 22, 23, 24, 25, 26, 27, or 28) nucleosides in length, wherein one or more nucleobases at the 3′ end of the first antisense sequence is complementary to one or more nucleobases at the 3′ end of the second target region, and/or one or more nucleobases at the 5′ end of the second antisense sequence is complementary to one or more nucleobases at the 5′ end of the first target region.

[0204] In some embodiments, an antisense oligonucleotide disclosed herein is 28 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0205] In some embodiments, an antisense oligonucleotide disclosed herein is 28 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0206] In some embodiments, an antisense oligonucleotide disclosed herein is 27 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0207] In some embodiments, an antisense oligonucleotide disclosed herein is 27 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0208] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2′-4′ bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-6, 10-17, and/or 22-26 (counting 5′.fwdarw.3′). In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0209] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 11-16 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0210] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-4 and 23-26 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0211] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-4, 12-15, and 23-26 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0212] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-3, 11-16, and 24-26 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0213] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 1-4 and 11-16 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0214] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at positions 11-16 and 23-26 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0215] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0216] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0217] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises two phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0218] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises five phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0219] In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0220] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0221] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0222] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0223] In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0224] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2′-4′ bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-5, 10-16, and/or 20-24. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0225] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 4, 8, 12, 16, 20, and 24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0226] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 10-15 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0227] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1-4, 11-14, and 21-24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0228] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1-4, 11-14, and 21-24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein the antisense oligonucleotide comprises two phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0229] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 4, 11-14, 21, and 24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0230] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 4, 12, 13, 21, and 24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0231] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 2, 12, 13, 23, and 24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0232] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 2, 7, 8, 17, 18, 23, and 24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0233] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 13-15 and 22-24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0234] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1-3 and 10-12 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0235] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at positions 1, 6, 12, 13, 19, and 24 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0236] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0237] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21 or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 or more) phosphodiester internucleoside linkages.

[0238] In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0239] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2′-4′ bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-5, 9-14, and/or 20-22. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0240] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at positions 1, 2, 10-13, 21, and 22 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0241] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at positions 1, 2, 10-13, 21, and 22 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and five or more (e.g., 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0242] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at positions 1, 2, 10-13, 21, and 22 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more) phosphorothioate internucleoside linkages and twelve or less (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12) phosphodiester internucleoside linkages.
[0243] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.
[0244] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or more) phosphodiester internucleoside linkages.
[0245] In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage. In some embodiments, an antisense oligonucleotide disclosed herein is 19 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 21 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 22 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 23 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 24 nucleosides in length, wherein the nucleosides at four or more (e.g., 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 25 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 26 nucleosides in length, wherein the nucleosides at two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or

12) positions of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides. In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length and comprises one or more (e.g., 1, 2, or 3) motifs of 2-6 (e.g., 2, 3, 4, 5, or 6) consecutive 2′-4′ bridged nucleoside (e.g., LNA, ENA, or cEt) within positions 1-5, 9-13, and/or 16-20. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, the antisense oligonucleotide comprises a mix of one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphodiester internucleoside linkages.

[0246] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 19, and 20 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0247] In some embodiments, an antisense oligonucleotide disclosed herein is 21 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) phosphodiester internucleoside linkages.

[0248] In some embodiments, an antisense oligonucleotide disclosed herein is 21 nucleosides in length, wherein the nucleosides at positions 1 and 21 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-O-NMA modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0249] In some embodiments, an antisense oligonucleotide disclosed herein is 21 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0250] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 19, and 20 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein the antisense oligonucleotide comprises two phosphodiester internucleoside linkages and all of the remaining internucleoside linkages are phosphorothioate internucleoside linkages.

[0251] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 9, and 10 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0252] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 9-12 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0253] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 11, 12, 19, and 20 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0254] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 10, 11, 19, and 20 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.

[0255] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1, 2, 4, and 5 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.
[0256] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 16, 17, 19, and 20 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-MOE modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.
[0257] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17 or more) phosphorothioate internucleoside linkages and two or more (e.g., 2, 3, 4, 5, 6, 7, 8, 9 or more) phosphodiester internucleoside linkages.
[0258] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein the nucleosides at positions 1 and 20 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-O-NMA modified nucleosides, wherein each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.
[0259] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.
[0260] In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. In some embodiments, the antisense oligonucleotide comprises one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) phosphorothioate internucleoside linkages. In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage. In some embodiments, an antisense oligonucleotide disclosed herein is 20 nucleosides in length, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage, wherein one or more nucleobases at the 3′ end of the first antisense sequence is complementary to one or more nucleobases at the 3′ end of the second target region, and/or one or more nucleobases at the 5′ end of the second antisense sequence is complementary to one or more nucleobases at the 5′ end of the first target region.
[0261] In some embodiments, an antisense oligonucleotide disclosed herein comprises any one of the nucleobase sequences described herein (e.g., listed in Table 3 and Table 9) and further comprises modified nucleosides selected from one of the following patterns (5′.fwdarw.3′):
TABLE-US-00020 MMMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMMMMMMMMMMMMM, MMMMMMMMMMMMMMMMMMMM,
MMMMMMMMMMLLLLLLMMMMMMMMMM,
MMMMMMMMMLLLLLLMMMMMMMMM,
MMMMMMMMMMLLLLLLMMMMMMLLLL,
LLLLMMMMMMLLLLLLMMMMMMMMMM, LLLLMMMMMMLLLLMMMMMMLLLL,
LLLMMMMMMMLLLLLLMMMMMMMLLL, LLLLMMMMMMMLLLLMMMMMMMLLLL,

LLLLMMMMMMMMMMMMMMMMMMMLLLL, MMMLMMMLMMMLMMMLMMMLMMML, LMMLMMMMMMMLLLMMMMMMMLMML, LMMLMMMMMMMLLMMMMMMMLMML, LLMMMMMMMMMMLLMMMMMMMMMMLL, LLMMMMLLMMMMMMMMLLMMMMLL, MMMMMMMMMMMMMLLLMMMMMMMLLL, LLLMMMMMMMLLLMMMMMMMMMMMMM, LMMMMLMMMMMLLMMMMMLMMMML, LLMMMMMMMLLLLMMMMMMMLL, LLMMMMMMMMMMMMMMMMMMMLL, LLMMMMMMLLMMMMMMMMMMM, MMMMMMMMLLLLMMMMMMMMM, MMMMMMMMMMLLMMMMMMMLL, LLMMMMMMMLLMMMMMMMMLL, LLMLLMMMMMMMMMMMMMMMM, and MMMMMMMMMMMMMMMMMMLLMLL, wherein M is a 2′-O-methoxyethyl (2′-MOE) modified nucleoside and L is an LNA.

[0262] In some embodiments, an antisense oligonucleotide disclosed herein comprises any one of the nucleobase sequences described herein (e.g., listed in Table 3 and Table 9) and further comprises modified nucleosides selected from one of the following patterns (5′.fwdarw.3′):

TABLE-US-00021 MLLMMMMMMMMMMMMMMMMMMMMMMLL, MMLLMMMMMMLLMMMMMMMMMMMMM, MMMMMLLMMMMMMMMMMMMMMMMMMM, MMMMMLLMMMMMMMMMMMMLLMMMMMM, MMMMMMMMMMMMLLLLMMMMMMMMMMM, MLLMMMMMMMMMMMMMMMMMMMMMMMM, LLMMMMMMMMMMMLLMMMMMMMMMLL, LLMMMMMMMMMLLLLMMMMMMMLLLL, LLLLMMMMMMLLLLMMMMMMMMLL, LLLLMMMMMMMMLLMMMMMMMMLLLL, LLLMMMMMMMMLLLLMMMMMMMMLLL, LLLMMMMMMMMMLLMMMMMMMMMLLL, LLMLLMMMMMMLLLLMMMMMMLLLL, LLMMMMMMMMMMMMMMMMMMMMMMMMM, MMLLMMMMMMMMMMMMMMMMMMMMMMM, MMMMLLMMMMMMMMMMMMMMMMMMMMM, MMMMMMLLMMMMMMMMMMMMMMMMMMM, MMMMMMMMLLMMMMMMMMMMMMMMMMM, MMMMMMMMMMLLMMMMMMMMMMMMMMM, MMMMMMMMMMMMLLMMMMMMMMMMMMM, MMMMMMMMMMMMMMLLMMMMMMMMMMM, MMMMMMMMMMMMMMMMLLMMMMMMMMM, MMMMMMMMMMMMMMMMMMLLMMMMMMM, MMMMMMMMMMMMMMMMMMMMLLMMMMM, MMMMMMMMMMMMMMMMMMMMMMLLMMM, MMMMMMMMMMMMMMMMMMMMMMMMMLL, MMLLMMMMMMMMMMMMMMMMMMLLMM, MMMMLLMMMMMMMMMMMMMMLLMMMM, MMMMMMMMLLMMMMMMLLMMMMMMMM, LMMMMMMMMMMMMMMMMMMMML, LLMMMMMMMMMMMMMMMMMMMMMMLL, MMMMMMMMLLMMMMMMMMMMM, LLMMMLLMMMMMMMMMMMMLL, LLMMMMMLLMMMMMMMMMLL, LLMMMMLLMMMMMMMMMMLL, LLMMMMMMMMMLLMMMMMLL, LMMMMMMMMMMMMMMMMMMMML, LMMMMMLLMMMMMMMMMMML, LLMMMMMMMMMMMMMMMMLL, LMMMMMMLLMMMMMMMMMML, LMMMMMLMMLMMMMMMMMML, MMLMMMLMMMMMMLMMMLMM, LMMMMLMMMMMMMLMMMMML, MMLLMMMMMMMMMMMMLLMM, MMMMMMLLMMMMLLMMMMMM, LLMMMMMMMMMMMMMMMMMM, MMLLMMMMMMMMMMMMMMMM, MMMMLLMMMMMMMMMMMMMM, MMMMMMLLMMMMMMMMMMMM, MMMMMMMMLLMMMMMMMMMM, MMMMMMMMMMLLMMMMMMMM,

MMMMMMMMMMMMLLMMMMMM, MMMMMMMMMMMMMMLLMMMM, MMMMMMMMMMMMMMMMLLMM, MMMMMMMMMMMMMMMMMMLL, LLMMMMMMMMMMMMMMMMMMM, MMLLMMMMMMMMMMMMMMMMM, MMMMLLMMMMMMMMMMMMMMM, MMMMMMLLMMMMMMMMMMMMM, MMMMMMMMMMLLMMMMMMMMM, MMMMMMMMMMMMLLMMMMMMM, MMMMMMMMMMMMMMLLMMMMM, MMMMMMMMMMMMMMMMLLMMM, MMMMMMMMMMMMMMMMMMLLM, MMMMMMMMMMMMMMMMMMMLL, MLMMMMMMMMMMMMMMMMMLM, MMMLMMMMMMMMMMMMMLMMM, MMMMMLMMMMMMMMMLMMMMM, MMMMMMMLMMMMMLMMMMMMM, MLMMMMMMMMMMMMMMMMLM, MMMLMMMMMMMMMMMMLMMM, MMMMMLMMMMMMMMLMMMMM, MMMMMMMLMMMMLMMMMMMM, LMMMMMMMMMMMMMMMMMMLM, MLMMMMMMMMMMMMMMMMMML, LMMMMMMMMMMMMMMMMMMM, MMMMMMMMMMMMMMMMMMML, LMMMMMMMMMMMMMMMMMMMM, MMMMMMMMMMMMMMMMMMMML, LMMMMMMMMMMMMMMMMMMMML, LMMMMMMMMMMMMMMMMMMMMML, LMMMMMMMMMMMMMMMMMMMMMML, LMMMMMLMMMMMMMMMMMMM, MMMMMMLMMMMMMMMMMMML, MMMMMMMLLMMMMMMMMMMM, LMMMMMMLMMMMMMMMMMMM, MMMMMMMLMMMMMMMMMMML, LMMMMMMMLMMMMMMMMMMM, LMMMMMMMMLMMMMMMMMMM, MMMMMMMMMLMMMMMMMMML, MMMMMMMMLMMMMMMMMMML, LMMMMMLMMMMMMMMMMMMMM, MMMMMMLMMMMMMMMMMMMML, MMMMMMMLLMMMMMMMMMMMM, LMMMMMMLMMMMMMMMMMMMM, MMMMMMMLMMMMMMMMMMMML, LMMMMMMMLMMMMMMMMMMMM, MMMMMMMMMLMMMMMMMMMML, MMMMMMMMLMMMMMMMMMMML, LMMMMMMMMLMMMMMMMMMMM, LMMMMMMMMMLMMMMMMMMMM, MMMMMMMMMMLMMMMMMMMML, L-nma-nma-nma-nma-nma-nma-nma-nma- nma-nma-nma-nma-nma-nma-nma-nma- nma-nma-L, L-nma-nma-nma-nma-nma-nma-nma-nma- nma-nma-nma-nma-nma-nma-nma-nma- nma-nma-nma-L, nma-nma-nma-nma-nma-nma-nma-nma-nma- nma-nma-nma-nma-nma-nma-nma-nma- nma-nma-nma-nma-nma-nma-nma-nma-nma, xdddMMMMMMMMMMMMMMMMMMMMMMMMMM, zzMMMMMMMzzzzMMMMMMMzL, zzzzMMMMMMzzzzMMMMMMzzzL, enaMMMMMMMMMMMMMMMMMMena, bMMMMMMMMMMMMMMMMMMb, enaMMMMMMMMMMMMMMMMMMMena, scpMMMMMMMMMMMMMMMMMMMscp, bMMMMMMMMMMMMMMMMMMMb, LMMMMMMMMMMMMMMMMMLM, MzMMMMMMMMMMMMMMMMzM, enaMMMMMMMMMMMMMMMMMenaM, MenaMMMMMMMMMMMMMMMMenaM, MenaMMMMMMMMMMMMMMMMMena, MzMMMMMMMMMMMMMMMMMzM, MscpMMMMMMMMMMMMMMMMMMscp, enaMMMMMMMMMMMMMMMMMMenaM, MenaMMMMMMMMMMMMMMMMMenaM, MenaMMMMMMMMMMMMMMMMMMena, xdddLMMMMMMMMMMMMMMMMMML, xdddLMMMMMMMMMMMMMMMMMMML, MMMMMMMMMMMMM[n]MMMMMMMMMMMMM, LMMMMMM[n]MMMMMMMMMMMML, LMMMMMM[n]MMMMMMMMMMMMML, LMMMMMMMM[n]MMMMMMMMMML, LMMMMMMMM[n]MMMMMMMMMMML, LMMMMMMM[n]MMMMMMMMMMML, and LMMMMMMM[n]MMMMMMMMMMMML, wherein M is a 2′-O-methoxyethyl (2′-MOE) modified nucleoside, L is an LNA, x is a C16 lipid, 6-[(1-oxohexadecyl)amino]hexyl] phosphoryl, [n] is a C3 or C6 spacer (e.g., a C3 or C6 spacer as shown in Table 16), dis DNA, ena is ENA, b is BNA(NC), z is AmNA, nma is 2′O-NMA modified nucleoside, and scp is scpBNA.

[0263] In some embodiments, an antisense oligonucleotide disclosed herein comprises modified

internucleoside linkages comprising one of the following patterns (5′.fwdarw.3′):
TABLE-US-00022 S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S- S-S-S-S, S-S-S-S-S-S-S-S-S-
S-S-S-S-S-S-S-S-S-S-S-S-S-S-S- S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-
S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S,
S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-O-O-O-
O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-O-O-S, S-O-O-O-O-O-S-S-S-S-S-O-O-O-S-S-S-S-S-O-
O-O-O-O-S, S-S-S-S-S-S-S-S-S-O-O-O-O-O-O-O-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-
S-S-S-S-S-S-S-O-O-O-O-O-S, S-O-O-O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-O-O-O-O-
S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-O-S, S-O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-
O-S, S-O-S-O-S-O-S-O-S-O-S-O-S-O-S-O-S-O-S-O-S-O-S-O-S, S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-
S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-
S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-S, S-S-S-
S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-
O-O-S, S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-
S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-
S-S-S-S-S-S-O-O-S, S-O-O-O-O-S-S-S-S-S-S-S-S-S-S-S-O-O-O-O-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-
S-O-O-O-O-O-S, S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-O-O-O-S, S-O-O-S-S-S-S-S-S-O-O-O-O-
S-S-S-S-S-S-O-O-S, S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-
S-S-S, S-S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S, and S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-S,
wherein s is a phosphorothioate internucleoside linkage and o is a phosphodiester internucleoside linkage.

[0264] In some embodiments, an antisense oligonucleotide disclosed herein comprises modified internucleoside linkages comprising one of the following patterns (5′.fwdarw.3′):
TABLE-US-00023 S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S-S-S-O-O-O-O-O-S, S-S-S-S-S-S-S-S-S-
S-O-O-S-S-O-O-S-S-S-S-S-S-S-S-S, S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-S, S-O-
O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-
S-S-S-S, S-S-S-S-S-O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-O-O-O-S-S-S-S-
S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-
S-S-S-O-O-O-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S-S-S-S-S-S-S-S-S, S-
S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-
S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-
S-S-S-S-S-S-O-O-O-O-O-S, S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-
S-S-O-O-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-
S-S-S-O-O-O-S-S-S-S-S-S-S-S-S-S-S, S-O-O-O-S-S-S-S-S-S-O-O-O-S-S-S-S-S-S-S-S-S-S-S-S, S-
O-O-O-S-S-S-S-S-S-S-S-S-S-O-O-O-S-S-S-S-S-S-S-S, S-O-O-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-
S-S-S-S-S-S, S-O-O-S-S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-
S-S-S-S-S-S-O-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-S-S-S-S, S-S-S-S-S-S-S-
S-S-S-S-S-S-S-S-S-S-O-O-O-O-O-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S, S-S-S-S-S-S-S-O-O-
S-S-S-S-S-S-S-S-S-S-S, S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-O-O-S-S-S-O-O-S-S-S-S-S-
S-S-S-S-S-S, S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S-O-O-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-
O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-O-O-O-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S, S-O-O-S-S-
S-S-S-S-S-S-S-S-S-S-S-O-O-S, S-S-S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-O-O-S-S-S-
S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-
S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-S-S, S-O-
O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-O-
O-S-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-O-
O-S-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-O-O-S-S-S-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-
O-S-S-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-S-S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-
S-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-
S-S, O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-O-O-S, S-S-S-S-S-S-S-S-S-S-S-S-S-

S-S-S-S-S-S-O-O-O-O-O-S, O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, O-O-O-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O-O-O-S, S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-O, and S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S-S, wherein s is a phosphorothioate internucleoside linkage and o is a phosphodiester internucleoside linkage.

[0265] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 1 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0266] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 2 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0267] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 2 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0268] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 16 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0269] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 17 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.

[0270] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of ten or more (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 or more) phosphorothioate internucleoside linkages and one or more (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more) phosphodiester internucleoside linkages.

[0271] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of 22 phosphorothioate internucleoside linkages and 3 phosphodiester internucleoside linkages.

[0272] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of 21 phosphorothioate internucleoside linkages and 4 phosphodiester internucleoside linkages.

[0273] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises a mix of 19 phosphorothioate internucleoside linkages and 6 phosphodiester internucleoside linkages.

[0274] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide

comprises modified internucleoside linkages comprising the following pattern (5′.fwdarw.3′): s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-o-o-o-o-s.
[0275] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises modified internucleoside linkages comprising the following pattern (5′.fwdarw.3′): s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-o-o-o-s.
[0276] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises modified internucleoside linkages comprising the following pattern (5′.fwdarw.3′): s-o-o-s-s-s-s-s-s-s-s-s-s-s-o-o-s-s-s-s-s-s-s-s-s.
[0277] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises modified internucleoside linkages comprising the following pattern (5′.fwdarw.3′): s-o-o-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-o-o-s.
[0278] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside and the antisense oligonucleotide comprises modified internucleoside linkages comprising the following pattern (5′.fwdarw.3′): s-o-o-o-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-o-o-o-s.
[0279] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and the antisense oligonucleotide comprises modified internucleoside linkages comprising the following pattern (5′.fwdarw.3′): s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-s-o-o-o-o-o-s.
[0280] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 9 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate internucleoside linkage.
[0281] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 2 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.
[0282] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 93 and comprises chemical modifications, wherein each nucleoside of the antisense oligonucleotide is a 2′-O-NMA modified nucleoside and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.
[0283] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 455 and comprises chemical modifications, wherein the nucleosides at positions 1 and 20 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-O-NMA modified nucleosides and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.
[0284] In some embodiments, an antisense oligonucleotide disclosed herein comprises a nucleobase sequence of SEQ ID NO: 461 and comprises chemical modifications, wherein the nucleosides at positions 1 and 21 (counting 5′.fwdarw.3′) of the antisense oligonucleotide are LNAs and all of the remaining nucleosides are 2′-O-NMA modified nucleosides and each internucleoside linkage of the antisense oligonucleotide is a phosphorothioate linkage.
[0285] In some embodiments, any one of the antisense oligonucleotides disclosed herein, can be

linked to a lipid (e.g., a C16 lipid). In some embodiments, the lipid (e.g., a C16 lipid) targets muscle. In some embodiments, the lipid (e.g., a C16 lipid) is linked to the 5′ nucleoside of the antisense oligonucleotide via a DNA linker. In some embodiments, the DNA linker is a phosphodiester d(TCA) linker, wherein each internucleoside linkage is a phosphodiester linkage.

[0286] In some embodiments, an antisense oligonucleotide comprises a first antisense sequence and a second antisense sequence, and comprises chemical modifications (e.g., one or more modified nucleosides and/or one or more modified internucleoside linkages). In some embodiments the antisense oligonucleotide comprises a structure provided in Table 5.

[0287] In some embodiments, an antisense oligonucleotide comprises a first antisense sequence and a second antisense sequence, and comprises chemical modifications (e.g., one or more modified nucleosides and/or one or more modified internucleoside linkages). In some embodiments the antisense oligonucleotide comprises a structure provided in Table 11.

TABLE-US-00024 TABLE 5 Exemplary antisense oligonucleotides SEQ SEQ ASO Nucleobase Sequence ID Antisense Oligonucleotide ID # (5′ .fwdarw. 3′) NO: Structures (5′ .fwdarw. 3′) NO: 1 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 574 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][i.sup.5Cs][iT] 2 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 571 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 3 CACGAATCTACCCCATCCATGTACTC 3 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 575 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5C] 4 CGAATCTACCCACGAGTTCTTTCCAG 4 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 576 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iGs][iAs][iGs][iTs][iTs][i.sup.5Cs][iT s][iTs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iG] 5 CGAATCTACCCACTAAGAGTTCTTTC 5 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 577 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iTs][iAs][iAs][iGs][iAs][iGs][iTs ][iTs][i.sup.5Cs][iTs][iTs][iTs][i.sup.5C] 6 CCACGAATCTACCGATAAGAGTTCTT 6 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][i 578 As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][iAs][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5Cs][iTs][iT] 7 CACGAATCTACCCGGATAAGAGTTCT 7 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 579 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5Cs][iT] 8 CCACGAATCTACCGGATAAGAGTTCT 8 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][i 580 As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5Cs][iT] 9 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 581 ][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i Gs][iGs][iGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iTs][i.sup.5C] 10 GAATCTACCCACCGGGATAAGAGTTC 10 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 582 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 11 CGAATCTACCCACGGGATAAGAGTTC 11 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 583 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 12 ACGAATCTACCCAGGGATAAGAGTTC 12 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 584 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iGs][iGs][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][i.sup.5C] 13 CACGAATCTACCCGGGATAAGAGTTC 13 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 585 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 14 CCACGAATCTACCGGGATAAGAGTTC 14 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][i 586 As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 15 CTAGCCACGAATCGGGATAAGAGTTC 15 [i.sup.5Cs][iTs][iAs][iGs][i.sup.5Cs][i.sup.5Cs][i 587 As][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs] [iGs][iGs][iGs][iAs][iTs][iAs][iAs] [iGs][iAs]

[iGs][iTs][iTs][i.sup.5C] 16 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5Cs]
[iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 588 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs]
[iGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 17
GCATTTATTCAACGGGGATAAGAGTT 17 [iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iAs 589 ][iTs]
[iTs][i.sup.5Cs][iAs][iAs][i.sup.5Cs][i Gs][iGs][iGs][iGs][iAs][iTs][iAs][i As][iGs][iAs][iGs][iTs]
[iT] 18 CCACCAACTCATCGGGGATAAGAGTT 18 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][ 590 iAs][i.sup.5Cs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5C s][iGs][iGs][iGs][iGs][iAs]
[iTs][iA s][iAs][iGs][iAs][iGs][iTs][iT] 19 GAATCTACCCACCGGGGATAAGAGTT 19 [iGs]
[iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 591 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs
][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs][iGs][iAs][iGs][iTs][iT] 20
CGAATCTACCCACGGGGATAAGAGTT 20 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 592 s]
[iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs]
[iGs][iAs][iGs][iTs][iT] 21 CACGAATCTACCCGGGGATAAGAGTT 21 [i.sup.5Cs][iAs]
[i.sup.5Cs][iGs][iAs][iAs][iT 593 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs]
[iGs][iGs][iAs][iTs][iAs ][iAs][iGs][iAs][iGs][iTs][iT] 22
CCACGAATCTACCGGGGATAAGAGTT 22 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][i
594 As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs]
[iGs][iAs][iGs][iTs][iT] 23 GCCACGAATCTACGGGGATAAGAGTT 23 [iGs][i.sup.5Cs]
[i.sup.5Cs][iAs][i.sup.5Cs][iGs][i 595 As][iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs] [iGs][iGs][iGs]
[iGs][iAs][iTs][iAs] [iAs][iGs][iAs][iGs][iTs][iT] 24 CGAATCTACCCACCCTGGGGATAAGA 24
[i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 596 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs]
[i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iTs][iGs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iGs][iA] 25
ACGAATCTACCCATCCTGGGGATAAG 25 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 597 s]
[iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iTs][i.sup.5Cs][i.sup.5Cs][iTs][iGs][iGs][iG s]
[iGs][iAs][iTs][iAs][iAs][iG] 26 CGAATCTACCCACGTTCCTGGGGATA 26 [i.sup.5Cs][iGs]
[iAs][iAs][iTs][i.sup.5Cs][iT 598 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iGs][iTs]
[iTs][i.sup.5Cs][i.sup.5Cs][iTs][i Gs][iGs][iGs][iGs][iAs][iTs][iA] 27
CACGAATCTACCCGTTCCTGGGGATA 27 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 599 s]
[i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iTs][iTs][i.sup.5Cs][i.sup.5Cs][iTs][i
Gs][iGs][iGs][iGs][iAs][iTs][iA] 28 CTAGCCACGAATCGTTCCTGGGGATA 28 [i.sup.5Cs][iTs]
[iAs][iGs][i.sup.5Cs][i.sup.5Cs][i 600 As][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs] [iGs][iTs][iTs]
[i.sup.5Cs][i.sup.5Cs][iTs][iG s][iGs][iGs][iGs][iAs][iTs][iA] 29
CGAATCTACCCACTAGTTCCTGGGGA 29 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 601 s]
[iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iTs][iAs][iGs][iTs][iTs][i.sup.5Cs][i.sup.5
Cs][iTs][iGs][iGs][iGs][iGs][iA] 30 CTAGCCACGAATCTAGTTCCTGGGGA 30 [i.sup.5Cs][iTs]
[iAs][iGs][i.sup.5Cs][i.sup.5Cs][i 602 As][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs] [iTs][iAs][iGs]
[iTs][iTs][i.sup.5Cs][i.sup.5C s][iTs][iGs][iGs][iGs][iGs][iA] 31
ACGAATCTACCCACTAGTTCCTGGGG 31 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 603 s]
[iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][iTs][iAs][iGs][iTs][iTs][i.sup.5C s]
[i.sup.5Cs][iTs][iGs][iGs][iGs][iG] 32 ATCTAGCCACGAACTAGTTCCTGGGG 32 [iAs][iTs]
[i.sup.5Cs][iTs][iAs][iGs][i.sup.5C 604 s][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][ i.sup.5Cs][iTs]
[iAs][iGs][iTs][iTs][i.sup.5Cs ][i.sup.5Cs][iTs][iGs][iGs][iGs][iG] 33
GAATCTACCCACTCTGTTCAATCA 33 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 605 ][i.sup.5Cs]
[i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iTs] [i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iA s][iAs][iTs]
[i.sup.5Cs][iA] 34 CGAATCTACCCATCTGTTCAATCA 34 [i.sup.5Cs][iGs][iAs][iAs][iTs]
[i.sup.5Cs][iT 606 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][iTs][iGs][iTs][iTs]
[i.sup.5Cs][iA s][iAs][iTs][i.sup.5Cs][iA] 35 GAATCTACCCACATCTGTTCAATC 35 [iGs][iAs]
[iAs][iTs][i.sup.5Cs][iTs][iAs 607 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs] [iTs]
[i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][i.sup.5C] 36
CGAATCTACCCAATCTGTTCAATC 36 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 608 s][iAs]

[i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][i.sup.5C] 37 GAATCTACCCACTCTGTTCAATCA 33 [iGs][iAs][iAs][lTs][i.sup.5Cs][iTs][iAs 609 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][l.sup.5Cs][iTs] [i.sup.5Cs][iTs][lGs][iTs][iTs][i.sup.5Cs][lA s][iAs][iTs][i.sup.5Cs][lA] 38 CGAATCTACCCATCTGTTCAATCA 34 [i.sup.5Cs][iGs][iAs][lAs][iTs][i.sup.5Cs][iT 610 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][1As][iTs][i.sup.5Cs][iTs][lGs][iTs][iTs][i.sup.5Cs][lA s][iAs][iTs][i.sup.5Cs][lA] 39 GAATCTACCCACATCTGTTCAATC 35 [iGs][iAs][iAs][lTs][i.sup.5Cs][iTs][iAs 611 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs] [iTs][i.sup.5Cs][lTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][l.sup.5C] 40 CGAATCTACCCAATCTGTTCAATC 36 [i.sup.5Cs][iGs][iAs][lAs][iTs][i.sup.5Cs][iT 612 s][lAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs][iAs] [iTs][i.sup.5Cs][lTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][l.sup.5C] 41 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iA][i.sup.5C][iG][iA][iA][iTs][i.sup.5 613 Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA][ iT][iG][iA][iTs][l.sup.5C] 42 CACGAATCTACCCCATCCATGTACTC 3 [i.sup.5Cs][iA][i.sup.5C][iG][iA][iA][iTs][i.sup.5 614 Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT][iG][iT][iA][i.sup.5C][iTs][i.sup.5C] 43 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5C][iA][iT][i.sup.5C][i.sup.5C][iAs][ 615 iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][iAs][iA][iG] [iA][iG][iT][iTs][i.sup.5C] 44 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iG][iT][iT][i.sup.5C][iA][iAs][iTs 616 ][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iA][iG][iA] [iG][iT][iTs][i.sup.5C] 45 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iA][i.sup.5C][iG][iA][iA][iTs][i.sup.5 617 Cs][iTs][iAs][i.sup.5Cs][i.sup.5C][i.sup.5C][i.sup.5C][ i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA][iT] [iG][iT][iA][i.sup.5Cs][iT] 46 CACGAATCTACCCCATCCATGTACTC 3 [i.sup.5Cs][iA][i.sup.5C][iG][iA][iA][iTs][i.sup.5 618 Cs][iTs][iAs][i.sup.5Cs][i.sup.5C][i.sup.5C][i.sup.5C][ iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT][iG][iT][iA][i.sup.5C][iTs][i.sup.5C] 47 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5C][iA][iT][i.sup.5C][i.sup.5C][iAs][ 619 iTs][i.sup.5Cs][iAs][iTs][i.sup.5C][i.sup.5C][iG][iGs][iGs][iAs][iTs][iAs][iA][iG][iA ][iG][iT][iTs][i.sup.5C] 48 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iG][iT][iT][i.sup.5C][iA][iAs][iTs 620 ][i.sup.5Cs][iAs][iTs][iT][i.sup.5C][iG][iGs] [iGs][iAs][iTs][iAs][iA][iG][iA][iG ][iT][iTs][i.sup.5C] 49 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 621 s][i.sup.5Cs][iTs][iA][i.sup.5C][i.sup.5C][i.sup.5C][i.sup.5 C][i.sup.5C][iA][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][i.sup.5Cs][iT] 50 CACGAATCTACCCCATCCATGTACTC 3 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 622 s][i.sup.5Cs][iTs][iA][i.sup.5C][i.sup.5C][i.sup.5C][i.sup.5 C][iA][iT][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iG s][iTs][iAs][i.sup.5Cs][iTs][i.sup.5C] 51 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 623 iAs][iTs][i.sup.5Cs][iA][iT][i.sup.5C][i.sup.5C][i G][iG][iG][iAs][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i.sup.5C] 52 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 624 ][iTs][i.sup.5Cs][iA][iT][iT][i.sup.5C][iG][i G][iG][iAs][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5C] 53 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 625 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][i.sup.5Cs][iT] 54 CACGAATCTACCCCATCCATGTACTC 3 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 626 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][lAs][lTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5C] 55 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 627 iAs][iTs][i.sup.5Cs][iAs][lTs][l.sup.5Cs][l.sup.5C s][lGs][lGs][lGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 56 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 628 ][iTs][i.sup.5Cs][iAs][lTs][lTs][l.sup.5Cs][l Gs][lGs][lGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iTs][i.sup.5C] 57 CACGAATCTACCCCCATCCATGTACT 1 [l.sup.5Cs]

[lAs][l.sup.5Cs][lGs][iAs][iAs][iT 629 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][lTs][lAs][l.sup.5Cs][lT] 58 CACGAATCTACCCCATCCATGTACTC 3 [l.sup.5Cs][lAs][l.sup.5Cs][lGs][iAs][iAs][iT 630 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][lAs][l.sup.5Cs][lTs][l.sup.5C] 59 CCATCCATCATCCGGGATAAGAGTTC 16 [l.sup.5Cs][l.sup.5Cs][lAs][lTs][i.sup.5Cs][i.sup.5Cs][ 631 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][lGs][lTs][lTs][l.sup.5C] 60 TGTTCAATCATTCGGGATAAGAGTTC 9 [lTs][lGs][lTs][lTs][i.sup.5Cs][iAs][iAs 632 ][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i Gs][iGs][iGs][iAs][iTs][iAs][iAs][i Gs][iAs][lGs][lTs][lTs][l.sup.5C] 61 CACGAATCTACCCCCATCCATGTACT 1 [l.sup.5Cs][lAs][l.sup.5Cs][lGs][iAs][iAs][iT 633 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][lTs][lAs][l.sup.5Cs][lT] 62 CACGAATCTACCCCATCCATGTACTC 3 [l.sup.5Cs][lAs][l.sup.5Cs][lGs][iAs][iAs][iT 634 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][lAs][l.sup.5Cs][lTs][l.sup.5C] 63 CCATCCATCATCCGGGATAAGAGTTC 16 [l.sup.5Cs][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs][ 635 iAs][iTs][i.sup.5Cs][iAs][iTs][l.sup.5Cs][l.sup.5C s][lGs][lGs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][lGs][lTs][lTs][l.sup.5C] 64 TGTTCAATCATTCGGGATAAGAGTTC 9 [lTs][lGs][lTs][lTs][i.sup.5Cs][iAs][iAs 636 ][iTs][i.sup.5Cs][iAs][iTs][lTs][l.sup.5Cs][l Gs][lGs][iGs][iAs][iTs][iAs][iAs][i Gs][iAs][lGs][lTs][lTs][l.sup.5C] 65 CACGAATCTACCCCCATCCATGTACT 1 [l.sup.5Cs][lAs][l.sup.5Cs][iGs][iAs][iAs][iT 637 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][lAs][l.sup.5Cs][lT] 66 CACGAATCTACCCCATCCATGTACTC 3 [l.sup.5Cs][lAs][l.sup.5Cs][iGs][iAs][iAs][iT 638 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][lAs][lTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][l.sup.5Cs][lTs][l.sup.5C] 67 CCATCCATCATCCGGGATAAGAGTTC 16 [l.sup.5Cs][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs][ 639 iAs][iTs][i.sup.5Cs][iAs][lTs][l.sup.5Cs][l.sup.5C s][lGs][lGs][lGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][lTs][lTs][l.sup.5C] 68 TGTTCAATCATTCGGGATAAGAGTTC 9 [lTs][lGs][lTs][iTs][i.sup.5Cs][iAs][iAs 640 ][iTs][i.sup.5Cs][iAs][lTs][lTs][l.sup.5Cs][l Gs][lGs][lGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][lTs][lTs][l.sup.5C] 69 CACGAATCTACCCCCATCCATGTACT 1 [l.sup.5Cs][lAs][l.sup.5Cs][lGs][iAs][iAs][iT 641 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][i.sup.5Cs][lT] 70 CACGAATCTACCCCATCCATGTACTC 3 [l.sup.5Cs][lAs][l.sup.5Cs][lGs][iAs][iAs][iT 642 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][lAs][lTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5C] 71 CCATCCATCATCCGGGATAAGAGTTC 16 [l.sup.5Cs][l.sup.5Cs][lAs][lTs][i.sup.5Cs][i.sup.5Cs][ 643 iAs][iTs][i.sup.5Cs][iAs][lTs][l.sup.5Cs][l.sup.5C s][lGs][lGs][lGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 72 TGTTCAATCATTCGGGATAAGAGTTC 9 [lTs][lGs][lTs][lTs][i.sup.5Cs][iAs][iAs 644 ][iTs][i.sup.5Cs][iAs][lTs][lTs][l.sup.5Cs][l Gs][lGs][lGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iTs][i.sup.5C] 73 CACGAATCTACCCCCATCCATGTACT 1 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 645 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][lTs][lAs][l.sup.5Cs][lT] 74 CACGAATCTACCCCATCCATGTACTC 3 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 646 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][lAs][lTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][lAs][l.sup.5Cs][lTs][l.sup.5C] 75 CCATCCATCATCCGGGATAAGAGTTC 16 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 647 iAs][iTs][i.sup.5Cs][iAs][lTs][l.sup.5Cs][l.sup.5C s][lGs][lGs][lGs][iAs][iTs][iAs][iA s][iGs][iAs][lGs][lTs][lTs][l.sup.5C] 76 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 648 ][iTs][i.sup.5Cs][iAs][lTs][lTs][l.sup.5Cs][l Gs][lGs][lGs][iAs][iTs][iAs][iAs][i Gs][iAs][lGs][lTs][lTs][l.sup.5C] 77 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs]

[i.sup.5Cs][iT 649 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][l.sup.5Cs ][lAs][lTs][l.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 78 CGAATCTACCCACATCTGTTCAATCA 2 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 650 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][lAs][lTs][l.sup.5Cs][lA] 79 CGAATCTACCCACATCTGTTCAATCA 2 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 651 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][l.sup.5Cs ][lAs][lTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][lAs][lTs][l.sup.5Cs][lA] 80 ACGAATCTACCCCCATCCATGTAC 37 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 652 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5C] 81 CACGAATCTACCCCATCCATGTACT 38 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 653 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][l.sup.5Cs][lT] 82 ACGAATCTACCCCCATCCATGTACT 39 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 654 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][iT] 83 CACGAATCTACCCCCATCCATGTAC 40 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 655 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][i.sup.5C] 84 CACGAATCTACCCTCCATCCATGTACT 41 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 656 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5Cs ][iT] 85 CACGAATCTACCCACCATCCATGTACT 42 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 657 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5Cs ][iT] 86 ACGAATCTACCCCATCCATGTACT 43 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 658 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i Gs][iTs][iAs][i.sup.5Cs][iT] 87 CACGAATCTACCCATCCATGTACTC 44 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 659 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i Gs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5C] 88 ACGAATCTACCCCATCCATGTACTC 45 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 660 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i Gs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5C] 89 CACGAATCTACCCCATCCATGTACTCA 46 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 661 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs ][iA] 90 CCATCCATCATCGGGATAAGAGTT 47 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 662 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iGs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iT] 91 CCATCCATCATCGGATAAGAGTTC 48 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 663 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iGs ][iGs][iAs][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5C] 92 CATCCATCATCCGGGATAAGAGTT 49 [i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i 664 Ts][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iGs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iT] 93 CATCCATCATCCGGATAAGAGTTC 50 [i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i 665 Ts][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iGs ][iGs][iAs][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5C] 94 CCATCCATCATCGGGATAAGAGTTC 51 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 666 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iGs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5C] 95 CATCCATCATCCGGGATAAGAGTTC 52 [i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i 667 Ts][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iGs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5C] 96 CCATCCATCATCCGGGATAAGAGTT 53 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 668 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][iTs][iT] 97 CCATCCATCATCCGGATAAGAGTTC 54 [i.sup.5Cs][i.sup.5Cs][iAs][iTs]

[i.sup.5Cs][i.sup.5Cs][ 669 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i.sup.5C] 98 CCATCCATCATCCGGGATAAGAGTTC 55 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 670 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iGs][iGs][iAs][iTs][iA s][iAs][iGs][iAs][iGs][iTs][iTs][i.sup.5 C] 99 CCATCCATCATCCGGGATAAGAGTTCT 56 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 671 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][iTs][iTs][i.sup.5Cs][i T] 100 TCCATCCATCATCCGGGATAAGAGTTC 57 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i 672 .sup.5Cs][iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][iA s][iAs][iGs][iAs][iGs][iTs][iTs][i.sup.5 C] 101 GTTCAATCATTCGGGATAAGAGTT 58 [iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs 673 ][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iT] 102 GTTCAATCATTCGGATAAGAGTTC 59 [iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs 674 ][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][i Gs][iAs][iTs][iAs][iAs][iGs][iAs][i Gs][iTs][iTs][i.sup.5C] 103 GTTCAATCATTCGGGATAAGAGTTC 60 [iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs 675 ][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iTs][i.sup.5C] 104 TGTTCAATCATTCGGATAAGAGTTC 61 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 676 ][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i Gs][iGs][iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iTs][i.sup.5C] 105 TGTTCAATCATTCGGGATAAGAGTTCT 62 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 677 ][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i Gs][iGs][iGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iTs][i.sup.5Cs][iT] 106 GAATCTACCCACATCTGTTCAATCA 63 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 678 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][iCs][iA] 107 CGAATCTACCCACATCTGTTCAATC 64 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 679 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5C] 108 CGAATCTACCCACCATCTGTTCAATCA 65 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 680 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][i Ts][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs] [iA] 109 CACGAATCTACCCCCCATCCATGTACT 66 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 681 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5C s][iT] 110 CACGAATCTACCCGCCATCCATGTACT 67 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 682 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5Cs ][iT] 111 CACGAATCTACCCTCATCCATGTACTC 68 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 683 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][iGs][iTs][iAs][i.sup.5Cs][iTs] [i.sup.5C] 112 CACGAATCTACCCGCATCCATGTACTC 69 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 684 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][iGs][iTs][iAs][i.sup.5Cs][iTs] [i.sup.5C] 113 TGTTCAATCATTCCGGGATAAGAGTTC 70 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 685 ][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i .sup.5Cs][iGs][iGs][iGs][iAs][iTs][iAs][ iAs][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 114 CCATCCATCATCCCGGGATAAGAGTTC 71 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 686 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][i As][iAs][iGs][iAs][iGs][iTs][iTs][i .sup.5C] 115 CACGAATCTACCCCACCATCCATGTACT 72 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 687 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][iTs][iAs ][i.sup.5Cs][iT] 116 CACGAATCTACCCTCCCATCCATGTACT 73 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 688 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iTs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][iTs][iAs ][i.sup.5Cs][iT] 117

CACGAATCTACCCCCCCATCCATGTACT 74 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 689 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][iTs][iA s][i.sup.5Cs][iT] 118 CACGAATCTACCCCTCATCCATGTACTC 75 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 690 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5Cs ][iTs][i.sup.5C] 119 CACGAATCTACCCTTCATCCATGTACTC 76 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 691 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iTs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5Cs] [iTs][i.sup.5C] 120 TGTTCAATCATTCCAGGGATAAGAGTTC 77 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 692 ][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i .sup.5Cs][iAs][iGs][iGs][iGs][iAs][iTs][ iAs][iAs][iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 121 TGTTCAATCATTCCTGGGATAAGAGTTC 78 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 693 ][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i .sup.5Cs][iTs][iGs][iGs][iGs][iAs][iTs][ iAs][iAs][iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 122 CCATCCATCATCCCAGGGATAAGAGTTC 79 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 694 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][iGs][iGs][iGs][iAs][i Ts][iAs][iAs][iGs][iAs][iGs][iTs][i Ts][i.sup.5C] 123 CCATCCATCATCCTCGGGATAAGAGTTC 80 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 695 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iTs][i.sup.5Cs][iGs][iGs][iGs][iAs][i Ts][iAs][iAs][iGs][iAs][iGs][iTs][i Ts][i.sup.5C] 124 CCATCCATCATCCTTGGGATAAGAGTTC 81 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 696 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iTs][iTs][iGs][iGs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs][iTs][iT s][i.sup.5C] 125 ACGAATCTACCCCCATCCATGTAC 37 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 697 s][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][lAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5C] 126 ACGAATCTACCCCATCCATGTACT 43 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 698 s][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][lAs][lTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i Gs][iTs][iAs][i.sup.5Cs][iT] 127 CCATCCATCATCGGGATAAGAGTT 47 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 699 iAs][iTs][i.sup.5Cs][lAs][lTs][l.sup.5Cs][lGs ][lGs][lGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][lT] 128 CCATCCATCATCGGATAAGAGTTC 48 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 700 iAs][iTs][i.sup.5Cs][lAs][lTs][l.sup.5Cs][lGs ][lGs][lAs][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5C] 129 CATCCATCATCCGGGATAAGAGTT 49 [l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i 701 Ts][i.sup.5Cs][iAs][ITs][l.sup.5Cs][l.sup.5Cs][lGs ][lGs][lGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][lT] 130 CATCCATCATCCGGATAAGAGTTC 50 [l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i 702 Ts][i.sup.5Cs][iAs][lTs][l.sup.5Cs][l.sup.5Cs][lGs ][lGs][lAs][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5C] 131 GTTCAATCATTCGGGATAAGAGTT 58 [iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs 703 ][i.sup.5Cs][iAs][iTs][lTs][l.sup.5Cs][lGs][l Gs][lGs][iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][lT] 132 GTTCAATCATTCGGATAAGAGTTC 59 [iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs 704 ][i.sup.5Cs][iAs][iTs][lTs][l.sup.5Cs][lGs][l Gs][lAs][iTs][iAs][iAs][iGs][iAs][i Gs][iTs][iTs][i.sup.5C] 133 CGAATCTACCCATCTGTTCAATCA 34 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 705 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lTs] [l.sup.5Cs][lTs][iGs][iTs][iTs][i.sup.5Cs][iA s][iAs][iTs][i.sup.5Cs][iA] 134 CGAATCTACCCAATCTGTTCAATC 36 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 706 s][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs][lAs][lAs] [lTs][l.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][i.sup.5C] 135 ACGAATCTACCCCCATCCATGTAC 37 [lAs][l.sup.5Cs][lGs][lAs][iAs][iTs][i.sup.5C 707 s][iTs][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][lGs][lTs][lAs][l.sup.5C] 136 ACGAATCTACCCCATCCATGTACT 43 [lAs][l.sup.5Cs][lGs][lAs][iAs][iTs][i.sup.5C 708 s][iTs][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][lAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i Gs][lTs][lAs][lT] 137 CCATCCATCATCGGGATAAGAGTT 47 [l.sup.5Cs][l.sup.5Cs][lAs][lTs]

[i.sup.5Cs][i.sup.5Cs][ 709 iAs][iTs][i.sup.5Cs][iAs][lTs][l.sup.5Cs][lGs ][lGs][iGs][iAs][iTs][iAs][iAs][iGs ][lAs][lGs][lTs][lT] 138 CCATCCATCATCGGATAAGAGTTC 48 [l.sup.5Cs][l.sup.5Cs][lAs][lTs][i.sup.5Cs][i.sup.5Cs][ 710 iAs][iTs][i.sup.5Cs][iAs][lTs][l.sup.5Cs][lGs ][lGs][iAs][iTs][iAs][iAs][iGs][iAs ][lGs][lTs][lTs][l.sup.5C] 139 CATCCATCATCCGGGATAAGAGTT 49 [l.sup.5Cs][lAs][lTs][l.sup.5Cs][i.sup.5Cs][iAs][i 711 Ts][i.sup.5Cs][iAs][iTs][l.sup.5Cs][l.sup.5Cs][lGs ][lGs][iGs][iAs][iTs][iAs][iAs][iGs ][lAs][lGs][lTs][lT] 140 CATCCATCATCCGGATAAGAGTTC 50 [l.sup.5Cs][lAs][lTs][l.sup.5Cs][i.sup.5Cs][iAs][i 712 Ts][i.sup.5Cs][iAs][iTs][l.sup.5Cs][l.sup.5Cs][lGs ][lGs][iAs][iTs][iAs][iAs][iGs][iAs ][lGs][lTs][lTs][l.sup.5C] 141 GTTCAATCATTCGGGATAAGAGTT 58 [lGs][lTs][lTs][l.sup.5Cs][iAs][iAs][iTs 713 ][i.sup.5Cs][iAs][iTs][lTs][l.sup.5Cs][lGs][l Gs][iGs][iAs][iTs][iAs][iAs][iGs][l As][lGs][lTs][lT] 142 GTTCAATCATTCGGATAAGAGTTC 59 [lGs][lTs][lTs][l.sup.5Cs][iAs][iAs][iTs 714 ][i.sup.5Cs][iAs][iTs][lTs][l.sup.5Cs][lGs][l Gs][iAs][iTs][iAs][iAs][iGs][iAs][l Gs][lTs][lTs][l.sup.5C] 143 CGAATCTACCCATCTGTTCAATCA 34 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 715 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lTs] [l.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iA s][lAs][lTs][l.sup.5Cs][lA] 144 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 716 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lAs] [lTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][lAs][lAs][lTs][l.sup.5C] 145 CGAATCTACCATCCATGTAC 82 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 717 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5C ] 146 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iA][iT][i.sup.5C][iTs][iA 718 s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][iTs][iGs][iTs][iT][i.sup.5C ][iA][iA][iT][i.sup.5Cs][iA] 147 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iA][iT][i.sup.5C][iTs][iA 719 s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iA][i.sup.5C][iA][i Ts][i.sup.5Cs][iTs][iGs][iTs][iT][i.sup.5C][i A][iA][iT][i.sup.5Cs][iA] 148 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 720 s][iAs][i.sup.5Cs][i.sup.5C][i.sup.5C][iA][i.sup.5C][iA ][iT][i.sup.5C][iTs][iGs][iTs][iTs][i.sup.5Cs ][iAs][iAs][iTs][i.sup.5Cs][iA] 149 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 721 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT ][i.sup.5C][iA][iA][iT][i.sup.5Cs][iA] 150 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iA][iT][i.sup.5C][iTs][iA 722 s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 151 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iA][iT][i.sup.5Cs][iTs][i 723 As][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i .sup.5C][iA][iA][iT][i.sup.5Cs][iA] 152 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iA][iTs][i.sup.5Cs][iTs][ 724 iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i As][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][ i.sup.5Cs][iA][iA][lT][i.sup.5Cs][iA] 153 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iAs][iA][iTs][i.sup.5C][iTs][ 725 iA][i.sup.5Cs][i.sup.5C][i.sup.5Cs][iA][i.sup.5Cs][iA][ iTs][i.sup.5C][iTs][iG][iTs][iT][i.sup.5Cs][i As][iAs][iTs][i.sup.5Cs][iA] 154 AATCTACCCACCTGTTCAATCA 83 [lAs][lAs][iTs][l.sup.5Cs][iTs][iAs][i.sup.5C 726 s][i.sup.5Cs][i.sup.5Cs][lAs][l.sup.5Cs][l.sup.5Cs][lTs ][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iT s][l.sup.5Cs][lA] 155 AATCTACCCACATCTGTTCAAT 84 [lAs][lAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5C 727 s][i.sup.5Cs][i.sup.5Cs][lAs][l.sup.5Cs][lAs][lTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iA s][lAs][lT] 156 CGAATCTACCCCTGTTCAATCA 85 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT 728 s][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs][lTs ][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iT s][l.sup.5Cs][lA] 157 CGAATCTACCCATCTGTTCAAT 86 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT 729 s][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs][lAs][lTs] [l.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iA s][lAs][lT] 158 ACGAATCTACCCAATCTGTTCAATCA 87 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 730 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs ][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 159 CACGAATCTACCCATCTGTTCAATCA 88 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 731 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 160 GAATCTACCCACCATCTGTTCAATCA 89 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 732 ] [i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 161 AATCTACCCACCAATCTGTTCAATCA 90 [iAs] [iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5C 733 s][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 162 CGAATCTACCCACTCTGTTCAATCAT 91 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 734 s] [iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iT] 163 CGAATCTACCCACTCATCTGTTCAAT 92 [i.sup.5Cs] [iGs][iAs][iAs][iTs][i.sup.5Cs][iT 735 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iTs] [i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][i Gs][iTs][iTs][i.sup.5Cs][iAs][iAs][iT] 164 ACGAATCTACCCACATCTGTTCAATC 93 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 736 s] [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][iT s][iTs] [i.sup.5Cs][iAs][iAs][iTs][i.sup.5C] 165 GAATCTACCCACCCATCTGTTCAATC 94 [iGs][iAs] [iAs][iTs][i.sup.5Cs][iTs][iAs 737 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs ] [i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][i Ts][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5C] 166 ACGAATCTACCCATCTGTTCAATCAT 95 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 738 s] [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs] [iAs][iTs][i.sup.5Cs][iAs][iT] 167 GAATCTACCCACCTCTGTTCAATCAT 96 [iGs][iAs][iAs] [iTs][i.sup.5Cs][iTs][iAs 739 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs ][iTs] [i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iT] 168 CCACGAATCTACCTCAATCATTCATT 97 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][i 740 As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs ][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][i As] [iTs][iTs][i.sup.5Cs][iAs][iTs][iT] 169 CGAATCTACCCACCTCTGTTCAATCA 98 [i.sup.5Cs] [iGs][iAs][iAs][iTs][i.sup.5Cs][iT 741 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ] [i.sup.5Cs][iTs][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 170 ACGAATCTACCCACTCTGTTCAATCA 99 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 742 s] [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s] [i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 171 CGAATCTACCCAATCTGTTCAATCAT 100 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 743 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][i.sup.5Cs][iAs][iT] 172 GAATCTACCCACATCTGTTCAATCAT 101 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 744 ] [i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][i.sup.5Cs][iAs][iT] 173 AATCTACCCACCTGTTCAATCA 83 [iAs][iAs][iTs] [i.sup.5Cs][iTs][iAs][i.sup.5C 745 s][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iTs ][iGs] [iTs][iTs][i.sup.5Cs][iAs][iAs][iT s][i.sup.5Cs][iA] 174 AATCTACCCACATCTGTTCAAT 84 [iAs] [iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5C 746 s][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iA s][iAs][iT] 175 AATCTACCCACTCTGTTCAATC 102 [iAs][iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5C 747 s][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iTs] [i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iA s][iTs][i.sup.5C] 176 CGAATCTACCCCTGTTCAATCA 85 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 748 s][iAs] [i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iTs ][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iT s] [i.sup.5Cs][iA] 177 CGAATCTACCCATCTGTTCAAT 86 [i.sup.5Cs][iGs][iAs][iAs][iTs] [i.sup.5Cs][iT 749 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][iTs][iGs][iTs][iTs] [i.sup.5Cs][iA s][iAs][iT] 178 CGAATCTACCCTCTGTTCAATC 103 [i.sup.5Cs][iGs][iAs][iAs] [iTs][i.sup.5Cs][iT 750 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iTs][i.sup.5Cs ][iTs][iGs][iTs][iTs] [i.sup.5Cs][iAs][iA s][iTs][i.sup.5C] 179 GAATCTACCCACTGTTCAATCA 104 [iGs][iAs][iAs] [iTs][i.sup.5Cs][iTs][iAs 751 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iTs] [iGs][iTs][iTs] [i.sup.5Cs][iAs][iAs][iTs ][i.sup.5Cs][iA] 180 GAATCTACCCAATCTGTTCAAT 105 [iGs][iAs]

[iAs][iTs][i.sup.5Cs][iTs][iAs 752 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iAs][iTs][ i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs ][iAs][iT] 181 GAATCTACCCATCTGTTCAATC 106 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 753 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs] [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs ][iTs][i.sup.5C] 182 AATCTACCCACTCTGTTCAATC 102 [lAs][lAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5C 754 s][i.sup.5Cs][i.sup.5Cs][lAs][l.sup.5Cs][lTs][l.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iA s][lTs][l.sup.5C] 183 CGAATCTACCCTCTGTTCAATC 103 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT 755 s][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs][lTs][l.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iA s][lTs][l.sup.5C] 184 GAATCTACCCACTGTTCAATCA 104 [lGs][lAs][iAs][iTs][i.sup.5Cs][iTs][iAs 756 ][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][l.sup.5Cs][lTs] [iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs ][l.sup.5Cs][lA] 185 GAATCTACCCAATCTGTTCAAT 105 [lGs][lAs][iAs][iTs][i.sup.5Cs][iTs][iAs 757 ][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lAs][lTs][ i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs ][lAs][lT] 186 GAATCTACCCATCTGTTCAATC 106 [lGs][lAs][iAs][iTs][i.sup.5Cs][iTs][iAs 758 ][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lTs][l.sup.5Cs] [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs ][lTs][l.sup.5C] 187 GAATCTACCCATCTGTTCAATC 106 [iGs][A][iA][lT][i.sup.5C][iTs][iAs][i.sup.5C 759 s][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs] [iGs][iTs][lT][i.sup.5C][iA][iA][iTs][i.sup.5 C] 188 GAATCTACCCATCTGTTCAATC 106 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 760 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs] [iTs][iGs][iT][iT][i.sup.5C][iA][iA][iTs ][i.sup.5C] 189 GAATCTACCCATCTGTTCAATC 106 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 761 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iA][iT][i.sup.5Cs][i Ts][iGs][iTs][iTs][i.sup.5C][iA][iA][iTs ][i.sup.5C] 190 GAATCTACCCATCTGTTCAATC 106 [iGs][iA][iA][iTs][i.sup.5Cs][iTs][iAs][ 762 i.sup.5Cs][i.sup.5Cs][i.sup.5C][iA][iT][i.sup.5C][iTs][ iGs][iTs][iTs][i.sup.5Cs][iA][iA][iTs][i .sup.5C] 191 GAATCTACCCATCTGTTCAATC 106 [lGs][A][iA][lT][i.sup.5C][iTs][iAs][i.sup.5C 763 s][i.sup.5Cs][l.sup.5Cs][lAs][lTs][l.sup.5Cs][iTs] [iGs][iTs][iT][i.sup.5C][iA][iA][lTs][i.sup.5 C] 192 CGAATCTACCCAATCTGTTCAATC 36 [i.sup.5Cs][iGs][iAs][lAs][iTs][i.sup.5Cs][iT 764 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lAs] [lTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][lAs][iAs][iTs][l.sup.5C] 193 CGAATCTACCCAATCTGTTCAATC 36 [i.sup.5Cs][iGs][iAs][lAs][iTs][i.sup.5Cs][iT 765 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs][lAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][lAs][iAs][iTs][i.sup.5C] 194 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT 766 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs][lAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][lTs][l.sup.5C] 195 CGAATCTACCCACTCATCTGTTCAAT 92 [i.sup.5Cs][iG][iA][iA][iT][i.sup.5Cs][iTs][i 767 As][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iT s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][ iT][iT][i.sup.5C][iA][iAs][iT] 196 CGAATCTACCCACTCATCTGTTCAAT 92 [i.sup.5Cs][iG][iA][iA][iTs][i.sup.5Cs][iTs][ 768 iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i Ts][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs] [iTs][iT][i.sup.5C][iA][iAs][iT] 197 ACGAATCTACCCACATCTGTTCAATC 93 [iAs][i.sup.5C][iG][iA][iA][iTs][i.sup.5Cs][i 769 Ts][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5C s][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][i T][i.sup.5C][iA][iTs][i.sup.5C] 198 ACGAATCTACCCACATCTGTTCAATC 93 [iAs][i.sup.5C][iG][iA][iAs][iTs][i.sup.5Cs][ 770 iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5 Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][ iTs][i.sup.5C][iA][iA][iTs][i.sup.5C] 199 CAAATCTACCCACATCTGTTCAATCA 107 [i.sup.5Cs][iAs][iAs][iAs][iTs][i.sup.5Cs][iT 771 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 200 CGAATCTACCCACATCTATTCAATCA 108 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 772 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iAs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 201 CAAATCTACCCACATCTATTCAATCA 109 [i.sup.5Cs][iAs][iAs][iAs][iTs][i.sup.5Cs][iT 773 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iAs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 202 CGAATCTACCCAATCTATTCAATC 110 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 774 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lAs] [lTs]

[i.sup.5Cs][iTs][iAs][iTs][iTs][i.sup.5C s][lAs][lAs][lTs][l.sup.5C] 203 CGAATCTACCCACATCTGTTCAAT 111 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 775 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][l.sup.5Cs ][lAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][l.sup.5Cs][lAs][lAs][lT] 204 CGAATCTACCCATCATCTGTTCAA 112 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 776 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lTs] [l.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][iT s][lTs][l.sup.5Cs][lAs][lA] 205 ACGAATCTACCCATCTGTTCAATC 113 [lAs][l.sup.5Cs][lGs][lAs][iAs][iTs][i.sup.5C 777 s][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs][lAs][lTs][i.sup.5Cs][iTs][iGs][iTs][iTs] [i.sup.5Cs][lAs][lAs][lTs][l.sup.5C] 206 ACGAATCTACCCCATCTGTTCAAT 114 [lAs][l.sup.5Cs][lGs][lAs][iAs][iTs][i.sup.5C 778 s][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][lAs][iTs][i.sup.5Cs][iTs][iGs][iTs][lT s][l.sup.5Cs][lAs][lAs][iT] 207 CACGAATCTACCCATCCATGTACT 115 [l.sup.5Cs][lAs][l.sup.5Cs][lGs][iAs][iAs][iT 779 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][lAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i Gs][lTs][lAs][l.sup.5Cs][lT] 208 CACGAATCTACCCCATCCATGTAC 116 [l.sup.5Cs][lAs][l.sup.5Cs][lGs][iAs][iAs][iT 780 s][i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5Cs][l.sup.5Cs ][l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][lGs][lTs][lAs][l.sup.5C] 209 CATTTATTCAACGGGATAAGAGTT 117 [l.sup.5Cs][lAs][lTs][lTs][iTs][iAs][iTs 781 ][iTs][i.sup.5Cs][iAs][lAs][l.sup.5Cs][lGs][l Gs][iGs][iAs][iTs][iAs][iAs][iGs][l As][lGs][lTs][lT] 210 CATTTATTCAACGGGGATAAGAGT 118 [l.sup.5Cs][lAs][lTs][lTs][iTs][iAs][iTs 782 ][iTs][i.sup.5Cs][iAs][lAs][l.sup.5Cs][lGs][l Gs][iGs][iGs][iAs][iTs][iAs][iAs][l Gs][lAs][lGs][lT] 211 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iAs][iTs][i.sup.5Cs][iTs] 783 [iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs] [i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 212 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 784 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C][iA][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 213 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 785 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iT][i.sup.5C][iTs][iGs][iTs][iTs] [i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] 214 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 786 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iA][iT][i.sup.5Cs][iA] 215 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][lT 787 s][lAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iAs] [iTs][i.sup.5Cs][iTs][iGs][lTs][iTs][i.sup.5C s][iAs][iAs][lTs][l.sup.5C] 216 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 788 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][lAs] [lTs][l.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][lTs][l.sup.5C] 217 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT 789 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs][iAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][iAs][iAs][iTs][i.sup.5C] 218 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][iGs][iAs][iAs][iTs][l.sup.5Cs][iT 790 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs][lAs] [iTs][i.sup.5Cs][iTs][iGs][iTs][lTs][i.sup.5C s][iAs][iAs][iTs][i.sup.5C] 219 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lG][lA][lAs][iTs][i.sup.5Cs][iTs] 791 [iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lAs][l Ts][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs] [lAs][lAs][lTs][l.sup.5C] 220 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 792 s][iAs][i.sup.5Cs][i.sup.5C][l.sup.5C][lAs][lAs][l Ts][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs] [lAs][lAs][lTs][l.sup.5C] 221 CGAATCTACCCAATCTGTTCAATC 36 [1.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 793 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lAs] [lT][i.sup.5C][iTs][iGs][iTs][iTs][i.sup.5Cs] [lAs][lAs][lTs][l.sup.5C] 222 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 794 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][lAs] [lTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5C s][lA][lA][lTs][l.sup.5C] 223 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iA][i.sup.5C][iGs][iAs][iAs][iTs] 795 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ i.sup.5Cs][i.sup.5Cs][iAs]

[iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5Cs][iT] 224 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 796 s][i.sup.5Cs][iTs][iAs][i.sup.5C][i.sup.5C][i.sup.5Cs][ i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5Cs][iT] 225 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 797 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5C][iA][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5Cs][iT] 226 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 798 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iT][iA][i.sup.5Cs][iT] 227 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5C][iA][iTs][i.sup.5Cs][i.sup.5Cs][iA 799 s][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iGs][iGs][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][i.sup.5C] 228 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 800 iAs][iTs][i.sup.5Cs][iAs][iT][i.sup.5C][i.sup.5Cs][iGs][iGs][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][i.sup.5C] 229 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 801 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iG][iG][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][i.sup.5C] 230 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 802 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iG][iT][iTs][i.sup.5C] 231 GCATTTATTCAACGGGGATAAGAGTT 17 [iGs][i.sup.5C][iA][iTs][iTs][iTs][iAs][ 803 iTs][iTs][i.sup.5Cs][iAs][iAs][i.sup.5Cs][iGs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iT] 232 GCATTTATTCAACGGGGATAAGAGTT 17 [iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iAs 804 ][iTs][iTs][i.sup.5Cs][iA][iA][i.sup.5Cs][iGs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iT] 233 GCATTTATTCAACGGGGATAAGAGTT 17 [iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iAs 805 ][iTs][iTs][i.sup.5Cs][iAs][iAs][i.sup.5Cs][i Gs][iG][iG][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iT] 234 GCATTTATTCAACGGGGATAAGAGTT 17 [iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iAs 806 ][iTs][iTs][i.sup.5Cs][iAs][iAs][i.sup.5Cs][i Gs][iGs][iGs][iGs][iAs][iTs][iAs][i As][iGs][iA][iG][iTs][iT] 235 CGAATCTACCATCCATGTAC 82 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 807 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5 C] 236 ATCCATCCATCATTGGGGATAAGAGT 119 [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 808 Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iAs][iTs] [iTs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iAs][iGs][iAs][iGs][iT] 237 TCCATCCATCATCGGGGATAAGAGTT 120 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i 809 .sup.5Cs][iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs][iGs][iAs][iGs][iTs][iT] 238 TCCATCCATCATCGGGATAAGAGTTC 121 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i 810 .sup.5Cs][iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 239 TCCATCCATCATCGGATAAGAGTTCT 122 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i 811 .sup.5Cs][iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5Cs][iT] 240 CCATCCATCATCCGATAAGAGTTCTT 123 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 812 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iAs][iTs][iAs][iAs][iGs][iA s][iGs][iTs][iTs][i.sup.5Cs][iTs][iT] 241 GCATTTATTCAACGGGATAAGAGTTC 124 [iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iAs 813 ][iTs][iTs][i.sup.5Cs][iAs][iAs][i.sup.5Cs][i Gs][iGs][iGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iTs][i.sup.5C] 242 GCATTTATTCAACGGATAAGAGTTCT 125 [iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iAs 814 ][iTs][iTs][i.sup.5Cs][iAs][iAs][i.sup.5Cs][i Gs][iGs][iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iTs][i.sup.5Cs][iT] 243 CGAATCTACCCACCCTTTTATCTACT 126 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 815 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iTs][iTs][iTs][iTs][i As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][iT] 244 CGAATCTACCCACCATGTACTCACCC 127 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 816 s]

[iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5 Cs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5 C] 245 CACGAATCTACCCCCATCCTTTTATC 128 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 817 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iTs][iTs][iTs][iTs][iAs][iTs][i.sup.5C] 246 CACGAATCTACCCCTGTTCAATCATT 129 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 818 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][i As][iAs][iTs][i.sup.5Cs][iAs][iTs][iT] 247 CACGAATCTACCCGTTCAATCATTCA 130 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 819 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iT s][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iA] 248 CGAATCTACCCACTCCTTTTATCTAC 131 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 820 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iTs][i.sup.5Cs][i.sup.5Cs][iTs][iTs][iTs][i Ts][iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5C] 249 CGAATCTACCCACCTGTTCAATCATT 132 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 821 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][i As][iAs][iTs][i.sup.5Cs][iAs][iTs][iT] 250 CGAATCTACCCACCTTTTATCTACTC 133 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 822 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][i.sup.5Cs][iTs][iTs][iTs][iTs][iAs][iT s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5C] 251 CGAATCTACCCACGTTCAATCATTCA 134 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 823 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iT s][i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][iA] 252 ACGAATCTACCCAGTACTCACCCATC 135 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 824 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs][iA s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C] 253 CGAATCTACCCACTGTTCAATCATTC 136 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 825 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iA s][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5C] 254 CACGAATCTACCCCATCTGTTCAATC 137 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 826 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][i Ts][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5C] 255 CACGAATCTACCCCCTTTTATCTACT 138 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 827 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iTs][iTs][iTs][iTs][i As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][iT] 256 CACGAATCTACCCCCATGTACTCACC 139 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 828 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][iTs][i As][i.sup.5Cs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5 C] 257 CGAATCTACCCACATTCACCAGCATT 140 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 829 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i .sup.5Cs][iAs][iGs][i.sup.5Cs][iAs][iTs][iT] 258 CGAATCTACCCACCATCTGTTCAATC 141 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 830 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][i Ts][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5C] 259 ACGAATCTACCCACCATCCTTTTATC 142 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 831 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ iTs][iTs][iTs][iTs][iAs][iTs][i.sup.5C] 260 CGAATCTACCCACGTACTCACCCATC 143 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 832 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs][i As][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 C] 261 CACGAATCTACCCGTACTCACCCATC 144 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 833 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs][i As][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 C] 262 ACGAATCTACCCAATTCACCAGCATT 145 [iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 834 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iAs][iTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5 Cs][iAs][iGs][i.sup.5Cs][iAs][iTs][iT] 263 CGAATCTACCCACGAAACCCAGGCAG 146 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 835 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs]

[i.sup.5Cs ][iGs][iAs][iAs][iAs][i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iGs][iGs][i.sup.5Cs][iAs]
[iG] 264 CCACGAATCTACCGGAAACCCAGGCA 147 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs]
[iGs][iAs][i 836 As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iAs][iAs][iAs]
[i.sup.5Cs][i.sup.5 Cs][i.sup.5Cs][iAs][iGs][iGs][i.sup.5Cs][iA] 265
CAATCATATATCCGGAAACCCAGGCA 148 [i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iT 837 s]
[iAs][iTs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ iGs][iGs][iAs][iAs][iAs][i.sup.5Cs][i.sup.5Cs ]
[i.sup.5Cs][iAs][iGs][iGs][i.sup.5Cs][iA] 266 CAATAGTTCAATCGGAAACCCAGGCA 149
[i.sup.5Cs][iAs][iAs][iTs][iAs][iGs][iTs 838 ][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][i Gs][iGs]
[iAs][iAs][iAs][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs][iAs][iGs][iGs][i.sup.5Cs][iA] 267
TCTAGCCACGAATTTCCAGGAAACCC 150 [iTs][i.sup.5Cs][iTs][iAs][iGs][i.sup.5Cs][i.sup.5
839 Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iTs][ iTs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iGs][iGs ][iAs]
[iAs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C] 268 GTACTCACCCATCGATAAGAGTTCTT 151 [iGs]
[iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs][iA 840 s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ]
[iGs][iAs][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5Cs][iTs][iT] 269
CCATGTACTCACCGGATAAGAGTTCT 152 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][iTs][iA 841 s]
[i.sup.5Cs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs]
[iTs][iTs][i.sup.5Cs][iT] 270 TGTCCATCCATCCGGATAAGAGTTCT 153 [iTs][iGs][iTs]
[i.sup.5Cs][i.sup.5Cs][iAs][iT 842 s][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs ][iGs]
[iGs][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5Cs][iT] 271
TTCATTCACCAGCGGGATAAGAGTTC 154 [iTs][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5C 843 s]
[iAs][i.sup.5Cs][i.sup.5Cs][iAs][iGs][i.sup.5Cs] [iGs][iGs][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs]
[iTs][iTs][i.sup.5C] 272 CTACTCATCACTCGGGATAAGAGTTC 155 [i.sup.5Cs][iTs][iAs]
[i.sup.5Cs][iTs][i.sup.5Cs][i 844 As][iTs][i.sup.5Cs][iAs][i.sup.5Cs][iTs][i.sup.5Cs ][iGs][iGs][iGs]
[iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 273
TCTCCATCCATCCGGGATAAGAGTTC 156 [iTs][i.sup.5Cs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i
845 Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iGs][iAs][iTs][iAs][iA s]
[iGs][iAs][iGs][iTs][iTs][i.sup.5C] 274 GTACTCACCCATCGGGATAAGAGTTC 157 [iGs][iTs]
[iAs][i.sup.5Cs][iTs][i.sup.5Cs][iA 846 s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ]
[iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 275
CATGTACTCACCCGGGATAAGAGTTC 158 [i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5C 847 s]
[iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ][iGs][iAs]
[iGs][iTs][iTs][i.sup.5C] 276 CATCCATCTATCCGGGATAAGAGTTC 159 [i.sup.5Cs][iAs][iTs]
[i.sup.5Cs][i.sup.5Cs][iAs][i 848 Ts][i.sup.5Cs][iTs][iAs][iTs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs]
[iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs][i.sup.5C] 277
CCATCCATCTGTCGGGATAAGAGTTC 160 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs]
[i.sup.5Cs][ 849 iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][i.sup.5Cs ][iGs][iGs][iGs][iAs][iTs][iAs][iAs ]
[iGs][iAs][iGs][iTs][iTs][i.sup.5C] 278 CCAGCATTTATTCGGGGATAAGAGTT 161 [i.sup.5Cs]
[i.sup.5Cs][iAs][iGs][i.sup.5Cs][iAs][i 850 Ts][iTs][iTs][iAs][iTs][iTs][i.sup.5Cs][ iGs][iGs][iGs]
[iGs][iAs][iTs][iAs] [iGs][iAs][iGs][iTs][iT] 279 TTCATTCACCAGCGGGGATAAGAGTT 162
[iTs][iTs][i.sup.5Cs][iAs][iTs][iTs][i.sup.5C 851 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iGs][i.sup.5Cs]
[iGs][iGs][iGs][iGs][iAs][iTs][iAs] [iAs][iGs][iAs][iGs][iTs][iT] 280
TGTTCAATCATTCGGGGATAAGAGTT 163 [iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs 852 ][iTs]
[i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][i Gs][iGs][iGs][iGs][iAs][iTs][iAs][i As][iGs][iAs][iGs][iTs]
[iT] 281 TTATCTACTCATCGGGGATAAGAGTT 164 [iTs][iTs][iAs][iTs][i.sup.5Cs][iTs][iAs 853
][i.sup.5Cs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][ iGs][iGs][iGs][iGs][iAs][iTs][iAs][ iAs][iGs][iAs]
[iGs][iTs][iT] 282 CACCCATCTCTCCGGGGATAAGAGTT 165 [i.sup.5Cs][iAs][i.sup.5Cs]
[i.sup.5Cs][i.sup.5Cs][iAs][ 854 iTs][i.sup.5Cs][iTs][i.sup.5Cs][iTs][i.sup.5Cs][i.sup.5 Cs][iGs]
[iGs][iGs][iGs][iAs][iTs][i As][iAs][iGs][iAs][iGs][iTs][iT] 283
GTACTCACCCATCGGGGATAAGAGTT 166 [iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs][iA 855 s]
[i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs][iGs]

[iAs][iGs][iTs][iT] 284 CATGTACTCACCCGGGGATAAGAGTT 167 [i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5C 856 s][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs][iGs][iAs][iGs][iTs][iT] 285 CCATCTATCCATCGGGGATAAGAGTT 168 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][i 857 As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs][iGs][iAs][iGs][iTs][iT] 286 CATCCATCTAGCCGGGGATAAGAGTT 169 [i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i 858 Ts][i.sup.5Cs][iTs][iAs][iGs][i.sup.5Cs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs][iGs][iAs][iGs][iTs][iT] 287 CTGTCCATCCATCGGGGATAAGAGTT 170 [i.sup.5Cs][iTs][iGs][iTs][i.sup.5Cs][i.sup.5Cs][i 859 As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][iGs][iGs][iGs][iGs][iAs][iTs][iAs ][iAs][iGs][iAs][iGs][iTs][iT] 288 CAATCATATATCCGGGGATAAGAGTT 171 [i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iAs][iT 860 s][iAs][iTs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][ iAs][iGs][iAs][iGs][iTs][iT] 289 CCATCTAGCCACGTTCCTGGGGATAA 172 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][i 861 As][iGs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs ][iTs][iTs][i.sup.5Cs][i.sup.5Cs][iTs][iGs][iGs][iGs][iGs][iAs][iTs][iAs][iA] 290 CATGTACTCACCCGTTCCTGGGGATA 173 [i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5C 862 s][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iGs][iTs][iTs][i.sup.5Cs][i.sup.5Cs][iTs][i Gs][iGs][iGs][iGs][iAs][iTs][iA] 291 CTAGCCACGAATCACTAGTTCCTGGG 174 [i.sup.5Cs][iTs][iAs][iGs][i.sup.5Cs][i.sup.5Cs][i 863 As][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs] [iAs][i.sup.5Cs][iTs][iAs][iGs][iTs][iTs ][i.sup.5Cs][i.sup.5Cs][iTs][iGs][iGs][iG] 292 CCACGAATCTACCGCATTTATTCAAC 175 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][i 864 As][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs ][iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iA s][iTs][iTs][i.sup.5Cs][iAs][iAs][i.sup.5C] 293 CTAGCCACGAATCGTACTCACCCATC 176 [i.sup.5Cs][iTs][iAs][iGs][i.sup.5Cs][i.sup.5Cs][i 865 As][i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs] [iGs][iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs][iA s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C] 294 GAATCTACCCACTTCACCAGCATT 177 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 866 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iTs] [iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][i Gs][i.sup.5Cs][iAs][iTs][iT] 295 CGAATCTACCCATTCACCAGCATT 178 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 867 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iGs][i.sup.5Cs][iAs][iTs][iT] 296 GAATCTACCCACATTCACCAGCAT 179 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 868 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][i As][iGs][i.sup.5Cs][iAs][iT] 297 CGAATCTACCCAATTCACCAGCAT 180 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 869 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iAs] [iTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iGs][i.sup.5Cs][iAs][iT] 298 GAATCTACCCACTTCACCAGCATT 177 [iGs][iAs][iAs][lTs][i.sup.5Cs][iTs][iAs 870 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iTs] [iTs][i.sup.5Cs][lAs][i.sup.5Cs][i.sup.5Cs][iAs][i Gs][i.sup.5Cs][iAs][iTs][lT] 299 CGAATCTACCCATTCACCAGCATT 178 [i.sup.5Cs][iGs][iAs][lAs][iTs][i.sup.5Cs][iT 871 s][lAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs][iTs] [iTs][i.sup.5Cs][lAs][i.sup.5Cs][i.sup.5Cs][iAs][lGs][i.sup.5Cs][iAs][iTs][lT] 300 GAATCTACCCACATTCACCAGCAT 179 [iGs][iAs][iAs][lTs][i.sup.5Cs][iTs][iAs 872 ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs] [iTs][iTs][l.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][l As][iGs][i.sup.5Cs][iAs][lT] 301 CGAATCTACCCAATTCACCAGCAT 180 [i.sup.5Cs][iGs][iAs][lAs][iTs][i.sup.5Cs][iT 873 s][lAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iAs] [iTs][iTs][l.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iGs][i.sup.5Cs][iAs][lT] 302 CGAATCTACCCACATTCACCAGCATT 140 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 874 s][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5Cs][lAs][l.sup.5Cs ][lAs][lTs][lTs][i.sup.5Cs][iAs][i.sup.5Cs][i .sup.5Cs][iAs][iGs][i.sup.5Cs][iAs][iTs][lT] 303 CGAATCTACCCACATTCACCAGCATT 140 [l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 875 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i .sup.5Cs][iAs][iGs][l.sup.5Cs][lAs][lTs][lT] 304 CGAATCTACCCACATTCACCAGCATT 140

[l.sup.5Cs][lGs][lAs][lAs][iTs][i.sup.5Cs][iT 876 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs]
[l.sup.5Cs ][lAs][lTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i .sup.5Cs][iAs][iGs][l.sup.5Cs][lAs][lTs][lT]
305 GAATCTACCCACATTCACCAGCATT 181 [iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs 877 ]
[i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs]
[i.sup.5Cs][i As][iGs][i.sup.5Cs][iAs][iTs][iT] 306 CGAATCTACCCACATTCACCAGCAT 182
[i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 878 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs]
[i.sup.5Cs ][iAs][iTs][iTs][i.sup.5Cs][iAs][i.sup.5Cs][i .sup.5Cs][iAs][iGs][i.sup.5Cs][iAs][iT] 307
CGAATCTACCCATTCACCAGCATT 178 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 879 s]
[iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5Cs][lAs][lTs] [lTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][i
Gs][l.sup.5Cs][lAs][lTs][lT] 308 CGAATCTACCCATTCACCAGCATT 178 [1.sup.5Cs][lGs][lAs]
[lAs][iTs][i.sup.5Cs][iT 880 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][lAs][lTs] [lTs][i.sup.5Cs][iAs]
[i.sup.5Cs][i.sup.5Cs][iAs][i Gs][1.sup.5Cs][lAs][lTs][lT] 309 CATATATCCATCTAGCCACG 183
[1.sup.5Cs][iAs][iTs][iAs][iTs][iAs][iTs 881 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][ iAs]
[iGs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs] [iG] 310 ACGAATCTACCCATCCATGT 184 [iAs]
[i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 882 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iTs]
[i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs] [iT] 311 TGTCCATCCACGAATCTACC 185 [iTs][iGs][iTs]
[i.sup.5Cs][i.sup.5Cs][iAs][iT 883 s][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs] [iAs][iTs]
[i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5 C] 312 CGAATCTACCAACTCATCCA 186 [i.sup.5Cs]
[iGs][iAs][iAs][iTs][i.sup.5Cs][iT 884 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iAs][i.sup.5Cs] [iTs]
[i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i A] 313 ACGAATCTACCATCCATGTA 187 [iAs]
[i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 885 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ i.sup.5Cs]
[i.sup.5Cs][iAs][iTs][iGs][iTs][iA] 314 CGAATCTACCCATCTCTCCA 188 [i.sup.5Cs][iGs][iAs]
[iAs][iTs][i.sup.5Cs][iT 886 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][iTs]
[i.sup.5Cs][iTs][i.sup.5Cs][i.sup.5Cs][ iA] 315 CACGAATCTACCCATCTCTC 189 [i.sup.5Cs]
[iAs][i.sup.5Cs][iGs][iAs][iAs][iT 887 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][iAs]
[iTs][i.sup.5Cs][iTs][i.sup.5Cs][iTs][i .sup.5C] 316 ACGAATCTACCCATCTCTCC 190 [iAs]
[i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5C 888 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs] [iTs]
[i.sup.5Cs][iTs][i.sup.5Cs][iTs][i.sup.5Cs][i .sup.5C] 317 CGAATCTACCATCCATGTAC 82
[l.sup.5Cs][lG][iA][iAs][iTs][i.sup.5Cs][iTs] 889 [iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i
.sup.5Cs][iAs][iTs][iGs][iTs][lAs][l.sup.5C] 318 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs]
[lGs][iAs][iAs][iTs][i.sup.5Cs][iT 890 s][iA][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][ i.sup.5Cs]
[iAs][iTs][iGs][iTs][lAs][l.sup.5C] 319 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lGs][iAs]
[iAs][iTs][i.sup.5Cs][iT 891 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iT][i.sup.5C][i .sup.5Cs][iAs][iTs]
[iGs][iTs][lAs][l.sup.5C] 320 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lGs][iAs][iAs][iTs]
[i.sup.5Cs][iT 892 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iG][iT]
[iAs][i.sup.5C] 321 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs]
[iT 893 s][iAs][i.sup.5Cs][l.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs][lAs]
[l.sup.5 C] 322 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT
894 s][iAs][i.sup.5Cs][l.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5
C] 323 CGAATCTACCATCCATGTAC 82 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 895 s][iAs]
[i.sup.5Cs][i.sup.5Cs][lAs][lTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs][lAs][l.sup.5 C] 324
CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT 896 s][iAs]
[i.sup.5Cs][l.sup.5Cs][lAs][lTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs][lAs][l.sup.5 C] 325
CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lGs][iAs][iAs][iTs][i.sup.5Cs][iT 897 s][iAs]
[i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs][iAs][i.sup.5 C] 326
CGAATCTACCATCCATGTAC 82 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 898 s][iAs]
[i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iGs][iTs][iTs][iAs][i.sup.5 C] 327
CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iT 899 s]
[iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s]
[i.sup.5Cs][iA][iA][iT][i.sup.5Cs][iA] 328 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs]

[iGs][iAs][iAs][iTs][i.sup.5Cs][iT 900 s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5C][iA][iA][iT][i.sup.5Cs][iA] 329 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs][iAs][iT 901 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C][ iA][iT][iG][iT][iA][i.sup.5Cs][iT] 330 CCATCCATCATCCGGGATAAGAGTTC 16 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ 902 iAs][iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C s][iGs][iGs][iGs][iAs][iTs][iAs][iA ][iG][iA][iG][iT][iTs][i.sup.5C] 331 GCATTTATTCAACGGGGATAAGAGTT 17 [iGs][i.sup.5Cs][iAs][iTs][iTs][iTs][iAs 903 ][iTs][iTs][i.sup.5Cs][iAs][iAs][i.sup.5Cs][i Gs][iGs][iGs][iGs][iAs][iTs][iA][iA ][iG][iA][iG][iTs][iT] iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; lA, lG, l.sup.5C, and lT are 2′-4′ methylene bridge (LNA) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage, and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage. †Each 5′-methyl cytidine (5C) in any of the sequences provided in Table 5 may independently and optionally be replaced with cytidine.

TABLE-US-00025 TABLE 11 Exemplary antisense oligonucleotides ASO Nucleobase Sequence SEQ Antisense Oligonucleotide SEQ # (5′ .fwdarw. 3′) ID NO: Structures (5′ .fwdarw. 3′) ID NO: 350 GCATTTATTCAACGGGGATAAGAGTT 17 [lGs][l.sup.5Cs][iAs][iTs][iTs] 904 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][i.sup.5Cs][iGs][iGs] [iGs][iGs][iAs][iTs][iAs][ iAs][iGs][iAs][iGs][lTs][l T] 351 GCATTTATTCAACGGGGATAAGAGTT 17 [iGs][l.sup.5C][lA][iTs][iTs][i 905 Ts][iAs][iTs][iTs][i.sup.5Cs][i As][iAs][i.sup.5Cs][iGs][iGs][i Gs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][lTs][lT] 352 GCATTTATTCAACGGGATAAGAGTTC 124 [iGs][i.sup.5C][iA][iTs][iTs][i 906 Ts][iAs][iTs][iTs][i.sup.5Cs][i As][iAs][i.sup.5Cs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i.sup.5 C] 353 GCATTTATTCAACGGGATAAGAGTTC 124 [iGs][i.sup.5Cs][iAs][iTs][iTs] 907 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iA][iA][i.sup.5Cs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i.sup.5 C] 354 GCATTTATTCAACGGGATAAGAGTTC 124 [iGs][i.sup.5Cs][iAs][iTs][iTs] 908 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][i.sup.5Cs][iGs][iG][ iG][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i.sup.5 C] 355 GCATTTATTCAACGGGATAAGAGTTC 124 [iGs][i.sup.5Cs][iAs][iTs][iTs] 909 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][ iGs][iAs][iG][iT][iTs][i.sup.5 C] 356 GCATTTATTCAACGGGATAAGAGTTC 124 [iGs][i.sup.5Cs][iAs][iTs][iTs] 910 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 357 GCATTTATTCAACGGATAAGAGTTCT 125 [iGs][i.sup.5Cs][iAs][iTs][iTs] 911 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][i.sup.5Cs][iGs][iGs] [iAs][iTs][iAs][iAs][iG][i A][iG][iT][iT][i.sup.5Cs][iT] 358 CATTTATTCAACGGATAAGAGTTC 338 [l.sup.5Cs][lAs][lTs][lTs][iTs] 912 [iAs][iTs][iTs][i.sup.5Cs][iAs] [lAs][l.sup.5Cs][lGs][lGs][iAs] [iTs][iAs][iAs][iGs][iAs][ lGs][lTs][lTs][l.sup.5C] 359 CATTTATTCAACGGATAAGAGTTC 338 [i.sup.5Cs][iAs][iTs][iTs][iTs] 913 [iAs][iTs][iTs][i.sup.5Cs][iAs] [iAs][i.sup.5Cs][iGs][iGs][iAs] [iTs][iAs][iAs][iGs][iAs][ iGs][iTs][iTs][i.sup.5C] 360 CATTTATTCAACGGGATAAGAGTT 117 [i.sup.5Cs][iAs][iTs][iTs][iTs] 914 [iAs][iTs][iTs][i.sup.5Cs][iAs] [iAs][i.sup.5Cs][iGs][iGs][iGs] [iAs][iTs][iAs][iAs][iGs][ iAs][iGs][iTs][iT] 361 CATTTATTCAACGGGGATAAGAGTT 339 [i.sup.5Cs][iAs][iTs][iTs][iTs] 915 [iAs][iTs][iTs][i.sup.5Cs][iAs] [iAs][i.sup.5Cs][iGs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][ iGs][iAs][iGs][iTs][iT] 362 GCATTTATTCAAGGGGATAAGAGTT 340 [iGs][i.sup.5Cs][iAs][iTs][iTs] 916 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][iGs][iGs][iGs][ iGs][iAs][iTs][iAs][iAs][i Gs][iAs][iGs][iTs][iT] 363 GCATTTATTCAACGGGATAAGAGTT 341 [iGs][i.sup.5Cs][iAs][iTs][iTs] 917 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][ iGs][iAs][iGs][iTs][iT] 364 GCATTTATTCAACGGGGATAAGAGT 342 [iGs][i.sup.5Cs][iAs][iTs][iTs] 918 [iTs][iAs][iTs][iTs][i.sup.5Cs] [iAs][iAs][i.sup.5Cs][iGs][iGs] [iGs][iGs][iAs][iTs][iAs][ iAs][iGs][iAs][iGs][iT] 365 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iG][iT][iTs][i.sup.5Cs][i 919 As][iAs]

[iTs][i.sup.5Cs][iAs][i Ts][iTs][i.sup.5Cs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i.sup.5 C] 366 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 920 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iT][iT][i.sup.5Cs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i.sup.5 C] 367 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 921 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iG][ iG][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iTs][iTs][i.sup.5 C] 368 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 922 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][ iGs][iAs][iG][iT][iTs][i.sup.5 C] 369 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 923 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 370 TGTTCAATCATTGGGATAAGAGTTC 343 [iTs][iGs][iTs][iTs][i.sup.5Cs] 924 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][iGs][iGs][iGs][ iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iTs][i.sup.5C] 371 TGTTCAATCATTCGGGATAAGAGTT 344 [iTs][iGs][iTs][iTs][i.sup.5Cs] 925 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][ iGs][iAs][iGs][iTs][lT] 372 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 569 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][ iG][iA][iG][iT][iTs][i.sup.5C] 373 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 570 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iAs][ iGs][iA][iG][iT][iTs][i.sup.5C] 374 TGTTCAATCATTCGGGATAAGAGTTC 9 [lTs][lGs][iTs][iTs][i.sup.5Cs] 927 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 375 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][lTs][lTs][i.sup.5Cs] 928 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 376 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 929 [lAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 377 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 930 [iAs][lAs][lTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][lT][iTs][i.sup.5C] 378 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 931 [iAs][iAs][iTs][i.sup.5Cs][lAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 379 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 932 [iAs][iAs][iTs][i.sup.5Cs][iAs] [lTs][lTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 380 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 933 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][l.sup.5Cs][lGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 381 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 934 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][lGs] [lGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 382 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 935 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][lAs][lTs][iAs][iA][i G][iA][iG][lT][iTs][i.sup.5C] 383 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 936 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][l G][lA][iG][lT][iTs][i.sup.5C] 384 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 937 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][l G][lA][iG][iT][iTs][i.sup.5C] 385 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 938 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][lG][lT][iTs][i.sup.5C] 386 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 939 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][lTs][l.sup.5C] 387 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][lTs][lTs][i.sup.5Cs] 940 [iAs][iAs][iTs][i.sup.5Cs][iAs] [lTs][lTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT][iTs][i.sup.5C] 388 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 941 [lAs][lAs][iTs][i.sup.5Cs][iAs ][iTs][iTs][i.sup.5Cs][iGs][iGs ][iGs][iAs][iTs][iAs][iA][ iG][iA]

[iG][iT][iTs][i.sup.5C] 389 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs] [i.sup.5Cs] 942 [lAs][lAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs] [iA][i G][iA][iG][iT][iTs][i.sup.5C] 390 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs] [iTs][i.sup.5Cs] 943 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iG][iG][i G][iAs][iTs][iAs] [iA][iG][ iA][iG][iT][iTs][i.sup.5C] 391 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs] [iTs][i.sup.5Cs] 944 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iT][iT][i.sup.5Cs][iGs][iG][iG ][iAs][iTs][iAs] [iAs][iGs] [iAs][iGs][iTs][iTs][i.sup.5C] 392 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iG] [iT][iTs][i.sup.5Cs][i 945 As][iAs][iTs][i.sup.5Cs][iAs][i T][iT][i.sup.5Cs][iGs][iGs][iGs ][iAs][iTs] [iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i.sup.5C] 393 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs] [iG][iT][iTs][i.sup.5Cs][i 946 As][iAs][iTs][i.sup.5Cs][iAs][i Ts][iTs][i.sup.5Cs][iGs][iG][iG ][iAs] [iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i.sup.5C] 394 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iG][iT][iTs][i.sup.5Cs][i 572 As][iAs][iTs][i.sup.5Cs][iAs][i Ts][iTs][i.sup.5Cs][iGs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iG][iT][iTs][i.sup.5C] 395 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iG][iT][iT][i.sup.5Cs][iA 573 s][iAs][iTs] [i.sup.5Cs][iAs][iT s][iTs][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ][iA][iG][iT][iTs] [i.sup.5C] 396 TGTTCAATCATTCGGGATAAGAGTTC 345 [x][dt][dc][da][iTs][iGs][ 947 iTs] [iTs][i.sup.5Cs][iAs][iAs][ iTs][i.sup.5Cs][iAs][iTs][iTs][ i.sup.5Cs][iGs][iGs][iGs][iAs][ iTs][iAs] [iA][iG][iA][iG][ iT][iTs][i.sup.5C] 397 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iG][iT] [iT][i.sup.5Cs][iA 948 s][iAs][iTs][i.sup.5Cs][iAs][iT s][iTs][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs] [iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5C] 398 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs] [iGs][iTs][iT][i.sup.5C][i 949 A][iAs][iTs][i.sup.5Cs][iAs][iT s][iTs][i.sup.5Cs][iGs][iGs][iG s] [iAs][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i.sup.5C] 399 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 950 [iA][iA][iT] [i.sup.5Cs][iAs][iT s][iTs][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs] [i.sup.5C] 400 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 951 [iAs] [iAs][iT][i.sup.5C][iA][iT s][iTs][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ][iAs][iGs] [iTs][iTs][i.sup.5C] 401 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 952 [iAs][iAs][iTs][i.sup.5Cs][iA][ iT][iT][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ] [iAs][iGs][iTs][iTs][i.sup.5C] 402 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs] [i.sup.5Cs] 953 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iT][i.sup.5C][iG][iGs][iG s][iAs][iTs][iAs] [iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5C] 403 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs] [iTs][iTs][i.sup.5Cs] 954 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iG][iG][i G][iAs][iTs] [iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5C] 404 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs] [iGs][iTs][iTs][i.sup.5Cs] 955 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iG] [iA][iT][iAs][iAs][iGs ][iAs][iGs][iTs][iTs][i.sup.5C] 405 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 956 [iAs][iAs][iTs] [i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iT][iA][iA][iGs ][iAs][iGs][iTs][iTs] [i.sup.5C] 406 TGTTCAATCATTCGGGATAAGAGTTC 9 [iTs][iGs][iTs][iTs][i.sup.5Cs] 957 [iAs] [iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iGs] [iTs][iTs][i.sup.5C] 407 TGTTCAATCATTC[n3]GGGATAAGA 194/253 [iTs][iGs][iTs][iTs] [i.sup.5Cs] 958/959 GTTC [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][n3s][iGs] [iGs][iGs] [iAs][iTs][iAs][ iA][iG][iA][iG][iT][iTs][i .sup.5C] 408 TGTTCAATCATTC[n6]GGGATAAGA 194/253 [iTs][iGs][iTs][iTs][i.sup.5Cs] 958/959 GTTC [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs] [i.sup.5Cs][n6s][iGs] [iGs][iGs][iAs][iTs][iAs][ iA][iG][iA][iG][iT][iTs] [i.sup.5C] 409 TCTGTTCAATCATTGGGGATAAGAGT 347 [iTs][i.sup.5Cs][iTs][iGs][iTs] 960 [iTs][i.sup.5Cs] [iAs][iAs][iTs] [i.sup.5Cs][iAs][iTs][iTs][iGs] [iGs][iGs][iGs][iAs][IT][i A][iA][iG][iA][iGs][IT] 410 TCTGTTCAATCATGGGGATAAGAGTT 348 [iTs][i.sup.5Cs][iTs][iGs][iTs] 961 [iTs] [i.sup.5Cs][iAs][iAs][iTs] [i.sup.5Cs][iAs][iTs][iGs][iGs] [iGs][iGs][iAs][iTs][iA][i A][iG][iA][iTs] [IT] 411 TCTGTTCAATCATGGGATAAGAGTTC 349 [iTs][i.sup.5Cs][iTs][iGs][iTs] 962 [iTs] [i.sup.5Cs][iAs][iAs][iTs] [i.sup.5Cs][iAs][iTs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT]

[iTs][i.sup.5C] 412 CTGTTCAATCATTTGGGGATAAGAGT 350 [i5Cs][iTs][iGs][iTs][iTs] 963 [i.sup.5Cs][iAs][iAs][iTs][i5Cs ][iAs][iTs][iTs][iTs][iGs] [iGs][iGs][iGs][iAs][iT][i A][iA][iG][iA] [iGs][iT] 413 CTGTTCAATCATTGGGGATAAGAGTT 351 [i5Cs][iTs][iGs][iTs][iTs] 964 [i.sup.5Cs][iAs][iAs][iTs][i5Cs ][iAs][iTs][iTs][iGs][iGs] [iGs][iGs][iAs][iTs][iA][i A][iG][iA][iG] [iTs][lT] 414 CTGTTCAATCATTGGGATAAGAGTTC 352 [i5Cs][iTs][iGs][iTs][iTs] 965 [i.sup.5Cs][iAs][iAs][iTs][i5Cs ][iAs][iTs][iTs][iGs][iGs] [iGs][iAs][iTs][iAs][iA][i G][iA][iG][iT] [iTs][i.sup.5C] 415 TGTTCAATCATTCTGGGGATAAGAGT 353 [iTs][iGs][iTs][iTs][i.sup.5Cs] 966 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iTs][iGs] [iGs][iGs][iGs][iAs][iT][i A][iA] [iG][iA][iGs][iT] 416 TGTTCAATCATTCGGGGATAAGAGTT 163 [iTs][iGs][iTs][iTs][i.sup.5Cs] 967 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i5Cs][iGs][iGs] [iGs][iGs][iAs][iTs][iA][i A][iG][iA] [iG][iTs][iT] 417 TGTTCAATCATTCGGATAAGAGTTCT 354 [iTs][iGs][iTs][iTs][i.sup.5Cs] 968 [iAs][iAs][iTs][i.sup.5Cs][iAs] [iTs][iTs][i.sup.5Cs][iGs][iGs] [iAs][iTs][iAs][iAs][iG][i A][iG][iT] [iT][i.sup.5Cs][iT] 418 TGTTCAATCATTCGGGATAAGAGTTC 9 [nmaTs][nmaGs][nmaTs] [nmaT 969 s][nma.sup.5Cs][nmaAs][nmaAs][n maTs][nma.sup.5Cs][nmaAs][nmaTs ][nmaTs] [nma.sup.5Cs][nmaGs][nm aGs][nmaGs][nmaAs][nmaTs][ nmaAs][nmaA][nmaG][nmaA][n maG] [nmaT][nmaTs][nma.sup.5C] 419 TCCATCCATCCTTGAGTTCTTTCCAG 355 [iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs 970 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][i.sup.5Cs][iTs][iTs][iGs][i As] [iGs][iTs][iTs][i5Cs][i Ts][iTs][iTs][i.sup.5Cs][i.sup.5Cs][ iAs][iG] 420 TCCATCCATCCTT[n6]GAGTTCTTT 497/249 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs 971/972 CCAG ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5 Cs][i5Cs][iTs][iTs][n6s][ iGs][iAs][iGs][iTs][iTs][i .sup.5Cs][iTs][iTs][iTs][i.sup.5Cs][ i.sup.5Cs][iAs][iG] 421 CTCTCCATCCATCTTCTTTCCAGGAA 357 [i5Cs][iTs][i.sup.5Cs][iTs][i5C 973 s][i.sup.5Cs] [iAs][iTs][i5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][ iTs][i.sup.5Cs][iTs][iTs][iTs][ i.sup.5Cs] [i.sup.5Cs][iAs][iGs][iGs] [iAs][iA] 422 TCCATCCATCCTTGAGTTCTTTCCAG 355 [iTs][i5Cs] [i.sup.5Cs][iAs][iTs 974 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][i.sup.5Cs][lTs][lTs][lGs][ lAs] [iGs][iTs][iTs][i5Cs][ iTs][iTs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iG] 423 CATCCATCTATCCGGGATAAGAGTTC 159 [i.sup.5Cs][iA][iT][i.sup.5Cs][i.sup.5Cs] 975 [iAs] [iTs][i.sup.5Cs][iTs][iAs] [iTs][i.sup.5Cs][i.sup.5Cs][iGs][iGs ][iGs][iAs][iTs][iAs][iAs] [iGs][iAs] [iGs][iTs][iTs][ i.sup.5C] 424 CATCCATCTATCCGGGATAAGAGTTC 159 [i5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5C 976 s][iAs][iTs][i.sup.5Cs][iTs][iA s][iT][i.sup.5C][i.sup.5Cs][iGs][iGs ][iGs] [iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 425 CATCCATCTATCCGGGATAAGAGTTC 159 [i5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C 977 s][iAs][iTs] [i.sup.5Cs][iTs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iGs][i G][iG][iAs][iTs][iAs][iAs] [iGs][iAs][iGs] [iTs][iTs][ i.sup.5C] 426 CATCCATCTATCCGGGATAAGAGTTC 159 [i5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5C 978 s][iAs][iTs][i.sup.5Cs][iTs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs] [iAs][iA s][iGs][iAs][iG][iT][iTs][ i.sup.5C] 427 CATCCATCTATCCGGGATAAGAGTTC 159 [i5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C 979 s][iAs][iTs][i.sup.5Cs][iTs][iA s][iTs][i.sup.5Cs][i.sup.5Cs] [iGs][i Gs][iGs][iAs][iTs][iAs][iA ][iG][iA][iG][iT][iTs][i.sup.5 C] 428 ATCCATCTATCCGGATAAGAGTTC 358 [lAs][lTs][i5Cs][i.sup.5Cs][iAs 980 ][iTs][i.sup.5Cs] [iTs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iA s][iTs][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i5C] 429 ATCCATCTATCCGGGATAAGAGTT 359 [iAs][iTs][i5Cs][i.sup.5Cs][iAs 981 ][iTs] [i.sup.5Cs][iTs][iAs][iTs ][i.sup.5Cs][i5Cs][lGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][lTs] [lT] 430 ATCCATCTATCCGGATAAGAGTTC 358 [iAs][iTs][i5Cs][i.sup.5Cs][iAs 982 ][iTs] [i.sup.5Cs][iTs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iA s][iTs][iAs][iAs][iGs][iAs ][iGs][iTs] [iTs][i.sup.5C] 431 ATCCATCTATCCGGGATAAGAGTT 359 [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs 983 ][iTs][i.sup.5Cs][iTs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ] [iAs][iGs][iTs][iT] 432 CAACTCATCCATCAGAGTTCTTTCCA 360 [i5Cs][iAs][iAs][i.sup.5Cs] [iTs 984 ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i5 Cs][iAs][iTs][i.sup.5Cs][iAs][i Gs][iAs][iGs][iTs][iTs] [i.sup.5 Cs][iTs][iTs][iTs][i.sup.5Cs][i .sup.5Cs][iA] 433 GCCACGAATCTACGGATAAGAGTTCT 361 [iGs][i.sup.5Cs][i5Cs][iAs][i.sup.5C 985 s][iGs]

[iAs][iAs][iTs][i5C s][iTs][iAs][i.sup.5Cs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iTs] [i.sup.5Cs] [iT] 434 CCACGAATCTACCGGGATAAGAGTTC 14 [i.sup.5Cs][i.sup.5C][iA] [i.sup.5Cs][iGs] 986 [iAs][iAs][iTs][i.sup.5Cs][iTs] [iAs][i5Cs][i.sup.5Cs][iGs][iGs ][iGs][iAs][iTs] [iAs][iAs] [iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 435 CCACGAATCTACCGGGATAAGAGTTC 14 [i5Cs][i5Cs][iAs][i.sup.5Cs][iG 987 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iA][i.sup.5C][i.sup.5Cs][iGs] [iGs ][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 436 CCACGAATCTACCGGGATAAGAGTTC 14 [i5Cs][i5Cs][iAs][i5Cs][iG 988 s][iAs][iAs][iTs] [i.sup.5Cs][iT s][iAs][i5Cs][i.sup.5Cs][iGs][i G][iG][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 437 CCACGAATCTACCGGGATAAGAGTTC 14 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 989 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iG][iT][iTs][ i.sup.5C] 438 CCACGAATCTACCGGGATAAGAGTTC 14 [i5Cs] [i.sup.5Cs][iAs][i.sup.5Cs][iG 990 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i5Cs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iA ][iG][iA][iG][iT][iTs][i.sup.5 C] 439 CCACGAATCTACCGGATAAGAGTTCT 8 [i5Cs][i.sup.5C][iA][i5Cs][iGs] 991 [iAs][iAs][iTs] [i.sup.5Cs][iTs] [iAs][i.sup.5Cs][i.sup.5Cs][iGs][iGs ][iAs][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs] [i.sup.5Cs] [iT] 440 CCACGAATCTACCGGATAAGAGTTCT 8 [i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][iG 992 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iA][i.sup.5C][i.sup.5Cs][iGs][iGs ][iAs][iTs] [iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i5Cs] [iT] 441 CCACGAATCTACCGGATAAGAGTTCT 8 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 993 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i5Cs][i.sup.5Cs] [iGs][i G][iA][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i.sup.5Cs] [iT] 442 CCACGAATCTACCGGATAAGAGTTCT 8 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 994 s][iAs][iAs] [iTs][i.sup.5Cs][iT s][iAs][i5Cs][i.sup.5Cs][iGs][i Gs][iAs][iTs][iAs][iAs][iG s][iAs][iGs][iT][iT] [i.sup.5Cs] [iT] 443 CCACGAATCTACCGGATAAGAGTTCT 8 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs] [iG 995 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][iGs][i Gs][iAs][iTs][iAs][iAs] [iG ][iA][iG][iT][iT][i.sup.5Cs][iT ] 444 CACGAATCTACCGATAAGAGTTCT 362 [i5Cs][iAs] [i.sup.5Cs][iGs][iAs 996 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i5Cs][i5Cs][iGs][iAs][iT s][iAs][iAs] [iGs][iAs][iGs ][iTs][iTs][i.sup.5Cs][iT] 445 CCACGAATCTACGATAAGAGTTCT 363 [i5Cs] [i.sup.5Cs][iAs][i.sup.5Cs][iG 997 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][iGs][iAs][iT s] [iAs][iAs][iGs][iAs][iGs ][iTs][iTs][i.sup.5Cs][iT] 446 CACGAATCTACCGGGGATAAGAGT 364 [i5Cs][iAs][i.sup.5Cs][iGs][iAs 998 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][iGs] [iGs][iG s][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iT] 447 CCACGAATCTACGGGGATAAGAGT 365 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 999 s][iAs][iAs] [iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][iGs][iGs][iG s][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iT] 448 CACGAATCTACCGGATAAGAGTTC 366 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1000 ][iAs] [iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iA s][iTs][iAs][iAs][iGs][iAs ][iGs][iTs] [iTs][i.sup.5C] 449 CACGAATCTACCGGGATAAGAGTT 367 [i.sup.5Cs][iAs][i.sup.5Cs][iGs] [iAs 1001 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs][iAs][iAs] [iGs ][iAs][iGs][iTs][iT] 450 CCACGAATCTACGGATAAGAGTTC 368 [i.sup.5Cs][i.sup.5Cs] [iAs][i.sup.5Cs][iG 1002 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][iGs][iGs][iA s][iTs][iAs] [iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5C] 451 CCACGAATCTACGGGATAAGAGTT 369 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1003 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs] [iGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iT] 452 CACGAATCTACCGATAAGAGTTCT 362 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1004 ][iAs][iTs] [i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][iGs][iAs][iT s][iAs][iAs][iGs][iAs][iGs ][iTs][iTs] [i.sup.5Cs][iT] 453 CCACGAATCTACGGGGATAAGAGT 365 [i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][iG 1005 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][iGs][iGs][iG s][iGs][iAs] [iTs][iAs][iAs ][iGs][iAs][iGs][iT] 454 CCACGAATCTACCGGGGATAAGAGTT 22 [i5Cs] [i.sup.5C][iA][i5Cs][iGs] 1006 [iAs][iAs][iTs][i.sup.5Cs][iTs] [iAs][i5Cs][i.sup.5Cs][iGs][iGs ] [iGs][iGs][iAs][iTs][iAs] [iAs][iGs][iAs][iGs][iTs][ iT] 455 CCACGAATCTACCGGGGATAAGAGTT 22 [i5Cs][i5Cs][iAs][i.sup.5Cs][iG 1007 s][iAs][iAs]

[iTs][i.sup.5Cs][iT s][iA][i.sup.5C][i.sup.5Cs][iGs][iGs ][iGs][iGs][iAs][iTs][iAs] [iAs][iGs][iAs] [iGs][iTs][ iT] 456 CCACGAATCTACCGGGGATAAGAGTT 22 [i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][iG 1008 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][iGs][i G][iG][iGs] [iAs][iTs][iAs] [iAs][iGs][iAs][iGs][iTs][ iT] 457 CCACGAATCTACCGGGGATAAGAGTT 22 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1009 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i5Cs][i.sup.5Cs] [iGs][i Gs][iGs][iGs][iAs][iTs][iA s][iAs][iGs][iA][iG][iTs][ iT] 458 CCACGAATCTACCGGGGATAAGAGTT 22 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1010 s][iAs] [iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][iGs][i Gs][iGs][iGs][iAs][iT][iA ][iA][iG][iA] [iG][iTs][iT] 459 ACGAATCTACCCAGGGGATAAGAGTT 370 [iAs][i5Cs][iGs][iAs][iAs] 1011 [iTs][i.sup.5Cs][iTs][iAs][i5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iGs][iG s][iGs][iGs][iAs][iTs][iAs ] [iAs][iGs][iAs][iGs][iTs] [iT] 460 CTAGCCACGAATCTAAGAGTTCTTTC 371 [i5Cs][iTs][iAs] [iGs][i.sup.5Cs 1012 ][i.sup.5Cs][iAs][i.sup.5Cs][iGs][iA s][iAs][iTs][i5Cs][iTs][iA s][iAs][iGs] [iAs][iGs][iTs ][iTs][i.sup.5Cs][iTs][iTs][iTs ][i.sup.5C] 461 CCACGAATCTACCATAAGAGTTCTTT 372 [i5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1013 s][iAs] [iAs][iTs][i.sup.5Cs][iT s][iAs][i5Cs][i.sup.5Cs][iAs][i Ts][iAs][iAs][iGs][iAs][iG s][iTs][iTs] [i.sup.5Cs][iTs][iT s][lT] 462 CCATCCATCATCCGGGATAAGAGTCC 373 [i5Cs][i.sup.5Cs][iAs] [iTs][i5C 1014 s][i.sup.5Cs][iAs][iTs][i5Cs][i As][iT][i5C][i.sup.5Cs][iGs][iG s][iGs][iAs][iTs] [iAs][iAs ][iGs][iAs][iGs][iTs][i.sup.5Cs ][i.sup.5C] 463 TCCATCCATCATCGGGGATAAGAGTT 120 [iTs][i.sup.5C][i.sup.5C][iAs][iTs][ 1015 i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5Cs ][iAs][iTs] [i.sup.5Cs][iGs][iGs ][iGs][iGs][iAs][iTs][iAs] [iAs][iGs][iAs][iGs][iTs][ iT] 464 TCCATCCATCATCGGGGATAAGAGTT 120 [iTs][i.sup.5Cs][i5Cs][iAs][iTs 1016 ][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][i.sup.5 Cs][iA][iT][i.sup.5Cs][iGs][iGs ][iGs][iGs][iAs][iTs][iAs] [iAs][iGs] [iAs][iGs][iTs][ iT] 465 TCCATCCATCATCGGGGATAAGAGTT 120 [iTs][i5Cs][i.sup.5Cs][iAs] [iTs 1017 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5 Cs][iAs][iTs][i.sup.5Cs][iGs][i G][iG][iGs][iAs][iTs] [iAs] [iAs][iGs][iAs][iGs][iTs][ iT] 466 TCCATCCATCATCGGGGATAAGAGTT 120 [iTs][i5Cs] [i.sup.5Cs][iAs][iTs 1018 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5 Cs][iAs][iTs][i.sup.5Cs][iGs][i Gs] [iGs][iGs][iAs][iTs][iA s][iAs][iGs][iA][iG][iTs][ iT] 467 TCCATCCATCATCGGGGATAAGAGTT 120 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs 1019 ] [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5 Cs][iAs][iTs][i.sup.5Cs][iGs][i Gs][iGs][iGs][iAs][iTs][iA ][iA] [iG][iA][iG][iTs][iT] 468 TCCATCCATCATCGGGATAAGAGTTC 121 [iTs][i5C][i5C][iAs][iTs][ 1020 i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5Cs ][iAs][iTs][i.sup.5Cs][iGs][iGs ][iGs][iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 469 TCCATCCATCATCGGGATAAGAGTTC 121 [iTs][i5Cs] [i.sup.5Cs][iAs][iTs 1021 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5 Cs][iA][iT][i.sup.5Cs][iGs][iGs ][iGs] [iAs][iTs][iAs][iAs] [iGs][iAs][iGs][iTs][iTs][ i.sup.5C] 470 TCCATCCATCATCGGGATAAGAGTTC 121 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs 1022 ] [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5 Cs][iAs][iTs][i.sup.5Cs][iGs][i G][iG][iAs][iTs][iAs][iAs] [iGs] [iAs][iGs][iTs][iTs][ i.sup.5C] 471 TCCATCCATCATCGGGATAAGAGTTC 121 [iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs 1023 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i5 Cs][iAs][iTs][i.sup.5Cs][iGs][i Gs] [iGs][iAs][iTs][iAs][iA s][iGs][iAs][iG][iT][iTs][ i.sup.5C] 472 TCCATCCATCATCGGGATAAGAGTTC 121 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs 1024 ] [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][iAs][iTs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iA ] [iG][iA][iG][iT][iTs][i.sup.5C ] 473 TCCATCCATCATCGGATAAGAGTTCT 122 [iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs 1025 ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][iAs][iTs][i.sup.5Cs][iGs][i Gs][iAs][iTs][iAs][iAs][iG ][iA][iG][iT][iT][i.sup.5Cs][iT ] 474 CCATCCATCATCCGATAAGAGTTCTT 123 [i.sup.5Cs][i.sup.5C][iA][iTs][i.sup.5Cs] 1026 [i.sup.5Cs][iAs][iTs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iGs][iA s][iTs][iAs][iAs][iGs][iAs ] [iGs][iTs][iTs][i.sup.5Cs][iTs ][iT] 475 CCATCCATCATCCGATAAGAGTTCTT 123 [i.sup.5Cs] [i.sup.5Cs][iAs][iTs][i.sup.5C 1027 s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i As][iTs][i.sup.5Cs] [i.sup.5Cs][iGs][ iA][iT][iAs][iAs][iGs][iAs ][iGs][iTs][iTs][i.sup.5Cs][iTs ][iT] 476 CCATCCATCATCCGATAAGAGTTCTT 123 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1028 s]

[i.sup.5Cs][iAs][iTs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iGs][ iAs][iTs][iAs][iAs][iGs][i As][iGs][iTs][iT][i.sup.5C][iTs ][iT] 477 CCATCCATCATCCGATAAGAGTTCTT 123 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1029 s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iGs][ iAs][iTs][iAs][iAs][iGs][i A][iG][iT][iT][i.sup.5C][iTs][i T] 478 CCATCCATCATCGGGGATAAGAGT 374 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1030 s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i As][iTs][i.sup.5Cs][iGs][iGs][i Gs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][iT] 479 CCATCCATCATCGGGGATAAGAGT 374 [l.sup.5Cs][l.sup.5Cs][iAs][iTs][i.sup.5C 1031 s][i.sup.5Cs][iAs][iTs][l.sup.5Cs][i As][iTs][l.sup.5Cs][lGs][iGs][ iGs][iGs][iAs][iTs][iAs][i As][iGs][iAs][lGs][lT] 480 CCATCCATCATCGGATAAGAGTTC 48 [l.sup.5Cs][l.sup.5Cs][iAs][iTs][i.sup.5C 1032 s][i.sup.5Cs][iAs][iTs][l.sup.5Cs][i As][iTs][l.sup.5Cs][lGs][iGs][i As][iTs][iAs][iAs][iGs][iA s][iGs][iTs][lTs][l.sup.5C] 481 TCATCCATCTAGCGGGATAAGAGTTC 375 [iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs 1033 ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iT s][iAs][iGs][i.sup.5Cs][iGs][iG s][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iTs][iTs] [i.sup.5C] 482 CATCCATCTAGCCGGGATAAGAGTTC 376 [i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5C 1034 s][iAs][iTs][i.sup.5Cs][iTs][iA s][iGs][i.sup.5Cs][i.sup.5Cs][iGs][i Gs][iGs][iAs][iTs][iAs][iA s][iGs][iAs][iGs][iTs][iTs ][i.sup.5C] 483 TCCATCTAGCCACGGGGATAAGAGTT 377 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs 1035 ][i.sup.5Cs][iTs][iAs][iGs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iGs][i Gs][iGs][iGs][iAs][iTs][iA s][iAs][iGs][iAs][iGs][iTs ][iT] 484 ATCCATCTAGCCGGGATAAGAGTT 378 [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs 1036 ][iTs][i.sup.5Cs][iTs][iAs][iGs ][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iG s][iAs][iTs][iAs][iAs][iGs ][iAs][iGs][iTs][iT] 485 ATCCATCTAGCCGGGGATAAGAGT 379 [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs 1037 ][iTs][i.sup.5Cs][iTs][iAs][iGs ][i.sup.5Cs][i.sup.5Cs][iGs][iGs][iG s][iGs][iAs][iTs][iAs][iAs ][iGs][iAs][iGs][iT] 486 CCATCCATCATCCGGGGATAAGAGTT 380 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1038 s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iGs][ iGs][iGs][iGs][iAs][iTs][i As][iAs][iGs][iAs][iGs][iT s][iT] 487 ATATATCCATCCACTAGTTCCTGGGG 381 [iAs][iTs][iAs][iTs][iAs][ 1039 iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iT s][iAs][iGs][iTs][iTs][i.sup.5C s][i.sup.5Cs][iTs][iGs][iGs][iG s][iG] 488 ATATATCCATCCA[n6]CTAGTTCCT 506/259 [iAs][iTs][iAs][iTs][iAs][ 1040/1041 GGGG iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iAs][n6s][i.sup.5C s][iTs][iAs][iGs][iTs][iTs ][i.sup.5Cs][i.sup.5Cs][iTs][iGs][iG s][iGs][iG] 489 ATATATCCATCCACTAGTTCCTGGGG 381 [iAs][iTs][iAs][iTs][iAs][ 1042 iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iT s][iAs][iGs][iTs][iTs][i.sup.5C s][i.sup.5Cs][iTs][iGs][iGs][iG s][iG] 490 TAGTTCAATCATAAGAGTTCTTTCCA 383 [iTs][iAs][iGs][iTs][iTs] 1043 [i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs] [iAs][iTs][iAs][iAs][iGs] [iAs][iGs][iTs][iTs][i.sup.5Cs] [iTs][iTs][iTs][i.sup.5Cs][i.sup.5Cs] [iA] 491 TCATCTGTTCAACAAACTAG 384 [iTs][i.sup.5Cs][iAs][iTs][i.sup.5Cs 1044 ][iTs][iGs][iTs][iTs][i.sup.5Cs ][iAs][iAs][i.sup.5Cs][iAs][iAs ][iAs][i.sup.5Cs][iTs][iAs][iG] 492 ATCTGTTCAATCAGCATTTATTCAAC 385 [iAs][iTs][i.sup.5Cs][iTs][iGs] 1045 [iTs][iTs][i.sup.5Cs][iAs][iAs] [iTs][i.sup.5Cs][iAs][iGs][i.sup.5Cs ][iAs][iTs][iTs][iTs][iAs] [iTs][iTs][i.sup.5Cs][iAs][iAs] [i.sup.5C] 493 ATCTACCCACCAGCATTTAT 386 [iAs][iTs][i.sup.5Cs][iTs][iAs] 1046 [i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5 Cs][i.sup.5Cs][iAs][iGs][i.sup.5Cs][ iAs][iTs][iTs][iTs][iAs][ iT] 494 CCACGAATCTACCGTTCAATCATTCA 387 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1047 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][iGs][i Ts][iTs][i.sup.5Cs][iAs][iAs][i Ts][i.sup.5Cs][iAs][iTs][iTs][i .sup.5Cs][iA] 495 CACGAATCTACCCATTCACCAGCATT 388 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1048 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iTs][i.sup.5Cs][iAs][i.sup.5Cs][ i.sup.5Cs][iAs][iGs][i.sup.5Cs][iAs] [iTs][iT] 496 ACGAATCTACCCAGTTCAATCATTCA 389 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1049 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iGs][iT s][iTs][i.sup.5Cs][iAs][iAs][iT s][i.sup.5Cs][iAs][iTs][iTs][i.sup.5 Cs][iA] 497 CCACGAATCTACCGTTCAATCATTCA 387

[i.sup.5Cs][i.sup.5C][iA][i.sup.5C][iGs][ 1050 iAs][iAs][iTs][i.sup.5Cs][iTs][ iAs][i.sup.5Cs] [i.sup.5Cs][iGs][iTs] [iTs][i.sup.5Cs][iAs][iAs][iTs] [i.sup.5Cs][iAs][iTs][iTs][i.sup.5Cs ][iA] 498 CCACGAATCTACCGTTCAATCATTCA 387 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1051 s] [iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][iGs][i Ts][iTs][i.sup.5Cs][iAs][iAs][i Ts] [i.sup.5Cs][iA][iT][iT][i.sup.5Cs ][iA] 499 CACGAATCTACCCATTCACCAGCATT 388 [i.sup.5Cs][iA][i.sup.5C][iG][iAs][i 1052 As][iTs][i.sup.5Cs][iTs][iAs][i .sup.5Cs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs] [iTs][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5C s][iAs][iGs][i.sup.5Cs][iAs][iT s][iT] 500 CACGAATCTACCCATTCACCAGCATT 388 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1053 ] [iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iTs][i.sup.5Cs][iAs] [i.sup.5Cs][ i.sup.5Cs][iAs][iG][i.sup.5C][iA][iT s][iT] 501 CCATCCATGTACTCAATCAT 390 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1054 s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5Cs] [iTs][i.sup.5Cs][i As][iAs][iTs][i.sup.5Cs][iAs][l T] 502 GAATCTACCCATCTGTTCAATC 106 [zGs][zAs][iAs][iTs][i.sup.5Cs] 1055 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][z.sup.5C s][zAs][zTs] [z.sup.5Cs][iTs][iG s][iTs][iTs][i.sup.5Cs][iAs][iA s][zTs][l.sup.5C] 503 CGAATCTACCCAATCTGTTCAATC 36 [z.sup.5Cs][zGs][zAs][zAs][iTs] 1056 [i.sup.5Cs][iTs] [iAs][i.sup.5Cs][i.sup.5C s][z.sup.5Cs][zAs][zAs][zTs][i.sup.5 Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs] [zAs][zAs][zTs][l.sup.5C] 504 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs] [iAs][iTs] 1057 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts] [i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iAs][iAs][iTs][i .sup.5Cs][lA] 505 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iAs][iTs][i 1058 .sup.5Cs][iTs] [iAs][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs][iAs][i.sup.5Cs][iAs][iTs ][i.sup.5Cs][iTs][iGs][iTs][iTs ] [i.sup.5Cs][iAs][iAs][iTs][i.sup.5C s][iA] 506 ACGAATCTACCCACATCTGTTCAATC 93 [iAs] [i.sup.5Cs][iGs][iAs][iAs] 1059 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][i As][iTs][i.sup.5Cs][iTs][iGs][i T][iT][i.sup.5C][iA][iA][iTs][i .sup.5C] 507 CGAATCTACCCACATCTGTTCAAT 111 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1060 [i.sup.5Cs][iTs] [iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts] [i.sup.5Cs][iAs][iAs][iT] 508 ACGAATCTACCCATCTGTTCAATC 113 [iAs][i.sup.5Cs][iGs][iAs] [iAs] 1061 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][iTs] [iGs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5C] 509 ACGAATCTACCCACATCTGTTCAATC 93 [iAs][i.sup.5C][iG][iAs][iAs][i 1062 Ts][i.sup.5Cs][iTs][iAs][i.sup.5Cs][ i.sup.5Cs][i.sup.5Cs] [iAs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][iTs][iGs][iTs ][iTs][i.sup.5Cs][iAs][iAs][iTs ][i.sup.5C] 510 ACGAATCTACCCACATCTGTTCAATC 93 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1063 [iTs][i.sup.5Cs] [iTs][iAs][i.sup.5Cs ][i.sup.5C][i.sup.5C][iAs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][iTs][iGs][iTs ][iTs] [i.sup.5Cs][iAs][iAs][iTs ][i.sup.5C] 511 ACGAATCTACCCACATCTGTTCAATC 93 [iAs] [i.sup.5Cs][iGs][iAs][iAs] 1064 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5Cs][i A][iT][i.sup.5Cs][iTs][iGs][iTs ][iTs][i.sup.5Cs][iAs][iAs][iTs ][i.sup.5C] 512 ACGAATCTACCCACATCTGTTCAATC 93 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1065 [iTs][i.sup.5Cs] [iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i As][iTs][i.sup.5Cs][iTs][iGs][i Ts][iTs] [i.sup.5Cs][iA][iA][iTs ][i.sup.5C] 513 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lG] [iA][iAs][iTs][i 1066 .sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs][lAs][lAs][iTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs ][iAs][iAs][lTs][l.sup.5C] 514 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1067 [i.sup.5Cs][iTs] [iAs][i.sup.5Cs][i.sup.5C ][i.sup.5C][lAs][lAs][iTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs ][iAs] [iAs][lTs][l.sup.5C] 515 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1068 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][lAs][lAs][iT][i.sup.5C ][iTs][iGs][iTs] [iTs][i.sup.5Cs ][iAs][iAs][lTs][l.sup.5C] 516 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs] [lGs][iAs][iAs][iTs] 1069 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][lAs][lAs][iTs] [i.sup.5 Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs][iA][iA][lTs][l.sup.5C] 517 CGAATCTACCCACCTGTTCAATCA 391 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1070 [l.sup.5Cs][iTs] [iAs][i.sup.5Cs][l.sup.5C s][i.sup.5Cs][iAs][l.sup.5Cs][l.sup.5Cs][ iTs][iGs][iTs][iTs][i.sup.5Cs][

iAs][iAs][iTs][l.sup.5Cs][lA] 518 CGAATCTACCCACATCTGTTCAAT 111 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1071 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][l.sup.5Cs][lAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iAs][lAs][lT] 519 CGAATCTACCCACTCTGTTCAATC 392 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1072 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][l.sup.5Cs][lTs][i .sup.5Cs][iTs][iGs][iTs][iTs][i .sup.5Cs][iAs][iAs][lTs][l.sup.5C] 520 CGAATCTACCCAATCTGTTCAATCA 393 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1073 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5 Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5Cs][i A] 521 CGAATCTACCCACTCTGTTCAATCA 394 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1074 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iTs][i .sup.5Cs][iTs][iGs][iTs][iTs][i .sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][ iA] 522 ACGAATCTACCCCATCTGTTCAATC 395 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1075 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iAs][iAs][iTs][i .sup.5C] 523 ACGAATCTACCCAATCTGTTCAATC 396 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1076 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iAs][iT s][i.sup.5Cs][iTs][iGs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs][i.sup.5 C] 524 CACGAATCTACCCTGTTCAATCATTC 397 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1077 ][iAs][iTs][i.sup.5Cs][iTsCs][i As][iAs][iTs][i.sup.5Cs][iAs][i Ts][iTs][i.sup.5C] 525 ACGAATCTACCCATGTTCAATCATTC 398 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1078 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iG s][iTs][iTs][i.sup.5Cs][iAs][iA s][iTs][i.sup.5Cs][iAs][iTs][iT s][i.sup.5C] 526 CCACGAATCTACCCTGTTCAATCATT 399 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1079 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ iTs][iGs][iTs][iTs][i.sup.5Cs][ iAs][iAs][iTs][i.sup.5Cs][iAs][ iTs][iT] 527 ACGAATCTACCCACTGTTCAATCATT 400 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1080 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i Ts][iGs][iTs][iTs][i.sup.5Cs][i As][iAs][iTs][i.sup.5Cs][iAs][i Ts][iT] 528 CCACGAATCTACCCATCTGTTCAATC 401 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1081 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][iTs][iGs][ iTs][iTs][i.sup.5Cs][iAs][iAs][ iTs][i.sup.5C] 529 CGAATCTACCCAATCTGTTCAATC 36 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1082 [i.sup.5Cs][iTs][iAs][i.sup.5C][i.sup.5C] [i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs ][iAs][iAs][iTs][i.sup.5C] 530 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iT 1083 s][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i .sup.5Cs][l.sup.5C][lA][lAs][lTs][i.sup.5 Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs][lAs][lAs][lTs][l.sup.5C] 531 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iTs] 1084 [i.sup.5Cs][iTs][iAs][i.sup.5C][l.sup.5C] [l.sup.5C][lAs][lAs][lTs][i.sup.5Cs] [iTs][iGs][iTs][iTs][i.sup.5Cs] [lAs][lAs][lTs][l.sup.5C] 532 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iTs] 1085 [i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5C ][l.sup.5C][lA][lAs][lTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs ][lAs][lAs][lTs][l.sup.5C] 533 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1086 [i.sup.5Cs][iTs][iAs][l.sup.5Cs][i.sup.5C ][l.sup.5C][lAs][lAs][lTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs ][lAs][lAs][lTs][l.sup.5C] 534 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iTs] 1087 [l.sup.5Cs][iTs][iAs][l.sup.5Cs][i.sup.5C ][l.sup.5C][lAs][lAs][lTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][l.sup.5Cs ][iAs][iAs][lTs][l.sup.5C] 535 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][lAs][iTs] 1088 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][l.sup.5C ][i.sup.5C][lAs][lAs][iTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs ][lAs][lAs][lTs][l.sup.5C] 536 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][iAs][iTs 1089 ][i.sup.5Cs][iTs][iAs][i.sup.5Cs][l.sup.5 C][l.sup.5C][lAs][lAs][lTs][i.sup.5C s][iTs][iGs][iTs][iTs][l.sup.5C s][iAs][lAs][lTs][l.sup.5C] 537 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][lAs][iAs][iTs] 1090 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C ][i.sup.5C][lAs][lAs][iTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][i.sup.5Cs ][iAs][lAs][lTs][l.sup.5C] 538 CGAATCTACCCAATCTGTTCAATC 36 [l.sup.5Cs][lGs][iAs][lAs][lTs] 1091 [i.sup.5Cs][iTs][iAs][l.sup.5Cs][i.sup.5C ][l.sup.5C][lAs][lAs][lTs][i.sup.5Cs ][iTs][iGs][iTs][iTs][l.sup.5Cs ][lAs]

[lAs][lTs][l.sup.5C] 539 AGCCACGAATCTGTTCAATC 402 [lAs][lGs][l.sup.5Cs][l.sup.5Cs][iAs 1092 ][i.sup.5Cs][iGs][iAs][iAs][iTs ][l.sup.5Cs][iTs][iGs][iTs][iTs ][i.sup.5Cs][iAs][iAs][lTs][l.sup.5C] 540 CCACCAACTCATTCATCTGT 403 [l.sup.5Cs][l.sup.5Cs][iAs][i.sup.5Cs][i.sup.5 1093 Cs][iAs][iAs][i.sup.5Cs][iTs][i .sup.5Cs][iAs][iTs][iTs][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][iTs][lGs][ lT] 541 CGAATCTACCCACATCTGTTCAATCA 2 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1094 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 542 CGAATCTACCCACATCTGTTCAATCA 2 [l.sup.5Cs][iGs][lAs][lAs][iTs 1095 ][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5 Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][iTs][iGs][iTs][ iTs][i.sup.5Cs][iA][iA][iT][i.sup.5C s][iA] 543 CGAATCTACCCACATCTGTTCAATCA 2 [l.sup.5Cs][iGs][iAs][iAs][lTs 1096 ][l.sup.5Cs][iTs][iAs][l.sup.5Cs][i.sup.5 Cs][l.sup.5Cs][iAs][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][iTs][iGs][iTs][ iTs][i.sup.5Cs][iA][iA][iT][i.sup.5C s][iA] 544 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1097 [i.sup.5Cs][lTs][lAs][i.sup.5Cs][i.sup.5 Cs][l.sup.5Cs][iAs][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][iTs][iGs][iTs][ iTs][i.sup.5Cs][iA][iA][iT][i.sup.5C s][iA] 545 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1098 [i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 546 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1099 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][l.sup.5Cs][lAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 547 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1100 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][l.sup.5Cs][lAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 548 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1101 [i.sup.5Cs][iTs][iAs]i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][l Ts][l.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 549 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1102 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][l.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][lTs][lGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 550 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1103 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][lTs][l Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 551 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1104 [i.sup.5Cs][iTs][iAs][l.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][lA][iT][i.sup.5Cs ][iA] 552 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1105 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][l.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][lA][lT][i.sup.5Cs ][iA] 553 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1106 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][l.sup.5Cs ][lA] 554 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][lAs][lAs][iTs] 1107 [i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i Ts][i.sup.5Cs][iA][lA][lT][i.sup.5Cs ][iA] 555 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][lTs 1108 ][l.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5 Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][iTs][iGs][iTs][ iTs][l.sup.5Cs][lA][iA][iT][i.sup.5C s][iA] 556 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1109 [i.sup.5Cs][iTs][iAs][l.sup.5Cs][l.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][lTs][lGs][iTs][i Ts][i.sup.5Cs][iA][iA][iT][i.sup.5Cs ][iA] 557 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iA][iTs][i.sup.5 1110 Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][ i.sup.5C][iA][i.sup.5C][iAs][iTs][i.sup.5 Cs][iTs][iGs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5Cs][i A] 558 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iA][iTs][i.sup.5 1111 Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][ i.sup.5Cs][iAs][i.sup.5Cs][iAs][iT][ i.sup.5C][iT][iGs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5Cs][i A] 559 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA]

[iAs][iTs][i 1112 .sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs] [i.sup.5C][iA][i.sup.5Cs][iAs][iTs][ i.sup.5Cs][iTs][iGs][iTs][iTs][ i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs] [iA] 560 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iG][iA][iAs][iTs][i 1113 .sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs][iAs][i.sup.5Cs][iAs][lT] [i.sup.5C][iTs][iGs][iTs][iTs][ i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs] [iA] 561 CGAATCTACCCACATCTGTTCAATCA 404 [x][dt][dc][da][i.sup.5Cs][iGs] 1114 [iAs][iAs][iTs][i.sup.5Cs][iTs] [iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iA s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i Ts][iGs][iTs][iTs][i.sup.5Cs][i A][iA][iT][i.sup.5Cs][iA] 562 CGAATCTACCCACATCTGTTCAATCA 2 [nma.sup.5Cs][nmaGs][nmaAs][nma 1115 As][nmaTs][nma.sup.5Cs][nmaTs][ nmaAs][nma.sup.5Cs][nma.sup.5Cs][nma .sup.5Cs][nmaAs][nma.sup.5Cs][nmaAs] [nmaTs][nma.sup.5Cs][nmaTs][nm aGs][nmaTs][nmaTs][nma.sup.5C s][nmaAs][nmaAs][nmaTs][nm a.sup.5Cs][nmaA] 563 ACGAATCTACCCACATCTGTTCAATC 93 [nmaAs][nma.sup.5Cs][nmaGs][nma 1116 As][nmaAs][nmaTs][nma.sup.5Cs][ nmaTs][nmaAs][nma.sup.5Cs][nm a.sup.5Cs][nma.sup.5Cs][nmaAs][nma.sup.5C s][nmaAs][nmaTs][nma.sup.5Cs][n maTs][nmaGs][nmaTs][nmaTs] [nma.sup.5Cs][nmaAs][nmaAs][nma Ts][nma.sup.5C] 564 CGAATCTACCCACATCTGTTCAATCA 2 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1117 [l.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][iTs][iGs][iTs][i T][i.sup.5Cs][iAs][iAs][iTs][i.sup.5 Cs][iA] 565 ACGAATCTACCCACATCTGTTCAATC 93 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1118 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i As][iTs][i.sup.5Cs][iTs][iGs][i Ts][iT][i.sup.5Cs][iAs][iAs][iT s][i.sup.5C] 566 CCATCCATCATCTGTTCAAT 405 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1119 s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i As][iTs][i.sup.5Cs][iTs][iGs][i Ts][iTs][i.sup.5Cs][iAs][iAs] [lT] 567 CCATCCATGTACTGTTCAAT 406 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1120 s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5Cs][iTs][iGs][iT s][iTs][i.sup.5Cs][iAs][iAs] [lT] 568 CCATCCATGTACTTTCAATC 407 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1121 s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][l.sup.5Cs][iTs][iTs][iT s][i.sup.5Cs][iAs][iAs][iTs] [l.sup.5C] 569 CCATCCATGTACTTCAATCA 408 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1122 s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][l.sup.5Cs][iTs][iTs][i.sup.5 Cs][iAs][iAs][iTs][i.sup.5Cs][l A] 570 ACGAATCTACCCAATCCTTTTATCTA 409 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1123 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iAs][iT s][i.sup.5Cs][i.sup.5Cs][iTs][iTs][i Ts][iTs][iAs][iTs][i.sup.5Cs][i Ts][iA] 571 CGAATCTACCCACATCCTTTTATCTA 410 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1124 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iTs][iTs][ iTs][iTs][iAs][iTs][i.sup.5Cs][ iTs][iA] 572 CACGAATCTACCCCTTTTATCTACTC 411 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1125 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ iTs][iTs][iTs][iTs][iAs][i Ts][i.sup.5Cs][iTs][iAs][i.sup.5Cs][ iTs][i.sup.5C] 573 ACGAATCTACCCATCCTTTTATCTAC 412 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1126 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][i.sup.5Cs][iTs][iTs][iTs][i Ts][iAs][iTs][i.sup.5Cs][iTs][i As][i.sup.5C] 574 ACGAATCTACCCACTTTTATCTACTC 413 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1127 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][l.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i Ts][iTs][iTs][iTs][iAs][iT s][l.sup.5Cs][iTs][iAs][i.sup.5Cs][i Ts][i.sup.5C] 575 CACGAATCTACCCATCCTTTTATCTA 414 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1128 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iTs][iTs][ iTs][iTs][iAs][iTs][i.sup.5Cs][ iTs][iA] 576 CCATCCATGTACTTTTTATC 415 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1129 s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5Cs][iTs][iTs][iT s][iTs][iTs][iAs][iTs] [l.sup.5C] 577 CCATCCATGTACTCTTTTAT 416 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1130 s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5Cs][iTs][i.sup.5Cs][i Ts][iTs][iTs][iTs][iAs][l T] 578 CCATCCATGTACTTGTACTC 417 [l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1131 s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5Cs][iTs][iTs][iG s][iTs][iAs][i.sup.5Cs][iTs][l.sup.5 C] 579 CGAATCTACCCACCATCCATGTACTC 418 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1132 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][l.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][i.sup.5Cs]

[iAs] [iTs][iGs][iTs][iAs][i.sup.5Cs] [iTs][i.sup.5C] 580 CGAATCTACCCACATCCATGTACTCA 419 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1133 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs] [iAs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iTs][ i.sup.5Cs][iA] 581 GCCACGAATCTACTCCATGTACTCAC 420 [iGs][i.sup.5Cs][i.sup.5Cs][iAs] [i.sup.5C 1134 s][iGs][iAs][iAs][iTs][i.sup.5C s][iTs][iAs][i.sup.5Cs][iTs][i.sup.5 Cs][i.sup.5Cs] [iAs][iTs][iGs][i Ts][iAs][i.sup.5Cs][iTs][i.sup.5Cs][ iAs][i.sup.5C] 582 ACGAATCTACCCACATCCATGTACTC 421 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1135 [iTs] [i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i As][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][ iTs][i.sup.5C] 583 ACGAATCTACCCAATCCATGTACTCA 422 [iAs][il.sup.5Cs][iGs][iAs][iAs] 1136 [iTs] [i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iAs][iT s][i.sup.5Cs][i.sup.5Cs][iAs] [iTs][i Gs][iTs][iAs][i.sup.5Cs][iTs][i .sup.5Cs][iA] 584 CGAATCTACCCACTCCATGTACTCAC 423 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1137 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs] [iAs][i.sup.5Cs][iTs][i .sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][ iTs][iAs][i.sup.5Cs][iTs][i.sup.5Cs] [iAs][i.sup.5C] 585 CACGAATCTACCCCATCCATGTACTC 3 [i.sup.5Cs][iAs][i.sup.5Cs][iGs] [iAs 1138 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iAs] [iT][iG][iT][iA][i.sup.5C][iTs] [i.sup.5C] 586 ACGAATCTACCCATCCATGTACTC 424 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1139 [iTs][i.sup.5Cs] [iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][i.sup.5Cs][iAs][iTs][iGs][i Ts] [iAs][i.sup.5Cs][iTs][i.sup.5C] 587 ACGAATCCATCCATGTACTC 425 [lAs][l.sup.5Cs][iGs][iAs] [iAs] 1140 [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iG s][iTs][iAs] [i.sup.5Cs][lTs][l.sup.5 C] 588 ACGAATCATCCATGTACTCA 426 [lAs][l.sup.5Cs][iGs][iAs][iAs] 1141 [iTs][i.sup.5Cs][iAs][iTs][l.sup.5Cs ][i.sup.5Cs][iAs][iTs][iGs][iTs ][iAs][i.sup.5Cs][iTs] [l.sup.5Cs][lA ] 589 CGAATCTCATCCATGTACTC 427 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1142 [i.sup.5Cs][iTs][i.sup.5Cs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iG s][iTs][iAs][i.sup.5Cs][lTs] [l.sup.5 C] 590 CGAATCTATCCATGTACTCA 428 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1143 [i.sup.5Cs][iTs][iAs][iTs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][iGs][iTs ][iAs][i.sup.5Cs][iTs][l.sup.5Cs] [l A] 591 GAATCTACATCCATGTACTC 429 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1144 [iTs][iAs] [i.sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs ][iTs][iAs][i.sup.5Cs][lTs][l.sup.5 C] 592 AATCTACCATCCATGTACTC 430 [lAs][lAs][iTs][i.sup.5Cs][iTs] 1145 [iAs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iG s][iTs][iAs][i.sup.5Cs][lTs][l.sup.5 C] 593 ACGAATCCATCCATGTACTC 425 [lAs][l.sup.5C][iG][iAs][iAs][i 1146 Ts][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][ i.sup.5Cs][i.sup.5Cs][iAs][iTs][iGs] [iTs][iAs][i.sup.5Cs][lTs][l.sup.5 C] 594 ACGAATCCATCCATGTACTC 425 [lAs][l.sup.5Cs][iGs][iAs][iAs] 1147 [iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iG s][iTs][iA][i.sup.5C][lTs][l.sup.5C] 595 GAATCTACCCATCTCTCCAT 431 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1148 [iTs][iAs][i.sup.5Cs] [i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][iTs][i.sup.5 Cs][iTs][i.sup.5Cs][l.sup.5Cs][lAs][ lT] 596 TCCATCATCCATGTACTCAC 432 [lTs][l.sup.5Cs][i.sup.5Cs][iAs][iTs 1149 ][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][i.sup.5Cs][iTs][i.sup.5Cs][lAs][l .sup.5C] 597 GAATCTACCCATCCATGTAC 433 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1150 [iTs][iAs][i.sup.5Cs] [i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][lAs][l.sup.5 C] 598 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][iG][iA][iAs][iTs][i 1151 .sup.5Cs][iTs][iAs] [i.sup.5Cs][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][iAs][i.sup.5C] 599 CGAATCTACCATCCATGTAC 82 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1152 [i.sup.5Cs][iTs][iA] [i.sup.5C][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][iAs][i.sup.5C] 600 CGAATCTACCATCCATGTAC 82 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1153 [i.sup.5Cs][iTs][iAs] [i.sup.5Cs][i.sup.5C s][iAs][iT][i.sup.5C][i.sup.5Cs][iAs ][iTs][iGs][iTs][iAs][i.sup.5C] 601 CGAATCTACCATCCATGTAC 82 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1154 [i.sup.5Cs][iTs][iAs] [i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iG][iT][iAs][i.sup.5C] 602 CCATCCATGTACTTCACCCA 434 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1155 s][i.sup.5Cs]

[iAs][iTs][iGs][iT s][iAs][i.sup.5Cs][iTs][iTs][i.sup.5 Cs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs] [lA] 603 CAATCATCCATCCATGTACT 435 [l.sup.5Cs][lAs][iAs][iTs][i.sup.5Cs 1156 ][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][l.sup.5Cs] [lT] 604 CAATCATCCATCCATGTACT 435 [l.sup.5Cs][lA][iA][iTs][i.sup.5Cs][ 1157 iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs] [iTs][l.sup.5Cs][l.sup.5Cs][iAs][iTs ][iGs][iTs][iAs][l.sup.5Cs][lT] 605 CAATCATCCATCCATGTACT 435 [l.sup.5Cs][lAs][iAs][iTs][i.sup.5Cs 1158 ][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iT][iA][l.sup.5Cs][lT] 606 GTTCAATCCATCCATGTACT 436 [lGs][lTs][iTs][i.sup.5Cs][iAs] 1159 [iAs][iTs][i.sup.5Cs] [l.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][l.sup.5Cs] [lT] 607 TTCAATCCCATCCATGTACT 437 [lTs][lTs][i.sup.5Cs][iAs][iAs] 1160 [iTs][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][l.sup.5Cs][l T] 608 TCAATCACCATCCATGTACT 438 [lTs][l.sup.5Cs][iAs][iAs][iTs] 1161 [i.sup.5Cs][iAs] [i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][l.sup.5Cs][l T] 609 TTTTATCCCATCCATGTACT 439 [lTs][lTs][iTs][iTs][iAs][ 1162 iTs][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][l.sup.5Cs][l T] 610 CTTTTATCCATCCATGTACT 440 [l.sup.5Cs][lTs][iTs][iTs][iTs] 1163 [iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][l.sup.5Cs][lT ] 611 TCACCCACCATCCATGTACT 441 [lTs][l.sup.5Cs][iAs][i.sup.5Cs][i.sup.5C 1164 s][i.sup.5Cs] [iAs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs] [iTs][iGs][iTs][iAs][l.sup.5Cs ][lT] 612 CCATCCATGTACTCCATCCATGTACT 442 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1165 s] [i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5Cs][iTs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ i.sup.5Cs][iT] 613 CCATCCATGTACT[n3]CCATCCATG 246/246 [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C 1166/1167 TACT s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5Cs][iTs][n3s][i.sup.5 Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs][iGs] [iTs][ iAs][i.sup.5Cs][iT] 614 CCATCCATCATCC[n6]GGGATAAGA 198/253 [i.sup.5Cs] [i.sup.5Cs][iAs][iTs][i.sup.5C 1168/1169 GTTC s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i As][iTs] [i.sup.5Cs][i.sup.5Cs][n6s] [iGs][iGs][iGs][iAs][iTs][ iAs][iAs][iGs][iAs][iGs][i Ts][iTs][i.sup.5C] 615 GCCACGAATCTACTCCATCCATGTAC 445 [iGs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5C 1170 s][iGs][iAs][iAs][iTs][i.sup.5C s][iTs][iAs][i.sup.5Cs][iTs][i.sup.5 Cs][i.sup.5Cs][iAs][iTs] [i.sup.5Cs][ i.sup.5Cs][iAs][iTs][iGs][iTs][ iAs][iC] 616 GCCACGAATCTACCCATCCATGTACT 446 [iGs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5C 1171 s][iGs][iAs][iAs][iTs][i.sup.5C s][iTs][iAs] [i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ i.sup.5Cs][iT] 617 CACGAATCTACCCTCCATCCATGTAC 447 [i.sup.5Cs][iAs][i.sup.5Cs][iGs] [iAs 1172 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iTs][i .sup.5Cs] [i.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][iTs][iGs][iTs] [iAs][i.sup.5C] 618 ACGAATCTACCCACCATCCATGTACT 448 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1173 [iTs] [i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][i .sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ i.sup.5Cs][iT] 619 CCACGAATCTACCCCATCCATGTACT 449 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1174 s] [iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs ][iAs][iTs][iGs][iTs][iAs] [i.sup.5Cs][iT] 620 CACGAATCTACCCCCATCCATGTACT 1 [l.sup.5Cs][lAs][i.sup.5Cs][iGs][iAs 1175 ][iAs][iTs] [i.sup.5Cs][iTs][iAs ][i.sup.5Cs][l.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs][i.sup.5Cs] [i.sup.5Cs ][iAs][iTs][iGs][iTs][iAs] [l.sup.5Cs][lT] 621 CACGAATCTACCCCCATCCATGTACT 1 [i.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1176 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs] [i.sup.5Cs][i.sup.5Cs][ l.sup.5C][lA][iTs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][iGs][iTs][iAs][i .sup.5Cs] [iT] 622 ACGAATCTACCCCCATCCATGTAC 37 [iAs][i.sup.5C][iG][iAs][iAs][i 1177 Ts] [i.sup.5Cs][iTs][iAs][i.sup.5Cs][ i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs] [i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][i.sup.5C] 623 ACGAATCTACCCCCATCCATGTAC 37 [iAs] [i.sup.5Cs][iGs][iAs][iAs] 1178 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5C] [i.sup.5C][i.sup.5Cs][i.sup.5Cs]

[i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][i.sup.5C] 624 ACGAATCTACCCCCATCCATGTAC 37 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1179 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5C][i A][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][i.sup.5C] 625 ACGAATCTACCCCCATCCATGTAC 37 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1180 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs] [iTs][iG][iT][iAs][i.sup.5C] 626 ACGAATCTACCCCCATCCATGTAC 37 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1181 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][i.sup.5Cs][ 627 ACGAATCTATCCATCCATGTAC 450 [iAs][i.sup.5Cs][iGs][iAs][iAs] 1182 [iTs][i.sup.5Cs][iTs][iAs][iTs] [i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5C s][i.sup.5Cs][iAs][iTs][iGs][iT s][iAs][i.sup.5C] 628 CGAATCTACCCACCCATCCATGTACT 451 [i.sup.5Cs][iGs][iAs][iAs][iTs] 1183 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][i.sup.5Cs][i.sup.5Cs][ i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][iGs][iTs][iAs] [i.sup.5Cs][iT] 629 ACGAATCCCATCCATGTACT 452 [lAs][l.sup.5Cs][iGs][iAs][iAs 1184 ][iTs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][l.sup.5Cs][ lT] 630 GCCACGACCATCCATGTACT 453 [lGs][l.sup.5Cs][i.sup.5Cs][iAs][i.sup.5C 1185 s][iGs][iAs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][l.sup.5Cs][ lT] 631 CGAATCTCCATCCATGTACT 454 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1186 [i.sup.5Cs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][l.sup.5Cs][l T] 632 GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1187 [iTs][iAs][i.sup.5Cs][l.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][l.sup.5Cs][l T] 633 AATCTACCCATCCATGTACT 456 [lAs][lAs][iTs][i.sup.5Cs][iTs] 1188 [iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][l.sup.5Cs][l T] 634 ACGAATCTCCATCCATGTAC 457 [lAs][l.sup.5Cs][iGs][iAs][iAs] 1189 [iTs][i.sup.5Cs][iTs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][lAs][l .sup.5C] 635 CGAATCTTCCATCCATGTAC 458 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1190 [i.sup.5Cs][iTs][iTs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][lAs][l .sup.5C] 636 GAATCTATCCATCCATGTAC 459 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1191 [iTs][iAs][iTs][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][lAs][l.sup.5C ] 637 GAATCTACCATCCATGTACT 455 [lGs][lA][iA][iTs][i.sup.5Cs][i 1192 Ts][iAs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][l.sup.5Cs][lT] 638 GAATCTACCATCCATGTACT 455 [lGs][lA][iA][iT][i.sup.5Cs][iT 1193 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][l.sup.5Cs][lT] 639 GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1194 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iT][iA][l.sup.5Cs][lT] 640 GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1195 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iG][iT][iA][l.sup.5Cs][lT] 641 GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1196 [lT][lA][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][l.sup.5Cs][lT] 642 GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1197 [iTs][iAs][l.sup.5C][l.sup.5C][iAs] [iTs][i.sup.5Cs][l.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][l.sup.5Cs] [lT] 643 GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1198 [iTs][lA][l.sup.5C][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][l.sup.5Cs][lT] 644 GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1199 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][l.sup.5C][l.sup.5C][iAs][iTs] [iGs][iTs][iAs][l.sup.5Cs][lT] 645 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lG][iA][iAs][iTs][i 1200 .sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs] [lAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][lAs][l.sup.5C ] 646 CGAATCTACCATCCATGTAC 82 [l.sup.5Cs][lG][iA][iA][iTs][i.sup.5 1201 Cs][iTs][iAs][i.sup.5Cs][l.sup.5Cs][ lAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs] [iTs][iGs][iTs][lAs][l.sup.5C] 647 ACGAATCTCCATCCATGTAC 457 [lAs][l.sup.5C][iG][iAs][iAs][i 1202 Ts][i.sup.5Cs][iTs]

[i.sup.5Cs][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][lAs][l.sup.5C] 648
ACGAATCTCCATCCATGTAC 457 [lAs][l.sup.5Cs][iGs][iAs][iAs] 1203 [iTs][i.sup.5Cs][iTs]
[i.sup.5Cs][l.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iG][lT][lAs][l.sup.5C] 649
ACGAATCCCATCCATGTACT 452 [lAs][l.sup.5C][iG][iAs][iAs][i 1204 Ts][i.sup.5Cs][i.sup.5Cs]
[i.sup.5Cs][iAs] [iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs ][iGs][iTs][iAs][l.sup.5Cs][lT] 650
CGAATCTCCATCCATGTACT 454 [l.sup.5Cs][G][iA][iAs][iTs][i.sup.5 1205 Cs][iTs][i.sup.5Cs]
[i.sup.5Cs][iAs][ iTs][l.sup.5Cs][l.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][l.sup.5Cs][lT] 651
CGAATCTCCATCCATGTACT 454 [l.sup.5Cs][lGs][iAs][iAs][iTs] 1206 [i.sup.5Cs][iTs]
[i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iT][iA][l.sup.5Cs][lT] 652
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1207 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [iT] 653
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1208 [iTs][lAs][l.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [lT] 654
GAATCTACCATCCATGTACT 455 [lGs][iA][iA][iTs][i.sup.5Cs][i 1209 Ts][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][i.sup.5Cs][lT] 655
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1210 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][lT][iA][i.sup.5Cs][lT] 656
GAATCTACCATCCATGTACT 455 [lGs][iA][iA][iTs][i.sup.5Cs][i 1211 Ts][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iT][iA][i.sup.5Cs][lT] **657**
GAATCTACCATCCATGTACT **455 [**lGs**][**iA**][**iA**][**iTs**][**i5Cs**][**i **1212** Ts][iA][i.sup.5C][i.sup.5Cs]
[iAs][iT s][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i Gs][iTs][iAs][i.sup.5Cs][lT] 658
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1213 [iTs][iA][i.sup.5C]
[i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iT][iA][i.sup.5Cs][lT] 659
GAATCTACCATCCATGTAC 460 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1214 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][lAs][l.sup.5C] 660
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1215 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][lT ] 661
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1216 [iTs][iAs][l.sup.5Cs]
[l.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][iCs] [lT] 662
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1217 [iTs][lAs][i.sup.5Cs]
[i.sup.5Cs][lAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [lT] 663
GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1218 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][l.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 664
CGAATCTACCATCCATGTACT 462 [l.sup.5Cs][iGs][iAs][iAs][iTs] 1219 [i.sup.5Cs][iTs][iAs]
[l.sup.5Cs][l.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 665
ACGAATCTACCATCCATGTACT 463 [lAs][i.sup.5Cs][iGs][iAs][iAs] 1220 [iTs][i.sup.5Cs][iTs]
[iAs][i.sup.5Cs ][l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][i.sup.5Cs][lT]
666 CACGAATCTACCATCCATGTACT 464 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1221 ][iAs][iTs]
[i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][i.sup.5Cs][iAs][iTs][iGs][i Ts]
[iAs][i.sup.5Cs][lT] 667 CCACGAATCTACCATCCATGTACT 465 [l.sup.5Cs][i.sup.5Cs][iAs]
[i.sup.5Cs][iG 1222 s][iAs][iAs][iTs][i.sup.5Cs][iT s][iAs][l.sup.5Cs][i.sup.5Cs][iAs][i Ts]
[i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 668
CCACGAATCTACCATCCATGTACT 465 [i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iG 1223 s][iAs]
[iAs][iTs][i.sup.5Cs][iT s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][
iGs][iTs][iAs][i.sup.5Cs][iT] 669 GAATCTACCATCCATGTACT 455 [lGs][lA][iA][iT][i.sup.5Cs]
[iT 1224 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs]
[i.sup.5Cs][iT] 670 GAATCTACCATCCATGTACT 455 [iGs][iA][lA][lT][l.sup.5Cs][iT 1225 s]
[iAs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][l.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT]
671 GAATCTACCATCCATGTACT 455 [iGs][iA][iA][lT][l.sup.5Cs][iT 1226 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 672

GAATCTACCATCCATGTACT 455 [iGs][iA][iA][iT][i.sup.5Cs][iT 1227 s][lAs][l.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 673
GAATCTACCATCCATGTACT 455 [iGs][iA][iA][iT][i.sup.5Cs][iT 1228 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 674
GAATCTACCATCCATGTACT 455 [iGs][iA][iA][iT][i.sup.5Cs][iT 1229 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 675
GAATCTACCATCCATGTACT 455 [iGs][iA][iA][iT][i.sup.5Cs][iT 1230 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 676
GAATCTACCATCCATGTACT 455 [iGs][iA][iA][iT][i.sup.5Cs][iT 1231 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][lTs][ lGs][iTs][iAs][i.sup.5Cs][iT] 677
GAATCTACCATCCATGTACT 455 [iGs][iA][iA][iT][i.sup.5Cs][iT 1232 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][lTs][lAs][i.sup.5Cs][lT] 678
GAATCTACCATCCATGTACT 455 [iGs][iA][iA][iT][i.sup.5Cs][iT 1233 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][l.sup.5Cs][lT] 679
GAATCTACCATCCATGTACT 455 [iGs][iA][lA][iT][i.sup.5Cs][iT 1234 s][lAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][lAs][iTs][ iGs][iTs][lAs][i.sup.5Cs][iT] 680
GAATCTACCATCCATGTACT 455 [lGs][iA][iA][lT][i.sup.5Cs][lT 1235 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][lAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 681
GAATCTACCATCCATGTACT 455 [iGs][iA][lA][lT][i.sup.5Cs][iT 1236 s][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][lTs][lAs][i.sup.5Cs][iT] 682
GAATCTACCATCCATGTACT 455 [iGs][iA][iA][lT][i.sup.5Cs][iT 1237 s][lAs][l.sup.5Cs]
[i.sup.5Cs][iAs][i Ts][i.sup.5Cs][l.sup.5Cs][lAs][iTs][ iGs][iTs][iAs][i.sup.5Cs][iT] 683
GAATCTACCATCCATGTACT 455 [lGs][lAs][iAs][iTs][i.sup.5Cs] 1238 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][i T] 684
GAATCTACCATCCATGTACT 455 [iGs][iAs][lAs][lTs][i.sup.5Cs] 1239 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][i T] 685
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][l.sup.5Cs] 1240 [lTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][i.sup.5Cs][i T] 686
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1241 [iTs][lAs][l.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][i T] 687
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1242 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][lAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][i T] 688
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1243 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][lTs][l.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][i T] 689
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1244 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][l.sup.5Cs][lAs][iT s][iGs][iTs][iAs][i.sup.5Cs][iT ] 690
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1245 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][lT s][lGs][iTs][iAs][i.sup.5Cs][i T] 691
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1246 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][lTs][lAs][i.sup.5Cs][l T] 692
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1247 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][l.sup.5Cs][l T] 693
GAATCTACCATCCATGTACT 455 [iGs][iAs][lAs][iTs][i.sup.5Cs] 1248 [iTs][lAs][i.sup.5Cs]
[i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][lAs][i Ts][iGs][iTs][lAs][i.sup.5Cs][i T] 694
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1249 [lTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][l T] 695
GAATCTACCATCCATGTACT 455 [iGs][iAs][lAs][lTs][i.sup.5Cs] 1250 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][lTs][lAs][i.sup.5Cs][iT ] 696
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1251 [iTs][lAs][l.sup.5Cs]
[i.sup.5Cs][iA s][iTs][i.sup.5Cs][l.sup.5Cs][lAs][i Ts][iGs][iTs][iAs][i.sup.5Cs][i T] 697

GAATCTACCATCCATGTACT 455 [lGs][lA][iA][iT][i.sup.5Cs][iT 1252 s][iAs][i.sup.5Cs] [i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iG][iT][iA][l.sup.5Cs][lT] [lGs][lA][iA][iT] [i.sup.5Cs][iT 698 GAATCTACCATCCATGTACT 455 s][iAs][i.sup.5Cs][i.sup.5Cs][iAs][i As][iTs] [i.sup.5Cs][i.sup.5Cs][iAs] 1253 [iTs][iGs][iT][iA][l.sup.5Cs][l T] 699 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iT][i.sup.5C][i 1254 Ts][iAs][i.sup.5Cs] [i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][i.sup.5Cs][lT] 700 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1255 [iT][iA][i.sup.5Cs] [i.sup.5Cs][iAs][ iTs][i.sup.5Cs][l.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][i.sup.5Cs][lT] 701 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1256 [iTs][iAs][i.sup.5C] [i.sup.5C][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][i.sup.5Cs][lT] 702 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1257 [iTs][iAs][i.sup.5Cs] [i.sup.5Cs][iA] [iT][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][i.sup.5Cs][lT] 703 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1258 [iTs][iAs][i.sup.5Cs] [i.sup.5Cs][iAs ][iTs][i.sup.5C][i.sup.5C][iAs][iTs] [iGs][iTs][iAs][i.sup.5Cs][lT] 704 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1259 [iTs][iAs][i.sup.5Cs] [i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iA][iT] [iGs][iTs][iAs][i.sup.5Cs][lT] 705 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1260 [iTs][iAs][i.sup.5Cs] [i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iG][iT][iAs][i.sup.5Cs][lT] 706 GAATCTACCCATCCATGTACT 461 [lGs][iA][iA][iTs][i.sup.5Cs][i 1261 Ts][iAs][i.sup.5Cs] [i.sup.5Cs][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][iAs][i.sup.5Cs ][lT] 707 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iT][i.sup.5C][i 1262 Ts][iAs][i.sup.5Cs] [i.sup.5Cs][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][iAs][i.sup.5Cs ][lT] 708 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1263 [iT][iA][i.sup.5Cs] [i.sup.5Cs][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][iAs][i.sup.5Cs ][lT] 709 GAATCTACCCATCCATGTACT 461 i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs 1264 ][iAs][iTs] [iGs][iTs][iAs] [i.sup.5Cs][lT] 710 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs] [i.sup.5Cs] 1265 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C ][iA][iTs][i.sup.5Cs][i.sup.5Cs][iAs ] [iTs][iGs][iTs][iAs][i.sup.5Cs ][lT] 711 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs] [i.sup.5Cs] 1266 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iT][i.sup.5C][i.sup.5Cs][iAs ] [iTs][iGs][iTs][iAs][i.sup.5Cs ][lT] 712 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs] [i.sup.5Cs] 1267 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5C][iA ][iTs] [iGs][iTs][iAs][l.sup.5Cs ][lT] 713 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs] [i.sup.5Cs] 1268 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As] [iT][iG][iTs][iAs][i.sup.5Cs ][lT] 714 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs] [i.sup.5Cs] 1269 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As] [iTs][iGs][iT][iAs][i.sup.5C s][lT] 715 GAATCTACCCATCCATGTACT 461 [lGs][lAs][iAs][iTs] [i.sup.5Cs] 1270 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As] [iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 716 GAATCTACCCATCCATGTACT 461 [iGs][iAs][lAs][lTs] [i.sup.5Cs] 1271 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As] [iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] **717** GAATCTACCCATCCATGTACT **461 [**iGs**][**iAs**][**iAs**][**iTs**]** **[**l**5**Cs**] 1272** [lTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs] [iGs][lTs][iAs][i.sup.5 Cs][iT] 718 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs] [i.sup.5Cs] 1273 [iTs][lAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As] [iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 719 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs] [i.sup.5Cs] 1274 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As] [iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 720 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs] [i.sup.5Cs] 1275 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][lAs][lTs][i.sup.5Cs][i.sup.5Cs][i As] [iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 721 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs] [i.sup.5Cs] 1276 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][l.sup.5Cs][l.sup.5Cs][i As] [iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 722 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs]

[i.sup.5Cs] 1277 [iTs][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][l As][lTs][iGs][lTs][iAs][l .sup.5Cs][lT] 723 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1278 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][lGs][lTs][iAs][i.sup.5 Cs][iT] 724 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1279 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][lAs][l.sup.5 Cs][lT] 725 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1280 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][l .sup.5Cs][lT] [iGs][iAs][iAs][iTs][i.sup.5Cs] 1281 726 GAATCTACCCATCCATGTACT 461 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C ][l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][l.sup.5Cs][iT] 727 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][lTs][i.sup.5Cs] 1282 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][l.sup.5C s][iAs][iTs][l.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][lTs][iAs][l.sup.5 Cs][lT] 728 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1283 [lTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][lTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 729 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1284 [iTs][iAs][l.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][l.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 730 GAATCTACCATCCATGTACT 455 [iGs][lAs][iAs][iTs][i.sup.5Cs] 1285 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][l.sup.5Cs][i T] 731 GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][lTs][i.sup.5Cs] 1286 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][lTs][iAs][i.sup.5Cs][iT] 732 GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1287 [lTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][lT s][iGs][iTs][iAs][i.sup.5Cs] [iT] 733 GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1288 [iTs][iAs][l.sup.5Cs][l.sup.5Cs][iAs ][iTs][i.sup.5Cs][l.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [iT] 734 GAATCTACCATCCATGTACT 455 [enaGs][iAs][iAs][iTs][i.sup.5C 1289 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][ enaT] 735 GAATCTACCATCCATGTACT 455 [bGs][iAs][iAs][iTs][i.sup.5Cs] 1290 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][b T] 736 GAATCTACCCATCCATGTACT 461 [enaGs][iAs][iAs][iTs][i.sup.5C 1291 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ i.sup.5Cs][enaT] 737 GAATCTACCCATCCATGTACT 461 [scpGs][iAs][iAs][iTs][i.sup.5C 1292 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][i.sup.5Cs][scpT] 738 GAATCTACCCATCCATGTACT 461 [bGs][iAs][iAs][iTs][i.sup.5Cs] 1293 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][bT] 739 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1294 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][l.sup.5Cs][i T] 740 GAATCTACCATCCATGTACT 455 [iGs][zAs][iAs][iTs][i.sup.5Cs] 1295 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][z.sup.5Cs][i T] 741 GAATCTACCATCCATGTACT 455 [enaGs][iAs][iAs][iTs][i.sup.5C 1296 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][ena.sup.5Cs ][iT] 742 GAATCTACCATCCATGTACT 455 [iGs][enaAs][iAs][iTs][i.sup.5C 1297 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][ena.sup.5Cs ][iT] 743 GAATCTACCATCCATGTACT 455 [iGs][enaAs][iAs][iTs][i.sup.5C 1298 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][ enaT] 744 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1299 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 745 GAATCTACCCATCCATGTACT 461 [iGs][lAs][iAs][iTs][i.sup.5Cs] 1300 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 746 GAATCTACCCATCCATGTACT 461 [iGs][zAs][iAs][iTs][l.sup.5Cs] 1301 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][l.sup.5Cs][i As][iTs][iGs][iTs][iAs][z.sup.5 Cs][lT] 747

GAATCTACCCATCCATGTACT 461 [iGs][scpAs][iAs][iTs][i.sup.5C 1302 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ i.sup.5Cs][scpT] 748 GAATCTACCCATCCATGTACT 461 [enaGs][iAs][iAs][iTs][i.sup.5C 1303 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ ena.sup.5Cs][iT] 749 GAATCTACCCATCCATGTACT 461 [iGs][enaAs][iAs][iTs][i.sup.5C 1304 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ ena.sup.5Cs][iT] 750 GAATCTACCCATCCATGTACT 461 [iGs][enaAs][iAs][iTs][i.sup.5C 1305 s][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][iGs][iTs][iAs][ i.sup.5Cs][enaT] 751 GAATCTACCATCCATGTACT 466 [x][dt][dc][da][lGs][iAs][ 1306 iAs][iTs][i.sup.5Cs][iTs][iAs][ l.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][iGs][iTs ][iAs][i.sup.5Cs][lT] 752 GAATCTACCCATCCATGTACT 467 [x][dt][dc][da][lGs][iAs][ 1307 iAs][iTs][i.sup.5Cs][iTs][iAs][ i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][iTs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][iG s][iTs][iAs][i.sup.5Cs][lT] 753 GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1308 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [iT] 754 GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1309 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][lT] 755 GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1310 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 756 GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1311 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 757 CGAATCTACCCATCCATGTACT 468 [l.sup.5Cs][iGs][iAs][iAs][iTs] 1312 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][iGs][iTs][i As][i.sup.5Cs][lT] 758 GAATCTACCCCATCCATGTACT 469 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1313 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][iGs][iTs][i As][i.sup.5Cs][lT] 759 ACGAATCTACCCATCCATGTACT 470 [lAs][i.sup.5Cs][iGs][iAs][iAs] 1314 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i.sup.5 Cs][i.sup.5Cs][iAs][iTs][iGs][i Ts][iAs][i.sup.5Cs][lT] 760 CGAATCTACCCCATCCATGTACT 471 [l.sup.5Cs][iGs][iAs][iAs][iTs] 1315 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i .sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][ iTs][iAs][i.sup.5Cs][lT] 761 GAATCTACCCCCATCCATGTACT 472 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1316 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][l.sup.5Cs][i.sup.5Cs][iAs][iTs][i .sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][ iTs][iAs][i.sup.5Cs][lT] 762 CACGAATCTACCCATCCATGTACT 115 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1317 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][iC][lT] 763 CGAATCTACCCCCATCCATGTACT 473 [l.sup.5Cs][iGs][iAs][iAs][iTs] 1318 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][i.sup.5Cs][i.sup.5Cs][iAs][iTs] [iGs][iTs][iAs][i.sup.5C][lT] 764 GAATCTACCCACCATCCATGTACT 474 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1319 [iTs][iAs][i.sup.5Cs][l.sup.5Cs][l.sup.5C s][iAs][l.sup.5Cs][i.sup.5Cs][iAs][i Ts][i.sup.5Cs][i.sup.5Cs][iAs][iTs][ iGs][iTs][iAs][i.sup.5C][lT] 765 CGAATCTACATCCATGTACT 475 [l.sup.5Cs][iGs][iAs][iAs][iTs] 1320 [i.sup.5Cs][iTs][iAs][i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [lT] 766 ACGAATCTACATCCATGTAC 476 [lAs][i.sup.5Cs][iGs][iAs][iAs] 1321 [iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][l.sup.5C] 767 GAATCTACCCCATCCATGTA 477 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1322 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5C s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][iGs][iTs][l A] 768 CACGAATCTATCCATGTACT 478 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1323 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][iTs][l.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][lT ] 769 CACGAATCTAATCCATGTAC 479 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1324 ][iAs][iTs][i.sup.5Cs][iTs][iAs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs] [l.sup.5C] 770 CACGAATCTACATCCATGTA 480 [l.sup.5Cs][iAs][i.sup.5Cs][iGs][iAs 1325 ][iAs][iTs]

[i.sup.5Cs][iTs][iAs ][l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][l A] 771
GAATCTACCCCCATCCATGT 481 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1326 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][l.sup.5C s][i.sup.5Cs][i.sup.5Cs][iAs][iTs][i .sup.5Cs][i.sup.5Cs][iAs][iTs][iGs][ lT] 772
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1327 [iTs][lAs][i.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [lT] 773
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1328 [iTs][lAs][i.sup.5Cs]
[l.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [lT] 774
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1329 [iTs][iAs][l.sup.5Cs]
[l.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [iT] 775
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1330 [iTs][iAs][l.sup.5Cs]
[l.sup.5Cs][iAs ][iTs][l.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [iT] 776
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1331 [iTs][iAs][l.sup.5Cs]
[i.sup.5Cs][iAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs] [lT] 777
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1332 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][iAs ][iTs][l.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][iT ] 778
GAATCTACCATCCATGTACT 455 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1333 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][lAs ][iTs][i.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][iT ] 779
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1334 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][lAs ][iTs][l.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][lT ] 780
GAATCTACCATCCATGTACT 455 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1335 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][iAs ][iTs][l.sup.5Cs][i.sup.5Cs][iAs][iT s][iGs][iTs][iAs][i.sup.5Cs][lT ] 781
GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1336 [iTs][lAs][i.sup.5Cs]
[i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 782
GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1337 [iTs][lAs][i.sup.5Cs]
[i.sup.5Cs][i.sup.5 Cs][iAs][iTs][i.sup.5Cs][i.sup.5Cs][ iAs][iTs][iGs][iTs][iAs][i .sup.5Cs][iT] 783
GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1338 [iTs][iAs][l.sup.5Cs]
[l.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 784
GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1339 [iTs][iAs][l.sup.5Cs]
[i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 785
GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1340 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 786
GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1341 [iTs][iAs][i.sup.5Cs]
[l.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][lTs][iAs][i.sup.5 Cs][lT] 787
GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1342 [iTs][iAs][l.sup.5Cs]
[i.sup.5Cs][i.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 788
GAATCTACACATCCATGTACT 482 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1343 [iTs][iAs][i.sup.5Cs]
[lAs][l.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][iT] 789
GAATCTACACATCCATGTACT 482 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1344 [iTs][iAs][i.sup.5Cs]
[lAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][iT] 790
GAATCTACACATCCATGTACT 482 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1345 [iTs][iAs][i.sup.5Cs]
[iAs][l.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][lT] 791
GAATCTACACATCCATGTACT 482 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1346 [iTs][iAs][i.sup.5Cs]
[lAs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][iT] 792
GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1347 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][l.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][iT] 793
GAATCTACCCATCCATGTACT 461 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1348 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][l.sup.5C s][lAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][i.sup.5Cs] [iT] 794
GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1349 [iTs][iAs][i.sup.5Cs]
[i.sup.5Cs][i.sup.5C s][lAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][lTs][iAs][i.sup.5 Cs][lT] 795
GAATCTACCCATCCATGTACT 461 [iGs][iAs][iAs][iTs][i.sup.5Cs] 1350 [iTs][iAs][i.sup.5Cs]

[i.sup.5Cs][l.sup.5C s][iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][i.sup.5 Cs][lT] 796 GAATCTA[n3]CCATCCATGTACT 246 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1351 [iTs][iAs][n3s][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][lT] 797 GAATCTA[n6]CCATCCATGTACT 246 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1351 [iTs][iAs][n6s][i.sup.5Cs][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][lT] 798 GAATCTA[n3]CCCATCCATGTACT 534 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1352 [iTs][iAs][n3s][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][i.sup.5Cs][lT] 799 GAATCTA[n6]CCCATCCATGTACT 534 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1352 [iTs][iAs][n6s][i.sup.5Cs][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][i.sup.5Cs][lT] 800 GAATCTACC[n3]ATCCATGTACT 535 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1353 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][n3s ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs 801 GAATCTACC[n6]ATCCATGTACT 535 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1353 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][n6s ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][lT] 802 GAATCTACC[n3]CATCCATGTACT 264 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1354 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][n3s ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][i.sup.5Cs][lT] 803 GAATCTACC[n6]CATCCATGTACT 264 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1354 [iTs][iAs][i.sup.5Cs][i.sup.5Cs][n6s ][l.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][l.sup.5Cs][lT] 804 GAATCTAC[n3]CATCCATGTACT 264 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1354 [iTs][iAs][i.sup.5Cs][n3s][i.sup.5Cs ][iAs][iTs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][iGs][iTs][iAs][i.sup.5C s][lT] 805 GAATCTAC[n3]CCATCCATGTACT 246 [l.sup.5Gs][iAs][iAs][iTs][i.sup.5Cs 1351 ][iTs][iAs][i.sup.5Cs][n3s][i.sup.5C s][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i .sup.5Cs][iAs][iTs][iGs][iTs][i As][i.sup.5Cs][lT] 806 GAATCTAC[n6]CCATCCATGTACT 246 [lGs][iAs][iAs][iTs][i.sup.5Cs] 1351 [iTs][iAs][i.sup.5Cs][n6s][i.sup.5Cs ][i.sup.5Cs][iAs][iTs][i.sup.5Cs][i.sup.5 Cs][iAs][iTs][iGs][iTs][iA s][i.sup.5Cs][lT] 807 GAATCTACCATCCATGTACT 455 [lGs][nmaAs][nmaAs][nmaTs] 1355 [nma.sup.5Cs][nmaTs][nmaAs][nma .sup.5Cs][nma.sup.5Cs][nmaAs][nmaTs] [nma.sup.5Cs][nma.sup.5Cs][nmaAs][nm aTs][nmaGs][nmaTs][nmaAs] [nma.sup.5Cs][lT] 808 GAATCTACCCATCCATGTACT 461 [lGs][nmaAs][nmaAs][nmaTs] 1356 [nma.sup.5Cs][nmaTs][nmaAs][nma .sup.5Cs][nma.sup.5Cs][nma.sup.5Cs][nmaAs ][nmaTs][nma.sup.5Cs][nma.sup.5Cs][n maAs][nmaTs][nmaGs][nmaTs] [nmaAs][nma.sup.5Cs][lT] 809 TCTACCCCCATCCATGTACT 489 [lTs][l.sup.5Cs][iTs][iAs][i.sup.5C 1357 s][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][iAs ][iTs][iGs][iTs][iAs][l.sup.5Cs] [lT] 810 ATCTACCCCATCCATGTACT 490 [lAs][lTs][i.sup.5Cs][iTs][iAs] 1358 [i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][l.sup.5Cs] [lT] 811 GTCGCCGCCATCCATGTACT 491 [lGs][lTs][i.sup.5Cs][iGs][i.sup.5Cs 1359 ][i.sup.5Cs][iGs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][l.sup.5Cs][ lT] 812 GGGGTCGCCATCCATGTACT 492 [lGs][lGs][iGs][iGs][iTs] 1360 [i.sup.5Cs][iGs][i.sup.5Cs][i.sup.5Cs] [iAs][iTs][i.sup.5Cs][i.sup.5Cs][i As][iTs][iGs][iTs][iAs][l.sup.5 Cs][lT] 813 GGGTCGCCCATCCATGTACT 493 [lGs][lGs][iGs][iTs][i.sup.5Cs] 1361 [iGs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iA s][iTs][i.sup.5Cs][i.sup.5Cs][iAs][i Ts][iGs][iTs][iAs][i.sup.5Cs][l T] 814 GGTCGCCCCATCCATGTACT 494 [lGs][lGs][iTs][i.sup.5Cs][iGs] 1362 [i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][i As][iTs][i.sup.5Cs][i.sup.5Cs][iAs][ iTs][iGs][iTs][iAs][i.sup.5Cs][ lT] iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; lA, lG, l.sup.5C, and lT are 2′-4′methylene bridge (LNA) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; nmaA, nmaG, nma5C, and nmaT are 2′O-NMA modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; s is a phosphorothioate internucleoside linkage, and the absence of s between two nucleosides indicate a phosphodiester internucleoside linkage; ena is ENA; b is BNA(NC), wherein R is Me; z is AmNA, wherein R is Me; scp is scpBNA; [dt][dc][da] is a phosphodiester DNA “d(TCA)” linker; n3 is a C3 spacer as shown in Table 16; n6 is a C6 spacer as shown in Table 16; and x is a C16 lipid, 6-[(1-oxohexadecyl)amino]hexyl]phosphoryl.

†Each 5′-methyl cytidine (.sup.5C) in any of the sequences provided in Table 5 may independently and optionally be replaced with cytidine.

## Secondary Structures

[0288] The term "secondary structures" as used herein conforms to its generally known meaning. i.e. stems or loops. In some embodiments. pharmaceutically secondary structures include, but are not limited to, hairpin, bulge. helix, internal loop (or interior loop), external loop (or exterior loop), multi-branched loop and pseudoknot. In some embodiment, the antisense oligonucleotide may induce a secondary structure between the first target region and the second target region in the UNC13A pre-mRNA. In some embodiment, the antisense oligonucleotide may induce a secondary structure between the first target region and the second target region in the UNC13A pre-mRNA by simultaneously binding to the first target region and the second target region. In some embodiment, a secondary structure induced by the oligonucleotide may inhibit inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA. In some embodiment, a secondary structure induced by the oligonucleotide may restore UNC13A expression.

## Salts

[0289] The term "salts" as used herein conforms to its generally known meaning, i.e. an ionic assembly of anions and cations. In some embodiments, the antisense oligonucleotide may be in the form of a pharmaceutically acceptable salt. In some embodiments, pharmaceutically acceptable salts comprise inorganic salts, such as monovalent or divalent inorganic salts. In some embodiments, pharmaceutically acceptable salts include, but are not limited to, sodium, potassium, calcium, magnesium, and ammonium salts. In some embodiments, the pharmaceutically acceptable salt is a sodium salt. In some embodiments, the pharmaceutically acceptable salt is a potassium salt. In some embodiments, the pharmaceutically acceptable salt is a calcium salt. In some embodiments, the pharmaceutically acceptable salt is a magnesium salt. In some embodiments, the pharmaceutically acceptable salt is an ammonium salt. In some embodiments the ammonium salt is represented by the formula: N(R).sub.3, wherein each R is H or an alkyl having from 1 to 6 carbon atoms.

## Pharmaceutical Compositions

[0290] Pharmaceutical compositions comprising one or more antisense oligonucleotides, either alone or in combination with prophylactic agents, therapeutic agents, and/or pharmaceutically acceptable carriers are provided. The pharmaceutical compositions comprising antisense oligonucleotides provided herein are for use in, but not limited to, diagnosing, detecting, or monitoring a disease, in preventing, treating, managing, or ameliorating a disease or one or more symptoms thereof, and/or in research. In some embodiments, a pharmaceutical composition may further comprise any other suitable therapeutic agent for treatment of a subject, e.g., a human subject having an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer's disease, or LATE). In some embodiments, the other therapeutic agents may enhance or supplement the effectiveness of the complexes disclosed herein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100%. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly, p.Ala141Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Ala90Val, p.Arg116Gly, p.Asn87Ser, p.Asp102Gly, p.Asp125Val, p.Asp91Ala, p.Gln23Lcu, p.Glu101Gly, p.Glu101Lys, p.Glu50Lys, p.Gly13Arg, p.Gly148Ser, p.Gly38Arg, p.Gly42Asp, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.Gly94Asp, p.Gly94Cys, p.Gly94Ser, p.His121Gln, p.His44Arg, p.His47Arg, p.Ile113Thr, p.Ile114Thr, p.Ile150Thr, p.Leu127Ser, p.Leu145Phe, p.Leu145Ser, p.Leu39Val, p.Leu85Phe, p.Phe21Ile, p.Phe65Leu, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Asn87Ser, p.Asp102Gly, p.Gly13Arg, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.His121Gln, p.His44Arg, p.Ile150Thr, p.Leu39Val, p.Leu85Phe, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly or any combination thereof. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

[0291] An aspect of the disclosure includes pharmaceutical compositions comprising any one of the antisense oligonucleotides disclosed herein. In some embodiments, the pharmaceutical composition comprises any one of the antisense oligonucleotides disclosed herein and a pharmaceutically acceptable carrier. The pharmaceutical compositions disclosed herein are formulated for administration to a subject. In some embodiments, formulations as disclosed herein comprise an excipient. In some embodiments, an excipient confers to a composition improved stability, improved absorption, improved solubility and/or (e.g., and) therapeutic enhancement of the active ingredient. In some embodiments, an excipient is a pH modifier (e.g., sodium citrate, sodium phosphate, a tris base, or sodium hydroxide), a vehicle (e.g., a buffered solution, petrolatum, dimethyl sulfoxide, or mineral oil), a tonicity agent (e.g., sodium chloride, calcium chloride, magnesium chloride, potassium chloride, dextrose or sucrose or D-mannitol), a solubilizing agent (e.g., polysorbate, cyclodextrin), a binder, a disintegrant or a lubricant. Any one of the disclosed antisense oligonucleotides, when added to pharmaceutically acceptable excipients, can be packaged into kits, containers, packs, or dispensers. The pharmaceutical compositions disclosed herein can be packaged in pre-filled syringes or vials. In some embodiments, such carriers enable pharmaceutical compositions to be formulated as, for example, tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspension and lozenges for the oral ingestion by a subject. In some embodiments, a pharmaceutically acceptable carrier or diluent is sterile water, sterile saline, sterile buffer solution or sterile artificial cerebrospinal fluid.

[0292] In some embodiments, the delivery vehicle can be used to deliver an antisense oligonucleotide to a cell or tissue. A delivery vehicle is a compound that improves delivery of an antisense oligonucleotide to a cell or tissue. A delivery vehicle can include, or consist of, but is not limited to a polymer, such as an amphipathic polymer, a membrane active polymer, a peptide, a melittin peptide, a melittin-like peptide (MLP), an anti-transferrin peptides, an anti-transferrin antibodies, a lipid, a reversibly modified polymer or peptide, or a reversibly modified membrane active polyamine. In some embodiments, any one of the antisense oligonucleotides or pharmaceutical compositions disclosed herein can be combined with lipids, nanoparticles, polymers, liposomes, micelles, DPCs, peptides, antibodies, virus vectors (e.g., AAV vector), or other delivery systems available in the art.

[0293] In some embodiments, a pharmaceutical composition is formulated to be compatible with its intended route of administration. Non-limiting examples of routes of administration include intravenous, intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, intrathecal, or intracerebroventricular routes. In some embodiments, the route of administration is intrathecal, or intracerebroventricular routes. In some embodiments, the route of administration is subcutaneous.

Kits

[0294] An aspect of the disclosure includes kits comprising any one of the antisense oligonucleotides disclosed herein or a pharmaceutical composition disclosed herein. In some embodiments, the kit is for treating an UNC13A-associated disease disclosed herein (e.g., ALS, FTD, Alzheimer's disease or LATE). In some embodiments, the kit further comprises an additional agent disclosed herein.

Methods

[0295] Some aspects of the present disclosure provide methods for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell. In some embodiments, a method for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell comprises contacting the cell with any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein, thereby inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in the cell.

[0296] Some aspects of the present disclosure provide methods for restoring UNC13A expression (e.g., mRNA level and/or protein level) in a cell. In some embodiments, a method for restoring UNC13A expression comprises contacting the cell with any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein, thereby restoring UNC13A expression (e.g., mRNA level and/or protein level) in the cell. In some embodiments, contacting the cell with any one of the antisense oligonucleotides disclosed herein restores the UNC13A expression level (e.g., mRNA level and/or protein level) by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 95%, relative to UNC13A expression in cells not contacted with an antisense oligonucleotide. In some embodiments, the cell is in vitro (e.g., in a cell culture). In some embodiments, the cell is in vivo (e.g., in a subject).

[0297] The disclosure also provides methods for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell of a subject. In some embodiments, a method for inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell comprises administering to the subject any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein, thereby inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in the cell.

[0298] The disclosure also provides methods for restoring UNC13A expression (e.g., mRNA level and/or protein level) in a subject compared to baseline pre-treatment levels. In some embodiments, methods comprise administering to the subject any one of the antisense oligonucleotides disclosed herein or any one of the pharmaceutical compositions disclosed herein. The UNC13A expression level (e.g., mRNA level and/or protein level) in the subject is restored in a cell, group of cells, tissue, blood, and/or other fluid of the subject. In some embodiments, the above-described methods further comprise determining UNC13A expression (e.g., mRNA level and/or protein level) in a sample(s) from the subject (e.g., UNC13A level in a blood or serum sample(s)). The level of UNC13A in a sample may be measured by general methods known in the art.

[0299] In some embodiments, in any one of the methods disclosed herein, a subject is non-human primate, or rodent. In some embodiments, a subject is a human. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100%, relative to expression of TDP-43 in a healthy subject or a subject that does not have an UNC13A-associated disease (e.g., ALS, FTD, Alzheimer's disease, or LATE). In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5 Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly, p.Ala141Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Ala90Val, p.Arg116Gly, p.Asn87Ser, p.Asp102Gly, p.Asp125Val, p.Asp91Ala, p.Gln23Leu, p.Glu101Gly, p.Glu101Lys, p.Glu50Lys, p.Gly13Arg, p.Gly148Ser, p.Gly38Arg, p.Gly42Asp, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.Gly94Asp, p.Gly94Cys, p.Gly94Ser, p.His121Gln, p.His44Arg, p.His47Arg, p.Ile113Thr, p.Ile114Thr, p.Ile150Thr, p.Leu127Ser, p.Leu145Phe, p.Leu145Ser, p.Leu39Val, p.Leu85Phe, p.Phe21Ile, p.Phe65Leu, p.Thr138Ile, p.

Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Asn87Ser, p.Asp102Gly, p.Gly13Arg, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.His121Gln, p.His44Arg, p.Ile150Thr, p.Leu39Val, p.Leu85Phe, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly or any combination thereof. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof. In some embodiments, a subject is a patient, e.g., a human patient that has or is suspected of having a disease. In some embodiments, the subject is a human patient who has or is suspected of having an UNC13A-associated disease and/or a disease associated with TDP-43 dysfunction in the subject. In some embodiments, the UNC13A-associated disease and/or a disease associated with TDP-43 dysfunction in the subject is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is ALS or FTD. In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is LATE. In some embodiments, the neurodegenerative disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0300] An aspect of the disclosure includes a method of treating a subject having an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression). In some embodiments, the method comprises administering to the subject any one of the antisense oligonucleotides disclosed herein, or a pharmaceutical composition disclosed herein. In some embodiments the method results in treating the subject having the UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression). In some embodiments, the disease is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is ALS. In some embodiments, the neurodegenerative disease is FTD. In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is LATE. In some embodiments, the neurodegenerative disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem protcinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0301] An aspect of the disclosure includes a method of treating a subject having a disease associated with TDP-43 dysfunction in the subject. In some embodiments, the subject has decreased expression of TDP-43 gene or protein. In some embodiments, the subject has decreased expression of TDP-43 gene or protein by at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 100% relative to expression of TDP-43 in a healthy subject. In some embodiments, the subject has dysfunction of TDP-43 protein. In some embodiments, the subject has cytoplasmic aggregation of

TDP-43 protein. In some embodiments, the subject has cytoplasmic mislocalization of TDP-43 protein. In some embodiments, the subject does not possess an SOD-1 gene mutation. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p.Val149Gly, p.Ala141Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Ala90Val, p.Arg116Gly, p.Asn87Ser, p.Asp102Gly, p.Asp125Val, p.Asp91Ala, p.Gln23Leu, p.Glu101Gly, p.Glu101Lys, p.Glu50Lys, p.Gly13Arg, p.Gly148Ser, p.Gly38Arg, p.Gly42Asp, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.Gly94Asp, p.Gly94Cys, p.Gly94Ser, p.His121Gln, p.His44Arg, p.His47Arg, p.Ile113Thr, p.Ile114Thr, p.Ile150Thr, p.Leu127Ser, p.Leu145Phe, p.Leu145Ser, p.Leu39Val, p.Leu85Phe, p.Phe21Ile, p.Phe65Leu, p. Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises intronic, p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly, p.Ala146Thr, p.Ala5Ser, p.Ala5Thr, p.Ala5Val, p.Ala90Thr, p.Asn87Ser, p.Asp102Gly, p.Gly13Arg, p.Gly42Ser, p.Gly94Ala, p.Gly94Arg, p.His121Gln, p.His44Arg, p.Ile150Thr, p.Leu39Val, p.Leu85Phe, p.Thr138Ile, p. Val149Gly or any combination thereof. In some embodiments, the SOD-1 gene mutation comprises p.Ala5Val, p.Ala5Thr, p.Leu39Val, p.Gly42Ser, p.His44Arg, p.Leu85Val, p.Gly94Ala, p.Leu107Val, p. Val149Gly or any combination thereof. In some embodiments, the subject possesses an UNC13A gene mutation in intron 20. In some embodiments, the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof. In some embodiments, the method comprises administering to the subject any one of the antisense oligonucleotides disclosed herein or a pharmaceutical composition disclosed herein. In some embodiments the method results in treating the subject having a disease associated with TDP-43 dysfunction. In some embodiments, the disease is a neurodegenerative disease. In some embodiments, the neurodegenerative disease is ALS. In some embodiments, the neurodegenerative disease is FTD. In some embodiments, the neurodegenerative disease is Alzheimer's disease. In some embodiments, the neurodegenerative disease is LATE. In some embodiments, the neurodegenerative disease is frontotemporal lobar degeneration (FTLD), Progressive supranuclear palsy (PSP), Primary lateral sclerosis (PLS), Progressive muscular atrophy (PMA), facial onset sensory and motor neuronopathy (FOSMN), cerebral age-related TDP-43 with sclerosis (CARTS), Parkinsons disease, Guam Parkinson-dementia complex (G-PDC), Multisystem proteinopathy (MSP), chronic traumatic encephalopathy (CTE), Autism, hippocampal sclerosis (HS), Hippocampal sclerosis dementia, Down syndrome, Huntington's disease, multiple sclerosis, Perry disease, peripheral myopathy, polyglutamine diseases (e.g., spinocerebellar ataxia 3, myopathies and Chronic Traumatic Encephalopathy), Rasmussen's encephalitis, attention deficit hyperactivity disorder, autism, central pain syndromes, anxiety or depression.

[0302] In some embodiments, the above-described methods further comprise administering an additional agent for the treatment of an UNC13A-associated disease (e.g., a disease associated with abnormal UNC13A expression) and/or a disease associated with TDP-43 dysfunction in a subject.

[0303] Having now described some embodiments in detail, practice of the invention will be more fully understood from the following examples, which are presented herein for illustration only and should not be construed as limiting the invention in any way.

## EXAMPLES

### Example 1: Synthesis of Oligonucleotides

[0304] The automated solid-support synthesis of 331 oligonucleotides was performed on nS8-II oligonucleotide synthesizer (GeneDesign). Antisense oligonucleotides correspond to ASOs 1-331 and 350-814 in Tables 3, 5, 9, and 11. All the oligonucleotides were synthesized using low-loaded Unylinker™ universal solid support (1 μmol scale, 42.6 μmol/g, ChemGenes, N-4000-10). The protocol for solid-support synthesis is shown in Table 6 below.

TABLE-US-00026 TABLE 6 Step Reaction Reagent Reaction time 1 Detritylation Dichloroacetic Acid-Toluene (3:97) 90 seconds 2 Coupling Activator solution:0.1M Amidite solution = 40:60 (12 eq.) 300 seconds 3 Oxidation or Iodine Solution (abt. 0.05 mol/L)][Pyridine:Water(9:1) (for 30 seconds Thiolation oxidation) or 0.1M ((dimethylamino-methylidene)amino)- (for oxidation) 3H-1,2,4-dithiazaoline-3-thione in pyridine (for thiolation) or 400 seconds (for thiolation) 4 Capping 1-Methylimidazole-Acetonitrile (2:8) + Acetic 90 seconds Anhydride-2,6-Lutidine-Acetonitrile (2:3:5)

[0305] At the end of the solid phase synthesis, cyanoethyl protecting groups were removed by a 10 minute treatment with 10% diethylamine in acetonitrile. The solid support was dried with an argon gas flush and placed into a 2 mL microtube. 500 μL of concentrated aqueous ammonia was added to the tube and the support was treated for 8 hours at 60° C. As for AmNA (R=Me)-containing oligonucleotides, 500 μL of tert-butylamine-MeOH-water (1/1/2) solution was added to the tube and the support was treated for 7 hours at 65° C. (Masaki Yamagami et. al., 1P-40 Synthesis and Properties of Antisense Oligonucleotides Containing AmNA and GuNA, The 46th International Symposium on Nucleic Acids Chemistry & The 3rd Annual Meeting of Japan Society of Nucleic Acids Chemistry on October 29th). The antisense oligonucleotides were isolated with the cartridge purification system (Glen-Pak™ DNA Purification Cartridge, #60-5200, according to the manual provided) to afford an aqueous solution containing the purified oligonucleotide. For distillation of volatile components, a Thermo Scientific SAVANT SPD2010 Speed Vac Concentrator was used. Quantification of the antisense oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μm, 2.1×50 mm) column. The eluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH). The sample was lyophilized to afford a white powder. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides. A subset of the antisense oligonucleotides was selected for testing in Example 2.

Alternative Method for Synthesis of Oligonucleotides

[0306] The automated solid-support synthesis of oligonucleotides was performed on nS8-II oligonucleotide synthesizer (GeneDesign). All the oligonucleotides were synthesized using low-loaded Unylinker™ universal solid support (1.5 μmol scale, 42.6 μmol/g, ChemGenes, N-4000-10). The detailed protocol for nS8-II is shown in Table 7 below.

TABLE-US-00027 TABLE 7 Step Reaction Reagent Reaction time 1 Detritylation Dichloroacetic Acid-Toluene (3:97) 90 sec 2 Coupling Activator solution:0.1M Amidite solution = 40:60 (12 eq.) 300 sec 3 Oxidation or Iodine Solution (abt. 0.05 mol/L) Pyridine:Water(9:1) (for 30 seconds Thiolation oxidation) or 0.1M ((dimethylamino-methylidene)amino)- (for oxidation) 3H-1,2,4-dithiazaoline-3-thione in pyridine (for thiolation) or 400 seconds (for thiolation) 4 Capping 1-Methylimidazole-Acetonitrile (2:8) + Acetic 90 sec Anhydride-2,6-Lutidine-Acetonitrile (2:3:5)

[0307] At the end of the solid phase synthesis, cyanoethyl protecting groups were removed by a 10 minute treatment with 10% diethylamine in acetonitrile. The synthesis was repeated three times on a 1.5 μmol scale, and those were combined into one, resulting in a total of 4.5 μmol scale. The subsequent experiment was then carried out using this combined amount. The solid support was dried with an argon gas flush and placed into a 2 mL microtube. 500 μL of conc. Aq. ammonia was added to the tube and the support was treated for 8 hours at 60° C. Oligonucleotides were isolated with the cartridge purification system (Glen-Pak™ DNA Purification Cartridge, #60-5200, according to the manual provided) to afford aqueous solution containing the purified oligonucleotide. For distillation of volatile components, a Thermo Scientific SAVANT SPD2010 SpeedVac Concentrator was used. The resulting solution was filtered through a 0.20 μm filter (ADVANTEC, DISMIC: 13CP020AS). UV absorbance of the solution was measured and then the sample was lyophilyzed to afford a white powder. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μm, 2.1×50 mm) column. The cluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100%

MeOH). The data represents the calculated values obtained through deconvolution processing using Waters MassLynx, such as MaxEntl because the mass spectrum was observed not as molecular ion peaks, but as multivalent ions. Quantification of oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides.

Synthesis of 2′-O-NMA-Modified Oligonucleotides

[0308] The automated solid-support synthesis of 2′-O-NMA-modified oligonucleotides was performed on nS8-II oligonucleotide synthesizer (GeneDesign). All the oligonucleotides were synthesized using low-loaded Unylinker™ universal solid support (1 μmol scale, 42.6 μmol/g, ChemGenes, N-4000-10). The protocol for solid-support synthesis is shown in Table 17.

TABLE-US-00028 TABLE 17 Step Reaction Reagent Reaction time 1 Detritylation Dichloroacetic Acid-Toluene (3:97) 90 seconds 2 Coupling Activator solution:0.1M Amidite solution = 40:60 (12 eq.) 300 seconds 3 Oxidation or Iodine Solution (abt. 0.05 mol/L) Pyridine:Water(9:1) (for 30 seconds Thiolation oxidation) or 0.1M ((dimethylamino-methylidene)amino)- (for oxidation) 3H-1,2,4-dithiazaoline-3-thione in pyridine (for thiolation) or 400 seconds (for thiolation) 4 Capping 1-Methylimidazole-Acetonitrile (2:8) + Acetic 90 seconds Anhydride-2,6-Lutidine-Acetonitrile (2:3:5)

[0309] At the end of the solid-phase synthesis, cyanoethyl protecting groups were removed by treating the solid support with 10% diethylamine in acetonitrile for 10 minutes. The solid support was then dried using an argon gas flush and transferred into a 2 mL microtube. Next, 500 μL of concentrated aqueous ammonia was added to the tube, and the mixture was incubated at 65° C. for 15 minutes. Subsequently, 500 μL of 40% methylamine was added, and the solution was further incubated for 15 minutes with an aqueous ammonia-to-methylamine ratio of 1:1. Volatile components were removed using a Thermo Scientific SAVANT SPD2010 SpeedVac Concentrator. The resulting solution was filtered through a 0.20 μm filter (ADVANTEC, DISMIC: 13CP020AS) and purified using a C18 reverse-phase column (Xbridge @Oligonucleotide BEH C18, 10 mm×50 mm, 130A, 2.5 μm) via HPLC, employing a gradient of 10%-70% of mobile phase B over 30 minutes. Mobile phase A consisted of 240 mM hexafluoroisopropanol and 7 mM triethylamine in an aqueous solution containing 5% methanol, while mobile phase B was 100% methanol. The purified DMT-ON fraction was collected and freeze-dried. The resulting freeze-dried powder was dissolved in 1 mL of distilled water and subjected to detritylation using a cartridge purification system (Glen-Pak™ DNA Purification Cartridge, #60-5200) according to the manufacturer's instructions, yielding an aqueous solution containing the purified oligonucleotide. Volatile components were again removed using the Thermo Scientific SAVANT SPD2010 SpeedVac Concentrator. The final solution was filtered through a 0.20 μm filter (ADVANTEC, DISMIC: 13CP020AS), and its UV absorbance was measured. The purified sample was then lyophilized to afford a white powder. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μm, 2.1×50 mm) column. The cluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH).

[0310] The data represents the calculated values obtained through deconvolution processing using Waters MassLynx, such as MaxEntl because the mass spectrum was observed not as molecular ion peaks, but as multivalent ionfragment ions. Quantification of oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides.

Synthesis of C16-Conjugated Oligonucleotides

[0311] Oligonucleotide was chemically synthesized using the deoxynucleoside phosphoramidite approach by AKTA oligopilot plus 10 System. (solid support: UnyLinker™ NittoPhase® 341 umol/g). 5′-TFA-Amino Modifier C6 CE-Phosphoramidite (Glen research, 10-1916-02) was subjected at the end of coupling cycle. The protocol for solid-support synthesis is shown in Table 18.

TABLE-US-00029 TABLE 18 Step Reaction Reagent Volume Reaction time 1 Detritylation Dichloroacetic Acid-Toluene (3:97) Until trityl UV disappears 2 Coupling Activator solution:0.1M Amidite 4 eq. 10 min (for solution = 40:60 DNA)/20 min (for MOE) 3 Oxidation or Iodine Solution (abt. 3.5 CV 10 min Thiolation 0.05 mol/L)][Pyridine:Water(9:1) (for (for oxidation) oxidation) or 0.1M ((dimethylamino- or 3 CV methylidene)amino)-3H-1,2,4- (for thiolation) dithiazaoline-3-thione in pyridine (for thiolation) 4 Capping 1-Methylimidazole-Acetonitrile (2:8) + 2CV 10 min Acetic Anhydride-2,6-Lutidine- Acetonitrile (2:3:5)

[0312] At the end of the solid phase synthesis, cyanoethyl protecting groups were removed by a 10 min treatment with 10% diethylamine in acetonitrile. After transferring the solid phase support from the column reactor to 15 mL Falcon tube, aq. ammonia (4 mL) was added to the tube and the support was treated for 8 hours at 60° C. After filtering the reaction solution, nitrogen gas was blown against the filtrate to remove ammonia. The solution was then lyophilized. The residue was dissolved in water (2 mL), and 300 μL of 3 M aq. Sodium acetate solution (pH=7.0) was added to the solution. EtOH was added to the solution, and the resulting white suspension was centrifuged. The supernatant was removed by decantation, and the residue was dissolved in water. Quantification of the antisense oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μm, 2.1×50 mm) column. The eluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH). The sample was lyophilized to afford a C6-amino conjugated oligonucleotide without further purification.

[0313] The crude C6 amino conjugated oligonucleotide was dissolved with water. N,N-diisopropylethylamine (100 eq.) and palmitic acid N-hydroxysuccinimide ester (10 eq.) dissolved in DMSO was added to the substrate solution. The reaction mixture was incubated at 45 degree for 10 h. 3 M aq. Sodium acetate solution (pH=7.0) was added to the solution. EtOH was added to the solution, and the resulting white suspension was centrifuged. The supernatant was removed by decantation, and the residue was dissolved in water. The mixture was purified by HPLC (XBridge, OST C18 column, 2.5 μm, 10×50 mm0.1M TEAA in DW/0.1M TEAA in MeCN, 10%-60% for 30 min). The fraction containing the target product was evaporated, and the residue was dissolved in water, and 3M NaOAc aq. was added to the solution for anion exchange. EtOH was added to the solution, and the resulting white suspension was centrifuged. The supernatant was removed by decantation, and the residue was dissolved with water. The solution was then lyophilized to give C16-conjugated oligonucleotide. Oligonucleotides were analyzed for characterization by a Waters Xevo G2-XS Qtof UPLC-MS system (ESI-Qtof) with a ACQUITY UPLC OST C18 (1.7 μm, 2.1×50 mm) column. The cluent was 240 mM hexafluoroisopropanol/7 mM triethylamine aqueous solution containing 5% MeOH and the methanolic solution (100% MeOH).

[0314] The data represents the calculated values obtained through deconvolution processing using Waters MassLynx, such as MaxEntl because the mass spectrum was observed not as molecular ion peaks, but as multivalent ionfragment ions. Quantification of oligonucleotides was conducted based on UV absorbance measurement on Unchaind Labs LUNATIC. Measured molecular weights by mass spectrometry confirmed the molecular weight of the antisense oligonucleotides.

Example 2: Activities of Antisense Oligonucleotides In Vivo

[0315] This example demonstrates that antisense oligonucleotides disclosed herein inhibited inclusion of the cryptic exon (CE) into UNC13A mRNA.

[0316] Single stranded Adeno-associated viral vector 9 (ssAAV9) encoding the UNC13A exon20-intron 20-exon 21 sequence and part of the intron 21 sequence with alternative primer binding site upstream and downstream of the UNC13A sequences driven by a human synapsin I promoter (ssAAV9-SYN1-hUNC13A (SEQ ID NO: 328)) was produced by VectorBuilder. ssAAV9-SYN1-hUNC13A (2 μL of 1.0×10.sup.13 genome copies/mL) was injected into the right dorsal hippocampus of 8-week-old mice. One week later, to knock down the endogenous mouse Tardbp

(TDP-43), the mice received an antisense oligonucleotide (2 μL of 2.5 mM) targeting mouse Tardbp (TDP-43) (5′-AAGGCttcatattgtACTTT-3′ (SEQ ID NO: 317) (hereinafter referred to as "Tardbp ASO") with a phosphorothioate backbone, lowercase letters indicating deoxyribonucleosides and uppercase letters indicating 2′-O-methoxyethyl-modified nucleosides, and bases of "C" or "c" indicating 5′-methyl cytosine, or phosphate-buffered saline (PBS) (2 μL) (control; Tardbp ASO (−)) by intrahippocampal injection. The antisense oligonucleotides targeting mouse Tardbp were synthesized by Gene Design Co., Ltd. (Osaka, Japan). One-week post-injection, test antisense oligonucleotides (10 μL of 1.0 mM) or PBS (10 μL) (control; test antisense oligonucleotide (−)) or were intracerebroventricularly injected. Test antisense oligonucleotides correspond to ASOs 1, 2, 8, 9, 14, 16, 17, 80, 144, 145, 149, 164, 193, 194, 214, 220, 222, 235, 238, 276, and 328 in Tables 3 and 5. Mice were euthanized at 14 days post-test antisense oligonucleotides injection and the right dorsal hippocampus was isolated. Total RNA was extracted using QIAzol Lysis Reagent (Qiagen, 79306) and RNeasy Mini Kit (Qiagen, 74106) and was reverse-transcribed with SuperSeript IV VILO Master Mix (Invitrogen, 11756050) following the manufacturers' protocol. RT-qPCR was performed with TaqPath qPCR Master Mix, CG (Applied Biosystems, A15297) on the QuantStudio 7 Flex Real-Time PCR System. Expression levels of mouse Tardbp ((TDP-43) and mouse Actb (Actb) were evaluated using Thermofisher TaqMan Gene Expression Assays: Tardbp (TDP-43) (ID: Mm01257504_g1), Actb (ID: Mm00607939_s1). For detecting the splicing of the UNC13A cryptic exon (CE) and normal splicing, the following sequences of primers and probes were synthesized at Integrated DNA technologies (IDT) and were used: UNC13A_CE FWD 5′-3′ CCCCGTACCATGTCCAGTACA (SEQ ID NO: 320), UNC13A_CE REV 5′-3′ CATTCACCAGCATTTATTCAACAAA (SEQ ID NO: 321), UNC13A_CE Probe 5′-3′/56-FAM/CTGCATGAG/ZEN/CTGCCT/3IABKFQ/; UNC13A minigene_Normal FWD 5′-3′ CTCACTAAAGGGAACAAGCGA (SEQ ID NO: 322), UNC13A minigene_Normal REV 5′-3′ GTGGAACAGGTTCTCATGC (SEQ ID NO: 323), UNC13A minigene_Normal Probe 5′-3′/56-FAM/CAGTGTGGA/ZEN/GATCAAAGGCGAGGA/3IABKFQ/(SEQ ID NO: 324).

[0317] Each target expression level was calculated using the absolute quantification methods. The expression level of mouse Tardbp was normalized by mouse Actb and the expression level of UNC13A cryptic exon (CE) was normalized by minigene-derived normal UNC13A (detected by using UNC13A minigene_Normal primers and probe) expression level. To calculate TDP-43 knock-down level in Tardbp ASO treated group, the average expression value of the control group (Tardbp ASO non-treated and test antisense oligonucleotide non-treated group) were used for the normalization (100%: Tardbp ASO (−), test antisense oligonucleotide (−) group). For the value of UNC13A CE normalized by minigene-derived normal UNC13A, the average expression value of the control group (Tardbp ASO treated and test antisense oligonucleotide non-treated group) were used for the normalization (100%: Tardbp ASO (+) test antisense oligonucleotide (−)). If multiple tests were performed, the average of those values was calculated.

[0318] Results are shown in Table 8 below.

TABLE-US-00030 TABLE 8 % TDP-43 level in mice % inclusion of UNC13A CE that received the in mice treated with the Tardbp ASO relative to indicated ASO relative to ASO # mice that received PBS mice treated with PBS 1 38.3 60.8 2 31.0 45.5 8 35.4 55.2 9 33.7 21.5 14 38.1 57.7 16 34.9 63.0 17 26.3 64.3 80 30.2 47.0 144 34.6 54.6 145 33.2 62.9 149 34.5 47.0 164 31.7 70.0 193 30.7 68.5 194 34.1 42.2 214 29.0 54.4 220 30.5 19.3 222 32.8 54.6 235 35.1 65.7 238 37.4 34.7 276 31.8 53.2 328 30.7 50.5

[0319] The results show that all antisense oligonucleotides tested in Table 8 showed inhibition of the inclusion of the CE into UNC13A mRNA.

[0320] The same experiment was run with additional test antisense oligonucleotides corresponding to ASOs 81, 107, 324, 327, 369, 373, 396, 486, 494, 495, 514, 521, 532, 561, 603, 629, 631, 632, 633, 650, 660, 663, 692, 726, and 328 in Tables 3, 9, 5, 11. Results are shown in Table 12 below.

TABLE-US-00031 TABLE 12 % TDP-43 % inclusion of level in mice UNC13A CE in that received mice treated with the Tardbp the indicated ASO relative ASO relative to mice that to mice treated ASO # received PBS with PBS Amount 81 28.5 37.3 10 nmol 107 26.4 45.1 10 nmol 324 27.4 40.8 10 nmol 327 33.1 35.3 10 nmol 369 36.7 46.8 3 nmol 373 30.3 21.2 10 nmol 396 33.1 48.8 3 nmol 486 33.1 51.5 10 nmol 494 27.3 63.8 10 nmol 495 28.6 47.2 10 nmol 514 33.8 48.6 10 nmol 521 33.2 49.4 10 nmol 532 46.7 35.3 10 nmol 561 47.7 21.1 3 nmol 603 25.7 49.4 10 nmol 629 26.1 51.4 10 nmol 631 34.7 56.9 10 nmol 632 42.0 39.0 10 nmol 633 24.5 63.1 3 nmol 650 27.0 62.5 3 nmol 660 23.9 51.9 3 nmol 663 26.0 42.6 3 nmol 692 34.4 60.2 10 nmol 726 43.9 57.8 10 nmol

[0321] The results show that all antisense oligonucleotides tested in Table 12 showed inhibition of the inclusion of the CE into UNC13A mRNA.

Example 3: Activities of Antisense Oligonucleotides In Vivo (hUNC13A BAC Tg Mice)

[0322] This example demonstrates that antisense oligonucleotides (ASO) disclosed herein inhibited inclusion of the cryptic exon (CE) into UNC13A mRNA in human UNC13A BAC transgenic (Tg) mouse model. This mouse model was generated as follows by Institute of Immunology Co., Ltd. Following a modified version of the method by Abe et al. (Exp Anim. 2004 53(4):311-320. Establishment of an efficient BAC transgenesis protocol and its application to functional characterization of the mouse Brachyury locus), human CH17-170L19 BAC clone (BACPAC GENOMICS, INC.) including human UNC13A gene was purified using a plasmid extraction kit, and linearized by reacting with PI-SceI. The linearized DNA was separated in an agarose gel and extracted using electroelution and dialyzed in TE buffer prepared for microinjection. The purified DNA fragments were assessed for high purification without fragmentation by applying them to pulse-field gel and determine the DNA concentration using a NanoDrop spectrophotometer (Asahi Techno Glass Corporation). Dilute the DNA solution to 0.5 ng/μl and store it at 4° C. until used for injection. For the verification of the Human UNC13A Locus contained in BAC clone, PCR using following primers were conducted: PCR-A FWD 5′-3′ GGTCTGCACAGGAGGAAACC (SEQ ID NO: 536), PCR-A REV 5′-3′ CCACACGACCCCTCCAGCAG (SEQ ID NO: 537). For the verification of purified BAC clone by PCR with following primers: PCR-B FWD 5′-3′ CTACGAGCATGTCATGAAGTTGC (SEQ ID NO: 538), PCR-B REV 5′-3′ AAGCCCATTCATTCCTAGTGCTG (SEQ ID NO: 539), PCR-C FWD 5′-3′ ACATCCCTCAGAGTGTTGTGTCA (SEQ ID NO: 540), PCR-C REV 5′-3′ CCTGGGGAGGAGAATGAGGTAAA (SEQ ID NO: 541). The DNA were injected into the fertilized egg pronucleus of C57BL/6J mice. After injection, the embryos were injected into the one-sided fallopian tubes of female mice. Genotyping and southern blotting analysis for the F0 mouse were performed by using primers and probes as follows: PCR-1 FWD 5′-3′ ACATCCCTCAGAGTGTTGTGTCA (SEQ ID NO: 540), PCR-1 REV 5′-3′ CCTGGGGAGGAGAATGAGGTAAA (SEQ ID NO: 541), PCR-2 FWD 5′-3′ TCCATTGAAGTGCATGTGTGTAG (SEQ ID NO: 542), PCR-2 REV 5′-3′ GGGACCATAAGAGCTAAAGTTGG (SEQ ID NO: 543), PCR-3 FWD 5′-3′ GCATTTGAAAGGTCTCAACTGTATT (SEQ ID NO: 544), PCR-3 REV 5′-3′ TTGTTGCTGATGAATCTTGTGTG (SEQ ID NO: 545), PCR-4 FWD 5′-3′ GGGAATTTATCTGAAGACCAAGC (SEQ ID NO: 546), PCR-4 REV 5′-3′ TCAGCTTTTGAATGGAGAATGAA (SEQ ID NO: 547), PCR-5 FWD 5′-3′ TACTCTCGCTGTCTCCTTGTGTC (SEQ ID NO: 548), PCR-5 REV 5′-3′ ACTGAAAGAGGCTCAGGAACACT (SEQ ID NO: 549), PCR-6 FWD 5′-3′ GACAGTGACTACCGCAGTGAAAC (SEQ ID NO: 550), PCR-6 REV 5′-3′ CCACAAGAAAATGTGAATTCTGC (SEQ ID NO: 551), PCR-7 FWD 5′-3′ AGTGGGCTTAATTTGCATTCATT (SEQ ID NO: 552), PCR-7 REV 5′-3′ GCACAGGTATAATCAGACCCACA (SEQ ID NO: 553), PCR-8 FWD 5′-3′ GGTTGGAAGAAGCACTGAGAGG (SEQ ID NO: 554), PCR-8 REV 5′-3′ AGAAGAAGGGGAAAGTCAGAGGT (SEQ ID NO: 555), PCR-9 FWD 5′-3′ AGTTACACCGTCGTCCCTATCAT (SEQ ID NO: 556), PCR-9 REV 5′-3′

ATGACCTTGAAGAAGTGGCTTTC (SEQ ID NO: 557), PCR-10 FWD 5′-3′ CCTCACAGTTGTAGACCATGGAA (SEQ ID NO: 558), PCR-10 REV 5′-3′ CACACCTCCTTCCAGATGTTGTA (SEQ ID NO: 559), PCR-11 FWD 5′-3′ CTACGAGCATGTCATGAAGTTGC (SEQ ID NO: 538), PCR-11 REV 5′-3′ AAGCCCATTCATTCCTAGTGCTG (SEQ ID NO: 539), PCR-12 FWD 5′-3′ CCCTGTCTCGTGCTCCCTCAGA (SEQ ID NO: 560), PCR-12 REV 5′-3′ CATGGGCAAGCAGTGTGTTCTA (SEQ ID NO: 561), PCR-Internal control (mouse Actb) FWD 5′-3′ AGAGAGCTCACCATTCACCATCT (SEQ ID NO: 562), REV 5′-3′ TCCTAGGCTCTCAAAACAAAACC (SEQ ID NO: 563), Southern blotting Probe 5′-3′ AGCCCAGGGTATCCCCATGGCCCGTCCACACCAGCCTGGCTATGCTAACTCCCAAA CCATGCACCAAGTCTGGCCCCACACATCCCCAGCCCAGCCTCCTTCCCAATGTCAGC CTGGGGTTTCTAGATCTAGAGGGAGGAGAGAAAACTAGGGCTGAAACTGGGCTGA AGTCATGGTTAGGGATCAAGGTCAGGGTTAGTTCAGCCATCAAAGATCAGAGGTCA GGGTTAGAATTGGTGTTAGGCAAAGGGGTCAGTGCTTAGGGCCAAATTGGAGGTCA TTGTCAGGGTTAGGGAAAGGTCTGGGGCCGGCTCAGTAGGGGAGTAGGAATCAAG GGAAGCAATCAGCATCAGGGGCAAGCTGACTGTTTAGATGGGCAGCAGGGTTG (SEQ ID NO: 564). The F0 mice and subsequent generations were backcrossed with C57BL/6J.

[0323] To knock down the endogenous mouse Tardbp (TDP-43), eight to eleven week old mice received a heteroduplex oligonucleotide (10 μL of 2.5 mM) targeting mouse Tardbp (TDP-43) (antisense strand: 5′-ATATGcacgctgattCCTTT-3′ (SEQ ID NO: 1363), with phosphorothioate backbone, lowercase letters indicating deoxyribonucleosides, uppercase letters with underlines indicating ENA-modified nucleosides and bases of “C” or “c” indicating 5′-methyl cytosine, complementary strand: 5′-aaaggaatcagcgtgcatat-3′ (SEQ ID NO: 1364) with phosphate backbone, lowercase letters indicating deoxyribonucleosides and bases of “c” indicating 5′-methyl cytosine (hereinafter referred to as “Tardbp HDO”) or 5% dextrose (5% Dex) (10 μL) as vehicle control (Tardbp HDO (−)) by intracerebroventricular (ICV) injection on day 0 and day 7. The antisense strand of Tardbp HDO was synthesized by KNC Laboratories Co., Ltd. (Kobe, Japan). The complementary strand of Tardbp HDO was synthesized by Gene Design Co., Ltd. (Osaka, Japan). On day 1 and day 21, test ASOs (10 μL of 0.3 mM) or DPBS (10 μL) as vehicle control (test ASO (−)) were injected by ICV. Test ASOs correspond to ASOs 211, 372, 391, 392, 393, 394, 395, 511, 564, 565, 658 in Tables 3, 5, 9, and 11. Mice were euthanized at 21 days after the last test ASO injection, and the cerebral cortex and the spinal cord were isolated. Total RNA was extracted using QIAzol Lysis Reagent (Qiagen, 79306) and QuickGene RNA Tissue Kit (Kurabo, RT-S2) with DNase (QIAGEN, 79254) treatment, and was reverse-transcribed with SuperPrep II Cell Lysis & RT Kit for qPCR (Toyobo, SCQ-401) following the manufacturers' protocol.

[0324] RT-qPCR was performed with THUNDERBIRD Probe qPCR Mix (Toyobo, QPS-101) on the Viia7, QuantStudio 7 Flex or QuantStudio 12K Flex Real-Time PCR System (Applied Biosystems). Expression levels of mouse Tardbp (TDP-43) was evaluated using Thermofisher TaqMan Gene Expression Assays: Tardbp (TDP-43) (ID: Mm01257504_g1). For detecting the splicing of the UNC13A cryptic exon (CE) and mouse Rbfox3, the following sequences of primers and probes were synthesized at Integrated DNA technologies (IDT) and were used: UNC13A_CE FWD 5′-3′ CCCCGTACCATGTCCAGTACA (SEQ ID NO: 320), UNC13A_CE REV 5′-3′ CATTCACCAGCATTTATTCAACAAA (SEQ ID NO: 321), UNC13A_CE Probe 5′-3′/56-FAM/CTGCATGAG/ZEN/CTGCCT/3IABKFQ/. Rbfox3 FWD 5′-3′ CCGCTCGTTAAAAATGATCTCC (SEQ ID NO: 565), Rbfox3 REV 5′-3′ CGACTACATGTCTCCAACATCC (SEQ ID NO: 566), Rbfox3 Probe 5′-3′/56-FAM/TTCCCGAAT/ZEN/TGCCCGAACATTTGC/3IABKFQ/(SEQ ID NO: 567). RT-qPCR protocol was as below; 95° C. for 60 sec, 45 cycles of 95° C. for 15 sec and 60° C. for 60 sec.

[0325] Each target expression level was calculated using the absolute quantification methods. The expression level of mouse Tardbp (TDP-43), UNC13A cryptic exon (CE) were normalized by mouse Rbfox3 expression level. To calculate TDP-43 knock-down level in Tardbp HDO treated group, the average expression value of the control group (Tardbp HDO (−) and test ASO (−) group (PBS only)) were used for the normalization (100%: Tardbp HDO (−) and test ASO (−)). For the value of UNC13A CE normalized by Rbfox3, the average expression value of the control group (Tardbp HDO (+) and test ASO (−) group) were used for the normalization (100%: Tardbp HDO (+) and test ASO (−)).

[0326] For statistical analysis, post-hoc analysis was performed for CE expression data of cerebral cortex and spinal cord. Aspin-Welch's test for Tardbp HDO (−) and test ASO (−) group vs. Tardbp HDO (+) and test ASO (−) was performed to confirm the success of the TDP-43 knockdown. Then, Dunnett's multiple comparison test for Tardbp HDO (+) and test ASO (−) group vs. Tardbp HDO (+) and test ASO (+) groups was performed. Test ASOs that showed significant decrease ($p<0.05$) in CE inclusion are shown in Table 13 below.

TABLE-US-00032 TABLE 13 Cerebral Cortex Spinal cord % of inclusion % of inclusion of UNC13A CE of UNC13A CE % of TDP-43 in mice treated % of TDP-43 in mice treated level in mice with the Tardbp level in mice with Tardbp that received HDO and that received HDO and the Tardbp indicated ASO the Tardbp indicated ASO HDO and PBS relative to mice HDO and PBS relative to mice relative to mice treated with relative to mice treated with that received Tardbp HDO that received Tardbp HDO PBS only and PBS PBS only and PBS Concentration (normalized by (normalized by (normalized by (normalized by ASO # (nmol) Rbfox3 level) Rbfox3 level) Rbfox3 level) Rbfox3 level) 211 3 + 3 26.6 28.1 23.1 40.4 372 3 + 3 24.1 6.7 24.9 5.7 391 3 + 3 37.5 24.6 30.5 18.5 392 3 + 3 18.0 53.1 20.1 16.8 393 3 + 3 19.3 31.1 20.9 15.7 394 3 + 3 25.3 10.5 23.1 9.0 395 3 + 3 19.8 30.7 21.3 22.6 511 3 + 3 33.0 47.3 29.4 43.3 564 3 + 3 30.1 29.7 26.4 34.5 565 3 + 3 36.9 23.2 41.4 36.5 658 3 + 3 24.3 44.3 23.3 41.4

Example 4: Activities of Antisense Oligonucleotides In Vitro (shTDP—SH-SY5Y Cells with LNP TF)

[0327] SH-SY5Y cell line (Parental SH-SY5Y Cell line from ATCC; CRL-2266, Lot: 58462094) with doxycycline (dox) inducible expression (Tet-on system) of TDP-43 specific shRNA for knock down was generated (hereinafter referred to as shTDP—SH-SY5Y). TurboRFP and microRNA30-adapted short hairpin RNA is expressed under the Tet operator and minimal CMV promoter and rtTA3 is expressed under the UbC promoter. microRNA30-adapted short hairpin RNA sequence is as below:
ttgaatgaggcttcagtactttacagaatcgttgcctgcacatcttggaaacacttgctgggattacttcttcaggttaacccaacagaaggctc
gagaaggtatattgctgttgacagtgagcgacacactacaattgatatcaaatagtgaagccacagatgtatttgatatcaattgtagtgtgttg
cctactgcctcggaattcaaggggctactttaggagcaattatcttgtttactaaaactgaataccttgctatctctttgatacatttttacaaagct
gaattaaaatggtataaattaaatcacttt (SEQ ID NO: 568). shTDP-SH-SY5Y cells were grown in 43.5% DMEM, high glucose, HEPES (Gibco, 12430054), 43.5% Ham's F-12 Nutrient Mix (Gibco, 11765047), 10% Fatal Bovine Serum (Gibco, A31606-01 or Clontech, 631105), 1×MEM NEAA Solution (Gibco, 11140050), 2 mM Sodium Pyruvate (Gibco, 11360070), and 2 μg/mL Blasticidin S HCl (Gibco, A1113903) at 37° C., 5% CO2. To evaluate UNC13A splicing abnormality and its correction by ASOs, around 60% TDP-43 knock-down was induced by treating cells with 4 μg/mL of doxycycline (Clontech, 631311) for 14 days.

Manufacturing Methods

[0328] (1) A lipid mixture (cationic lipid:DPPC:cholesterol:GM-020=60:10.6:28:1.4, in mole ratio) was dissolved in 90% EtOH/10% RNase-free water to afford an approximately 14.4 mg/mL lipid solution. The cationic lipid was 3-{[4-(dimethylamino) butanoyl]oxy}-2,2-bis {[(9Z)-tetradec-9-enoyloxy]methyl}propyl (9Z)-tetradec-9-enoate, described in WO2019131839, the contents of which, related to cationic lipids, are incorporated herein by reference. The lipid solution and 10 mM Citrate buffer, pH3 were mixed by using the apparatus NanoAssemblr (Precision Nanosystems) at room temperature with a flow rate ratio of 3 mL/min:9 mL/min to afford a dispersion containing a lipid particle. By using a Slyde-A-Lyzer (molecular weight cutoff: 20k, Thermo Fisher Scientific), the dispersion obtained was dialyzed with water at 4° C. for 1 hour and with 10 mM MES/0.9% NaCl buffer, pH5 for 2 days. Subsequently, filtration was performed with a 0.2 μm syringe filter (IWAKI CO., LTD.) to prepare a composition for nucleic acid introduction, and thereafter the composition was stored at 4° C. The lipid concentration of the composition was analyzed using HPLC and the size was measured using Zetasizer Nano ZS (Malvern Instruments). The results are shown in Table 14. [0329] (2) At culturing day 11 with doxycycline treatment, shTDP-SH-SY5Y cells were seeded at 60,000 viable cells/well in 96-well plate (Corning, 3598). 0.853 uM of the composition of (1) diluted by saline were gently mixed with 2 uM of test compound in equal amount to form a mixed composition, diluted by 1 mM MES, pH 5.5 in saline and/or DPBS (−), and incubated for 10 min at room temperature. The mixed composition was added to the culture media at the final concentration of 50 nM test compound within 30 min after mixing lipid-based nanoparticles and test compound. The cells were harvested on day 14 after doxycycline treatment by using FastLane Cell Multiplex Kit (QIAGEN, 216513) according to the manufacturer's instructions. TABLE-US-00033 TABLE 14 Size Polydispersity Cationic lipid (nm) index (PDI) concentration Nanoparticle 1 69.1 0.065 3.73 mM

[0330] RT-qPCRs were run on a QuantStudio™ 7 Flex Real-Time PCR System (Applied Biosystems) or ViiA™ 7 Real-Time PCR System (Applied Biosystems). The experiments were run in duplicate. The following PCR probe assay for UNC13A CE synthesized at Integrated DNA technologies (IDT) were used: UNC13A_CE FWD 5′-3′ CCCCGTACCATGTCCAGTACA (SEQ ID NO: 320), UNC13A_CE REV 5′-3′ CATTCACCAGCATTTATTCAACAAA (SEQ ID NO: 321), UNC13A_CE Probe 5′-3′/56-FAM/CTGCATGAG/ZEN/CTGCCT/3IABKFQ/. For housekeeping gene GAPDH expression analysis was performed by using Thermofisher® TaqMan Gene Expression Assay Hs99999905_m1. PCR protocol was as below; 50° C. for 30 min, 95° C. for 15 min, 45 cycles of 94° C. for 15 sec and 60° C. for 60 sec.

[0331] Relative expression of UNC13A CE was calculated by normalizing to GAPDH expression (deltaCt), then the average expression value of the control group (saline) without ASO treatment were used for the normalization (100%: dox (+) ASO (−)). Test ASOs correspond to ASOs 1-331 and 350-814 in Tables 3, 5, 9, and 11. Results are shown in Table 15 below.

TABLE-US-00034 TABLE 15 Relative expression ASO # of UNC13A CE 1 25 2 30 3 25 4 96 5 80 6 54 7 46 8 43 9 8 10 52 11 51 12 33 13 62 14 20 15 43 16 14 17 27 18 63 19 52 20 51 21 56 22 28 23 27 24 64 25 57 26 90 27 46 28 72 29 44 30 62 31 41 32 75 33 22 34 5 35 18 36 15 37 31 38 42 39 42 40 22 41 5 42 5 43 2 44 4 45 4 46 3 47 2 48 3 49 5 50 7 51 7 52 7 53 48 54 47 55 65 56 37 57 43 58 36 59 9 60 13 61 76 62 63 63 33 64 27 65 65 66 62 67 42 68 28 69 56 70 49 71 59 72 47 73 71 74 74 75 38 76 26 77 31 78 34 79 33 80 25 81 16 82 25 83 23 84 23 85 31 86 18 87 19 88 26 89 30 90 21 91 30 92 36 93 57 94 11 95 16 96 12 97 25 98 8 99 8 100 8 101 5 102 5 103 3 104 7 105 2 106 12 107 24 108 18 109 21 110 38 111 27 112 44 113 5 114 68 115 27 116 32 117 33 118 26 119 16 120 6 121 13 122 31 123 6 124 10 125 38 126 40 127 51 128 48 129 81 130 83 131 34 132 50 133 40 134 36 135 70 136 56 137 25 138 43 139 41 140 69 141 18 142 11 143 47 144 18 145 36 146 7 147 11 148 23 149 11 150 27 151 11 152 15 153 14 154 74 155 41 156 65 157 77 158 31 159 24 160 32 161 31 162 28 163 34 164 46 165 41 166 30 167 38 168 96 169 55 170 37 171 19 172 21 173 95 174 95 175 89 176 85 177 91 178 85 179 85 180 95 181 40 182 54 183 54 184 78 185 60 186 20 187 51 188 54 189 61 190 47 191 37 192 35 193 32 194 20 195 48 196 43 197 34 198 47 199 47 200 95 201 97 202 78 203 61 204 74 205 42 206 61 207 58 208 75 209 15 210 43 211 33 212 35 213 32 214 30 215 39 216 24 217 46 218 27 219 26 220 17 221 24 222 27 223 25 224 35 225 27 226 29 227 25 228 17 229 27 230 15 231 19 232 33 233 27 234 26 235 13 236 51 237 20 238 17 239 24 240 18 241 22 242 15 243 76 244 64 245 67 246 61 247 58 248 81 249 65 250 75 251 48 252 72 253 43 254 40 255 73 256 87 257 28 258 65 259 96 260 78 261 81 262 45 263 85 264 43 265 82 266 56 267 67 268 98 269 86 270 23 271 11 272 61 273 67 274 88 275 100 276 20 277 20 278 40 279 42 280 36 281 86 282 78 283 106 284 105 285 62 286 27 287 31 288 46 289 73 290 89 291 68 292 18 293 70 294 34 295 16 296 18 297 16 298 77 299 21 300 57 301 42 302 41

303 19 304 6 305 29 306 19 307 31 308 3 309 93 310 97 311 99 312 99 313 99 314 59 315 99 316 77 317 12 318 13 319 12 320 17 321 31 322 45 323 28 324 14 325 22 326 100 327 18 328 12 329 21 330 6 331 26 350 12 351 26 352 12 353 18 354 24 355 12 356 16 357 8 358 24 359 32 360 32 361 46 362 41 363 32 364 24 365 13 366 11 367 12 368 10 369 8 370 29 371 23 372 3 373 2 374 3 375 6 376 4 377 3 378 5 379 4 380 4 381 12 382 14 383 10 384 5 385 3 386 4 387 8 388 3 389 23 390 9 391 11 392 12 393 8 394 7 395 5 396 14 397 8 398 11 399 15 400 11 401 12 402 13 403 13 404 17 405 16 406 14 407 10 408 12 409 45 410 40 411 12 412 28 413 40 414 20 415 23 416 30 417 7 418 2 419 31 420 14 421 38 422 26 423 16 424 21 425 28 426 12 427 15 428 57 429 51 430 68 431 48 432 56 433 30 434 25 435 25 436 22 437 17 438 16 439 26 440 33 441 36 442 33 443 47 444 62 445 58 446 14 447 24 448 51 449 33 450 37 451 28 452 31 453 38 454 29 455 36 456 40 457 23 458 31 459 63 460 61 461 54 462 68 463 14 464 8 465 18 466 8 467 4 468 18 469 8 470 11 471 7 472 4 473 3 474 14 475 13 476 9 477 10 478 38 479 15 480 47 481 20 482 29 483 37 484 67 485 56 486 27 487 27 488 24 489 45 490 49 491 2 492 36 493 55 494 24 495 37 496 38 497 15 498 16 499 16 500 17 501 25 502 32 503 46 504 15 505 31 506 32 507 62 508 56 509 37 510 34 511 31 512 28 513 19 514 17 515 15 516 16 517 68 518 42 519 28 520 46 521 44 522 61 523 67 524 27 525 43 526 48 527 48 528 49 529 27 530 22 531 36 532 20 533 31 534 16 535 40 536 16 537 30 538 26 539 43 540 42 541 15 542 15 543 18 544 14 545 22 546 15 547 18 548 26 549 15 550 18 551 11 552 6 553 8 554 20 555 19 556 29 557 14 558 15 559 16 560 20 561 23 562 19 563 20 564 27 565 42 566 36 567 55 568 47 569 46 570 59 571 69 572 51 573 54 574 65 575 69 576 49 577 36 578 62 579 28 580 35 581 69 582 36 583 45 584 58 585 21 586 20 587 10 588 38 589 40 590 31 591 29 592 12 593 6 594 5 595 38 596 11 597 11 598 42 599 36 600 35 601 27 602 27 603 16 604 13 605 11 606 19 607 26 608 22 609 28 610 22 611 36 612 35 613 27 614 15 615 18 616 30 617 19 618 31 619 28 620 23 621 23 622 20 623 26 624 22 625 26 626 19 627 41 628 41 629 23 630 37 631 27 632 12 633 17 634 21 635 37 636 20 637 20 638 25 639 23 640 21 641 22 642 26 643 26 644 31 645 17 646 19 647 7 648 9 649 18 650 16 651 32 652 19 653 10 654 11 655 13 656 7 657 8 658 8 659 10 660 21 661 15 662 9 663 24 664 9 665 7 666 5 667 6 668 7 669 14 670 14 671 20 672 27 673 15 674 27 675 27 676 20 677 14 678 16 679 22 680 19 681 13 682 28 683 13 684 18 685 21 686 26 687 21 688 40 689 34 690 21 691 23 692 19 693 29 694 14 695 18 696 32 697 14 698 16 699 18 700 15 701 13 702 18 703 23 704 18 705 20 706 23 707 23 708 25 709 22 710 20 711 18 712 24 713 20 714 23 715 26 716 31 717 34 718 38 719 29 720 40 721 31 722 40 723 33 724 32 725 38 726 16 727 16 728 24 729 26 730 16 731 24 732 23 733 34 734 16 735 32 736 13 737 14 738 13 739 14 740 13 741 12 742 14 743 13 744 16 745 13 746 18 747 17 748 22 749 19 750 14 751 16 752 18 753 19 754 25 755 22 756 28 757 24 758 34 759 25 760 23 761 27 762 17 763 27 764 21 765 37 766 59 767 48 768 49 769 62 770 65 771 69 772 23 773 35 774 42 775 27 776 38 777 22 778 21 779 30 780 32 781 24 782 34 783 45 784 30 785 24 786 27 787 36 788 64 789 68 790 70 791 66 792 24 793 22 794 36 795 38 796 53 797 60 798 54 799 58 800 50 801 47 802 33 803 45 804 58 805 43 806 66 807 10 808 8 809 36 810 28 811 57 812 69 813 67 814 69

Additional Embodiments

[0332] 1. An antisense oligonucleotide comprising a first antisense sequence targeting a first target region and a second antisense sequence targeting a second target region, wherein the first target region and/or the second target region each comprises an UNC13A sequence, wherein the antisense oligonucleotide modulates splicing of an UNC13A cryptic exon. [0333] 2. The antisense oligonucleotide of embodiment 1, wherein the UNC13A cryptic exon is in intron 20 of an UNC13A pre-mRNA. [0334] 3. The antisense oligonucleotide of embodiment 1 or embodiment 2, wherein the first target region and the second target region are in intron 20 of an UNC13A pre-mRNA. [0335] 4. The antisense oligonucleotide of any one of embodiments 1-3, wherein the first target region is adjacent to the second target region in an UNC13A pre-mRNA. [0336] 5. The antisense oligonucleotide of any one of embodiments 1-4, wherein the first target region is downstream of the second target region in an UNC13A pre-mRNA. [0337] 6. The antisense oligonucleotide of any one of embodiments 1-5, wherein the first target region is more than 1 nucleoside downstream of the second target region. [0338] 7. The antisense oligonucleotide of any one of embodiments 1-6, wherein the first target region and the second target region are both in the cryptic exon. [0339] 8. The antisense oligonucleotide of any one of embodiments 1-6, wherein the first target region is downstream of the cryptic exon and the second target region is in the cryptic exon. [0340] 9. The antisense oligonucleotide of any one of embodiments 1-6, wherein the first target region and the second target region are both downstream of the cryptic exon. [0341] 10. The antisense oligonucleotide of any one of embodiments 1-8, where in the first target region and/or the second target region comprises a nucleotide sequence of any one of SEQ ID NOs: 313-316. [0342] 11. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 315. [0343] 12. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 314. [0344] 13. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 315, and the second target region comprises the nucleotide sequence of SEQ ID NO: 313. [0345] 14. The antisense oligonucleotide of any one of embodiments 1-10, wherein the first target region comprises the nucleotide sequence of SEQ ID NO: 313, and the second target region comprises the nucleotide sequence of SEQ ID NO: 313. [0346] 15. The antisense oligonucleotide of any one of embodiments 1-14, wherein the first antisense sequence and/or the second antisense sequence is 6-15 nucleosides in length, optionally wherein the first antisense sequence and/or the second antisense sequence is 8-15 nucleosides in length. [0347] 16. The antisense oligonucleotide of any one of embodiments 1-15, wherein the first antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOS: 191-245, and/or wherein the second antisense sequence comprises at least 8 contiguous nucleobases of any one of SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. [0348] 17. The antisense oligonucleotide of any one of embodiments 1-16, wherein the first antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: 191-245, and/or wherein the second antisense sequence comprises the nucleobase sequence of any one of SEQ ID NOs: SEQ ID NOs: 194, 199, 212, 226, 227, and 246-306. [0349] 18. The antisense oligonucleotide of any one of embodiments 1-17, wherein the first antisense sequence and the second antisense sequence are immediately adjacent to each other. [0350] 19. The antisense oligonucleotide of any one of embodiments 1-18, wherein one or more nucleobases are between the first antisense sequence and the second antisense sequence. [0351] 20. The antisense oligonucleotide of any one of embodiments 1-17, wherein one or more nucleobases at the 3′ end of the first antisense sequence is complementary to one or more nucleobases at the 3′ end of the second target region, and/or one or more nucleobases at the 5′ end of the second antisense sequence is complementary to one or more nucleobases at the 5′ end of the first target region. [0352] 21. The antisense oligonucleotide of any one of embodiments 1-20, comprising the nucleobase sequence of any one of SEQ ID NOs: 1-190. [0353] 22. The antisense oligonucleotide of any one of embodiments 1-21, comprising one or more modified nucleosides. [0354] 23. The antisense oligonucleotide of embodiment 22, wherein the one or more modified nucleosides comprise a non-bicyclic 2′ modified nucleoside, a 2′-4′ bridged nucleoside, or combinations thereof. [0355] 24. The antisense oligonucleotide of embodiment 23, wherein the non-bicyclic 2′-modified nucleoside is a 2′-O-methoxyethyl (2′-MOE) modified nucleoside, a 2′-O-Methyl (2′-O-Me) nucleoside, or a 2′-fluoro (2′-F) modified nucleoside, and/or the 2′-4′ bridged nucleoside is a locked nucleic acid (LNA, 2′-4′ methylene bridge), an ethylene-bridged nucleic acid (ENA, 2′-4′ ethylene bridge), or a constrained ethyl nucleic acid (cEt, 2′-4′ ethylene bridge). [0356] 25. The antisense oligonucleotide of any one of embodiments 1-24, wherein the antisense oligonucleotide comprises one or more 2′-MOE modified nucleosides and one or more LNAs. [0357] 26. The antisense oligonucleotide of any one of embodiments 1-24, wherein each nucleoside of the antisense oligonucleotide is a 2′-MOE modified nucleoside. [0358] 27. The antisense oligonucleotide of any one of embodiments 1-26, comprising one or more modified internucleoside linkages. [0359] 28. The antisense oligonucleotide of any one of embodiments 1-27, wherein each internucleoside linkage of the oligonucleotide is a modified internucleoside linkage. [0360] 29. The antisense oligonucleotide of embodiment 27 or embodiment 28, wherein the modified internucleoside linkage is a phosphorothioate linkage. [0361] 30. The antisense oligonucleotide of any one of embodiments 1-29, wherein each internucleoside linkage of the oligonucleotide is a phosphorothioate linkage. [0362] 31. The antisense oligonucleotide of any one of embodiments 1-29, wherein the antisense oligonucleotide comprises a mix of phosphodiester linkages and phosphorothioate linkages. [0363] 32. The antisense oligonucleotide of any one of embodiments 1-22, wherein the antisense oligonucleotide is a phosphorodiamidate oligomer (PMO). [0364] 33. The antisense oligonucleotide of any one of embodiments 1-32, wherein each cytosine of the oligonucleotide is not a 5-methyl-cytosine. [0365] 34. The antisense oligonucleotide of any one of embodiments 1-32, wherein each cytosine of the oligonucleotide is a 5-methyl-cytosine. [0366] 35. The antisense oligonucleotide of any one of embodiments 1-32, wherein one or more of the cytosines of the oligonucleotide is a 5-methyl-cytosine. [0367] 36. The antisense oligonucleotide of any one of embodiments 1-35, comprising a structure as provided in Table 5. [0368] 37. The antisense oligonucleotide of any one of embodiments 1-36, wherein the antisense oligonucleotide inhibits inclusion of the UNC13A cryptic exon into an UNC13A mature mRNA. [0369] 38. The antisense oligonucleotide of any one of embodiments 1-37, wherein the antisense oligonucleotide is in the form of a pharmaceutically acceptable salt. [0370] 39. A composition comprising the antisense oligonucleotide of any one of embodiments 1-38. [0371] 40. The composition of embodiment 39, further comprising a pharmaceutically acceptable carrier. [0372] 41. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell, comprising contacting the cell with the antisense oligonucleotide of any one of embodiments 1-38, or the composition of embodiment 39 or embodiment 40. [0373] 42. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, comprising administering to the subject the antisense oligonucleotide of any one of embodiments 1-38, or the composition of embodiment 39 or embodiment 40. [0374] 43. A method of treating a disease associated with abnormal UNC13A expression in a subject, comprising administering to the subject the antisense oligonucleotide of any one of claims **1-38**, or the composition of embodiment 39 or embodiment 40. [0375] 44. A method of treating a disease associated with TDP-43 dysfunction in a subject, comprising administering to the subject the antisense oligonucleotide of any one of claims **1-38**, or the composition of embodiment 39 or embodiment 40. [0376] 45. The method of embodiment 43 or embodiment 44, wherein the disease is a neurodegenerative disease. [0377] 46. The method of any one of embodiments 43-45, wherein the neurodegenerative disease is amyotrophic lateral sclerosis (ALS). [0378] 47. The method of any one of embodiments 43-45, wherein the neurodegenerative disease is frontotemporal dementia (FTD). [0379] 48. The method of any one of embodiments 42-47, wherein the subject is human. [0380] 49. The method of any one of embodiments 42-48, wherein the subject has decreased expression of TDP-43 gene or protein. [0381] 50. The method of any one of embodiments 42-49, wherein the subject does not possess an SOD-1 gene mutation. [0382] 51. The method of any one of embodiments 42-50, wherein the subject possesses an UNC13A gene mutation in intron 20. [0383] 52. The method of embodiment 51, wherein the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof. [0384] 53. A pharmaceutical composition for use in a method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA and/or restoring UNC13A expression in a cell of a subject, comprising the antisense oligonucleotide of any one of embodiments 1-38. [0385] 54. A pharmaceutical composition for use in a method of treating a disease associated with abnormal UNC13A expression in a subject comprising the antisense oligonucleotide of any one of embodiments 1-38. [0386] 55. A pharmaceutical composition for use in a method of treating a disease associated with TDP-43 dysfunction in a subject comprising the antisense oligonucleotide of any one of embodiments 1-38. [0387] 56. The pharmaceutical composition of any one of embodiments 53-55, further comprising a pharmaceutically acceptable carrier. [0388] 57. The pharmaceutical composition of any one of embodiments 53-56, wherein the subject is human. [0389] 58. The pharmaceutical composition of any one of embodiments 55-57, wherein the subject has decreased expression of TDP-43 gene or protein. [0390] 59. The pharmaceutical composition of any one of embodiments 55-58, wherein the subject does not possess an SOD-1 gene mutation. [0391] 60. The pharmaceutical composition of any one of embodiments 55-59, wherein the subject possesses an UNC13A gene mutation in intron 20. [0392] 61. The pharmaceutical composition of embodiment 60, wherein the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof. [0393] 62. An antisense oligonucleotide of any one of embodiments 1-38 for use as a medicament. [0394] 63. The antisense oligonucleotide of embodiment 62, wherein the medicament is administered to a subject. [0395] 64. An antisense oligonucleotide of any one of embodiments 1-38 for use in a method of treating a disease associated with abnormal UNC13A expression in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. [0396] 65. An antisense oligonucleotide of any one of embodiments 1-38 for use in a method of treating a disease associated with TDP-43 dysfunction in a subject, the method comprising administering to the subject an antisense oligonucleotide provided herein. [0397] 66. The antisense oligonucleotide of any one of embodiments 63-65, wherein the subject is human. [0398] 67. The antisense oligonucleotide of any one of embodiments 63-66, wherein the subject has decreased expression of TDP-43 gene or protein. [0399] 68. The antisense oligonucleotide of any one of embodiments 63-67, wherein the subject does not possess an SOD-1 gene mutation. [0400] 69. The antisense oligonucleotide of any one of embodiments 63-68, wherein the subject possesses an UNC13A gene mutation in intron 20. [0401] 70. The antisense oligonucleotide of embodiment 69, wherein the UNC13A gene mutation comprises rs12608932, rs12973192, rs56041637, rs116169349, rs62121687, or any combination thereof.

EQUIVALENTS

[0402] The disclosure may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting of the disclosure. Scope of the disclosure is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are therefore intended to be embraced herein.

## Claims

**1-75**. (canceled)

**76**. An antisense oligonucleotide comprising the following structure: [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] (SEQ ID NO: 571), wherein iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage, or a pharmaceutically acceptable salt thereof.

**77**. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell of a human subject, comprising administering to the subject the antisense oligonucleotide of claim 76.

**78**. The method of claim 77, wherein the method restores UNC13A expression in the cell.

**79**. A method of treating a neurodegenerative disease in a human subject, comprising administering to the subject the antisense oligonucleotide of claim 76.

**80**. The method of claim 79, wherein the neurodegenerative disease is amyotrophic lateral sclerosis

(ALS).
**81**. The method of claim 79, wherein the neurodegenerative disease is frontotemporal dementia (FTD).
**82**.-**87**. (canceled)
**88**. An antisense oligonucleotide consisting of the following structure: [i.sup.5Cs][iGs][iAs][iAs][iTs][i.sup.5Cs][iTs][iAs][i.sup.5Cs][i.sup.5Cs][i.sup.5Cs][iAs][i.sup.5Cs][iAs][iTs][i.sup.5Cs][iTs][iGs][iTs][iTs][i.sup.5Cs][iAs][iAs][iTs][i.sup.5Cs][iA] (SEQ ID NO: 571), wherein iA, iG, i.sup.5C, and iT are 2′-O-methoxyethyl (2′-MOE) modified adenosine, guanosine, 5′-methyl cytidine, and thymidine, respectively; and s is a phosphorothioate internucleoside linkage, or a pharmaceutically acceptable salt thereof.
**89**. A method of inhibiting inclusion of an UNC13A cryptic exon into an UNC13A mature mRNA in a cell of a human subject, comprising administering to the subject the antisense oligonucleotide of claim 88.
**90**. The method of claim 89, wherein the method restores UNC13A expression in the cell.
**91**. A method of treating a neurodegenerative disease in a human subject, comprising administering to the subject the antisense oligonucleotide of claim 88.
**92**. The method of claim 91, wherein the neurodegenerative disease is amyotrophic lateral sclerosis (ALS).
**93**. The method of claim 91, wherein the neurodegenerative disease is frontotemporal dementia (FTD).

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